

RTL Design Sherpa

RAPIDS Micro-Architecture Specification

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1 Rapids Mas Index

Generated: 2026-10-24

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2 Document Information

This document describes RAPIDS, the byte-granular Rapid AXI Programmable In-band Descriptor System. RAPIDS provides network-to-memory and memory-to-network data transfer using descriptor-based DMA with 8 or 16 channels. A descriptor length is a byte count, source and destination addresses are byte addresses, and the sink write master drives byte enables. The earlier “Beats” architecture, which counted everything in beats, was the stepping stone to this design and is documented separately in the RAPIDS Beats MAS.

2.1 Revision History

Table 0.1: Revision History

Version	Date	Author	Description
0.6	2026-10-17	RTL Design Sherpa	New Section 2.10 Shared Engines (`descriptor_engine_shared`, `ctrlrd_engine_shared`, `ctrlwr_engine_shared`: parameters, ports, per-channel state, operation, measured cost). `SHARED_ENGINES` and `TILE_HDR` added to the core and top parameter tables; 8- and 16-channel top variants; scheduler group marked per-channel-build only; scheduler group array overview corrected (one engine of each kind per array, not per channel)
0.5	2026-10-17	RTL Design Sherpa	New Section 3.14 Tile Packet Header (`TILE_HDR`: header build and decode, cell-issued descriptors, status registers). Scheduler group array: shared descriptor / control-read / control-write engines per half (`SHARED_ENGINES`, default 1), with block diagram
0.4	2026-10-16	RTL Design Sherpa	Shared Sections 3.13 and 4.4 point at the routing and payload-type proposal in the tile's TILE_SPEC Section

Version	Date	Author	Description
			11
0.3	2026-10-16	RTL Design Sherpa	Network planes (rapids TASK-024/025): `rapids_top` / `rapids_cxs_top` generated from `rapids_top_base`; links the shared Network Planes chapter and CXS Interface Specification; block diagrams updated for the planes; the CXS ID mapping is marked provisional.
0.2	2026-09-30	RTL Design Sherpa	Channel reset recovers every error class: `cfg_channel_reset` routed into the AXI read and write engines, both data paths and the SRAM wrappers; scheduler clears sticky flags on the reset.
0.1	2026-09-30	RTL Design Sherpa	First byte-granular MAS. Rewrites the chapters that changed from RAPIDS Beats and links the rest.
0.7	2026-10-20	RTL Design Sherpa	Block diagrams redrawn to the repo element key (v3): class fills, double-bordered registered blocks, green return paths and a generated key under each figure. No technical content changed.
0.8	2026-10-20	RTL Design Sherpa	Section (h2) headings now open a new page, the pipeline default. No

Version	Date	Author	Description
			technical content changed.
0.81	2026-10-21	RTL Design Sherpa	New Chapter 5, rapids-g: gatling-sequenced schedulers (`scheduler_g`, `rapids_seq_g`, firmware, the generated tree, verification, measured cost). Scheduler: split into the FSM and `scheduler_core`; MonBus events on state entry (rapids BUG-016); run-base generators flushed by the channel reset (rapids BUG-017).
0.82	2026-10-22	RTL Design Sherpa	rapids-g at one hart per channel (NUM_HARTS = NUM_CHANNELS): Section 5.3 describes the one-slot tight loop, with flow charts of the poll loop and `service` (Figures 5.3.1 and 5.3.2), and no longer says the firmware computes beat totals (the hardware does); Section 5.5 gains the Genesys 2 gcompare table for rapids-8ch and rapids-g-8ch (45 points, interleaved sink), sim/board match and the chunk warm-up, the firmware-cost history, and the 8-hart bench and out-of-context results.
0.83	2026-10-23	RTL Design	rapids-g firmware as an

Version	Date	Author	Description
		Sherpa	executive over a policy table (rapids TASK-037): Section 5.3 rewritten (the poll and `step`, `ptab`, the policy table, the never-reached states and `fw_fault`), with new flow charts (Figures 5.3.1 and 5.3.2); the next-address selector and Z-order (TASK-035) in Sections 5.2 and 5.3; 16 channels (TASK-036), with 8 harts x 2 slots the measured default. Section 5.5: the mutation table re-run on the new image, the executive and 16-channel bench tables, out-of-context results at 8 and 16 channels, and the board run of the new image.
0.84	2026-10-23	RTL Design Sherpa	rapids-g firmware loadable live (rapids TASK-038): `FW_WRITABLE`, the `FW_*` registers, the load and its self-test, ROM once final (Section 5.3, Table 5.3.8). The register mailbox section rewritten as the serialized register path: one agent owns the path from command to response, with both arbitration levels and their signals (Tables 5.3.3-5.3.5). The two-slot poll's chain

Version	Date	Author	Description
			hook and why a chain is serial. Section 5.5: rapids-16ch and rapids-g-16ch on the Genesys 2 (fit, sim == board, MB/s at 16 channels) and the size sweep at 8 and 16 channels (Tables 5.5.8-5.5.10). Table 5.5.6 re-measured on the committed image. Table 5.5.10 extended to 128 KB per descriptor, with the source rows; new Table 5.5.11, the bytes per descriptor at which rapids-g is within 10%, 5% and 1% of rapids.

2.2 References

2.2.1 Related Documents

Table 0.2: Related Documents and Specifications

Source	Title	Version
RTL Design Sherpa	RAPIDS Beats MAS (shared chapters)	0.7
RTL Design Sherpa	RAPIDS HAS	0.93
RTL Design Sherpa	RAPIDS Product Requirements Document	1.0
ARM	AMBA AXI and ACE Protocol Specification	IHI0022H
ARM	AMBA AXI-Stream Protocol Specification	IHI0051A
ARM	AMBA CXS Protocol Specification	IHI0079D

2.3 Terminology

Beat

One data transfer of an AXI burst or an AXI-Stream transfer. One beat is `BYTE_LANES` bytes, where `BYTE_LANES = DATA_WIDTH/8`. The board design uses 256 bits, so a beat is 32 bytes.

Byte offset

The low `OFF_W` bits of a byte address, where `OFF_W = $clog2(BYTE_LANES)`. It is the lane at which a transfer begins inside its first memory beat.

Memory beat

A beat as it appears on the AXI side, aligned to `BYTE_LANES`. A packet that starts at a non-zero offset occupies one more memory beat than a stream of the same length would.

Stream beat

A beat as it appears on AXI-Stream, packed from lane 0 with no offset.

Packet record

The pair `{bytes, offset}` that the scheduler issues once per DATA descriptor, so the AXIS shifter knows how the memory-side and stream-side layouts relate.

Spill

The bytes of a stream beat that did not fit in the current memory beat during ingress shifting. They are held for the next memory beat.

Channel

One of 8 independent DMA channels. Each has its own descriptor chain, scheduler and SRAM allocation.

Descriptor

A 256-bit structure holding the source address, destination address, length in bytes, next pointer, and control flags.

TYPE=EXT

The descriptor type that describes a row/column transfer. It stays beat-aligned by design: addresses are aligned and lengths are multiples of the beat size.

DMA

Direct Memory Access. Data movement without processor involvement.

MonBus

The 64-bit monitor bus that carries event packets out of the design.

2.4 Document Conventions

- Shared chapters are linked from the RAPIDS Beats MAS and marked “(shared with RAPIDS Beats)” in the index. Module names in those chapters carry a `_beats` suffix that this design drops.
- Requirement priorities follow the RAPIDS PRD.
- No emojis appear in this document.

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3 Architecture Overview

Module: `rapids_core.sv` **Location:** `projects/components/dma-ip/rapids/rtl/macro/`
Status: Implemented **Last Updated:** 2026-09-30

3.1 Overview

RAPIDS moves data between AXI-Stream and system memory in units of bytes. A descriptor's length is a byte count in the existing 32-bit field, and its source and destination are byte addresses. A linear descriptor may start at any byte on either side. The design keeps the streaming, no-FSM data path of RAPIDS Beats and adds three things around it: byte-to-beat math in the scheduler, byte enables through the sink SRAM to the write master, and a shifter at each AXI-Stream edge that converts between packed stream beats and memory-aligned beats.

3.1.1 Key Features

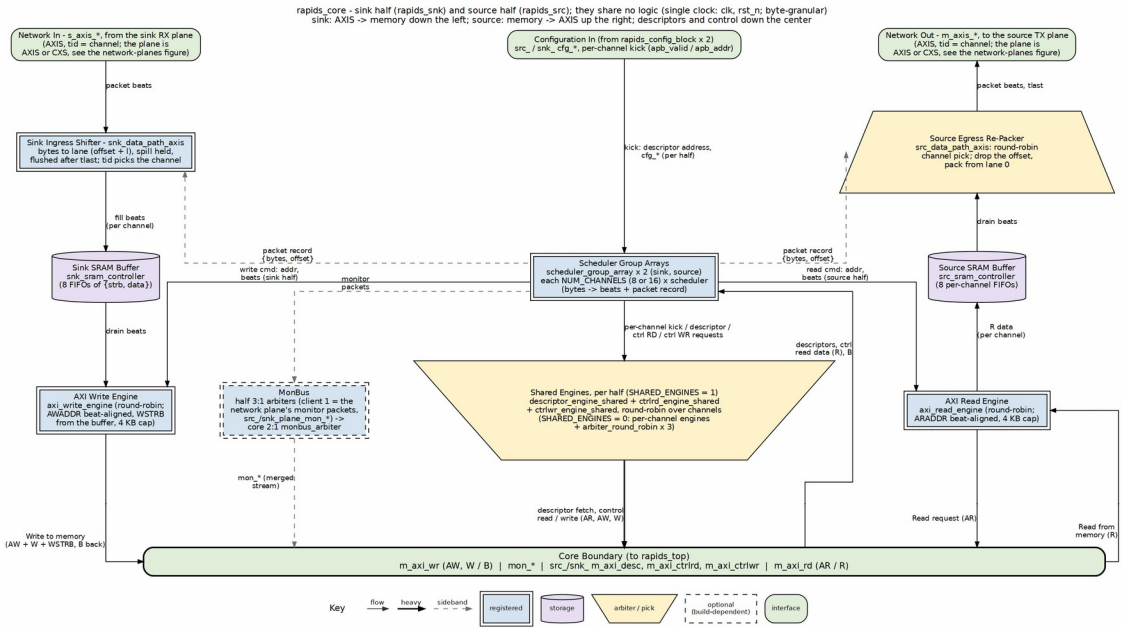
- **8 or 16 Independent Channels:** Each channel has its own descriptor chain and SRAM allocation. The 16-channel tops use the 16-channel register map.
- **Shared Engines:** By default each half has one descriptor engine, one control-read engine and one control-write engine for all its channels (`SHARED_ENGINES = 1`).
- **Separate Data Paths:** Sink (network to memory) and source (memory to network).
- **Byte Granularity:** Length in bytes, byte addresses, any starting offset for linear descriptors.
- **Byte Enables:** The sink SRAM entry is `{strb, data}`; the write engine drives `WSTRB` from it.
- **Packet Records:** The scheduler issues one `{bytes, offset}` record per DATA descriptor to the AXIS shifter.
- **One AXIS Packet per Descriptor:** The source path emits exactly one packet, with `TLAST` on the final byte.
- **Safe Bursts:** AXI bursts never cross a 4 KB boundary; `AxADDR` is beat-aligned.
- **MonBus Integration:** Same monitor packets as RAPIDS Beats.

3.1.2 What Is Shared with RAPIDS Beats

The SRAM controllers, the register block and config block, and the MonBus specification are unchanged. Those chapters are linked from the RAPIDS Beats MAS. The per-channel descriptor, control-read and control-write engines are also unchanged and are built when `SHARED_ENGINES = 0`. The default build replaces them with the Shared Engines, which keep their per-channel behaviour.

3.2 Block Diagram

3.2.1 Figure 1.1.1: RAPIDS Architecture Block Diagram



RAPIDS core block diagram

Source: 04_rapids_core.dot

Source: `rapids_core.sv`, `rapids_snk.sv`, `rapids_src.sv`

Table 1.1.1: Block Roles

Block	Role
scheduler	Loads a descriptor, computes beats from offset and length, issues the packet record, tracks completion
scheduler_group_array	Eight scheduler groups sharing one descriptor AXI master; per-channel packet-record ports
snk_data_path_axis	Converts packed AXIS beats into memory-aligned beats with byte enables; checks the packet length
snk_data_path	SRAM of `{strb, data}` entries plus the AXI write engine
src_data_path	SRAM of data entries plus the AXI

Block	Role
	read engine; unchanged from Beats
src_data_path_axis	Converts memory-aligned beats into packed stream beats with a partial last beat

3.3 Data Flow Overview

3.3.1 Sink Path (Network to Memory)

3.3.2 Figure 1.1.2: Sink Path Data Flow

```
1. Scheduler loads a descriptor, computes beats_total, pulses
sched_wr_pkt_valid
        |   (record {bytes, offset} enters the per-channel record
queue)
        v
2. s_axis packet arrives; tready is held low until the channel's
record is known
        |
        v
3. Ingress shifter: stream beat << (offset*8), OR the spill of the
previous beat
        |   low DATA_WIDTH bits = this memory beat, high bits become
the new spill
        v
4. Fill logic writes {strb, data} into the sink SRAM
(snk_sram_controller)
        |
        v
5. axi_write_engine: aligned AWADDR, capped burst, WSTRB from the SRAM
entry
        |
        v
6. System memory; B response commits the beats and completes the
descriptor
```

3.3.3 Source Path (Memory to Network)

3.3.4 Figure 1.1.3: Source Path Data Flow

```
1. Scheduler loads a descriptor, computes beats_total, pulses
sched_rd_pkt_valid
        |
        v
2. axi_read_engine: aligned ARADDR, capped burst, beats into the
source SRAM
        |
        v
3. src_data_path_axis reserves drain blocks (round-robin,
cfg_drain_size beats)
        |
        v
4. Egress shifter: memory beat >> (offset*8), OR the next beat's low
lanes
```

```
        |   the first pop only primes the hold; the last beat is
partial
        v
5. m_axis: TDATA packed from lane 0, TSTRB contiguous, TLAST on the
final byte
```

3.4 Byte-Granular Design Points

Table 1.1.2: Design Points

Point	Rule
Length unit	Bytes, in the existing 32-bit descriptor field
Beats moved	`(offset + length + BYTE_LANES - 1) >> OFF_W`, zero when length is zero
First burst	May begin mid-beat; the address it presents is beat-aligned and the offset is carried by the record
Later bursts	Beat-aligned
4 KB rule	A burst is truncated at the next 4 KB boundary
Sink lanes not written	`WSTRB` bit is 0
Source last beat	Partial; `TSTRB` marks the valid low lanes, `TLAST` set
TYPE=EXT	Beat-aligned by design (permanent): aligned addresses, beat-multiple lengths

The 4 KB rule and the byte math are specified in Scheduler and AXI Read Engine.

3.5 Chapter Map

Topic	Chapter
Scheduler byte math and record issue	Scheduler
Record queue interface	Packet Record Interface
Ingress shifter	Sink Data Path AXIS
Egress shifter	Source Data Path AXIS
Everything else	Linked from the RAPIDS Beats MAS via the index

4 Top-Level Port List

Module: `rapids_core_beats.sv` **Location:** `projects/components/dma-ip/rapids/rtl/macro_beats/` **Status:** Implemented **Last Updated:** 2026-10-16

4.1 Parameters

Every parameter of `rapids_core_beats`, as declared. The short aliases at the end are what the port widths below are written in (`NC`, `AW`, `DW`, `IW`, `SCW`, `CIW`, `SW`).

Table 1.2.1: Parameters (from `rapids_core_beats.sv`)

Parameter	Type	Default	Notes
`NUM_CHANNELS`	int	`8`	-
`CHAN_WIDTH`	int	`$clog2(NUM_CHANNELS)`	-
`ADDR_WIDTH`	int	`64`	-
`DATA_WIDTH`	int	`512`	-
`AXI_ID_WIDTH`	int	`8`	-
`SRAM_DEPTH`	int	`512`	-
`SEG_COUNT_WIDTH`	int	`$clog2(SRAM_DEPTH) + 1`	-
`PIPELINE`	int	`1`	write-engine pipelining; 1 is the perf design point (see rapids_beats_top)
`AR_MAX_OUTSTANDING`	int	`8`	-
`AW_MAX_OUTSTANDING`	int	`8`	-
`W_PHASE_FIFO_DEPTH`	int	`64`	-
`B_PHASE_FIFO_DEPTH`	int	`16`	-
`AXIS_ID_WIDTH`	int	`8`	-
`AXIS_DEST_WIDTH`	int	`4`	-
`AXIS_USER_WIDTH`	int	`1`	-
`DESC_MON_BASE_AGENT_ID`	int	`16`	0x10 - Descriptor Engines (16-23)
`SCHED_MON_BASE_AGENT_ID`	int	`48`	0x30 - Schedulers (48-55)
`DESC_AXI_MON_AGE`	int	`8`	0x08 - Descriptor

Parameter	Type	Default	Notes
NT_ID			AXI Master Monitor
MON_UNIT_ID	int	1	0x1
MON_MAX_TRANSACTIONS	int	16	-
USE_AXI_MONITORS	int	1	-
USE_ROW_COL_MAJOR_ADDRESSING	int	1	-
GEN_MON	bit	1'b1	-
NC	int	NUM_CHANNELS	-
AW	int	ADDR_WIDTH	-
DW	int	DATA_WIDTH	-
IW	int	AXI_ID_WIDTH	-
SD	int	SRAM_DEPTH	-
SCW	int	SEG_COUNT_WIDTH	-
CIW	int	(NC > 1) ? $clog2(NC) : 1	-
SW	int	DW / 8	-

4.2 Port List (324 ports)

The module is two independent halves. Ports both halves carry are prefixed `src_` / `snk_`; the direction-unique ports (the source's AXI read master and AXIS egress, the sink's AXI write master and AXIS ingress) carry no prefix. Sections follow the declaration order in the RTL, so a reader can walk this page and the module side by side. Widths use the parameter aliases.

4.2.1 Clock and Reset (shared - single clock/reset for the whole wrapper)

Table 1.2.2: Clock and Reset (shared - single clock/reset for the whole wrapper)

Signal	Direction	Width	Description
`clk`	input	1	-
`rst_n`	input	1	-

4.3 SOURCE HALF (u_src) - shared-infrastructure ports (src_ prefixed)

4.3.1 APB Programming Interface

Table 1.2.3: APB Programming Interface

Signal	Direction	Width	Description
src_apb_valid	input	NC	-
src_apb_ready	output	NC	-
src_apb_addr	input	NC x AW	-

4.3.2 Per-channel configuration

Table 1.2.4: Per-channel configuration

Signal	Direction	Width	Description
src_cfg_channel_enable	input	NC	-
src_cfg_channel_reset	input	NC	-

4.3.3 Scheduler Configuration (global)

Table 1.2.5: Scheduler Configuration (global)

Signal	Direction	Width	Description
src_cfg_sched_enable	input	1	-
src_cfg_sched_timeout_cycles	input	32	-
src_cfg_sched_timeout_limit	input	8	-
src_cfg_sched_timeout_enable	input	1	-
src_cfg_sched_err_enable	input	1	-
src_cfg_sched_compl_enable	input	1	-

Signal	Direction	Width	Description
src_cfg_sched_perf_enable	input	1	-

4.3.4 Descriptor Engine Configuration (global)

Table 1.2.6: Descriptor Engine Configuration (global)

Signal	Direction	Width	Description
src_cfg_descEng_enable	input	1	-
src_cfg_descEng_prefetch	input	1	-
src_cfg_descEng_fifo_thresh	input	4	-
src_cfg_descEng_addr0_base	input	AW	-
src_cfg_descEng_addr0_limit	input	AW	-
src_cfg_descEng_addr1_base	input	AW	-
src_cfg_descEng_addr1_limit	input	AW	-

4.3.5 Control Engine Configuration (Phase 2, global)

Table 1.2.7: Control Engine Configuration (Phase 2, global)

Signal	Direction	Width	Description
src_cfg_ctrlrd_max_try	input	9	-
src_tick_1us	input	1	-

4.3.6 Descriptor AXI Monitor Configuration

Table 1.2.8: Descriptor AXI Monitor Configuration

Signal	Direction	Width	Description
src_cfg_desc_mon_enable	input	1	-
src_cfg_desc_mon_err_enable	input	1	-

Signal	Direction	Width	Description
src_cfg_desc_mon_perf_enable	input	1	-
src_cfg_desc_mon_perf_run	input	1	-
src_cfg_desc_mon_timeout_enable	input	1	-
src_cfg_desc_mon_timeout_cycles	input	32	-
src_cfg_desc_mon_latency_thresh	input	32	-
src_cfg_desc_mon_pkt_mask	input	16	-
src_cfg_desc_mon_err_select	input	4	-
src_cfg_desc_mon_err_mask	input	8	-
src_cfg_desc_mon_timeout_mask	input	8	-
src_cfg_desc_mon_compl_mask	input	8	-
src_cfg_desc_mon_thresh_mask	input	8	-
src_cfg_desc_mon_perf_mask	input	8	-
src_cfg_desc_mon_addr_mask	input	8	-
src_cfg_desc_mon_debug_mask	input	8	-

4.3.7 Network-plane monitor input (rapids TASK-024)

The network endpoint and its monitor-lite live in the plane (`rtl/plane/`) in the top. Its monbus packets enter here as client 1 of this half's monbus arbiter; the plane's monitor configuration and status come from this half's register block in the top.

Table 1.2.9: Network-plane monitor input (rapids TASK-024)

Signal	Direction	Width	Description
src_plane_mon_valid	input	1	-
src_plane_mon_ready	output	1	-
src_plane_mon_packet	input	128	monitor_packet_t
src_plane_mon_timestamp	input	64	monbus_timestamp_t

4.3.8 Status

Table 1.2.10: Status

Signal	Direction	Width	Description
src_system_idle	output	1	-
src_descriptor_engine_idle	output	NC	-
src_scheduler_idle	output	NC	-
src_scheduler_state	output	NC x 7	-
src_sched_error	output	NC	-

4.3.9 Descriptor AXI Monitor Status

Table 1.2.11: Descriptor AXI Monitor Status

Signal	Direction	Width	Description
src_cfg_sts_desc_mon_busy	output	1	-
src_cfg_sts_desc_mon_active_txns	output	8	-
src_cfg_sts_desc_mon_error_count	output	16	-
src_cfg_sts_desc_mon_txn_count	output	32	-
src_cfg_sts_d	output	1	-

Signal	Direction	Width	Description
esc_mon_conflict_error			

4.3.10 Descriptor AXI Monitor perf window (feeds SRC_.MON.DAXMON_PERF_*).

Table 1.2.12: Descriptor AXI Monitor perf window (feeds SRC_.MON.DAXMON_PERF_*).

Signal	Direction	Width	Description
src_sts_desc_mon_win_active	output	1	-
src_sts_desc_mon_win_cycles	output	32	-
src_sts_desc_mon_prod_cycles	output	32	-
src_sts_desc_mon_bp_cycles	output	32	-
src_sts_desc_mon_starv_cycles	output	32	-
src_sts_desc_mon_idle_cycles	output	32	-
src_sts_desc_mon_beat_count	output	32	-
src_sts_desc_mon_byte_count	output	64	-
src_sts_desc_mon_burst_count	output	32	-

4.3.11 Descriptor Fetch AXI Master

Table 1.2.13: Descriptor Fetch AXI Master

Signal	Direction	Width	Description
src_m_axi_desc_arvalid	output	1	-

Signal	Direction	Width	Description
src_m_axi_desc_arready	input	1	-
src_m_axi_desc_araddr	output	AW	-
src_m_axi_desc_arlen	output	8	-
src_m_axi_desc_arsize	output	3	-
src_m_axi_desc_arburst	output	2	-
src_m_axi_desc_arid	output	IW	-
src_m_axi_desc_arlock	output	1	-
src_m_axi_desc_arcache	output	4	-
src_m_axi_desc_arprot	output	3	-
src_m_axi_desc_arqos	output	4	-
src_m_axi_desc_arregion	output	4	-
src_m_axi_desc_rvalid	input	1	-
src_m_axi_desc_rready	output	1	-
src_m_axi_desc_rdata	input	256	-
src_m_axi_desc_rresp	input	2	-
src_m_axi_desc_rlast	input	1	-
src_m_axi_desc_rid	input	IW	-

4.3.12 Control Read AXI Master (32-bit) [Phase 2]

Table 1.2.14: Control Read AXI Master (32-bit) [Phase 2]

Signal	Direction	Width	Description
src_m_axi_ctrlrd_arvalid	output	1	-

Signal	Direction	Width	Description
src_m_axi_ctrlrd_arready	input	1	-
src_m_axi_ctrlrd_araddr	output	AW	-
src_m_axi_ctrlrd_arlen	output	8	-
src_m_axi_ctrlrd_arsize	output	3	-
src_m_axi_ctrlrd_arburst	output	2	-
src_m_axi_ctrlrd_arid	output	IW	-
src_m_axi_ctrlrd_arlock	output	1	-
src_m_axi_ctrlrd_arcache	output	4	-
src_m_axi_ctrlrd_arprot	output	3	-
src_m_axi_ctrlrd_arqos	output	4	-
src_m_axi_ctrlrd_arregion	output	4	-
src_m_axi_ctrlrd_rvalid	input	1	-
src_m_axi_ctrlrd_rready	output	1	-
src_m_axi_ctrlrd_rdata	input	32	-
src_m_axi_ctrlrd_rresp	input	2	-
src_m_axi_ctrlrd_rlast	input	1	-
src_m_axi_ctrlrd_rid	input	IW	-

4.3.13 Control Write AXI Master (32-bit) [Phase 2]

Table 1.2.15: Control Write AXI Master (32-bit) [Phase 2]

Signal	Direction	Width	Description
src_m_axi_ctrlwr_awvalid	output	1	-

Signal	Direction	Width	Description
src_m_axi_ctrlwr_awready	input	1	-
src_m_axi_ctrlwr_awaddr	output	AW	-
src_m_axi_ctrlwr_awlen	output	8	-
src_m_axi_ctrlwr_awsize	output	3	-
src_m_axi_ctrlwr_awburst	output	2	-
src_m_axi_ctrlwr_awid	output	IW	-
src_m_axi_ctrlwr_awlock	output	1	-
src_m_axi_ctrlwr_awcache	output	4	-
src_m_axi_ctrlwr_awprot	output	3	-
src_m_axi_ctrlwr_awqos	output	4	-
src_m_axi_ctrlwr_awregion	output	4	-
src_m_axi_ctrlwr_wvalid	output	1	-
src_m_axi_ctrlwr_wready	input	1	-
src_m_axi_ctrlwr_wdata	output	32	-
src_m_axi_ctrlwr_wstrb	output	4	-
src_m_axi_ctrlwr_wlast	output	1	-
src_m_axi_ctrlwr_bvalid	input	1	-
src_m_axi_ctrlwr_bready	output	1	-
src_m_axi_ctrlwr_bid	input	IW	-
src_m_axi_ctrlwr_bresp	input	2	-

4.4 SOURCE HALF (u_src) - direction-unique ports (no prefix)

4.4.1 AXI Transfer Configuration (source-only)

Table 1.2.16: AXI Transfer Configuration (source-only)

Signal	Direction	Width	Description
cfg_axi_rd_xfer_beats	input	8	-
cfg_drain_size	input	8	source: beats drained per AXIS packet

4.4.2 Source Path - AXIS Master Interface (SRAM -> Network); tid = channel id

Table 1.2.17: Source Path - AXIS Master Interface (SRAM -> Network); tid = channel id

Signal	Direction	Width	Description
m_axis_tdata	output	DW	-
m_axis_tstrb	output	SW	-
m_axis_tlast	output	1	-
m_axis_tid	output	AXIS_ID_WIDTH	-
m_axis_tdest	output	AXIS_DEST_WIDTH	-
m_axis_tuser	output	AXIS_USER_WIDTH	-
m_axis_tvalid	output	1	-
m_axis_tready	input	1	-

4.4.3 AXI4 Master - Data Read (Memory -> Source SRAM)

Table 1.2.18: AXI4 Master - Data Read (Memory -> Source SRAM)

Signal	Direction	Width	Description
m_axi_rd_arid	output	IW	-
m_axi_rd_araddr	output	AW	-
m_axi_rd_arle	output	8	-

Signal	Direction	Width	Description
n			
m_axi_rd_arsize	output	3	-
m_axi_rd_arburst	output	2	-
m_axi_rd_arvalid	output	1	-
m_axi_rd_arready	input	1	-
m_axi_rd_rid	input	IW	-
m_axi_rd_rdata	input	DW	-
m_axi_rd_rresp	input	2	-
m_axi_rd_rlast	input	1	-
m_axi_rd_rvalid	input	1	-
m_axi_rd_rready	output	1	-

4.5 SINK HALF (u_snk) - shared-infrastructure ports (snk_ prefixed)

4.5.1 APB Programming Interface

Table 1.2.19: APB Programming Interface

Signal	Direction	Width	Description
snk_apb_valid	input	NC	-
snk_apb_ready	output	NC	-
snk_apb_addr	input	NC x AW	-

4.5.2 Per-channel configuration

Table 1.2.20: Per-channel configuration

Signal	Direction	Width	Description
snk_cfg_channel_enable	input	NC	-
snk_cfg_channel_reset	input	NC	-

4.5.3 Scheduler Configuration (global)

Table 1.2.21: Scheduler Configuration (global)

Signal	Direction	Width	Description
snk_cfg_sched_enable	input	1	-
snk_cfg_sched_timeout_cycles	input	32	-
snk_cfg_sched_timeout_limit	input	8	-
snk_cfg_sched_timeout_enable	input	1	-
snk_cfg_sched_err_enable	input	1	-
snk_cfg_sched_compl_enable	input	1	-

Signal	Direction	Width	Description
snk_cfg_sched_perf_enable	input	1	-

4.5.4 Descriptor Engine Configuration (global)

Table 1.2.22: Descriptor Engine Configuration (global)

Signal	Direction	Width	Description
snk_cfg_descneng_enable	input	1	-
snk_cfg_descneng_prefetch	input	1	-
snk_cfg_descneng_fifo_thresh	input	4	-
snk_cfg_descneng_addr0_base	input	AW	-
snk_cfg_descneng_addr0_limit	input	AW	-
snk_cfg_descneng_addr1_base	input	AW	-
snk_cfg_descneng_addr1_limit	input	AW	-

4.5.5 Control Engine Configuration (Phase 2, global)

Table 1.2.23: Control Engine Configuration (Phase 2, global)

Signal	Direction	Width	Description
snk_cfg_ctrlrd_max_try	input	9	-
snk_tick_1us	input	1	-

4.5.6 Descriptor AXI Monitor Configuration

Table 1.2.24: Descriptor AXI Monitor Configuration

Signal	Direction	Width	Description
snk_cfg_desc_mon_enable	input	1	-
snk_cfg_desc_mon_err_enable	input	1	-

Signal	Direction	Width	Description
snk_cfg_desc_mon_perf_enable	input	1	-
snk_cfg_desc_mon_perf_run	input	1	-
snk_cfg_desc_mon_timeout_enable	input	1	-
snk_cfg_desc_mon_timeout_cycles	input	32	-
snk_cfg_desc_mon_latency_thresh	input	32	-
snk_cfg_desc_mon_pkt_mask	input	16	-
snk_cfg_desc_mon_err_select	input	4	-
snk_cfg_desc_mon_err_mask	input	8	-
snk_cfg_desc_mon_timeout_mask	input	8	-
snk_cfg_desc_mon_compl_mask	input	8	-
snk_cfg_desc_mon_thresh_mask	input	8	-
snk_cfg_desc_mon_perf_mask	input	8	-
snk_cfg_desc_mon_addr_mask	input	8	-
snk_cfg_desc_mon_debug_mask	input	8	-

4.5.7 Network-plane monitor input (rapids TASK-024)

The network endpoint and its monitor-lite live in the plane (`rtl/plane/`) in the top. Its monbus packets enter here as client 1 of this half's monbus arbiter; the plane's monitor configuration and status come from this half's register block in the top.

Table 1.2.25: Network-plane monitor input (rapids TASK-024)

Signal	Direction	Width	Description
snk_plane_mon_valid	input	1	-
snk_plane_mon_ready	output	1	-
snk_plane_mon_packet	input	128	monitor_packet_t
snk_plane_mon_timestamp	input	64	monbus_timestamp_t

4.5.8 Status

Table 1.2.26: Status

Signal	Direction	Width	Description
snk_system_idle	output	1	-
snk_descriptor_engine_idle	output	NC	-
snk_scheduler_idle	output	NC	-
snk_scheduler_state	output	NC x 7	-
snk_sched_error	output	NC	-

4.5.9 Descriptor AXI Monitor Status

Table 1.2.27: Descriptor AXI Monitor Status

Signal	Direction	Width	Description
snk_cfg_sts_desc_mon_busy	output	1	-
snk_cfg_sts_desc_mon_active_txns	output	8	-
snk_cfg_sts_desc_mon_error_count	output	16	-
snk_cfg_sts_desc_mon_txn_count	output	32	-
snk_cfg_sts_d	output	1	-

Signal	Direction	Width	Description
esc_mon_conflict_error			

4.5.10 Descriptor AXI Monitor perf window (feeds SNK_.MON.DAXMON_PERF_*).

Table 1.2.28: Descriptor AXI Monitor perf window (feeds SNK_.MON.DAXMON_PERF_*).

Signal	Direction	Width	Description
snk_sts_desc_mon_win_active	output	1	-
snk_sts_desc_mon_win_cycles	output	32	-
snk_sts_desc_mon_prod_cycles	output	32	-
snk_sts_desc_mon_bp_cycles	output	32	-
snk_sts_desc_mon_starv_cycles	output	32	-
snk_sts_desc_mon_idle_cycles	output	32	-
snk_sts_desc_mon_beat_count	output	32	-
snk_sts_desc_mon_byte_count	output	64	-
snk_sts_desc_mon_burst_count	output	32	-

4.5.11 Descriptor Fetch AXI Master

Table 1.2.29: Descriptor Fetch AXI Master

Signal	Direction	Width	Description
snk_m_axi_desc_arvalid	output	1	-

Signal	Direction	Width	Description
snk_m_axi_desc_arready	input	1	-
snk_m_axi_desc_araddr	output	AW	-
snk_m_axi_desc_arlen	output	8	-
snk_m_axi_desc_arsize	output	3	-
snk_m_axi_desc_arburst	output	2	-
snk_m_axi_desc_arid	output	IW	-
snk_m_axi_desc_arlock	output	1	-
snk_m_axi_desc_arcache	output	4	-
snk_m_axi_desc_arprot	output	3	-
snk_m_axi_desc_arqos	output	4	-
snk_m_axi_desc_arregion	output	4	-
snk_m_axi_desc_rvalid	input	1	-
snk_m_axi_desc_rready	output	1	-
snk_m_axi_desc_rdata	input	256	-
snk_m_axi_desc_rresp	input	2	-
snk_m_axi_desc_rlast	input	1	-
snk_m_axi_desc_rid	input	IW	-

4.5.12 Control Read AXI Master (32-bit) [Phase 2]

Table 1.2.30: Control Read AXI Master (32-bit) [Phase 2]

Signal	Direction	Width	Description
snk_m_axi_ctrlrd_arvalid	output	1	-

Signal	Direction	Width	Description
snk_m_axi_ctrlrd_arready	input	1	-
snk_m_axi_ctrlrd_araddr	output	AW	-
snk_m_axi_ctrlrd_arlen	output	8	-
snk_m_axi_ctrlrd_arsize	output	3	-
snk_m_axi_ctrlrd_arburst	output	2	-
snk_m_axi_ctrlrd_arid	output	IW	-
snk_m_axi_ctrlrd_arlock	output	1	-
snk_m_axi_ctrlrd_arcache	output	4	-
snk_m_axi_ctrlrd_arprot	output	3	-
snk_m_axi_ctrlrd_arqos	output	4	-
snk_m_axi_ctrlrd_arregion	output	4	-
snk_m_axi_ctrlrd_rvalid	input	1	-
snk_m_axi_ctrlrd_rready	output	1	-
snk_m_axi_ctrlrd_rdata	input	32	-
snk_m_axi_ctrlrd_rresp	input	2	-
snk_m_axi_ctrlrd_rlast	input	1	-
snk_m_axi_ctrlrd_rid	input	IW	-

4.5.13 Control Write AXI Master (32-bit) [Phase 2]

Table 1.2.31: Control Write AXI Master (32-bit) [Phase 2]

Signal	Direction	Width	Description
snk_m_axi_ctrlwr_awvalid	output	1	-

Signal	Direction	Width	Description
snk_m_axi_ctrlwr_awready	input	1	-
snk_m_axi_ctrlwr_awaddr	output	AW	-
snk_m_axi_ctrlwr_awlen	output	8	-
snk_m_axi_ctrlwr_awsize	output	3	-
snk_m_axi_ctrlwr_awburst	output	2	-
snk_m_axi_ctrlwr_awid	output	IW	-
snk_m_axi_ctrlwr_awlock	output	1	-
snk_m_axi_ctrlwr_awcache	output	4	-
snk_m_axi_ctrlwr_awprot	output	3	-
snk_m_axi_ctrlwr_awqos	output	4	-
snk_m_axi_ctrlwr_awregion	output	4	-
snk_m_axi_ctrlwr_wvalid	output	1	-
snk_m_axi_ctrlwr_wready	input	1	-
snk_m_axi_ctrlwr_wdata	output	32	-
snk_m_axi_ctrlwr_wstrb	output	4	-
snk_m_axi_ctrlwr_wlast	output	1	-
snk_m_axi_ctrlwr_bvalid	input	1	-
snk_m_axi_ctrlwr_bready	output	1	-
snk_m_axi_ctrlwr_bid	input	IW	-
snk_m_axi_ctrlwr_bresp	input	2	-

4.6 Monitor Bus (SINGLE aggregated stream for the whole core)

The two halves' monitor outputs are merged through a top-level monbus_arbiter, so the core exposes exactly one monitor stream.

4.6.1 Monitor Bus

Table 1.2.32: Monitor Bus

Signal	Direction	Width	Description
mon_valid	output	1	-
mon_ready	input	1	-
mon_packet	output	ocelus_common_pkg::monitor_packet_t	-
mon_timestamp	output	ocelus_common_pkg::monbus_timestamp_t	-

4.7 SINK HALF (u_snk) - direction-unique ports (no prefix)

4.7.1 AXI Transfer Configuration (sink-only)

Table 1.2.33: AXI Transfer Configuration (sink-only)

Signal	Direction	Width	Description
cfg_axi_wr_xfer_beats	input	8	-
cfg_alloc_size	input	8	sink: SRAM alloc size per AXIS fill

4.7.2 Sink Path - AXIS Slave Interface (Network -> SRAM); tid = channel id

Table 1.2.34: Sink Path - AXIS Slave Interface (Network -> SRAM); tid = channel id

Signal	Direction	Width	Description
s_axis_tdata	input	DW	-
s_axis_tstrb	input	SW	-
s_axis_tlast	input	1	-
s_axis_tid	input	AXIS_ID_WIDTH	-
s_axis_tdest	input	AXIS_DEST_WIDTH	-
s_axis_tuser	input	AXIS_USER_WIDTH	-
s_axis_tvalid	input	1	-
s_axis_tready	output	1	-

4.7.3 AXI4 Master - Data Write (Sink SRAM -> Memory)

Table 1.2.35: AXI4 Master - Data Write (Sink SRAM -> Memory)

Signal	Direction	Width	Description
m_axi_wr_awid	output	IW	-
m_axi_wr_awaddr	output	AW	-
m_axi_wr_awlen	output	8	-

Signal	Direction	Width	Description
m_axi_wr_awsize	output	3	-
m_axi_wr_awburst	output	2	-
m_axi_wr_awlock	output	1	-
m_axi_wr_awcache	output	4	-
m_axi_wr_awprot	output	3	-
m_axi_wr_awqos	output	4	-
m_axi_wr_awregion	output	4	-
m_axi_wr_awvalid	output	1	-
m_axi_wr_awready	input	1	-
m_axi_wr_wdata	output	DW	-
m_axi_wr_wstrb	output	(DW/8)	-
m_axi_wr_wlast	output	1	-
m_axi_wr_wvalid	output	1	-
m_axi_wr_wready	input	1	-
m_axi_wr_bid	input	IW	-
m_axi_wr_bresp	input	2	-
m_axi_wr_bvalid	input	1	-
m_axi_wr_bready	output	1	-

4.8 Debug Interface (per half, src_dbg_ / snk_dbg_ prefixed)

4.8.1 Source half debug

Table 1.2.36: Source half debug

Signal	Direction	Width	Description
src_dbg_rd_all_complete	output	NC	-
src_dbg_r_beats_rcvd	output	32	-
src_dbg_sram_writes	output	32	-
src_dbg_arb_request	output	NC	-
src_dbg_src_sram_bridge_pending	output	NC	-
src_dbg_src_sram_bridge_out_valid	output	NC	-
src_dbg_axis_beats_sent	output	32	-
src_dbg_axis_packets_sent	output	32	-

4.8.2 Sink half debug

Table 1.2.37: Sink half debug

Signal	Direction	Width	Description
snk_dbg_snk_sram_bridge_pending	output	NC	-
snk_dbg_snk_sram_bridge_out_valid	output	NC	-
snk_dbg_axis_beats_received	output	32	-
snk_dbg_axis_packets_received	output	32	-

4.8.3 Active-channel sideband for per-channel bus instrumentation

(axi_bus_meter). The W bus carries no wid, so the write engine's channel index must travel out of band to reach the meter at the top.

Table 1.2.38: Active-channel sideband for per-channel bus instrumentation

Signal	Direction	Width	Description
`snk_active_channel_id`	output	CIW	-
`snk_active_channel_valid`	output	1	-

Verified: every row above is a declared port of `rapids_core_beats` and every declared port has a row (324 of 324), generated from the module declaration on 2026-09-28. Regenerate rather than hand-edit when the interface changes.

5 Clocks and Reset

Module: `rapids_core_beats.sv` **Location:** `projects/components/dma-ip/rapids/rtl/macro_beats/` **Status:** Implemented **Last Updated:** 2025-01-10

5.1 Clock Domain

RAPIDS Beats operates in a single clock domain for simplified design and maximum throughput.

5.1.1 Clock Specifications

Table 1.3.1: Clock Specifications

Parameter	Specification
Clock Signal	`clk` (also `axi_aclk` in some submodules)
Frequency Range	100 - 500 MHz
Duty Cycle	45% - 55%
Clock Jitter	< 100 ps peak-to-peak

5.1.2 Clock Distribution

```
clk (one clock domain; rst_n active-low, async assert)
├── scheduler_group_array_beats (one per half)
│   └── descriptor_engine_beats, scheduler_beats, ctrlrd_engine,
ctrlwr_engine (x8)
├── snk_data_path_axis_beats
│   └── snk_sram_controller_beats, axi_write_engine_beats
└── src_data_path_axis_beats
    └── axi_read_engine_beats, src_sram_controller_beats
```

5.2 Reset Specification

RAPIDS uses asynchronous assert, synchronous deassert reset methodology.

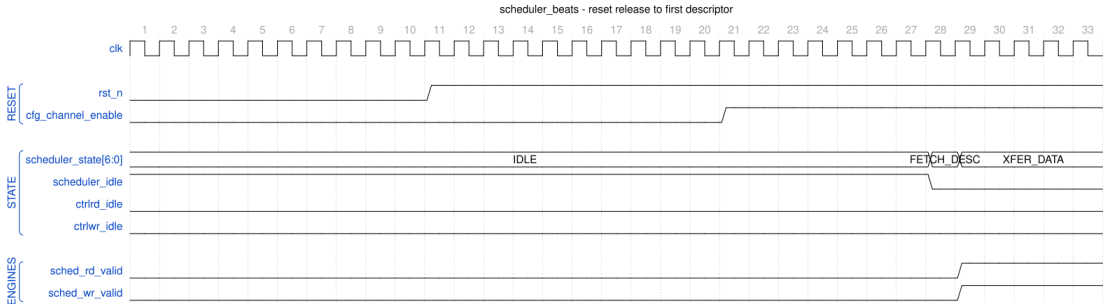
5.2.1 Reset Signal

Table 1.3.2: Reset Specifications

Parameter	Specification
Reset Signal	`rst_n` (also `axi_aresetn` in some submodules)
Polarity	Active-low
Assert	Asynchronous (immediate)
Deassert	Synchronous (on rising clock edge)
Minimum Duration	10 clock cycles

5.2.2 Reset Timing Diagram

5.2.3 Figure 1.3.1: Reset Timing



scheduler_beats - reset release to first descriptor

Source: scheduler_reset.json, captured from `dv/tests/fub_beats/test_scheduler_beats.py` (basic transfer, the same run as Figure 2.1.3) with `WAVES=1`.

Reading it: `rst_n` is held low for ten cycles and released on a clock edge. Every state register is already at its reset value while reset is asserted – the FSM sits in `IDLE`, `scheduler_idle` is high, both control engines report idle and neither engine request is raised. Nothing changes at the release edge itself; the scheduler stays in `IDLE` until the testbench raises `cfg_channel_enable` ten cycles later and a descriptor becomes available, at which point it steps through `FETCH_DESC` to `XFER_DATA` and raises the read and write requests together.

5.3 Reset Behavior

5.3.1 On Reset Assert

All modules initialize to known states:

Table 1.3.3: Reset States

Component	Reset State
Scheduler FSM	IDLE
Descriptor Engine	IDLE, FIFOs empty
AXI Read Engine	Idle, no outstanding transactions
AXI Write Engine	Idle, no outstanding transactions
SRAM Pointers	All zeros
Alloc/Drain Controllers	Empty, full space available
MonBus	No packets pending

5.3.2 Reset Sequence

```
1. External reset asserted (rst_n = 0)
   - All flip-flops async reset
   - All outputs driven to safe states
   - No AXI transactions

2. System stabilization (minimum 10 cycles)
   - Clock running and stable
   - Configuration registers programmed (optional)

3. Reset deasserted (rst_n = 1, sync to clk rising edge)
   - FSMs begin operation
   - Ready to accept descriptors/data
```

5.4 Configuration During Reset

Some configuration signals MUST be stable before reset deassertion:

Table 1.3.4: Configuration Timing Requirements

Signal	Requirement
`src_cfg_channel_enable` / `snk_cfg_channel_enable`	Stable before rst_n=1 (one per half)
`src_cfg_sched_timeout_cycles` / `snk_cfg_sched_timeout_cycles`	Stable before rst_n=1 (one per half)
`cfg_axi_rd_xfer_beats`	Stable before rst_n=1
`cfg_axi_wr_xfer_beats`	Stable before rst_n=1

5.5 Reset Macros

RAPIDS uses standardized reset macros from `reset_defs.svh`:

```
`include "reset_defs.svh"

// Standard reset pattern
`ALWAYS_FF_RST(clk, rst_n,
    if (`RST_ASSERTED(rst_n)) begin
        r_state <= IDLE;
        r_counter <= '0;
    end else begin
        r_state <= w_next_state;
        r_counter <= w_next_counter;
    end
)
```

5.6 Power-On Reset Considerations

- External reset should be held for minimum 10 clock cycles after clock is stable
- Configuration registers should be programmed before or during reset
- No AXI transactions until reset is deasserted and channels are enabled

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Last Updated: 2025-01-10

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6 Scheduler

Module: `scheduler.sv` **Location:** `projects/components/dma-ip/rapids/rtl/fub/` **Status:** Implemented **Last Updated:** 2026-10-21

6.1 Overview

The scheduler coordinates one channel. It accepts a 256-bit descriptor from the descriptor engine, turns the descriptor's byte length and byte addresses into beat counts for the AXI engines, tells the data paths how the packet is laid out, and tracks progress until every source beat is read and every destination beat is committed.

Compared with the RAPIDS Beats scheduler, the state machine, the timeout logic, the control-descriptor path and the MonBus events are the same. Three things change: the beat math, the address advance, and the packet record.

6.1.1 Structure: the FSM around scheduler_core

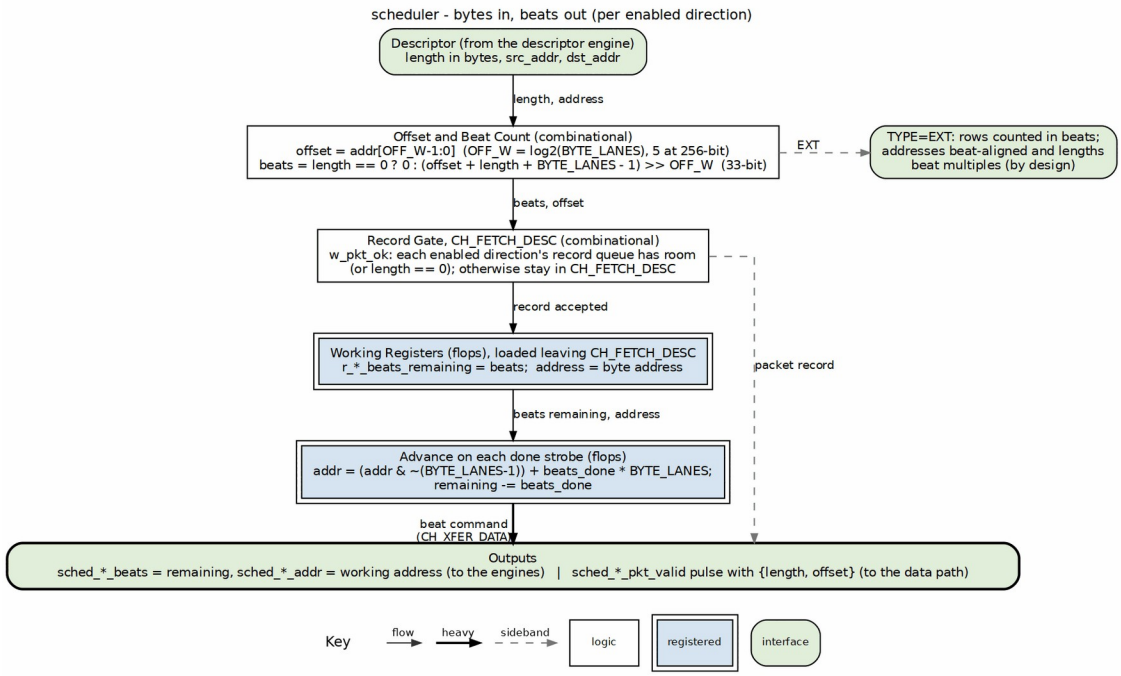
Since rapids TASK-034 the scheduler is two modules. `scheduler_core.sv` holds the datapath and nothing that decides: the descriptor skid and latch, the strobe-driven counters, the address advance, run accounting, timeouts, sticky errors, control-engine issue, tile checks, packet records and MonBus packets. Its state, its descriptor take, its beat totals and its EXT run bases are inputs. `scheduler.sv` keeps the port list it always had and supplies those inputs: the finite state machine below, the byte-to-beat ceiling and the two `stream_run_addr_gen` run-base generators. The gatling-sequenced `scheduler_g` (Chapter 5) wraps the same core and takes the four from firmware, so there is one copy of the datapath for both.

6.1.2 Key Features

- **Byte length:** the descriptor's 32-bit length is a byte count.
- **Byte addresses:** the offset of a transfer inside its first beat is `addr[OFF_W-1:0]`.
- **Beat math:** beats moved are derived from offset and length, in a 33-bit intermediate.
- **Packet record:** one `{bytes, offset}` pulse per DATA descriptor, per enabled direction, held back while the data path cannot take it.
- **Concurrent read and write:** both directions run inside one XFER_DATA state.
- **Recoverable write-progress timeout:** strikes escalate to a sticky error only when a limit is set.
- **Control descriptors:** CTRL_READ and CTRL_WRITE drive the control engines, unchanged.
- **Directional halves:** `EN_READ` and `EN_WRITE` select the source or sink half.

6.1.3 Block Diagram

6.1.4 Figure 2.1.1: Scheduler Byte Math



Scheduler byte math

Source: `assets/graphviz/03_scheduler_byte_math.dot`

6.2 Parameters

Table 2.1.1: Scheduler Parameters

Parameter	Type	Default	Description
CHANNEL_ID	int	0	Channel index of this scheduler
NUM_CHANNELS	int	8	Channel count
CHAN_WIDTH	int	$clog2(NUM_CHANNELS)	Channel field width
ADDR_WIDTH	int	64	Address width
DATA_WIDTH	int	512	Data width in bits; the board design uses 256
MON_AGENT_ID	logic [15:0]	16'h0040	MonBus agent ID
MON_UNIT_ID	logic [7:0]	8'h01	MonBus unit ID
MON_CHANNEL_ID	logic [8:0]	9'h000	MonBus base channel ID
DESC_WIDTH	int	256	Descriptor width; fixed at 256 (elaboration fails otherwise)
EN_READ	bit	1	Enable the read direction for DATA descriptors
EN_WRITE	bit	1	Enable the write direction for DATA descriptors
USE_ROW_COL_MAJOR_ADDRESSING	int	1	Enable TYPE=EXT run-base generators
BYTE_LANES	int	DATA_WIDTH/8	Bytes per beat
OFF_W	int	(BYTE_LANES>1) ? $clog2(BYTE_LANES) : 1	Offset field width

BYTE_LANES and OFF_W are the two parameters that do not exist in the Beats scheduler. A directional half sets exactly one of EN_READ and EN_WRITE: the source half reads only, the sink half writes only.

6.3 Port List

6.3.1 Clock, Reset and Configuration

Table 2.1.2: Clock, Reset and Configuration Ports

Signal	Dir	Width	Description
clk	in	1	Clock
rst_n	in	1	Active-low reset
cfg_channel_enable	in	1	Channel enable; IDLE leaves only when set
cfg_channel_reset	in	1	Channel reset; forces IDLE and clears sticky state
cfg_sched_timeout_cycles	in	32	Write-progress timeout window in cycles
cfg_sched_timeout_limit	in	8	Consecutive windows before escalation; 0 = never
cfg_sched_timeout_enable	in	1	Enable timeout detection

6.3.2 Status

Table 2.1.3: Status Ports

Signal	Dir	Width	Description
scheduler_idle	out	1	High in CH_IDLE with no channel reset pending and the descriptor skid empty
scheduler_state	out	7	One-hot state (see Table 2.1.12)
sched_error	out	1	Sticky, high in CH_ERROR
dbg_descriptor_error	out	1	Latched fatal descriptor-level error
dbg_read_error_sticky	out	1	Latched read-engine error
dbg_write_error_sticky	out	1	Latched write-engine error
dbg_timeout_expired	out	1	Live: the timeout window has elapsed

6.3.3 Descriptor Engine Interface

Table 2.1.4: Descriptor Engine Interface

Signal	Dir	Width	Description
descriptor_valid	in	1	A descriptor is available
descriptor_ready	out	1	The descriptor skid has room (see Descriptor Skid Buffer)
descriptor_packet	in	256	Descriptor (layout in Table 2.1.10)
descriptor_ext_packet	in	256	Extended chunk; used only when TYPE=EXT
descriptor_error	in	1	Descriptor fetch or parse error

6.3.4 Descriptor Skid Buffer

Every scheduler takes its descriptors through a two-deep skid buffer (gaxi_skid_buffer, DEPTH 2) on {descriptor_packet, descriptor_ext_packet}, in front of the descriptor engine's per-channel FIFO, which stays (rapids TASK-028). The FSM reads the descriptor from the skid's flops rather than straight off the engine's 256-bit output, one engine for every channel under SHARED_ENGINES, and a running channel holds two more descriptors.

- descriptor_ready is the skid's wr_ready: high while it has room, whatever the FSM state. The FSM's own take (CH_IDLE while cfg_channel_enable is set, CH_NEXT_DESC) pops the skid.
- A disabled channel's descriptor waits in the skid (or behind it in the engine FIFO) until the channel is enabled; it is not consumed and dropped.
- scheduler_idle stays low while the skid holds a descriptor, so the engine and the scheduler never both read idle with work pending.
- Channel reset: the engine masks descriptor_valid during its reset, so nothing is written during one, and the skid drains while the reset is active. Each entry carries the reset epoch it was written in, and the epoch flips as a reset starts, so an entry a reset too short to drain it left behind is dropped at the head, never run.

6.3.5 Data Read Interface (to AXI Read Engine)

Table 2.1.5: Data Read Interface

Signal	Dir	Width	Description
sched_rd_valid	out	1	Read request
sched_rd_addr	out	ADDR_WIDTH	Source byte address (working register)
sched_rd_beats	out	32	Beats remaining to read
sched_rd_pkt_valid	out	1	Packet record pulse to the source egress
sched_rd_pkt_ready	in	1	Source egress record queue has room
sched_rd_pkt_bytes	out	32	Descriptor length in bytes
sched_rd_pkt_offset	out	OFF_W	Source byte offset inside the first beat

6.3.6 Data Write Interface (to AXI Write Engine)

Table 2.1.6: Data Write Interface

Signal	Dir	Width	Description
sched_wr_valid	out	1	Write request
sched_wr_ready	in	1	Write engine can take the channel
sched_wr_addr	out	ADDR_WIDTH	Destination byte address (working register)
sched_wr_beats	out	32	Beats remaining to issue
sched_wr_pkt_valid	out	1	Packet record pulse to the sink ingress
sched_wr_pkt_ready	in	1	Sink ingress record queue has room
sched_wr_pkt_bytes	out	32	Descriptor length in bytes
sched_wr_pkt_offset	out	OFF_W	Destination byte offset inside the first beat

6.3.7 Completion Interface

Table 2.1.7: Completion and Error Interface

Signal	Dir	Width	Description
sched_rd_done_strobe	in	1	A read burst completed
sched_rd_beats_done	in	32	Beats in that burst
sched_wr_done_strobe	in	1	A write burst was issued (AW handshake)
sched_wr_beats_done	in	32	Beats in that burst
sched_wr_commit_strobe	in	1	A write burst was committed (B response)
sched_wr_commit_beats	in	32	Beats in that burst
sched_rd_error	in	1	Read engine error
sched_wr_error	in	1	Write engine error, including the sink packet length error

6.3.8 Control Engine Interface

Table 2.1.8: Control Engine Interface

Signal	Dir	Width	Description
ctrlrd_valid	out	1	CTRL_READ request
ctrlrd_ready	in	1	Engine accepted the request
ctrlrd_addr	out	ADDR_WIDTH	Poll address (descriptor source slot)
ctrlrd_data	out	32	Expected value (dst_addr[31:0])
ctrlrd_mask	out	32	Compare mask (dst_addr[63:32])
ctrlrd_error	in	1	Engine error (retries exhausted or AXI error)
ctrlrd_idle	in	1	Engine idle
ctrlwr_valid	out	1	CTRL_WRITE request
ctrlwr_ready	in	1	Engine accepted the request

Signal	Dir	Width	Description
ctrlwr_addr	out	ADDR_WIDTH	Doorbell address (descriptor source slot)
ctrlwr_data	out	32	Doorbell data (dst_addr[31:0])
ctrlwr_error	in	1	Engine error
ctrlwr_idle	in	1	Engine idle

6.3.9 Monitor Bus Interface

Table 2.1.9: Monitor Bus Interface

Signal	Dir	Width	Description
i_mon_time	in	timestamp	Time base for packets
mon_valid	out	1	Packet valid
mon_ready	in	1	Consumer ready
mon_packet	out	128	Monitor packet
mon_timestamp	out	64	Side-band timestamp

The control interface tables do not change from RAPIDS Beats. The packet-record ports in Tables 2.1.5 and 2.1.6 are the new ports.

6.4 Descriptor Format

Table 2.1.10: Descriptor Layout

Bits	Field	Notes
[63:0]	`src_addr`	Byte address
[127:64]	`dst_addr`	Byte address
[159:128]	`length`	Bytes
[191:160]	`next_descriptor_ptr`	Chain pointer
[192]	`valid`	Must be 1
[193]	`gen_irq`	Emit an IRQ event on completion
[194]	`last`	End of chain
[195]	`error`	Error flag
[199:196]	`channel_id`	
[207:200]	`priority`	
[209:208]	`opcode`	DATA = 0, CTRL_READ = 1, CTRL_WRITE = 2
[212:210]	`type`	LEGACY = 0, EXT = 1
[216:213]	tile `op`, `ptype`	`TILE_HDR` builds
[224:217]	`next_addr_sel`	must be 0 here; any other value is an invalid descriptor (rapids-g runs 1, HAS Section 5.1)

For CTRL_READ and CTRL_WRITE the address and data slots are reinterpreted as in Tables 2.1.8; the length field does not describe a byte count and no packet record is issued.

6.5 Byte-to-Beat Math

For each enabled direction the scheduler forms the offset from the low `OFF_W` bits of that side's address and derives the number of memory beats the transfer touches.

```
w_rd_off        = src_addr[OFF_W-1:0]
w_wr_off        = dst_addr[OFF_W-1:0]
w_rd_span       = 33'(w_rd_off) + 33'(length) + 33'(BYTE_LANES-1)
w_wr_span       = 33'(w_wr_off) + 33'(length) + 33'(BYTE_LANES-1)
w_rd_beats_total = (length == 0) ? 0 : 32'(w_rd_span >> OFF_W)
w_wr_beats_total = (length == 0) ? 0 : 32'(w_wr_span >> OFF_W)
```

The span is 33 bits wide so a length near 4 GiB does not overflow. The read and write beat totals are computed separately because the two addresses can have different offsets; a 60-byte transfer read from offset 5 and written at offset 0 touches three source beats and two destination beats on a 32-byte design.

Table 2.1.11: Beat Totals for a 32-byte Beat

Length (bytes)	Offset	Beats
0	any	0
1	0	1
32	0	1
32	1	2
60	5	3
60	0	2
64	0	2

The values load in CH_FETCH_DESC:

- `r_beats_remaining` loads `EN_READ ? rd_total : wr_total`.
- `r_read_beats_remaining` loads `EN_READ ? rd_total : 0`.
- `r_write_beats_remaining` and `r_write_beats_to_commit` load `EN_WRITE ? wr_total : 0`.

A disabled direction therefore loads zero, so its valid never asserts and its completion term is true from the start. Completion collapses onto the enabled direction.

For TYPE=EXT the run counters load `min(run_size, beats_total)` instead of the whole total, and `stream_run_addr_gen.cfg_total_beats` takes the beats total. TYPE=EXT is beat-aligned by design: addresses are aligned and lengths are beat multiples, so the offset is zero and the math reduces to the Beats result.

6.6 Address Advance

The working address registers start at the descriptor's byte addresses. On each done strobe the address moves forward by the beats reported:

```
r_src_addr <= {r_src_addr[AW-1:OFF_W], {OFF_W{1'b0}}} +
(AW'(sched_rd_beats_done) << OFF_W)   // on sched_rd_done_strobe
r_dst_addr <= {r_dst_addr[AW-1:OFF_W], {OFF_W{1'b0}}} +
(AW'(sched_wr_beats_done) << OFF_W)   // on sched_wr_done_strobe
```

The low bits are cleared before the add. Only the first burst of a descriptor starts inside a beat; every later burst starts on a beat boundary, so aligning down first is correct for both. The write address advances on the AW handshake (issue), not on the B response. The B response decrements `r_write_beats_to_commit` instead.

The scheduler reports the working address to the engines unmodified. The engines align it (see AXI Read Engine); the offset reaches the data path through the packet record, not through the address bus.

6.7 Packet Records

A DATA descriptor with a non-zero length causes one record pulse per enabled direction, on the cycle the scheduler leaves CH_FETCH_DESC:

```
w_pkt_ok = !w_is_data || length == 0 ||
           ((!EN_READ  || sched_rd_pkt_ready) &&
            (!EN_WRITE || sched_wr_pkt_ready))

sched_{rd,wr}_pkt_valid = w_state_fetch_desc && r_descriptor.valid &&
w_is_data &&
                          EN_{READ,WRITE} && w_pkt_ok && (length != 0)
sched_{rd,wr}_pkt_bytes  = r_descriptor.length
sched_{rd,wr}_pkt_offset = w_{rd,wr}_off
```

Both directions share `w_pkt_ok`, so each enabled direction sees exactly one pulse. `w_pkt_ok` also gates the FETCH_DESC to XFER_DATA transition: while a record queue is full, the scheduler holds in CH_FETCH_DESC and the pulse does not fire. Control descriptors and zero-length transfers carry no record. See Packet Record Interface.

6.8 Finite State Machine

The state is one-hot and is `rapids_pkg::channel_state_t`.

Table 2.1.12: State Encoding

State	Encoding	Description
CH_IDLE	7'b0000001	Waiting for a descriptor and `cfg_channel_enable`
CH_FETCH_DESC	7'b0000010	Descriptor latched; working registers and counters load; record pulses
CH_XFER_DATA	7'b0000100	Read and write engines run concurrently; control ops issue
CH_COMPLETE	7'b0001000	Descriptor done; chain decision
CH_NEXT_DESC	7'b0010000	Waiting for the chained descriptor
CH_ERROR	7'b0100000	Sticky; only channel reset leaves it
CH_RESERVED	7'b1000000	Unused

Transition priority is: registered channel reset to IDLE, then a hard error or timeout escalation to ERROR, then the normal case.

Table 2.1.13: Transitions

From	To	Condition
IDLE	FETCH_DESC	`descriptor_valid && cfg_channel_enable`
FETCH_DESC	XFER_DATA	`r_descriptor.valid && w_pkt_ok`
FETCH_DESC	FETCH_DESC	`r_descriptor.valid` and `w_pkt_ok` false (record queue full)
FETCH_DESC	ERROR	descriptor invalid: `valid` bit clear, or a next-address selector `[224:217]` other than 0 (rapids TASK-035)
XFER_DATA	COMPLETE	`w_exec_complete`
COMPLETE	NEXT_DESC	`next_descriptor_ptr != 0 && !last`
COMPLETE	IDLE	otherwise

From	To	Condition
NEXT_DESC	FETCH_DESC	`descriptor_valid`
ERROR	ERROR	always; channel reset is the only exit
any other	ERROR	default

`w_exec_complete` is `w_transfer_complete` for a DATA descriptor and `w_ctrl_complete` for a control descriptor. `w_transfer_complete` is the read remaining count reaching zero together with the commit remaining count reaching zero. `w_ctrl_complete` is the control request having been accepted and the control engine having returned to idle.

`scheduler_idle` is `CH_IDLE && !r_channel_reset_active && !w_desc_skid_valid`. `sched_error` is the ERROR state.

6.9 Operation

6.9.1 Data Request Outputs

sched_rd_valid is high in XFER_DATA for a DATA descriptor while !w_read_complete, !w_sched_rd_completing_this_cycle and, for TYPE=EXT, !w_rd_need_base. sched_rd_beats is the read remaining count, or for EXT the current run's remainder. sched_wr_valid is high in XFER_DATA for a DATA descriptor while r_write_beats_remaining != 0, the commit count has not reached zero, !w_sched_wr_completing_this_cycle and !w_wr_need_base.

The look-ahead terms drop the valid on the same cycle the last done strobe arrives. Without them the engine would see a still-asserted valid for one more cycle and issue an extra burst.

6.9.2 Write Completion

The write direction has two counters. r_write_beats_remaining decrements on the AW done strobe and drives burst sizing and the destination address. r_write_beats_to_commit decrements on the B commit strobe and gates w_write_complete. The channel therefore completes only after every issued burst has been acknowledged.

6.9.3 Control Descriptors

CTRL_READ and CTRL_WRITE do not use the data engines. ctrlrd_valid and ctrlwr_valid assert in XFER_DATA until the engine accepts (r_ctrl_issued), and completion is the engine returning to idle after acceptance.

6.10 Error Handling

`w_hard_error` is the OR of:

- `descriptor_error`, `sched_rd_error`, `sched_wr_error`
- the sticky read and write error flags
- `ctrlrd_error` while the descriptor is CTRL_READ
- `ctrlwr_error` while the descriptor is CTRL_WRITE

A hard error, or a timeout escalation, moves the FSM to CH_ERROR. The sink packet length check reports through `sched_wr_error` (see Sink Data Path AXIS), so a stream packet whose byte count differs from the descriptor length is a hard error for that channel. Sticky read and write flags and the descriptor error clear in CH_IDLE and also on the registered channel reset, in every state. Clearing on the reset matters: the data engines and the data paths clear their own error levels on the same channel reset, so the scheduler no longer re-enters CH_ERROR from a stale level when it reaches CH_IDLE. A channel reset therefore recovers a data read or write response error and a sink packet length error, and the channel accepts a good descriptor afterwards. The reset also clears the descriptor-loaded flag, the remaining beat counts and the control issued flag, so nothing of the failed descriptor carries into the next one.

6.11 Recoverable Write-Progress Timeout

The timeout counter increments while `sched_wr_valid && !sched_wr_ready`. It resets on `sched_wr_done_strobe` or `sched_wr_commit_strobe`, and also when the condition is false. When the counter reaches `cfg_sched_timeout_cycles` with `cfg_sched_timeout_enable` set, `w_timeout_expired` pulses and the counter re-arms.

Each expiry adds one strike, in an 8-bit counter that saturates. Strikes clear on write progress, in CH_IDLE, and on channel reset. Escalation is:

```
w_timeout_escalate = (cfg_sched_timeout_limit != 0) && (strikes >=
cfg_sched_timeout_limit)
```

A limit of zero never escalates: the timeout is then purely informational.

6.12 MonBus Events

Packets are `PktTypeCompletion` from `PROTOCOL_CORE` unless noted.

Table 2.1.14: MonBus Events

State	Event	Data
entry to CH_FETCH_DESC	`RAPIDS_EVENT_DESC_START`	descriptor length (bytes)
entry to CH_COMPLETE	`RAPIDS_EVENT_IRQ` when `gen_irq`, else `RAPIDS_EVENT_DESC_COMPLETE`	
CH_ERROR	`RAPIDS_EVENT_ERROR` (`PktTypeError`), once per error episode	`{29'h0, wr_sticky, rd_sticky, 33'h0}`

There are no events during XFER_DATA. The only difference from RAPIDS Beats is that the DESC_START data field is a byte count.

DESC_START and the completion events fire on the first cycle of their state, from an entry pulse in `scheduler_core`. Until 2026-10-21 DESC_START fired on every cycle in CH_FETCH_DESC; the byte tree holds that state while the record queue is full or the tile advertisement window is open, and one held descriptor sent 62 packets in the record-backpressure test (rapids BUG-016).

6.13 Run-Base Generator Flush

A channel reset flushes both `stream_run_addr_gen` instances (`flush` = the registered channel reset): generation stops, results still in `dma_address_gen`'s pipeline are discarded, and the base FIFO (a `gaxi_drop_fifo_sync`) is dropped whole. Before 2026-10-21 nothing emptied it: a generator started for an EXT descriptor kept producing that descriptor's bases through the reset, and the next EXT descriptor began its second run at the reset descriptor's base (rapids BUG-017; the same in RAPIDS Beats and STREAM, fixed in all three).
`test_scheduler_ext_channel_reset` covers it.

6.14 Waveforms

6.14.0.1 Waveform 2.1: Byte Descriptor Start

The beat-count arithmetic and the record pulse are combinational functions of the latched descriptor, so the start of a descriptor is short:

```
cycle                    : 0     1        2        3
state                    : IDLE  FETCH    XFER     XFER
descriptor_valid         : 1     0        0        0
sched_wr_pkt_ready       : 1     1        1        1
sched_wr_pkt_valid       : 0     1        0        0     (one pulse,
the last FETCH cycle)
r_write_beats_remaining  : 0     0        N        N     (loads at the
end of FETCH)
sched_wr_valid           : 0     0        1        1
```

If `sched_wr_pkt_ready` is low in FETCH, the state holds and the pulse is delayed until it rises.

7 Descriptor Engine Specification

Module: `descriptor_engine_beats.sv` **Location:** `projects/components/dma-ip/rapids/rtl/fub_beats/` **Status:** Implemented **Last Updated:** 2025-01-10

7.1 Overview

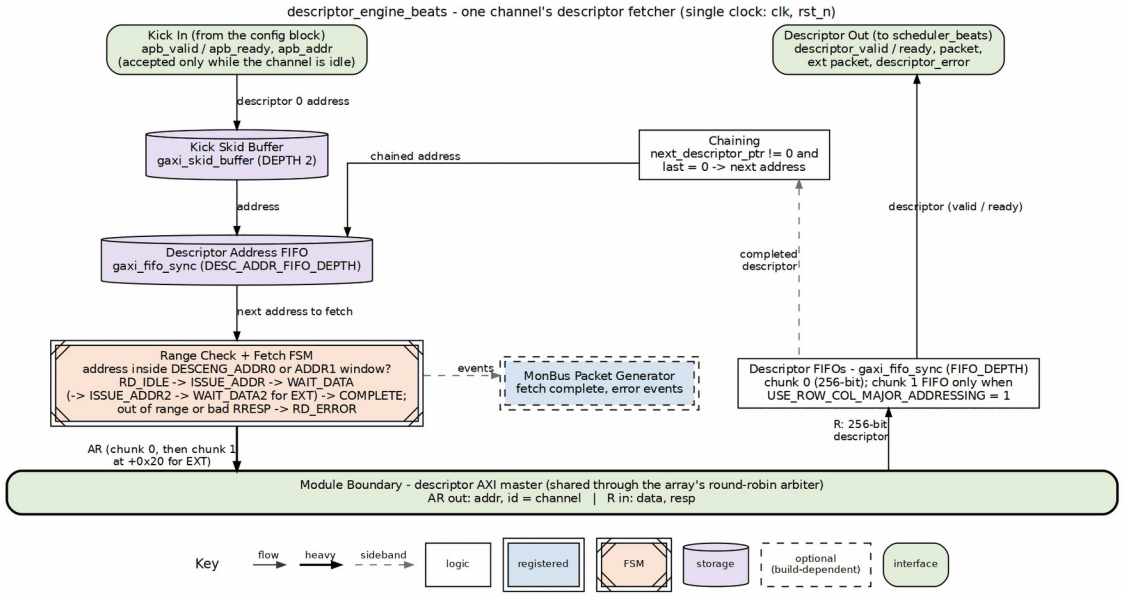
The Descriptor Engine fetches and manages 256-bit descriptors from memory via AXI4. It maintains a prefetch FIFO to hide memory latency and provides parsed descriptors to the scheduler.

7.1.1 Key Features

- **AXI4 Descriptor Fetch:** Single-beat 256-bit reads
- **Prefetch FIFO:** Configurable depth for latency hiding
- **Address Range Checking:** Validates descriptor addresses
- **Error Detection:** Address violations, AXI errors
- **MonBus Integration:** Fetch events and error reporting

7.1.2 Block Diagram

7.1.3 Figure 2.2.1: Descriptor Engine Block Diagram



Descriptor engine block diagram

Source: 02_descriptor_engine.dot

7.2 Parameters

```
parameter int CHANNEL_ID = 0;                    // Channel identifier
parameter int NUM_CHANNELS = 8;                  // Total channels
parameter int ADDR_WIDTH = 64;                   // Address bus width
parameter int FIFO_DEPTH = 4;                    // Prefetch FIFO
depth

// Monitor Bus Parameters
parameter logic [7:0] MON_AGENT_ID = 8'h10;      // Descriptor Engine
Agent ID
parameter logic [3:0] MON_UNIT_ID = 4'h1;        // Unit identifier
```

: Table 2.2.1: Descriptor Engine Parameters

7.3 Port List

7.3.1 Clock and Reset

Table 2.2.2: Clock and Reset

Signal	Direction	Width	Description
clk	input	1	System clock
rst_n	input	1	Active-low asynchronous reset

7.3.2 APB Programming Interface

Table 2.2.3: APB Programming Interface

Signal	Direction	Width	Description
apb_valid	input	1	Kick-off valid (start descriptor chain)
apb_ready	output	1	Ready to accept kick-off
apb_addr	input	AW	First descriptor address

7.3.3 Configuration Interface

Table 2.2.4: Configuration Interface

Signal	Direction	Width	Description
cfg_prefetch_enable	input	1	Enable prefetch chaining (0 = on-demand, ~1 ahead)
cfg_fifo_threshold	input	4	Prefetch depth (descriptors buffered ahead when enabled)
cfg_addr0_base	input	AW	Valid address range 0 base
cfg_addr0_limit	input	AW	Valid address range 0 limit

Signal	Direction	Width	Description
cfg_addr1_base	input	AW	Valid address range 1 base
cfg_addr1_limit	input	AW	Valid address range 1 limit

7.3.4 AXI4 Read Master Interface

Table 2.2.5: AXI4 Read Master Interface

Signal	Direction	Width	Description
ar_valid	output	1	Address valid
ar_ready	input	1	Address ready
ar_addr	output	ADDR_WIDTH	Descriptor fetch address
ar_len	output	8	Burst length - 1
ar_size	output	3	Burst size (log2 bytes)
ar_burst	output	2	Burst type (INCR)
ar_id	output	AXI_ID_WIDTH	Transaction ID
ar_lock	output	1	Lock type
ar_cache	output	4	Cache attributes
ar_prot	output	3	Protection attributes
ar_qos	output	4	Quality of service
ar_region	output	4	Region identifier
r_valid	input	1	Data valid
r_ready	output	1	Data ready
r_data	input	256	Descriptor payload (fixed 256-bit)
r_resp	input	2	Read response

Signal	Direction	Width	Description
r_last	input	1	Last beat
r_id	input	AXI_ID_WIDTH	Response ID

7.3.5 Scheduler Interface

Table 2.2.6: Scheduler Interface

Signal	Direction	Width	Description
descriptor_valid	output	1	Descriptor valid
descriptor_ready	input	1	Scheduler ready
descriptor_packet	output	256	Parsed descriptor (fixed 256-bit)
descriptor_ext_packet	output	256	Extended chunk 1 (0 when unused)
descriptor_error	output	1	Error flag
descriptor_eos	output	1	End of Stream
descriptor_eol	output	1	End of Line
descriptor_eod	output	1	End of Data
descriptor_type	output	2	Packet type
channel_idle	input	1	Scheduler idle (enables APB kick-off)
descriptor_engine_idle	output	1	Engine idle

7.4 FSM States

7.4.1 Figure 2.2.2: Descriptor Engine FSM

States and transitions of the `descriptor_engine_beats` fetch FSM.

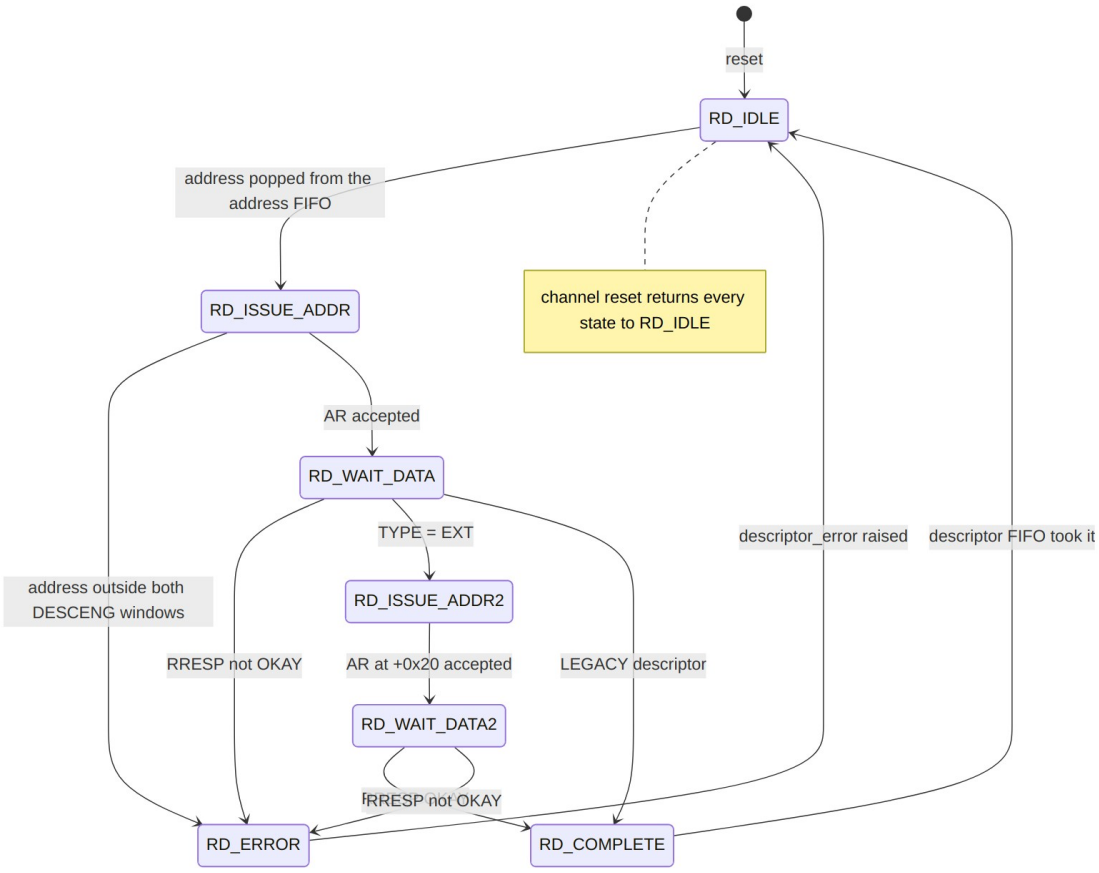
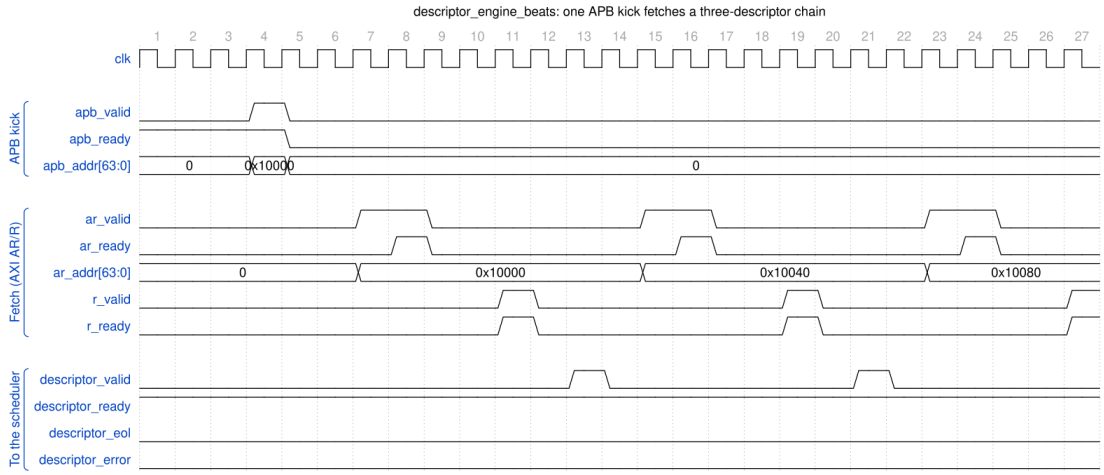


Diagram 1

7.5 Operation

7.5.1 Descriptor Chain Processing

7.5.2 Figure 2.2.3: Descriptor Chain Timing



Descriptor Engine - three-descriptor chain

Source: descriptor_engine_chain.json, captured from `dv/tests/fub_beats/test_descriptor_engine_beats.py` (`descriptor_chaining`, 32 channels, seed 7) with `WAVES=1`.

Reading it: one `apb_valid` kick carries the head address (0x10000). The engine issues an AXI read for it (`ar_valid`, `ar_addr`), the data returns on `r_valid`, and the parsed descriptor is presented on `descriptor_valid` to a ready scheduler. Because the descriptor's `next_ptr` points at 0x10040 and `last` is clear, the engine fetches the next one on its own – and the one after that at 0x10080 – with no further APB traffic; `descriptor_eol` marks the last of the chain. Each fetch costs the same AR -> R -> present latency, so a chain streams at one descriptor per fetch round trip.

7.6 Address Range Checking

Descriptors must fall within configured valid address ranges:

```
Valid if: (addr >= cfg_addr0_base && addr <= cfg_addr0_limit)
       || (addr >= cfg_addr1_base && addr <= cfg_addr1_limit)
```

Invalid addresses generate an error and halt the descriptor chain.

7.7 Prefetch and Autonomous Chaining

`cfg_prefetch_enable` and `cfg_fifo_threshold` are functional prefetch controls (previously declared-but-unused, which left chaining permanently greedy). They throttle how far ahead the engine fetches chained descriptors by comparing the output descriptor FIFO occupancy (`w_desc_fifo_count`) against a computed `w_prefetch_limit`:

```
if      (!cfg_prefetch_enable)      w_prefetch_limit = 1;   // on-
demand
else if (cfg_fifo_threshold == 0)   w_prefetch_limit = 1;   // guard
(0 would stall)
else                                w_prefetch_limit =
cfg_fifo_threshold;

w_prefetch_allows = (w_desc_fifo_count < w_prefetch_limit);
```

- **Prefetch OFF (`cfg_prefetch_enable = 0`):** limit forced to 1, so a chained fetch is only allowed when the output FIFO has drained – the engine stays roughly one descriptor ahead of the scheduler (on-demand).
- **Prefetch ON:** the engine buffers up to `cfg_fifo_threshold` descriptors ahead.

7.7.1 Chaining Decision

A chain fetch issues immediately when all of the following hold:

```
w_should_chain = w_chain_eligible     // valid non-last descriptor, in
range, no error
               && w_desc_committed      // current descriptor accepted
into output FIFO
               && w_prefetch_allows     // occupancy below prefetch
limit
               && w_desc_addr_fifo_wr_ready;  // address FIFO has a
slot
```

7.7.2 Deferred (Throttle-Blocked) Chain Fetch

If a descriptor commits and is chain-eligible but the fetch cannot issue this cycle (throttled by `w_prefetch_allows`, or the address FIFO is full), the owed fetch is recorded rather than dropped:

- `r_chain_pending` is set and the next-descriptor pointer is held in `r_pending_chain_addr`.
- The deferred fetch is reissued (`w_pending_push_fire`) once the output FIFO drains enough that `w_prefetch_allows` is true again and the address FIFO has a slot, clearing `r_chain_pending`.

Last Updated: 2026-07-02

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8 AXI Read Engine

Module: `axi_read_engine.sv` **Location:** `projects/components/dma-ip/rapids/rtl/fub/`
Status: Implemented **Last Updated:** 2026-09-30

8.1 Overview

The AXI read engine turns per-channel read requests from the schedulers into AXI4 read bursts and routes the returning data to the SRAM controller. It is a streaming pipeline with no state machine. The channel identity travels in the AXI ID.

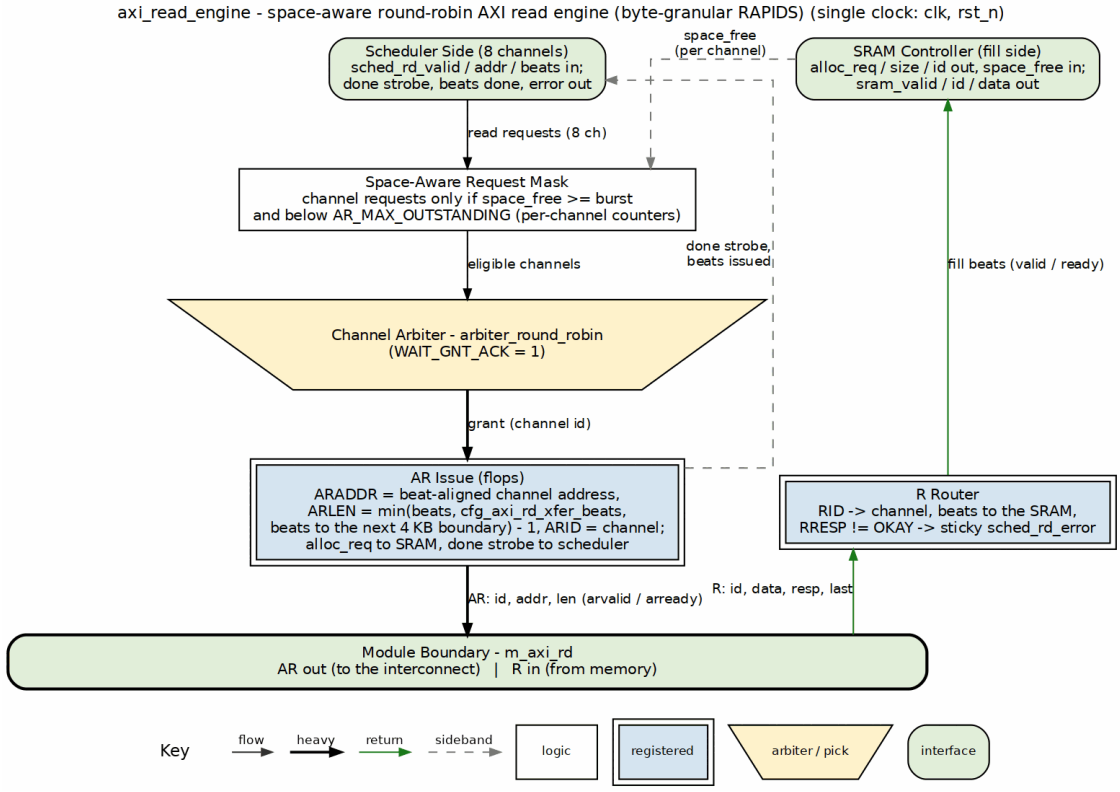
The engine is beat based, as in RAPIDS Beats. The scheduler has already converted the descriptor's byte length into a beat count. Two things change for byte-granular RAPIDS: the engine must cope with a channel address that is not beat aligned, and it must never let a burst cross a 4 KB boundary.

8.1.1 Key Features

- **Streaming pipeline:** no FSM, round-robin arbitration across channels.
- **Aligned AR address:** `ARADDR` is the channel address rounded down to a beat boundary. The lanes below the offset are read and ignored by the data path.
- **4 KB burst cap:** the burst length is limited so that no burst crosses a 4 KB boundary, whatever the start address.
- **Configurable burst length:** `cfg_axi_rd_xfer_beats`, clamped to what the buffer can hold (rapids BUG-009).
- **Space-checked:** a channel is only granted when the SRAM controller has room for the whole burst.
- **Sticky error:** SLVERR or DECERR on R sets `sched_rd_error` for the channel.

8.1.2 Block Diagram

8.1.3 Figure 2.3.1: AXI Read Engine Block Diagram



AXI read engine block diagram

Source: 05_axi_read_engine.dot

8.2 Parameters

Table 2.3.1: AXI Read Engine Parameters

Parameter	Default	Description
NUM_CHANNELS	8	Number of channels. Channel id is carried in the AXI ID.
ADDR_WIDTH	64	AXI address width.
DATA_WIDTH	512	AXI data width in bits. The Genesys 2 build uses 256.
ID_WIDTH	8	AXI ID width.
SEG_COUNT_WIDTH	8	Width of the per-channel free-space count from the SRAM controller. Sets the burst clamp.
PIPELINE	1	1 allows the next AR before the previous burst's data has returned. 0 waits for the burst to complete.
AR_MAX_OUTSTANDING	8	Maximum outstanding AR transactions.
STROBE_EVERY_BEAT	0	Declared for compatibility with STREAM. The completion strobe in this engine is generated on the AR handshake regardless of its value.

The module also accepts the short aliases NC, AW, DW, IW, SCW and CIW (the channel id width, $clog2(NUM_CHANNELS)).

8.2.1 Derived Constants

Table 2.3.2: AXI Read Engine Derived Constants

Constant	Definition	Meaning
BYTES_PER_BEAT	DW / 8	Bytes per beat.
AXSIZE	$clog2(BYTES_PER_BEAT)	Value driven on ARSIZE. Full-width

Constant	Definition	Meaning
		beats.
SD_BEATS	1 << (SCW - 1)	Buffer depth in beats implied by SEG_COUNT_WIDTH.
XFER_MAX	SD_BEATS < 256 ? SD_BEATS - 1 : 254	Largest burst length value the engine will use.

8.3 Port List

8.3.1 Clock, Reset and Configuration

Table 2.3.3: Clock, Reset and Configuration

Signal	Direction	Width	Description
clk	input	1	Clock.
rst_n	input	1	Active-low reset.
cfg_axi_rd_xfer_beats	input	8	Configured burst length (ARLEN value, 0 to 255). Clamped to XFER_MAX.
cfg_channel_reset	input	NC	Per-channel reset, level or pulse. Stops the channel's ARs, drains and discards its in-flight R data, and clears its error flag.

8.3.2 Scheduler Interface

Table 2.3.4: Scheduler Interface

Signal	Direction	Width	Description
sched_rd_valid	input	NC	Channel has a read request.
sched_rd_addr	input	NC x AW	Working source byte address.
sched_rd_beats	input	NC x 32	Beats still to read for the channel.
sched_rd_done_strobe	output	NC	Registered pulse, one cycle after the AR handshake for the channel.
sched_rd_beats_done	output	NC x 32	Beats issued by that AR (ARLEN + 1).
sched_rd_error	output	NC	Sticky read error for the channel, set by any non-OKAY R response.

The address is a byte address and may carry an offset in its low bits. The scheduler advances it to the next beat boundary after each done strobe, so only the first burst of a descriptor can see a non-zero offset.

8.3.3 SRAM Allocation and Fill Interface

Table 2.3.5: SRAM Allocation and Fill Interface

Signal	Direction	Width	Description
`axi_rd_alloc_req`	output	1	Registered pulse, one cycle after the AR handshake, reserving space for the burst.
`axi_rd_alloc_size`	output	8	Beats to reserve.
`axi_rd_alloc_id`	output	IW	Channel that owns the reservation.
`axi_rd_alloc_space_free`	input	NC x SCW	Free space per channel. A channel is granted only if it fits the burst.
`axi_rd_sram_valid`	output	1	Follows `m_axi_rvalid`.
`axi_rd_sram_ready`	input	1	SRAM controller accepts the beat. Drives `m_axi_rready`.
`axi_rd_sram_id`	output	IW	Follows `m_axi_rid`.
`axi_rd_sram_data`	output	DW	Follows `m_axi_rdata`.

8.3.4 AXI4 Read Master Interface

Table 2.3.6: AXI4 Read Master Interface

Signal	Direction	Width	Description
`m_axi_arid`	output	IW	Read address ID (the channel).
`m_axi_araddr`	output	AW	Beat-aligned read address.
`m_axi_arlen`	output	8	Burst length

Signal	Direction	Width	Description
			minus one.
m_axi_arsize	output	3	AXSIZE, full-width beats.
m_axi_arburst	output	2	INCR.
m_axi_arvalid	output	1	Address valid.
m_axi_arready	input	1	Address ready.
m_axi_rid	input	IW	Response ID.
m_axi_rdata	input	DW	Read data.
m_axi_rresp	input	2	Read response.
m_axi_rlast	input	1	Last beat of the burst.
m_axi_rvalid	input	1	Data valid.
m_axi_rready	output	1	Data ready.

8.3.5 Debug

Table 2.3.7: Debug Outputs

Signal	Direction	Width	Description
dbg_rd_all_complete	output	NC	Channel has no outstanding reads.
dbg_r_beats_rcvd	output	32	Total R beats received.
dbg_sram_writes	output	32	Total beats written to the SRAM controller.
dbg_arb_request	output	NC	Arbiter request vector.

8.4 Operation

8.4.1 Burst Length Cap (rapids BUG-009)

`cfg_axi_rd_xfer_beats` is an ARLEN value, but a burst must fit in the buffer that stages it. The engine clamps the configured value to `XFER_MAX`, giving the working limit `w_xfer_cfg`. Without the clamp an ARLEN of 255 wraps the 8-bit `ARLEN + 1` and a channel is granted with no space reserved. This is the same fix as in RAPIDS Beats.

8.4.2 Byte-Granular Burst Sizing

The engine sizes each burst from the channel's current byte address and remaining beats. Three steps produce `ARLEN`:

1. Distance to the next 4 KB boundary, in beats: `w_beats_to_4k = (4096 - {sched_rd_addr[11:AXSIZE], AXSIZE'b0}) >> AXSIZE`. The offset bits are dropped first, so the distance is measured from the aligned beat.
2. Burst cap: `w_cap_beats = min(w_xfer_cfg + 1, w_beats_to_4k)`.
3. Burst length: `w_transfer_size = (sched_rd_beats <= w_cap_beats) ? sched_rd_beats - 1 : w_cap_beats - 1`.

`m_axi_araddr` is `{sched_rd_addr[grant][AW-1:AXSIZE], AXSIZE'b0}`. The offset never reaches the bus. The first beat of the burst carries the bytes before the offset, and the data path discards them.

Because the cap always rounds toward the boundary, a burst can be shorter than the configured length, and a later burst restarts on the boundary. The scheduler needs no knowledge of this. It only sees `beats_done` per burst.

8.4.3 Streaming Pipeline

An arbiter picks a channel that has `sched_rd_valid` set and enough `axi_rd_alloc_space_free` for its burst. When the AR handshakes, the engine pulses `axi_rd_alloc_req` one cycle later with the burst size so the SRAM controller reserves the space, and pulses `sched_rd_done_strobe` for the channel. R beats pass straight through: `axi_rd_sram_valid`, `axi_rd_sram_id` and `axi_rd_sram_data` follow the R channel, and `m_axi_rready` is `axi_rd_sram_ready`.

With `PIPELINE = 1` the next AR for a channel can issue while earlier data is still in flight. With `PIPELINE = 0` the channel waits for the burst to finish. The top and core default to 1.

8.4.4 Figure 2.3.2: Burst Segmentation with a Mid-Beat Start

The example uses 32-byte beats (OFF_W 5) and a configured burst of 16 beats. A descriptor reads 8,200 bytes from address 0xFF8. The scheduler computes offset 0x18 (24) and `beats_total = (24 + 8200 + 31) >> 5 = 257`.

Table 2.3.8: Burst Segmentation Example

Burst	Channel address	ARADDR	Beats to 4 KB	ARLEN	Beats left after
1	0x0FF8	0x0FE0	1	0	256
2	0x1000	0x1000	128	15	240
3	0x1200	0x1200	112	15	224
…				15	continues

The first burst is one beat because the aligned start is 32 bytes below the 4 KB boundary. From then on each burst is the configured 16 beats until the remainder runs out. The scheduler advances its address to the next beat boundary after each done strobe, which is why burst 2 starts at 0x1000, not 0x0FF8 plus a beat.

8.4.5 Done Strobe

The engine pulses `sched_rd_done_strobe[ch]` once per AR handshake, with `sched_rd_beats_done[ch]` equal to `ARLEN + 1`. The scheduler subtracts this from its remaining count and advances the address. The strobe therefore reports beats issued, not beats returned.

8.5 Error Handling

Table 2.3.9: Error Handling

Error	Detection	Response
AXI SLVERR	`m_axi_rresp == 2'b10` on an accepted beat	Set sticky `sched_rd_error` for the channel
AXI DECERR	`m_axi_rresp == 2'b11` on an accepted beat	Set sticky `sched_rd_error` for the channel
Any other non-zero response	`m_axi_rresp != 2'b00` on an accepted beat	Same: the test is not-OKAY, and the channel is taken from `RID`

The scheduler turns the sticky error into its ERROR state and a MonBus event. The engine does not abort the burst: the remaining R beats still pass through, so the interconnect is not left mid-burst.

8.5.1 Channel Reset

`cfg_channel_reset[ch]` clears the sticky `sched_rd_error[ch]`, so a channel reset recovers a read error without `aresetn`. The reset also has to settle AR bursts that are already on the bus, because an AXI master cannot recall them:

- A channel in reset, or flushing, issues no new AR and does not enter the allocation pipeline.
- `r_rd_flush[ch]` is set by the reset and holds until the channel has no burst outstanding. While it holds, each R beat whose `RID` names the channel is accepted (`m_axi_rready` is forced high) and dropped: `axi_rd_sram_valid` stays low, so nothing reaches the SRAM, and its `RRESP` does not set the error flag.
- The completion strobe and pending allocation of the channel are cleared, so the scheduler does not see a stale done pulse after the reset.
- Other channels are not affected: every term is indexed by channel, and R beats of other channels pass as usual.

The channel's SRAM partition is cleared by the SRAM controller wrapper (see Sink Data Path and Source Data Path AXIS). The discard is what guarantees that no beat of the reset channel reaches the partition after it has been cleared.

8.6 Differences from RAPIDS Beats

Table 2.3.10: Read Engine Delta

Item	RAPIDS Beats	RAPIDS
ARADDR	Channel address as given (always aligned)	Aligned down to the beat
Burst cap	Configured length and remaining beats	Also capped at the 4 KB boundary
Everything else		Unchanged

The beat-count chapter for the beats engine is AXI Read Engine (RAPIDS Beats). The scheduler that feeds this engine is described in Scheduler.

Last Updated: 2026-09-30

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9 AXI Write Engine

Module: `axi_write_engine.sv` **Location:** `projects/components/dma-ip/rapids/rtl/fub/`
Status: Implemented **Last Updated:** 2026-09-30

9.1 Overview

The AXI write engine drains data from the SRAM controller and issues AXI4 write bursts to system memory. It is a streaming pipeline with no state machine. It also tracks the write responses, which is how the scheduler learns that data has been committed.

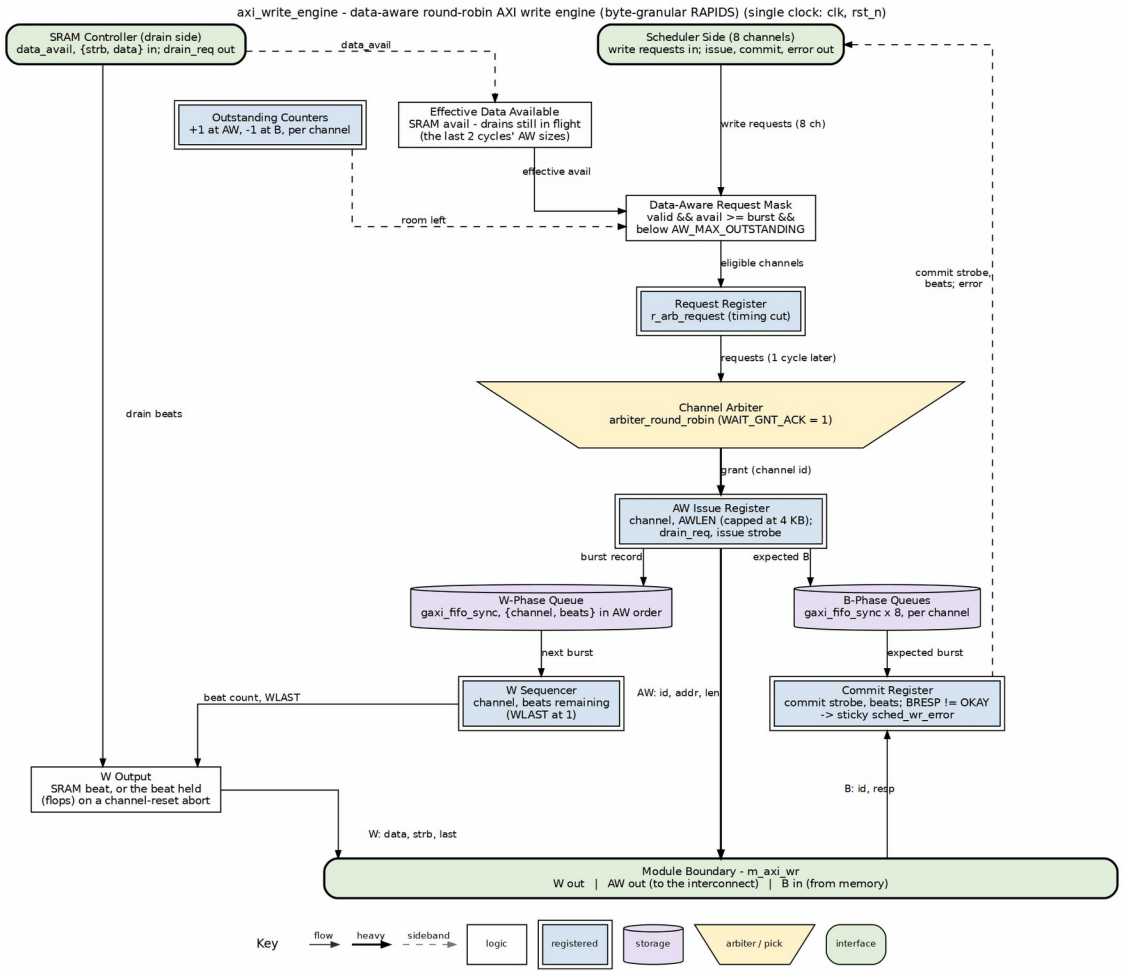
Byte-granular RAPIDS changes the engine in three places. Write strobes now come from the SRAM instead of being all ones, so a partial first or last beat writes only the bytes that belong to the packet. Burst lengths are capped at the 4 KB boundary. `AWADDR` is beat aligned.

9.1.1 Key Features

- **Streaming pipeline:** no FSM, round-robin arbitration across channels.
- **Real write strobes:** `m_axi_wstrb` is the strobe word stored beside each data beat (`axi_wr_sram_strb`). Lanes outside the packet are not written.
- **Aligned AW address:** `AWADDR` is the channel address rounded down to a beat boundary.
- **4 KB burst cap:** the same cap as the read engine, applied to both the burst length and the final-burst test.
- **Configurable burst length:** `cfg_axi_wr_xfer_beats`, clamped to what the buffer can hold (rapids BUG-009).
- **Two completion strobes:** done on the AW handshake, commit on the B response.
- **Sticky error:** a non-OKAY B response sets `sched_wr_error`.

9.1.2 Block Diagram

9.1.3 Figure 2.4.1: AXI Write Engine Block Diagram



AXI write engine block diagram

Source: 06_axi_write_engine.dot

9.2 Parameters

Table 2.4.1: AXI Write Engine Parameters

Parameter	Default	Description
NUM_CHANNELS	8	Number of channels. Channel id is carried in the AXI ID.
ADDR_WIDTH	64	AXI address width.
DATA_WIDTH	512	AXI data width in bits. The Genesys 2 build uses 256.
ID_WIDTH	8	AXI ID width.
USER_WIDTH	8	Width of `m_axi_wuser`, which carries the active channel.
SEG_COUNT_WIDTH	8	Width of the per-channel available-data count from the SRAM controller. Sets the burst clamp.
PIPELINE	1	1 allows further AWs before earlier B responses return. 0 waits for the B.
AW_MAX_OUTSTANDING	8	Maximum outstanding AW transactions per channel.
W_PHASE_FIFO_DEPTH	64	Depth of the W-phase transaction FIFO, kept in order with the AW issue.
B_PHASE_FIFO_DEPTH	16	Depth of the B-phase transaction FIFO used to match responses.

The module also accepts the short aliases `NC`, `AW`, `DW`, `IW`, `UW`, `SCW` and `CIW` (the channel id width).

9.2.1 Derived Constants

Table 2.4.2: AXI Write Engine Derived Constants

Constant	Definition	Meaning
BYTES_PER_BEAT	`DW / 8`	Bytes per beat.
AXSIZE	`$clog2(BYTES_PER_BEAT)`	Value driven on

Constant	Definition	Meaning
XFER_MAX	min(2^(SCW-1) - 1, 254)	AWSIZE. Largest burst length value the engine will use.

9.3 Port List

9.3.1 Clock, Reset and Configuration

Table 2.4.3: Clock, Reset and Configuration

Signal	Direction	Width	Description
`clk`	input	1	Clock.
`rst_n`	input	1	Active-low reset.
`cfg_axi_wr_xfer_beats`	input	8	Configured burst length (AWLEN value, 0 to 255). Clamped to `XFER_MAX`.
`cfg_channel_reset`	input	NC	Per-channel reset, level or pulse. Stops new bursts, completes open bursts with null beats, and clears the channel's error flag.

9.3.2 Scheduler Interface

Table 2.4.4: Scheduler Interface

Signal	Direction	Width	Description
`sched_wr_valid`	input	NC	Channel has a write request.
`sched_wr_ready`	output	NC	Pulses when the B response of the last burst of a descriptor arrives.
`sched_wr_addr`	input	NC x AW	Working destination byte address.
`sched_wr_beats`	input	NC x 32	Beats still to write for the channel.
`sched_wr_burst_len`	input	NC x 8	Requested burst length. Present on the port; the engine sizes bursts from `cfg_axi_wr_xfer_beats` and the 4 KB cap.
`sched_wr_don`	output	NC	Pulses on the AW

Signal	Direction	Width	Description
e_strobe			handshake for the channel.
sched_wr_beats_done	output	NC x 32	Beats issued by that AW (AWLEN + 1).
sched_wr_commit_strobe	output	NC	Pulses when the B response for the channel arrives.
sched_wr_commit_beats	output	NC x 32	Beats confirmed by that B response, recovered from the B-phase FIFO.
sched_wr_error	output	NC	Sticky write error for the channel.

The done strobe advances the scheduler's destination address, which lets the next AW issue without waiting for a response. The commit strobe is what the scheduler counts when it decides the descriptor is complete. Both strobes are per channel and report beat counts, not bytes.

9.3.3 SRAM Reservation and Drain Interface

Table 2.4.5: SRAM Drain Interface

Signal	Direction	Width	Description
axi_wr_drain_req	output	NC	Pulses for the granted channel on the AW handshake.
axi_wr_drain_size	output	NC x 8	Beats to reserve (AWLEN + 1).
axi_wr_drain_data_avail	input	NC x SCW	Data available per channel after reservations.
axi_wr_sram_valid	input	NC	Per-channel valid, registered. Used for arbitration.
axi_wr_sram_valid_comb	input	NC	Per-channel valid, combinational. Gates m_axi_wvalid.
axi_wr_sram_	output	1	Drain request. Equal to

Signal	Direction	Width	Description
drain			wvalid && wready.
axi_wr_sram_id	output	CIW	Channel selected for the drain.
axi_wr_sram_data	input	DW	Data from the selected channel.
axi_wr_sram_strb	input	DW/8	Byte strobes stored with that data beat. New in RAPIDS.

9.3.4 AXI4 Write Master Interface

Table 2.4.6: AXI4 Write Master Interface

Signal	Direction	Width	Description
m_axi_awid	output	IW	Write address ID (the channel).
m_axi_awaddr	output	AW	Beat-aligned write address.
m_axi_awlen	output	8	Burst length minus one.
m_axi_awsize	output	3	AXSIZE, full-width beats.
m_axi_awburst	output	2	INCR.
m_axi_awvalid	output	1	Address valid.
m_axi_awready	input	1	Address ready.
m_axi_wdata	output	DW	Write data.
m_axi_wstrb	output	DW/8	Write strobes, equal to axi_wr_sram_strb.
m_axi_wlast	output	1	Last beat of the burst.
m_axi_wuser	output	UW	Channel of the beat being written.
m_axi_wvalid	output	1	Data valid.
m_axi_wready	input	1	Data ready.
m_axi_bid	input	IW	Response ID.
m_axi_bresp	input	2	Write response.

Signal	Direction	Width	Description
m_axi_bvalid	input	1	Response valid.
m_axi_bready	output	1	Response ready.

9.3.5 Debug and Status

Table 2.4.7: Debug and Status Outputs

Signal	Direction	Width	Description
dbg_wr_all_complete	output	NC	Channel has no outstanding writes.
dbg_aw_transactions	output	32	Total AW handshakes.
dbg_w_beats	output	32	Total W beats.
o_active_channel_id	output	CIW	Channel whose burst is driving, or about to drive, the W bus.
o_active_channel_valid	output	1	o_active_channel_id is meaningful.

9.4 Operation

9.4.1 Burst Length Cap (rapids BUG-009)

`cfg_axi_wr_xfer_beats` is an AWLEN value, but a burst cannot be larger than the SRAM that stages it: the engine waits for the whole burst to be present before issuing the AW, and a burst the buffer cannot hold would wait forever. The engine clamps the configured value to `XFER_MAX`. Without the clamp an AWLEN of 255 wraps the 8-bit `AWLEN + 1` and the engine issues an AW on an empty buffer. This is the same fix as in RAPIDS Beats.

9.4.2 Byte-Granular Burst Sizing

The sizing follows the read engine:

1. `w_beats_to_4k` is the distance from the aligned channel address to the next 4 KB boundary, in beats.
2. `w_cap_beats = min(w_xfer_cfg + 1, w_beats_to_4k)`.
3. `w_transfer_size` (the AWLEN) is `sched_wr_beats - 1` if `sched_wr_beats <= w_cap_beats`, otherwise `w_cap_beats - 1`.

A channel may issue when the buffer holds at least `w_transfer_size + 1` beats (`w_has_data`), or when this is the final burst. The final-burst test is `sched_wr_beats > 0`, `sched_wr_beats <= w_cap_beats`, and the available data covers all remaining beats. The 4 KB cap therefore applies to both tests, so a descriptor that ends just past a boundary still finishes on a short final burst.

`m_axi_awaddr` is `{sched_wr_addr[ch][AW-1:AXSIZE], AXSIZE'b0}`. The offset of a descriptor that starts mid-beat never reaches the bus. The first W beat of that descriptor carries `WSTRB` bits only for the bytes at and above the offset, and the last carries only the bytes up to the end of the packet.

9.4.3 Write Strobes

Each SRAM word holds a data beat and its strobe word side by side (see Sink Data Path). The write engine takes both from the drain port and presents them together, so the strobe always matches the data. In RAPIDS Beats `WSTRB` was tied high.

9.4.4 Streaming Pipeline

Each channel with `sched_wr_valid` and a burst that passes the data test requests arbitration. On grant the AW is issued, and on the AW handshake the engine pulses `axi_wr_drain_req` with the burst size so the drain controller reserves the beats. W beats stream from the drain port that `axi_wr_sram_id` selects. `axi_wr_sram_drain` equals `m_axi_wvalid && m_axi_wready`. `m_axi_wlast` closes each burst, and `m_axi_wuser` carries the channel.

With `PIPELINE = 1` a channel may issue further AWs before earlier B responses return, up to `AW_MAX_OUTSTANDING`. With `PIPELINE = 0` the channel waits for each B. That costs throughput, so the top and core default to 1.

9.4.5 Two Completion Strobes

Table 2.4.8: Completion Strobes

Strobe	Raised when	Scheduler use
`sched_wr_done_strobe`	AW handshake for the channel	Advance the destination address and the issued-beat count.
`sched_wr_commit_strobe`	B response for the channel	Count committed beats. The descriptor is complete when all beats are committed.

`sched_wr_ready` is a third, coarser pulse: it fires when the B response belongs to the last burst of a descriptor.

9.5 Error Handling

Table 2.4.9: Error Handling

Error	Detection	Response
AXI SLVERR	`m_axi_bresp == 2'b10`	Set sticky `sched_wr_error` for the channel
AXI DECERR	`m_axi_bresp == 2'b11`	Set sticky `sched_wr_error` for the channel
Any other non-zero response	`m_axi_bresp != 2'b00`	Same: the test is not-OKAY, and the channel is taken from `BID`

The sink data path ORs `sched_wr_error` with its own packet-length error before passing it to the scheduler.

9.5.1 Channel Reset

`cfg_channel_reset[ch]` clears the sticky `sched_wr_error[ch]`, so a channel reset recovers a B-response error without `aresetn`. A burst whose AW has been accepted must still receive all of its W beats, so the engine settles in-flight work rather than dropping it:

- `w_kill[ch]` is the reset level ORed with `r_wr_flush[ch]`. The flush bit is set by the reset and holds until the channel has no open burst and no B response due.
- No new AW is arbitrated for a channel with `w_kill` set. `AWADDR` comes from `sched_wr_addr`, which the scheduler does not clear, so the value on the bus stays stable across the reset.
- An open burst is completed with null W beats: `WSTRB` is zero, `WDATA` repeats the last captured value and the SRAM drain is not popped, so the null beats modify no byte at the destination. A beat that was presented on W and not accepted when the reset hit is the one exception: it is replayed unchanged, with its own strobes, from a captured copy, so the W channel does not change its payload while `WVALID` is high. At most one beat per reset is replayed this way.
- B responses of a channel with `w_kill` set are consumed. The commit strobe, `sched_wr_ready`, the done strobe and the error flag are suppressed for them.
- Other channels are not affected.

The reset therefore leaves no partial state in the engine: the B-phase FIFO drains through the flush, and the channel restarts with clean counters when the scheduler next raises `sched_wr_valid`.

9.6 Differences from RAPIDS Beats

Table 2.4.10: Write Engine Delta

Item	RAPIDS Beats	RAPIDS
`WSTRB`	All ones	`axi_wr_sram_strb`
SRAM drain port	Data only	Data plus strobe word
`AWADDR`	Channel address as given (always aligned)	Aligned down to the beat
Burst cap and final-burst test	Configured length and remaining beats	Also capped at the 4 KB boundary
Everything else		Unchanged

The beats chapter is AXI Write Engine (RAPIDS Beats). The read-side counterpart is AXI Read Engine.

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10 Beats Alloc Control Specification

Module: `alloc_ctrl_beats.sv` **Location:** `projects/components/dma-ip/rapids/rtl/fub_beats/` **Status:** Implemented **Last Updated:** 2025-01-10

> **No longer in the SRAM datapath (2026-09-26).** RAPIDS' SRAM path now uses STREAM's `stream_alloc_ctrl` / `stream_drain_ctrl` / `stream_latency_bridge`, reached through the `snk_/src_sram_controller_beats` naming wrappers (`bdf4e0dff`). This FUB is still built and still verified – it keeps its own test, its own filelist and its `rapids_all.f` entry – it simply is not instantiated by the SRAM controllers any more. Treat this page as the FUB's own specification, not as a description of the current SRAM path.

10.1 Overview

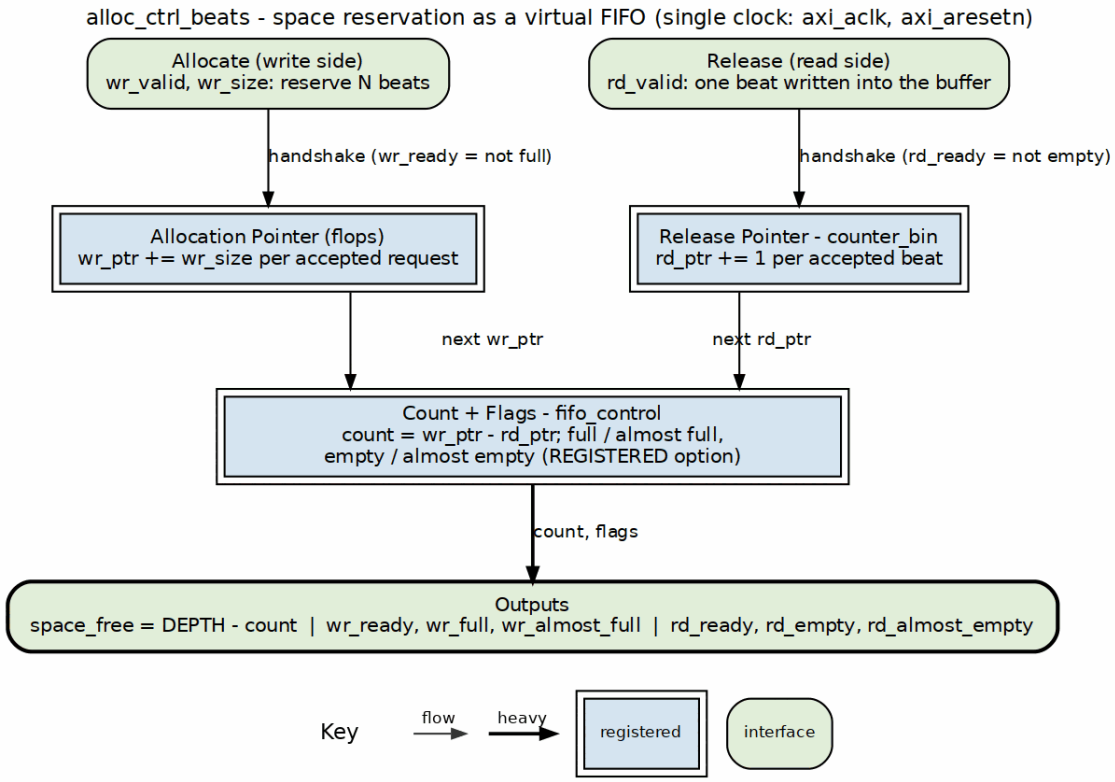
The Beats Alloc Control is a “virtual FIFO” that tracks space allocation without storing actual data. It uses FIFO pointer logic to manage pre-allocation for AXI engines, preventing race conditions between allocation and actual data arrival.

10.1.1 Key Features

- **Virtual FIFO:** Pointer tracking without data storage
- **Variable-Size Allocation:** Supports multi-beat allocation requests
- **Single-Beat Release:** Data writes release one beat at a time
- **Status Flags:** Full, almost full, empty, almost empty
- **Space Tracking:** Reports available space for flow control

10.1.2 Block Diagram

10.1.3 Figure 2.5.1: Beats Alloc Control Block Diagram



Alloc control block diagram

Source: 05_alloc_ctrl.dot

10.2 Concept: Virtual FIFO

Unlike a traditional FIFO that stores data, alloc_ctrl_beats only tracks **space reservations**:

	Traditional FIFO	Virtual FIFO (alloc_ctrl_beats)
Write side	`wr_data` into storage	`wr_size` advances a pointer
Read side	`rd_data` out of storage	one beat advances a pointer
Output	data	`space_free`

10.2.1 Use Case: Pre-Allocation for AXI Reads

```
1. AXI engine requests N beats of space
   - alloc_ctrl.wr_valid = 1
   - alloc_ctrl.wr_size = N
   - wr_ptr advances by N

2. As AXI read data arrives (1 beat at a time)
   - alloc_ctrl.rd_valid = 1 (each beat)
   - rd_ptr advances by 1

3. Space becomes available again
   - space_free reflects available capacity
```

10.3 Parameters

```
parameter int DEPTH = 512;                    // Virtual FIFO depth
parameter int ALMOST_WR_MARGIN = 1;           // Almost full margin
parameter int ALMOST_RD_MARGIN = 1;           // Almost empty margin
parameter int REGISTERED = 1;                 // Registered outputs

// Derived
parameter int AW = $clog2(DEPTH);             // Address width
```

: Table 2.5.1: Beats Alloc Control Parameters

10.4 Port List

10.4.1 Clock and Reset

Table 2.5.2: Clock and Reset

Signal	Direction	Width	Description
`axi_aclk`	input	1	System clock
`axi_aresetn`	input	1	Active-low reset

10.4.2 Write Interface (Allocation Requests)

Table 2.5.3: Write Interface

Signal	Direction	Width	Description
`wr_valid`	input	1	Allocate space
`wr_size`	input	8	Number of beats to allocate
`wr_ready`	output	1	Space available for allocation

10.4.3 Read Interface (Data Written)

Table 2.5.4: Read Interface

Signal	Direction	Width	Description
`rd_valid`	input	1	One beat of data written
`rd_ready`	output	1	Not full (can accept)

10.4.4 Status Outputs

Table 2.5.5: Status Outputs

Signal	Direction	Width	Description
`space_free`	output	AW+1	Available space in beats
`wr_full`	output	1	No space

Signal	Direction	Width	Description
			available
wr_almost_full	output	1	Near full
rd_empty	output	1	No allocations pending
rd_almost_empty	output	1	Near empty

10.5 Operation

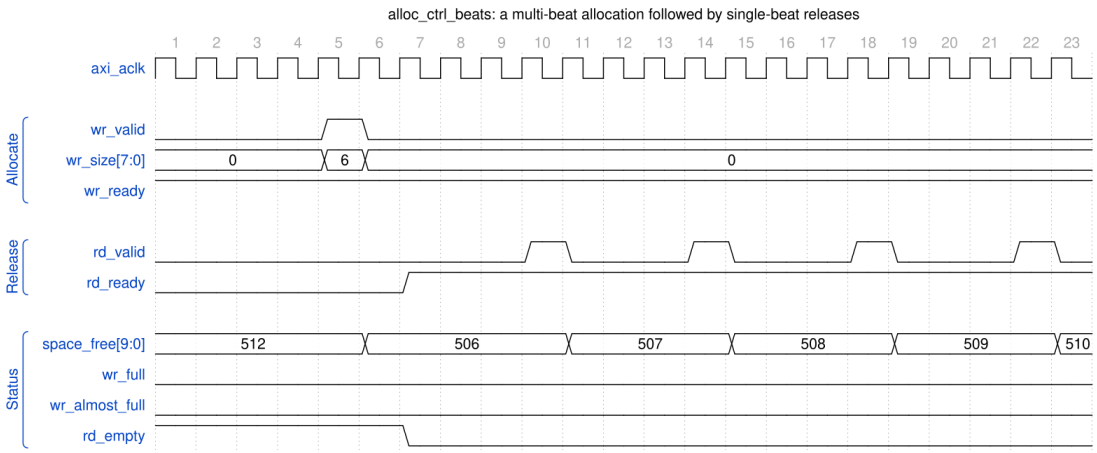
10.5.1 Pointer Arithmetic

```
space_free = DEPTH - (wr_ptr - rd_ptr)

Full condition:    wr_ptr - rd_ptr >= DEPTH
Empty condition:   wr_ptr == rd_ptr
```

10.5.2 Timing Diagram

10.5.3 Figure 2.5.2: Allocation and Release Timing



Allocation control - allocate then release

Source: alloc_ctrl_beats_alloc_release.json, captured from `dv/tests/fub_beats/test_alloc_ctrl_beats.py` (`basic_alloc_drain`, depth 512, seed 7) with `WAVES=1`.

Reading it: one `wr_valid` handshake with `wr_size = 6` reserves six beats and `space_free` drops 512 to 506 the next cycle; `rd_empty` falls because reserved space is now outstanding. Each later `rd_valid` handshake releases one beat and `space_free` climbs back one at a time (507, 508, ...). `wr_full` and `wr_almost_full` stay low: the virtual FIFO is nowhere near its depth.

10.6 Integration Example

```
// Source path: Pre-allocate SRAM space before AXI read
alloc_ctrl_beats #(
    .DEPTH(SRAM_DEPTH),
    .ALMOST_WR_MARGIN(8)
) u_src_alloc (
    .axi_aclk       (clk),
    .axi_aresetn    (rst_n),

    // Scheduler allocates space before issuing AXI read
    .wr_valid       (sched_rd_valid),
    .wr_size        (sched_rd_beats[7:0]),
    .wr_ready       (alloc_ready),

    // AXI read data arrival releases allocation
    .rd_valid       (m_axi_rvalid && m_axi_rready),
    .rd_ready       (),  // Not used for single-beat

    // Status
    .space_free     (sram_space_available),
    .wr_full        (sram_full),
    .wr_almost_full (sram_almost_full)
);
```

Last Updated: 2025-01-10

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11 Beats Drain Control Specification

Module: `drain_ctrl_beats.sv` **Location:** `projects/components/dma-ip/rapids/rtl/fub_beats/` **Status:** Implemented **Last Updated:** 2025-01-10

> **No longer in the SRAM datapath (2026-09-26).** RAPIDS' SRAM path now uses STREAM's `stream_alloc_ctrl` / `stream_drain_ctrl` / `stream_latency_bridge`, reached through the `snk_/src_sram_controller_beats` naming wrappers (`bdf4e0dff`). This FUB is still built and still verified – it keeps its own test, its own filelist and its `rapids_all.f` entry – it simply is not instantiated by the SRAM controllers any more. Treat this page as the FUB's own specification, not as a description of the current SRAM path.

11.1 Overview

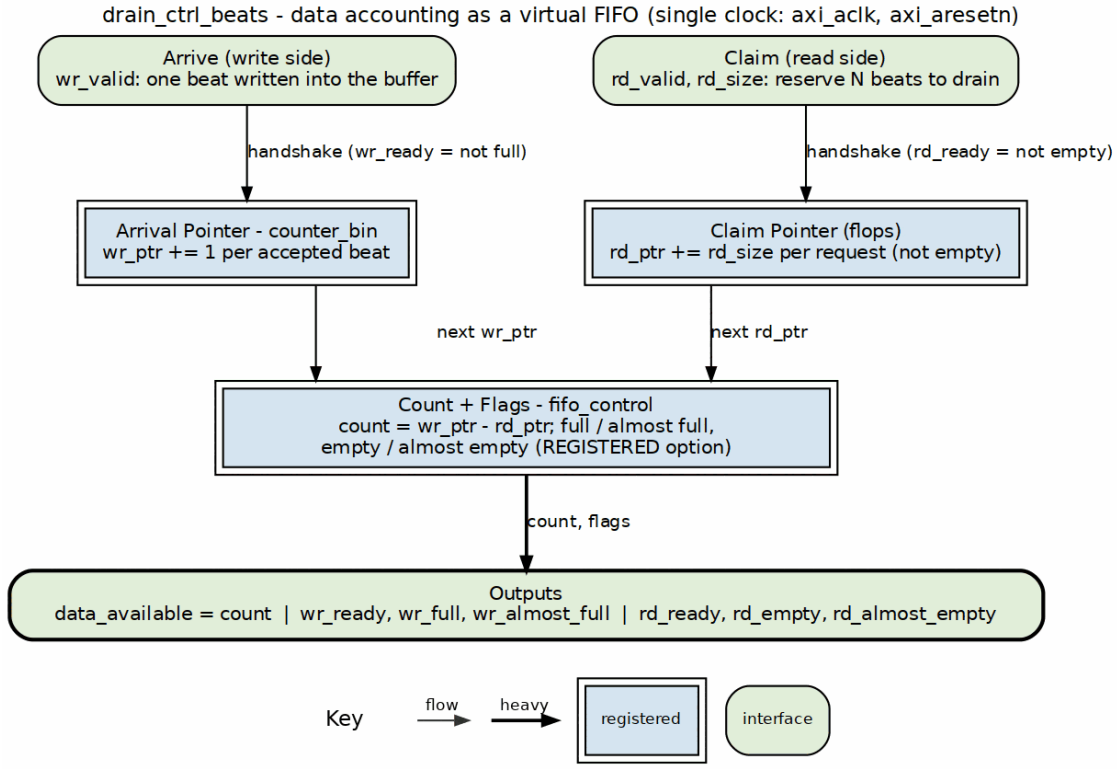
The Beats Drain Control is a “virtual FIFO” that tracks data availability without storing actual data. It complements alloc_ctrl_beats by tracking when data has been written to SRAM and is available for draining.

11.1.1 Key Features

- **Virtual FIFO:** Pointer tracking without data storage
- **Single-Beat Write:** Data arrivals tracked one beat at a time
- **Variable-Size Drain:** Supports multi-beat drain requests
- **Status Flags:** Full, almost full, empty, almost empty
- **Data Availability:** Reports beats available for drain

11.1.2 Block Diagram

11.1.3 Figure 2.6.1: Beats Drain Control Block Diagram



Drain control block diagram

Source: 06_drain_ctrl.dot

11.2 Concept: Data Availability Tracking

The drain_ctrl tracks when data has been written to SRAM:

```
alloc_ctrl:  Tracks SPACE reservations (allocation -> actual write)
drain_ctrl:  Tracks DATA availability (data write -> drain request)
```

11.2.1 Complementary Roles

```
Source Path:
  1. Scheduler allocates space (alloc_ctrl.wr)
  2. AXI reads data into SRAM
  3. Data written to SRAM (drain_ctrl.wr)
  4. Drain consumer reads (drain_ctrl.rd)

Sink Path:
  1. Fill allocates space (alloc_ctrl.wr)
  2. Fill writes to SRAM (alloc_ctrl.rd)
  3. Data available in SRAM (drain_ctrl.wr)
  4. AXI write drains SRAM (drain_ctrl.rd)
```

11.3 Parameters

```
parameter int DEPTH = 512;                    // Virtual FIFO depth
parameter int ALMOST_WR_MARGIN = 1;           // Almost full margin
parameter int ALMOST_RD_MARGIN = 1;           // Almost empty margin
parameter int REGISTERED = 1;                 // Registered outputs

// Derived
parameter int AW = $clog2(DEPTH);             // Address width
```

: Table 2.6.1: Beats Drain Control Parameters

11.4 Port List

11.4.1 Clock and Reset

Table 2.6.2: Clock and Reset

Signal	Direction	Width	Description
`axi_aclk`	input	1	System clock
`axi_aresetn`	input	1	Active-low reset

11.4.2 Write Interface (Data Written to SRAM)

Table 2.6.3: Write Interface

Signal	Direction	Width	Description
`wr_valid`	input	1	One beat written to SRAM
`wr_ready`	output	1	Can track more data

11.4.3 Read Interface (Drain Requests)

Table 2.6.4: Read Interface

Signal	Direction	Width	Description
`rd_valid`	input	1	Drain request
`rd_size`	input	8	Number of beats to drain
`rd_ready`	output	1	Data available for drain

11.4.4 Status Outputs

Table 2.6.5: Status Outputs

Signal	Direction	Width	Description
`data_available`	output	AW+1	Beats available to drain
`wr_full`	output	1	Tracking full

Signal	Direction	Width	Description
wr_almost_full	output	1	Near full
rd_empty	output	1	No data to drain
rd_almost_empty	output	1	Near empty

11.5 Operation

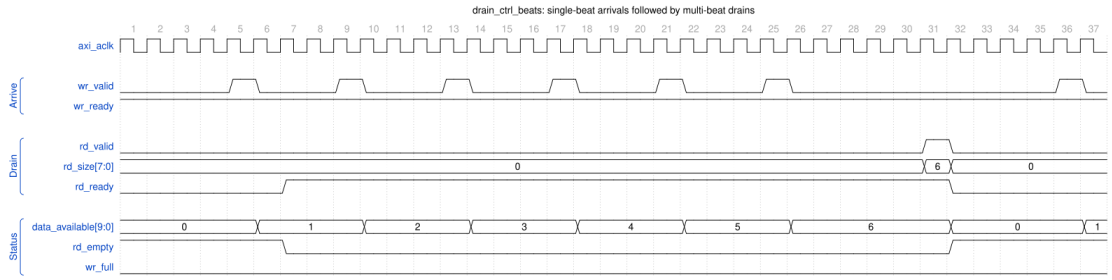
11.5.1 Pointer Arithmetic

```
data_avail = wr_ptr - rd_ptr

Full condition:    wr_ptr - rd_ptr >= DEPTH
Empty condition:   wr_ptr == rd_ptr
```

11.5.2 Timing Diagram

11.5.3 Figure 2.6.2: Data Arrival and Drain Timing



Drain control - single-beat arrivals then a multi-beat drain

Source: drain_ctrl_beats_arrive_drain.json, captured from `dv/tests/fub_beats/test_drain_ctrl_beats.py` (`basic_write_drain`, depth 512, seed 7) with `WAVES=1`.

Reading it: six `wr_valid` handshakes, one beat each, raise `data_available` 1, 2, ... 6 and drop `rd_empty` on the first arrival. A single `rd_valid` handshake with `rd_size = 6` then reserves all six at once: `data_available` returns to 0 and `rd_empty` rises the next cycle. The counter tracks beats, not bytes, and never touches the data itself.

11.6 Integration Example

```
// Sink path: Track data availability for AXI write engine
drain_ctrl_beats #(
    .DEPTH(SRAM_DEPTH),
    .ALMOST_RD_MARGIN(8)
) u_snk_drain (
    .axi_aclk       (clk),
    .axi_aresetn    (rst_n),

    // Fill writes data to SRAM
    .wr_valid       (fill_valid && fill_ready),
    .wr_ready       (),  // Single-beat, always ready

    // AXI write engine drains data
    .rd_valid       (axi_wr_drain_req),
    .rd_size        (axi_wr_burst_len),
    .rd_ready       (data_ready_for_drain),

    // Status
    .data_available (beats_to_write),
    .rd_empty       (no_data_to_write)
);
```

11.7 Relationship: alloc_ctrl vs drain_ctrl

Table 2.6.6: alloc_ctrl vs drain_ctrl Comparison

Aspect	alloc_ctrl_beats	drain_ctrl_beats
Tracks	Space reservations	Data availability
Write port	Multi-beat allocation	Single-beat arrival
Read port	Single-beat release	Multi-beat drain
Output	`space_free`	`data_available`
Use case	Pre-allocate before fill	Know when to drain

Last Updated: 2025-01-10

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12 Latency Bridge (beats) Specification

Module: `latency_bridge_beats.sv` **Location:** `projects/components/dma-ip/rapids/rtl/fub_beats/` **Status:** Implemented, tested, not in the SRAM path **Last Updated:** 2026-09-27

> **Not in the SRAM datapath since 2026-09-26.** RAPIDS' SRAM path is STREAM's `sram_controller`, which carries its own `stream_latency_bridge` (the same design), reached through the `snk_/src_sram_controller_beats` naming wrappers (`bdf4e0dff`). This FUB is still built and verified – it keeps its test, its filelist and its `rapids_all.f` entry – but nothing in `rapids_core_beats` instantiates it. This page is the FUB's own specification.

12.1 Overview

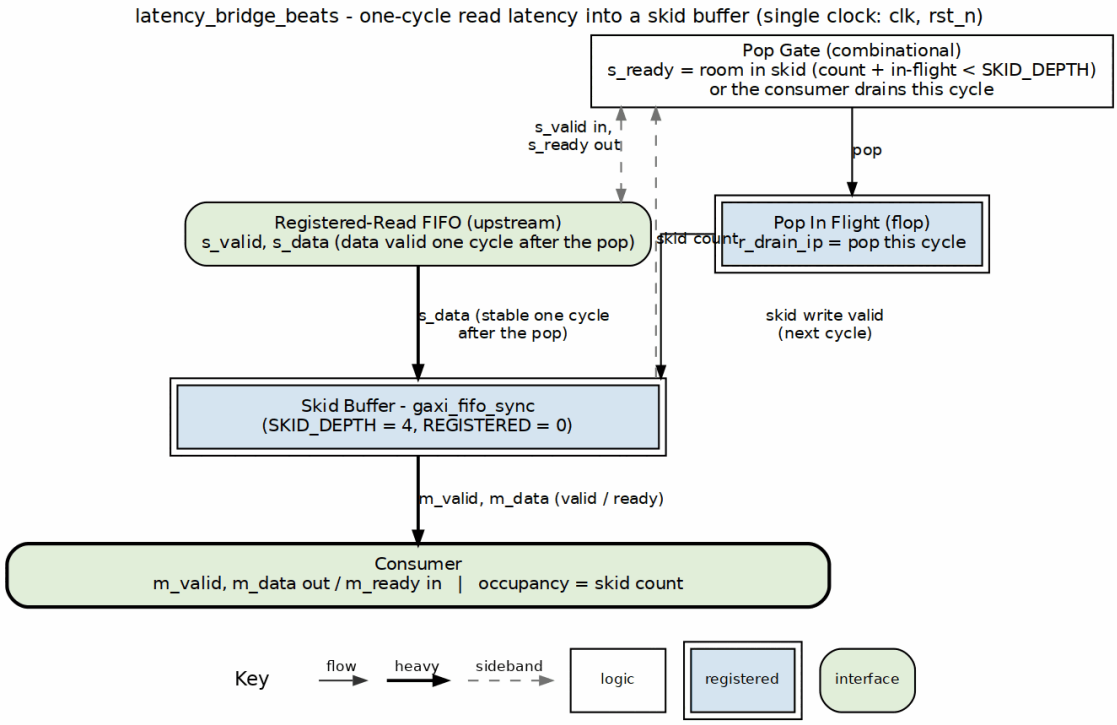
`latency_bridge_beats` turns a **registered-read FIFO** – one whose data appears the cycle *after* the read handshake – into a plain **valid/ready stream** the consumer can stall at will. It is a data-width bridge: it moves `DATA_WIDTH` bits per beat and carries nothing else. There is no beat count, no channel ID and no programmable delay in it; an earlier version of this page described such an interface, and it never existed in this module.

12.1.1 Key Features

- **One-cycle glue, one skid buffer:** a single flop (`r_drain_ip`) remembers that a FIFO read was accepted, so the data that lands the next cycle is pushed into a `gaxi_fifo_sync` skid of depth `SKID_DEPTH`
- **Full throughput when the consumer is ready:** back-to-back reads are accepted as long as the skid keeps draining
- **Backpressure absorbed in the skid:** `s_ready` drops only when a skid write is actually stalled, so the FIFO is never read into a slot that does not exist
- **Occupancy reported:** `occupancy` counts the beats the bridge holds (the one in flight plus the skid contents), which is what an occupancy-based data-available counter needs
- **Debug taps:** `dbg_r_pending` and `dbg_r_out_valid` expose the in-flight flop and the skid output valid, for catching stuck data from a test

12.1.2 Block Diagram

12.1.3 Figure 2.7.1: Latency Bridge Block Diagram



Latency bridge block diagram

Source: 07_latency_bridge.dot

12.2 Concept: bridging a registered read

A FIFO with `REGISTERED=1` read (the SRAM style this repo uses so BRAM infers cleanly) answers a read handshake with data **one cycle later**. A consumer speaking valid/ready expects data *with* valid, and may deassert ready at any time. Without a bridge the FIFO has already popped a beat the consumer did not take.

```
cycle 0   s_valid && s_ready         FIFO read accepted
r_drain_ip <= 1
cycle 1   data arrives on s_data     skid_wr_valid = r_drain_ip, data
pushed
cycle N   m_valid && m_ready         consumer drains the skid at its
own pace
```

The bridge never guesses about downstream readiness: `s_ready` is derived from the skid's room, counting only a write that is *stalled* this cycle as pending (a write that completes this cycle frees its slot at the same edge). That is why full throughput holds while writes complete immediately, and why the FIFO is never over-read when they do not.

12.3 Parameters

```
parameter int DATA_WIDTH = 64;   // Beat width
parameter int SKID_DEPTH = 4;    // Skid buffer depth (2-4
recommended)
parameter int DW = DATA_WIDTH;   // Short alias
```

: Table 2.7.1: Latency Bridge Parameters

`occupancy` is 3 bits wide and counts 0..5 at the default depth (one beat in flight plus four in the skid); a larger `SKID_DEPTH` needs the port widened.

12.4 Port List

12.4.1 Clock and Reset

Table 2.7.2: Clock and Reset

Signal	Direction	Width	Description
`clk`	input	1	System clock
`rst_n`	input	1	Active-low reset

12.4.2 Upstream Interface (from the registered FIFO)

Table 2.7.3: Upstream Interface

Signal	Direction	Width	Description
`s_valid`	input	1	FIFO has a beat to read (not empty)
`s_ready`	output	1	Bridge accepts a read this cycle; the data lands NEXT cycle
`s_data`	input	DW	Read data, valid one cycle after the `s_valid && s_ready` handshake

12.4.3 Downstream Interface (to the consumer)

Table 2.7.4: Downstream Interface

Signal	Direction	Width	Description
`m_valid`	output	1	Beat available
`m_ready`	input	1	Consumer takes the beat
`m_data`	output	DW	Beat data, valid with `m_valid`

12.4.4 Status and Debug

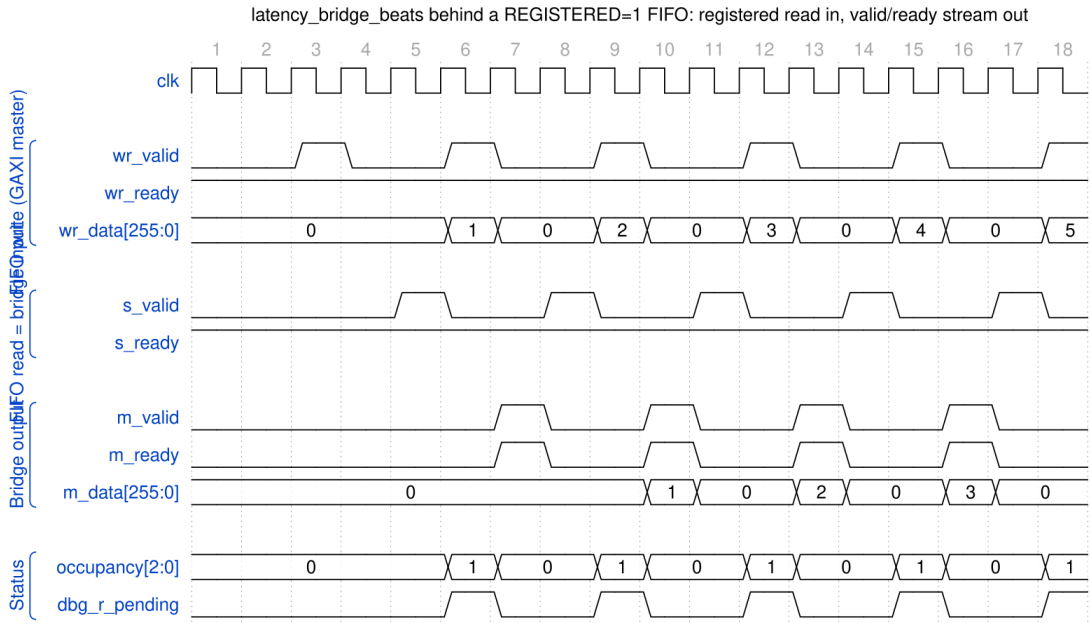
Table 2.7.5: Status and Debug

Signal	Direction	Width	Description
`occupancy`	output	3	Beats held by the bridge:

Signal	Direction	Width	Description
			the in-flight read plus the skid contents (0..5)
dbg_r_pending	output	1	r_drain_ip: a read was accepted and its data is due next cycle
dbg_r_out_valid	output	1	Skid output valid (mirrors m_valid)

12.5 Operation

12.5.1 Figure 2.7.2: Latency Bridge Timing (consumer always ready)



Latency bridge behind a registered-read FIFO

Source: latency_bridge_beats_streaming.json, captured from `dv/tests/fub_beats/test_latency_bridge_beats.py` (`streaming`, 256-bit, seed 7) with `WAVES=1`. The test drives a `REGISTERED = 1` `gaxi_fifo_sync` in front of the bridge (`dv/tb/latency_bridge_beats_tb_top.sv`), because that is the only way to produce the bridge's upstream contract: a valid/ready master presents data *with* valid, a registered FIFO presents it one cycle *after* the read handshake.

Reading it: each `wr_valid` beat lands in the FIFO and `s_valid` (FIFO not empty) rises a cycle later; the bridge reads it at once (`s_ready` stays high) and `dbg_r_pending` marks the cycle the data is in flight. Two cycles after the write the beat is on `m_valid`/`m_data` with its payload intact (1, 2, 3, ...), and `occupancy` shows exactly one beat inside the bridge per transfer. With a consumer that is always ready the bridge adds the FIFO's read latency and no more.

An earlier revision of this test drove `s_*` directly from the BFM and never compared `m_data`; every beat came out 0 and the test passed (rapids TASK-003). (`dv/tests/fub_beats/test_latency_bridge_beats.py`).

12.6 Integration Context

The bridge sits between a `gaxi_fifo_sync` with `REGISTERED=1` and whatever consumes the stream. The same arrangement, with STREAM's twin `stream_latency_bridge`, is what `sram_controller_unit` uses inside the shared SRAM controller today; `occupancy` there is what the drain accounting must allow for (stream BUG-011).

```
latency_bridge_beats #(
    .DATA_WIDTH (512),
    .SKID_DEPTH (4)
) u_bridge (
    .clk             (clk),
    .rst_n           (rst_n),
    // registered-read FIFO side
    .s_valid         (fifo_rd_valid),
    .s_ready         (fifo_rd_ready),
    .s_data          (fifo_rd_data),
    // consumer side
    .m_valid         (drain_valid),
    .m_ready         (drain_read),
    .m_data          (drain_data),
    // status / debug
    .occupancy       (bridge_occupancy),
    .dbg_r_pending   (dbg_bridge_pending),
    .dbg_r_out_valid (dbg_bridge_out_valid)
);
```

12.7 Design Considerations

Table 2.7.6: Skid Depth Selection

SKID_DEPTH	Behaviour
2	Minimum: one stall of the consumer costs a bubble on the next read
4	Default: absorbs a short consumer stall with no upstream bubble
8	Deeper stall absorption; `occupancy` must be widened past 3 bits

The skid does not need to cover the FIFO's read latency (that is the one `r_drain_ip` flop's job); it needs to cover how long the *consumer* stalls without the bridge stopping upstream reads.

Last Updated: 2026-09-27

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13 Control-Read Engine Specification

Module: `ctrlrd_engine.sv` **Location:** `projects/components/dma-ip/rapids/rtl/fub/`
Status: Implemented **Last Updated:** 2026-07-12

13.1 Overview

The Control-Read Engine implements the CTRL_READ descriptor opcode (DESC_OP_CTRL_READ = 2'b01) as a **consumer gate**. When the scheduler decodes a CTRL_READ descriptor, it hands the engine a poll address, an expected value, and a compare mask. The engine issues an AXI4 read to the poll address, applies the mask to the returned data, and compares (read_data & mask) against the expected value. If they match, the gate is satisfied and the descriptor chain proceeds; if not, the engine retries. Retries are bounded so a gate that never matches cannot hang the channel.

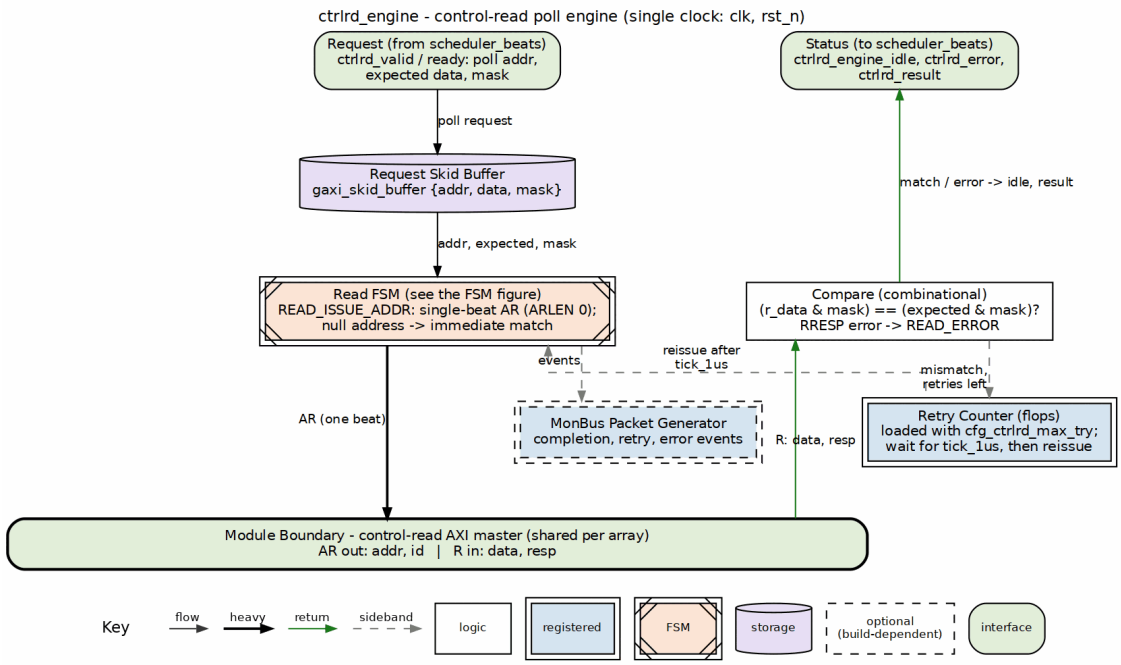
One Control-Read Engine is instantiated per channel in scheduler_group_beats.sv (u_ctrlrd_engine), configured for 32-bit AXI reads.

13.1.1 Key Features

- **Consumer Gate Semantics:** Polls memory until (read_data & mask) == expected
- **Masked Comparison:** Only the masked bits of the polled word participate in the match
- **Bounded Retry:** cfg_ctrlrd_max_try (0-511) caps the number of poll attempts
- **1 microsecond Retry Pacing:** Retries wait for tick_1us from the scheduler group
- **Error on Exhaustion:** Asserts ctrlrd_error if the gate never matches within budget
- **Null Address Fast-Path:** A poll address of zero completes immediately as a match
- **Channel Reset Support:** cfg_channel_reset returns the FSM to idle and drains any read still on the fabric
- **MonBus Integration:** Completion, retry, and error events

13.1.2 Block Diagram

13.1.3 Figure 2.8.1: Control-Read Engine Block Diagram



Control-read engine block diagram

Source: 08_ctrlrd_engine.dot

13.2 Parameters

```
parameter int CHANNEL_ID = 0;                    // Channel identifier
parameter int NUM_CHANNELS = 32;                 // Total channels
parameter int CHAN_WIDTH = $clog2(NUM_CHANNELS); // Channel index
width (derived)
parameter int ADDR_WIDTH = 64;                   // Address bus width
parameter int AXI_DATA_WIDTH = 64;               // AXI read data
width (>= 32)
parameter int AXI_ID_WIDTH = 8;                  // AXI ID width (>=
CHAN_WIDTH)

// Monitor Bus Parameters
parameter logic [15:0] MON_AGENT_ID  = 16'h0030; // Control-Read
Engine Agent ID
parameter logic [7:0]  MON_UNIT_ID   = 8'h02;    // Unit identifier
parameter logic [8:0]  MON_CHANNEL_ID = 9'h000;  // Base channel ID
```

: Table 2.8.1: Control-Read Engine Parameters

Parameter validation enforces `AXI_ID_WIDTH >= CHAN_WIDTH` and `AXI_DATA_WIDTH >= 32` (the engine consumes only the lower 32 bits of the read data). The per-channel instance in `scheduler_group_beats.sv` overrides `AXI_DATA_WIDTH` to 32.

13.3 Port List

13.3.1 Clock and Reset

Table 2.8.2: Clock and Reset

Signal	Direction	Width	Description
`clk`	input	1	System clock
`rst_n`	input	1	Active-low asynchronous reset

13.3.2 Scheduler Interface

Table 2.8.3: Scheduler Interface

Signal	Direction	Width	Description
`ctrlrd_valid`	input	1	Gate request valid from scheduler
`ctrlrd_ready`	output	1	Engine ready to accept a request
`ctrlrd_pkt_addr`	input	AW	Poll address (descriptor `poll_addr[63:0]`)
`ctrlrd_pkt_data`	input	32	Expected value (descriptor `expected[95:64]`)
`ctrlrd_pkt_mask`	input	32	Compare mask (descriptor `mask[127:96]`)
`ctrlrd_error`	output	1	Gate failed (retries exhausted or AXI error)
`ctrlrd_result`	output	32	Last read value (lower 32 bits)
`ctrlrd_engine_idle`	output	1	Engine idle (post-issue completion indicator)

13.3.3 Configuration Interface

Table 2.8.4: Configuration Interface

Signal	Direction	Width	Description
`cfg_ctrlrd_max_try`	input	9	Poll retry budget, 0-511 (from `CTRL_CONFIG.CTRLRD_MAX_TRY`)
`cfg_channel_reset`	input	1	Per-channel reset request
`tick_1us`	input	1	1 microsecond tick that paces retries

13.3.4 AXI4 Read Master Interface

Table 2.8.5: AXI4 Read Master Interface

Signal	Direction	Width	Description
`ar_valid`	output	1	Read address valid
`ar_ready`	input	1	Read address ready
`ar_addr`	output	AW	Read address (poll address)
`ar_len`	output	8	Burst length (`8'h00`, single beat)
`ar_size`	output	3	Burst size (`3'b010`, 4 bytes)
`ar_burst`	output	2	Burst type (`2'b01`, INCR)
`ar_id`	output	ID	Transaction ID (channel-derived)
`ar_lock`	output	1	Lock (`1'b0`, normal access)
`ar_cache`	output	4	Cache (`4'b0010`, non-cacheable bufferable)
`ar_prot`	output	3	Protection (`3'b000`)
`ar_qos`	output	4	Quality of service (`4'h0`)
`ar_region`	output	4	Region (`4'h0`)
`r_valid`	input	1	Read data valid (shared

Signal	Direction	Width	Description
			channel)
r_ready	output	1	Read data ready
r_data	input	ADW	Read data (lower 32 bits used)
r_id	input	ID	Read response ID (matched to our request)
r_resp	input	2	Read response
r_last	input	1	Read last beat

13.3.5 MonBus Interface

Table 2.8.6: MonBus Interface

Signal	Direction	Width	Description
i_mon_time	input	64	Side-band monitor timestamp
mon_valid	output	1	Monitor packet valid
mon_ready	input	1	Monitor consumer ready
mon_packet	output	128	Monitor packet (monitor_packet_t)
mon_timestamp	output	64	Monitor packet timestamp

13.4 FSM States

13.4.1 Figure 2.8.2: Control-Read Engine FSM

States and transitions of the `ctrlrd_engine` poll FSM.

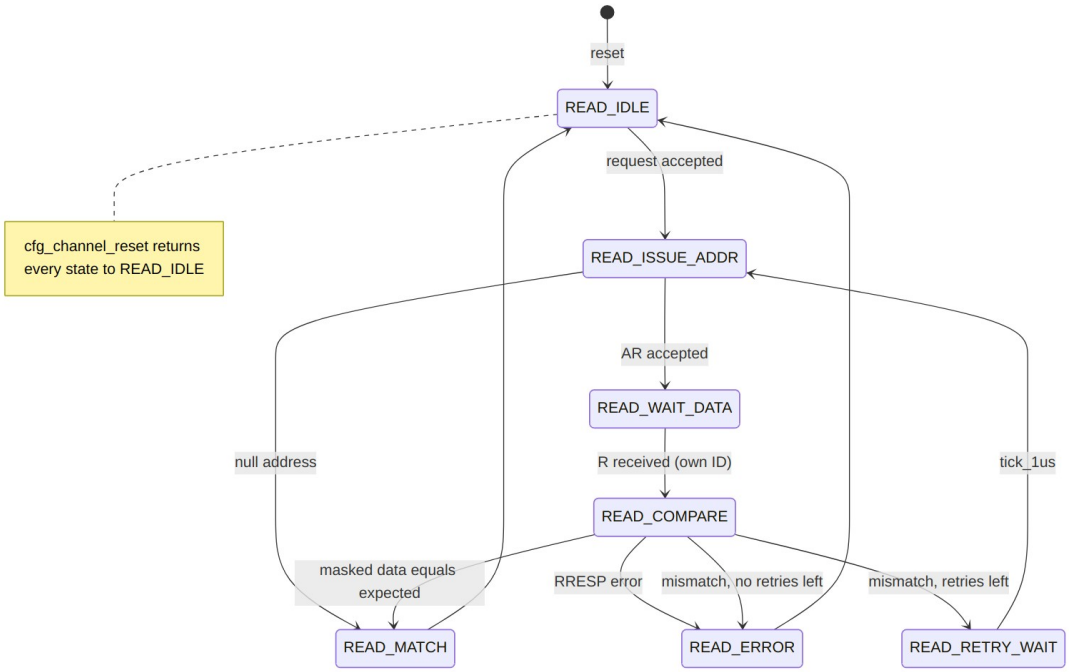


Diagram 2

State summary:

- **READ_IDLE:** Waits for a request from the scheduler (via a two-deep skid buffer). Latches poll address, expected value, mask, and the AXI ID; initializes the retry counter from `cfg_ctrlrd_max_try`.
- **READ_ISSUE_ADDR:** Drives `ar_valid`. A null address (`64'h0`) shortcuts to `READ_MATCH`; otherwise, once `ar_ready` handshakes, advances to `READ_WAIT_DATA`.
- **READ_WAIT_DATA:** Accepts the read response whose `r_id` matches the channel ID, capturing the lower 32 bits and the response code.
- **READ_COMPARE:** On an AXI error response, goes to `READ_ERROR`. On a masked match, goes to `READ_MATCH`. Otherwise, if retries remain, decrements the counter and goes to `READ_RETRY_WAIT`; if the budget is exhausted, goes to `READ_ERROR`.
- **READ_RETRY_WAIT:** Waits for `tick_1us`, then reissues the read.

- **READ_MATCH:** Gate satisfied. Clears the error flag and returns to `READ_IDLE`.
- **READ_ERROR:** Sets `ctrlrd_error` and returns to `READ_IDLE`.

13.5 Operation

13.5.1 Masked Compare

The engine gates on a masked equality check:

```
w_masked_expected = ctrlrd_pkt_data & ctrlrd_pkt_mask
w_masked_actual   = r_data[31:0]    & ctrlrd_pkt_mask
w_data_match      = (w_masked_expected == w_masked_actual)
```

Only the bits set in `mask` participate. A mask of `32'hFFFFFFFF` compares the full word; a mask of a single bit turns the gate into a flag-poll.

13.5.2 Retry Budget and Pacing

The retry counter is loaded from `cfg_ctrlrd_max_try` when the request is accepted. On each non-matching, non-error compare with retries remaining, the counter decrements and the engine parks in `READ_RETRY_WAIT` until the next `tick_1us`, which paces successive polls at roughly 1 microsecond intervals. When the counter reaches zero without a match, the engine transitions to `READ_ERROR` and asserts `ctrlrd_error`, so a never-matching gate cannot stall the channel indefinitely.

`cfg_ctrlrd_max_try` is driven by the `CTRL_CONFIG` register at offset `0x240`, field `CTRLRD_MAX_TRY[8:0]` (0-511, reset 16).

13.5.3 Null Address Fast-Path

If the latched poll address is `64'h0`, the engine treats the gate as already satisfied: `READ_ISSUE_ADDR` transitions directly to `READ_MATCH` without issuing an AXI read. This lets software emit an unconditional (no-op) gate.

13.5.4 AXI Read Transaction

Polls are single-beat, 4-byte INCR reads (`ar_len = 0`, `ar_size = 3'b010`, `ar_burst = 2'b01`). The `ar_id` is derived from `CHANNEL_ID`, and only responses whose `r_id` matches are consumed on the shared read data channel. `ctrlrd_result` exposes the last captured 32-bit value; in the per-channel instantiation the scheduler gates on match/error rather than the raw value, so `ctrlrd_result` is left unconnected there.

13.5.5 Channel Reset

`cfg_channel_reset` is registered into `r_channel_reset_active`. While asserted, the engine refuses new requests (`ctrlrd_ready` is forced low), the retry and error state is cleared, and the FSM is driven back to `READ_IDLE`.

What the reset cannot do is cancel a read already on the fabric, because AXI has no cancel. The engine therefore drains it (rapids TASK-014, 2026-09-27):

- an AR that was raised and not yet accepted stays raised until the slave takes it (`r_drain_ar`), then
- the R beat that the accepted AR is owed is taken and discarded (`r_drain_pending`, which alone lifts `r_ready` outside `READ_WAIT_DATA`).

Until the beat is gone `ctrlrd_engine_idle` stays low and the request path stays closed, so the next read on this ID cannot be answered by the stale beat. Before this the reset dropped the FSM to idle with `r_ready` low, the beat sat on the bus, and a re-kicked channel took it as its own answer.

`ctrlrd_engine_idle` asserts only in `READ_IDLE` with no pending request, no active channel reset and nothing left to drain.

13.6 MonBus Reporting

The engine emits monitor packets on notable events:

Table 2.8.7: MonBus Events

Event	Packet Type	Code	Payload
Gate satisfied (non-null)	PktTypeCompletion	CORE_COMPL_CTRLRD_COMPLETED	Poll address
Retry attempt	PktTypePerf	CORE_PERF_CTRLRD_RETRY	Remaining retry count
Retries exhausted	PktTypeError	CORE_ERR_CTRLRD_MAX_RETRIES	Poll address
AXI error response	PktTypeError	AXI_ERR_RESP_SLVERR / AXI_ERR_RESP_DECERR	Response code

All packets carry MON_AGENT_ID, MON_UNIT_ID, and MON_CHANNEL_ID, and are timestamped from i_mon_time.

13.7 Descriptor Field Mapping

The CTRL_READ opcode reinterprets the shared 256-bit descriptor (the descriptor engine's existing field extraction is reused):

Table 2.8.8: CTRL_READ Descriptor Field Mapping

Descriptor Field	Bits	Engine Port
poll_addr	[63:0]	ctrlrd_pkt_addr
expected	[95:64]	ctrlrd_pkt_data
mask	[127:96]	ctrlrd_pkt_mask
max_try	[143:128]	Per-descriptor budget (0 = use cfg_ctrlrd_max_try default)

Last Updated: 2026-07-12

14 Control-Write Engine Specification

Module: `ctrlwr_engine.sv` **Location:** `projects/components/dma-ip/rapids/rtl/fub/`
Status: Implemented **Last Updated:** 2025-01-10

14.1 Overview

The Control-Write Engine executes a CTRL_WRITE control descriptor (DESC_OP_CTRL_WRITE = 2'b10). A CTRL_WRITE is a producer doorbell: the engine performs a single-beat AXI4 write of the descriptor's 32-bit wr_data value to its 64-bit wr_addr target, then completes so the descriptor chain continues. One instance is dedicated per channel.

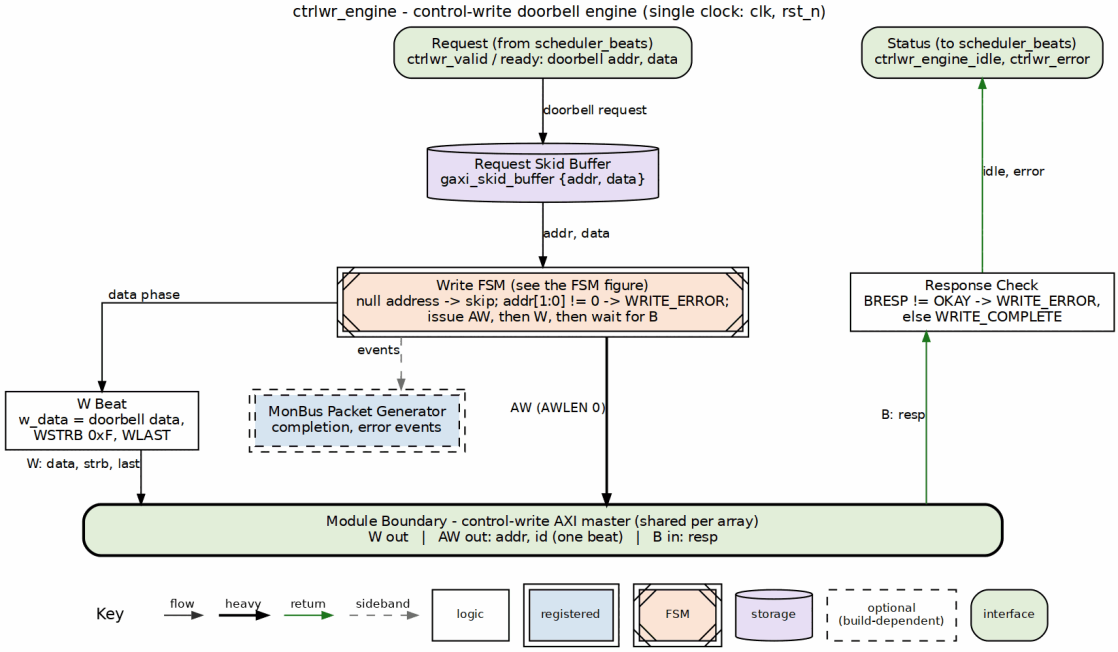
The write target address must be 4-byte aligned because the transfer is a 32-bit (aw_size = 3'b010) single-beat write. A null address (wr_addr == 0) is treated as a no-op and the request completes without issuing an AXI transaction.

14.1.1 Key Features

- **Single-Beat AXI4 Write:** One 32-bit doorbell write per request (aw_len = 0, w_last = 1)
- **Producer Doorbell Semantics:** Write value to target, then release the chain (no poll, no retry)
- **Address Alignment Check:** Requires 4-byte-aligned wr_addr; misalignment raises an error
- **Null-Address Skip:** wr_addr == 0 completes as a no-op
- **Channel Reset Support:** cfg_channel_reset returns the FSM to idle and drains any write still on the fabric
- **MonBus Integration:** Completion and error event reporting

14.1.2 Block Diagram

14.1.3 Figure 2.9.1: Control-Write Engine Block Diagram



Control-write engine block diagram

Source: 09_ctrlwr_engine.dot

14.2 Parameters

```
parameter int CHANNEL_ID = 0;                    // Channel identifier
parameter int NUM_CHANNELS = 32;                 // Total channels
parameter int CHAN_WIDTH = $clog2(NUM_CHANNELS); // Derived channel
index width
parameter int ADDR_WIDTH = 64;                   // Address bus width
parameter int AXI_ID_WIDTH = 8;                  // AXI transaction ID
width


// Monitor Bus Parameters
parameter logic [15:0] MON_AGENT_ID  = 16'h0020; // Ctrlwr Engine
Agent ID
parameter logic [7:0]  MON_UNIT_ID   = 8'h01;    // Unit identifier
parameter logic [8:0]  MON_CHANNEL_ID = 9'h000;  // Base channel ID
```

: Table 2.9.1: Control-Write Engine Parameters

Note: `AXI_ID_WIDTH` must be `>= CHAN_WIDTH` (checked by an elaboration-time `$fatal`). When instantiated in `scheduler_group_beats.sv` as `u_ctrlwr_engine`, `MON_AGENT_ID` is overridden to `16'h0021` (`CTRLWR_MON_AGENT_ID`) to distinguish it from the Control-Read Engine's `16'h0020`.

14.3 Port List

14.3.1 Clock and Reset

Table 2.9.2: Clock and Reset

Signal	Direction	Width	Description
clk	input	1	System clock
rst_n	input	1	Active-low asynchronous reset

14.3.2 Configuration Interface

Table 2.9.3: Configuration Interface

Signal	Direction	Width	Description
cfg_channel_reset	input	1	Per-channel reset: blocks new requests, drains transaction, forces idle

14.3.3 Scheduler Interface

Table 2.9.4: Scheduler Interface

Signal	Direction	Width	Description
ctrlwr_valid	input	1	Request valid (doorbell write requested)
ctrlwr_ready	output	1	Ready to accept request
ctrlwr_pkt_addr	input	AW	Doorbell target address (wr_addr, 4-byte aligned)
ctrlwr_pkt_data	input	32	Doorbell write value (wr_data)
ctrlwr_error	output	1	Error flag (alignment or AXI response error)
ctrlwr_engine_idle	output	1	Engine idle (in WRITE_IDLE, no active transaction, not in channel reset)

14.3.4 AXI4 Write Master Interface

Table 2.9.5: AXI4 Write Master Interface

Signal	Direction	Width	Description
`aw_valid`	output	1	Write address valid
`aw_ready`	input	1	Write address ready
`aw_addr`	output	AW	Write address (= `wr_addr`)
`aw_len`	output	8	Burst length (`0` = single beat)
`aw_size`	output	3	Transfer size (`3'b010` = 4 bytes / 32-bit)
`aw_burst`	output	2	Burst type (`2'b01` = INCR)
`aw_id`	output	AXI_ID_WIDTH	Transaction ID (derived from `CHANNEL_ID`)
`aw_lock`	output	1	Lock type (`0` = normal)
`aw_cache`	output	4	Cache type (`4'b0010` = non-cacheable bufferable)
`aw_prot`	output	3	Protection type (`3'b000`)
`aw_qos`	output	4	Quality of service (`0`)
`aw_region`	output	4	Region identifier (`0`)
`w_valid`	output	1	Write data valid
`w_ready`	input	1	Write data ready
`w_data`	output	32	Write data (= `wr_data`)
`w_strb`	output	4	Write strobes (`4'b1111` = all bytes)
`w_last`	output	1	Last beat (`1` = single beat)
`b_valid`	input	1	Write response valid
`b_ready`	output	1	Write response ready
`b_id`	input	AXI_ID_WIDTH	Response ID (matched against expected AXI ID)
`b_resp`	input	2	Write response code

14.3.5 MonBus Interface

Table 2.9.6: MonBus Interface

Signal	Direction	Width	Description
`i_mon_time`	input	64	Monitor timestamp side-band input
`mon_valid`	output	1	Monitor packet valid
`mon_ready`	input	1	Monitor packet ready
`mon_packet`	output	128	Monitor packet (`monitor_packet_t`)
`mon_timestamp`	output	64	Monitor packet timestamp

14.4 FSM States

14.4.1 Figure 2.9.2: Control-Write Engine FSM

States and transitions of the `ctrlwr_engine` doorbell FSM.

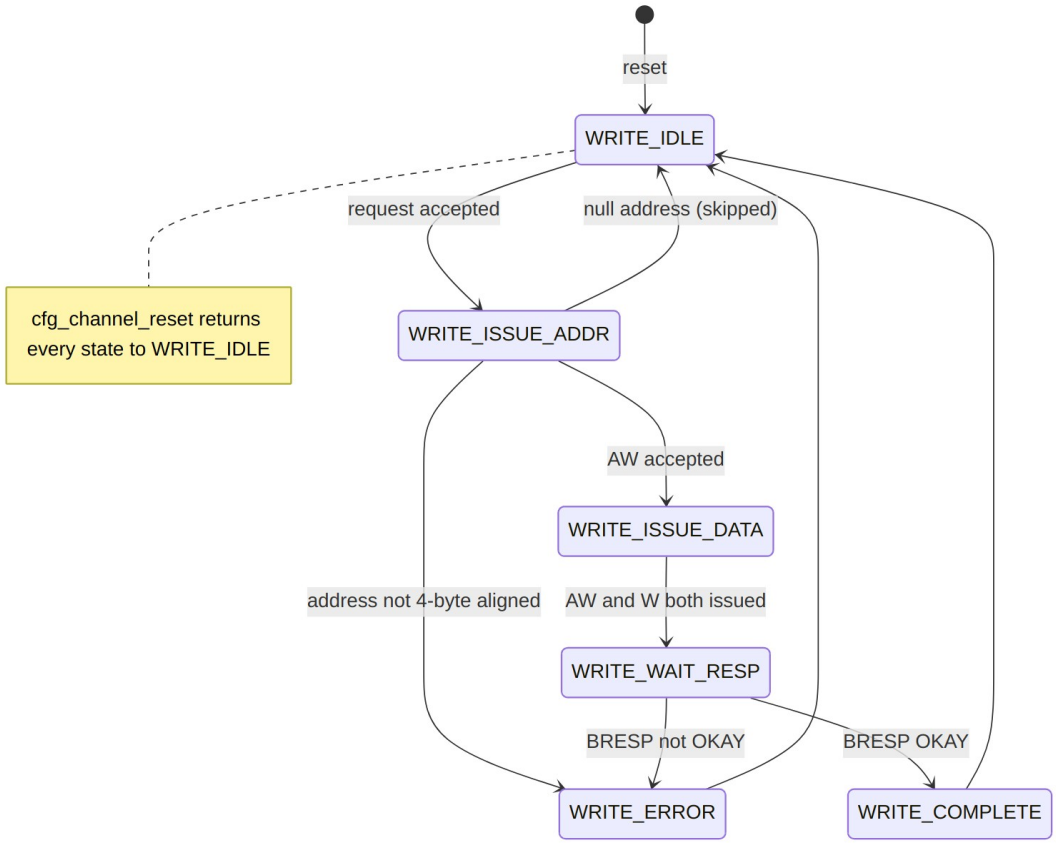


Diagram 3

State summary:

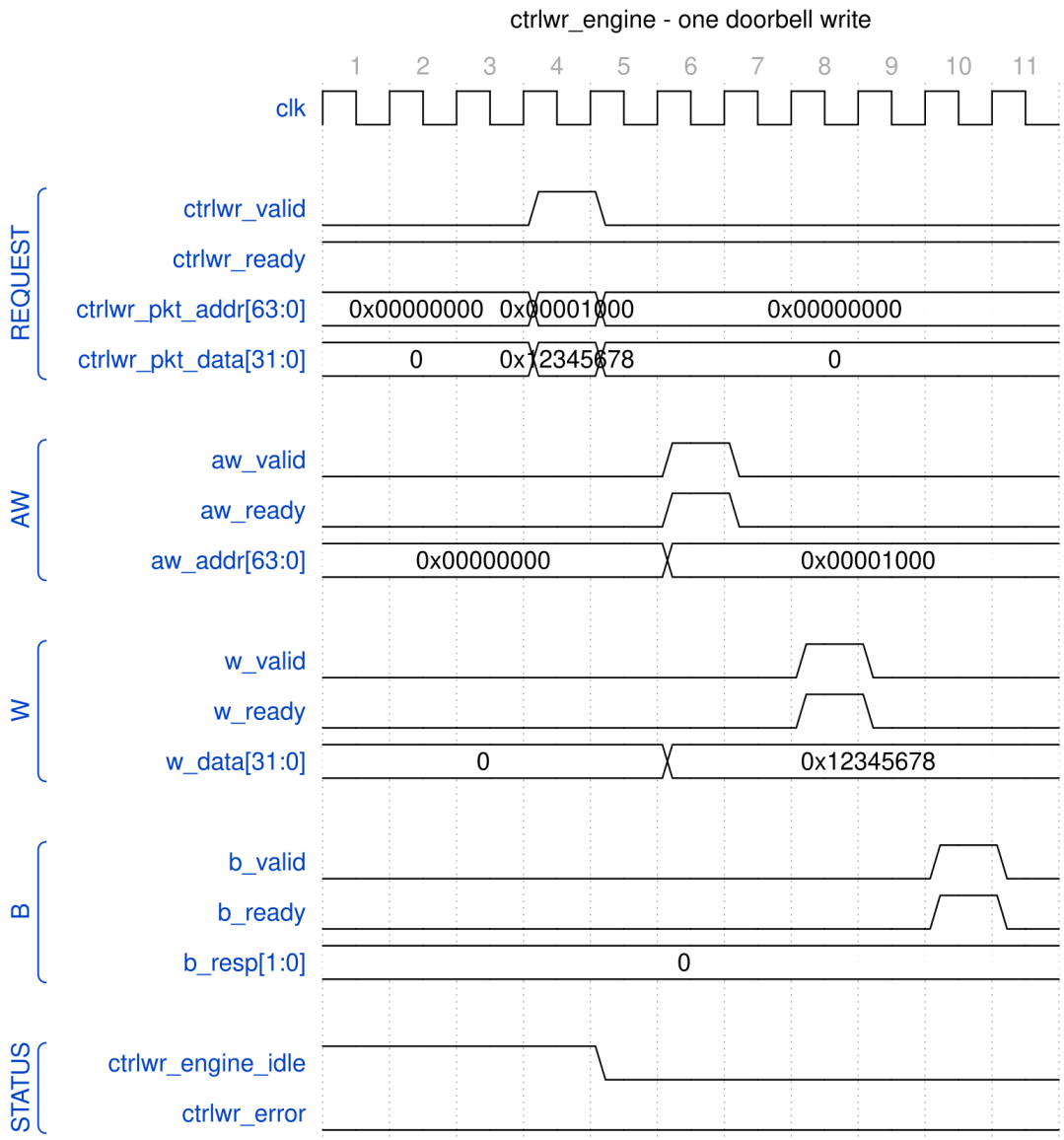
- **WRITE_IDLE:** Wait for a request from the request skid buffer; latch `wr_addr`/`wr_data` and the expected AXI ID.
- **WRITE_ISSUE_ADDR:** Drive `aw_valid` until `aw_ready`. A misaligned address routes to `WRITE_ERROR`; a null address returns to `WRITE_IDLE` (no-op skip).
- **WRITE_ISSUE_DATA:** Drive `w_valid` until `w_ready`; advance once both address and data phases have been issued.
- **WRITE_WAIT_RESP:** Accept the matching `b`-channel response; a non-OKAY `b_resp` routes to `WRITE_ERROR`, otherwise to `WRITE_COMPLETE`.

- **WRITE_COMPLETE:** Emit a completion MonBus packet (unless null address) and return to `WRITE_IDLE`.
- **WRITE_ERROR:** Latch `ctrlwr_error`, emit an error MonBus packet, and return to `WRITE_IDLE`. The error flag persists until reset or `cfg_channel_reset`.

14.5 Operation

14.5.1 Doorbell Write Sequence

14.5.2 Figure 2.9.3: Control-Write Doorbell Timing



ctrlwr_engine - one doorbell write

Source: ctrlwr_engine_doorbell.json, captured from `dv/tests/fub/test_ctrlwr_engine.py` (`basic_write`, channel 0, 64-bit address, `REG_LEVEL=GATE`) with `WAVES=1`; the AXI side is the framework write slave.

Reading it: the request handshakes in one cycle (ctrlwr_ready is held high by the skid buffer) with the doorbell address 0x1000 and data 0x12345678, and the engine leaves idle the next cycle. AW goes out two cycles after the request, W two cycles after AW, and the slave answers B (OKAY) two cycles after W; ctrlwr_engine_idle returns the cycle after the B handshake, eight cycles after the request. Nothing is pipelined across doorbells: the engine takes one request at a time.

14.5.3 Request Handshake and Skid Buffer

Incoming requests are captured through a depth-2 gaxi_skid_buffer. The engine accepts a request (ctrlwr_ready) only when not in an active channel reset. The FSM pops the buffered request in WRITE_IDLE, latching {wr_addr, wr_data} and computing the expected AXI ID from CHANNEL_ID:

```
aw_id = {{(AXI_ID_WIDTH-CHAN_WIDTH){1'b0}}, CHANNEL_ID[CHAN_WIDTH-
1:0]};
```

The b-channel response is matched against this expected ID (b_id == r_expected_axi_id) so a shared write master can be demultiplexed per channel.

14.5.4 Address Validation

```
Null (skip)  : wr_addr == 64'h0
Align error  : wr_addr[1:0] != 2'b00  (and not null)
```

A null address completes immediately as a no-op with no AXI activity and no completion packet. A misaligned address never issues on AXI; it transitions to WRITE_ERROR, asserts ctrlwr_error, and emits an error MonBus packet.

14.5.5 Error Reporting

ctrlwr_error is a registered, sticky flag. It is set on an alignment error or on a non-OKAY AXI write response (b_resp != 2'b00), and it persists until rst_n or cfg_channel_reset. The MonBus error packet carries either the offending address (alignment error) or the captured b_resp code (AXI response error).

14.5.6 Channel Reset

Asserting cfg_channel_reset registers r_channel_reset_active, which blocks new requests (ctrlwr_ready low), clears the error flag and the phase-issued state, and drives the FSM back to WRITE_IDLE.

A write already on the fabric cannot be cancelled, so the engine drains it (rapids TASK-014, 2026-09-27): a raised AW stays up until accepted (r_drain_aw), an accepted AW is followed by its W beat with the latched data (r_drain_w), and the B response is taken and discarded (r_drain_b, which alone lifts b_ready outside WRITE_WAIT_RESP). The write that was already issued therefore lands. Until B is taken ctrlwr_engine_idle stays low and the request path stays closed. Before this the reset abandoned whatever phase was in flight and left a dangling response on the channel's ID.

`ctrlwr_engine_idle` asserts only in `WRITE_IDLE` with no pending request, no active channel reset and nothing left to drain.

14.6 Contrast with the Control-Read Engine

The Control-Write Engine is the producer counterpart to the Control-Read Engine (Section 2.8). Both are per-channel control-descriptor executors, but they implement opposite half of the producer/consumer handshake:

Aspect	Control-Read Engine (2.8)	Control-Write Engine (2.9)
Opcode	`DESC_OP_CTRL_READ` (`2'b01`)	`DESC_OP_CTRL_WRITE` (`2'b10`)
Role	Consumer gate	Producer doorbell
AXI direction	Read (poll)	Single-beat write
Semantics	Poll until `(rd & mask) == expected`	Write `wr_data` to `wr_addr`, then continue
Retry budget	`max_try` retry budget (config default)	None (single unconditional write)
Chain effect	Holds off the chain until the gate passes	Releases the chain after the write completes
Descriptor fields	`poll_addr` / `expected` / `mask` / `max_try`	`wr_addr` / `wr_data`

A `CTRL_WRITE` therefore has no maximum-try configuration: it is one unconditional 32-bit write, whereas `CTRL_READ` polls with a bounded retry budget before either passing the gate or timing out.

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15 Shared Engines

Modules: `descriptor_engine_shared.sv`, `ctrlrd_engine_shared.sv`, `ctrlwr_engine_shared.sv` **Location:** `projects/components/dma-ip/rapids/rtl/fub/`
Status: Implemented **Last Updated:** 2026-10-17

15.1 Overview

With `SHARED_ENGINES = 1` (the default) each scheduler group array, one per half, builds one descriptor engine, one control-read engine and one control-write engine for all of its channels. These replace the `NUM_CHANNELS` copies of each that `SHARED_ENGINES = 0` builds. Each shared engine picks the next channel to serve round-robin on the scheduler side and owns its AXI master outright, so the array's AXI-side arbiters are gone. ARID and AWID carry the channel.

Each scheduler sees the same per-channel interface as with a dedicated engine. The schedulers, the array's ports and everything above the array are the same in both builds. The Scheduler Group Array chapter describes how the array instantiates the engines and the MonBus sources they add. This chapter describes the engines themselves.

15.1.1 Key Features

- **One fetch path per half:** a single AR issuer, one set of address-range comparators and one R path for every channel.
- **Per-channel ordering:** each channel keeps one descriptor fetch in flight, so its descriptors arrive in order. Fetches for different channels overlap on the bus under their own ARIDs.
- **No R-channel blocking:** a fetch issues only when its channel's FIFO has room, and an R beat for a channel reset mid-fetch is accepted and dropped.
- **Owner-register control engines:** one `ctrlrd_engine` and one `ctrlwr_engine`, each wrapped with a round-robin pick and an owner register.
- **Descriptor injection (`INJ_EN`):** a second writer into the descriptor FIFOs for the tile's cell-issued descriptors. See Tile Packet Header.

15.1.2 Cost

On the Genesys 2 board build (8 channels per half, 256 bits) the three per-channel engine sets took 15,283 LUTs and 19,308 flip-flops across both halves:

- descriptor engines: 11,009 LUTs and 7,360 flip-flops for 16 engines;
- control-read engines: 2,388 LUTs and 7,038 flip-flops;
- control-write engines: 1,886 LUTs and 4,910 flip-flops.

Sharing removed 11,024 LUTs (20%) and 16,483 flip-flops (37%) from RAPIDS. The HAS Resource Estimates chapter has the full before and after.

15.2 descriptor_engine_shared

15.2.1 Parameters

Table 2.10.1: descriptor_engine_shared Parameters

Parameter	Default	Description
NUM_CHANNELS	8	Channels served.
CHAN_WIDTH	$clog2(NUM_CHANNELS) (1 minimum)	Channel ID width. AXI_ID_WIDTH must be at least this; elaboration fails otherwise.
ADDR_WIDTH	64	Address width.
AXI_ID_WIDTH	8	ARID and RID width. ARID = channel.
FIFO_DEPTH	8	Descriptor FIFO depth per channel.
USE_ROW_COL_MAJOR_ADDRESSING	1	Enables the EXTENDED descriptor's second fetch and its chunk-1 FIFO.
MON_AGENT_BASE	0x10	MonBus agent is MON_AGENT_BASE + channel.
MON_UNIT_ID	0x01	MonBus unit ID.
MON_FIFO_DEPTH	4	Depth of the MonBus packet FIFO.
INJ_EN	0	1 builds the injection slot (set by the array when TILE_HDR = 1).

15.2.2 Port List

Every per-channel port is a packed [NUM_CHANNELS-1:0] vector of the signal a dedicated descriptor_engine has, so channel c sees bit or element c.

Table 2.10.2: descriptor_engine_shared Ports

Group	Signals	Description
Clock and reset	clk, rst_n	Single clock, active-low asynchronous reset.

Group	Signals	Description
Kick (per channel)	apb_valid, apb_ready, apb_addr	Staged descriptor address from the configuration block.
Scheduler (per channel)	channel_idle, descriptor_valid, descriptor_ready, descriptor_packet, descriptor_ext_packet, descriptor_error, descriptor_eos, descriptor_eol, descriptor_eod, descriptor_type	The same handshake and flags as descriptor_engine.
AXI4 read master	ar_*, r_* (256-bit r_data)	Owned outright; one master per half.
Configuration	cfg_prefetch_enable, cfg_fifo_threshold, cfg_addr0_base/limit, cfg_addr1_base/limit, cfg_channel_reset	Shared by every channel, as before; the reset is per channel.
Injection	inj_valid, inj_ch, inj_data, inj_dropped, inj_dropped_clr	Unused (tied off) unless INJ_EN = 1.
Status	descriptor_engine_idle (per channel)	Channel has no fetch pending or in flight.
MonBus	i_mon_time, mon_valid, mon_ready, mon_packet, mon_timestamp	One stream for every channel.

15.2.3 Per-Channel State

Each channel keeps only what a fetch needs between the kick and the FIFO write:

Table 2.10.3: Per-Channel Fetch States

State	Meaning
S_IDLE	Nothing to fetch.

State	Meaning
S_PEND	Chunk-0 address pending (from a kick or from chaining).
S_ISSUE	Chunk-0 request is in the shared issue slot.
S_WAIT	Chunk-0 AR accepted, waiting for R.
S_PEND2, S_ISSUE2, S_WAIT2	The same three steps for an EXTENDED descriptor's chunk 1.

Beside the state code each channel has one pending fetch address, a kick-in-progress flag, an owed-beat flag (r_owe), its descriptor FIFO and, when EXTENDED descriptors are built, a chunk-1 FIFO kept in lockstep with it. An injection slot is added when INJ_EN = 1.

15.2.4 Operation

Issue. A round-robin pick among channels in S_PEND or S_PEND2 loads one request into the issue slot. A channel is eligible only while its FIFO has a free entry, so a channel that has stopped consuming never takes an R beat it cannot store. The range check runs at the slot: an address outside both configured ranges is refused with no AR and raises the channel's descriptor error. Once ARVALID is high, the address and ID hold until accepted.

Return. Each R beat is routed by RID to its channel and written straight into that channel's FIFO, so there is no 256-bit holding register per channel. The exception is an EXTENDED descriptor's chunk 0, which waits for chunk 1. A non-OKAY response, or a beat that does not parse as a valid descriptor, raises the channel's descriptor error.

Chaining and prefetch. The next-pointer decision and the prefetch throttle are the per-channel engine's: autonomous chaining while the descriptor allows it, up to the FIFO threshold plus one buffered.

Channel reset. A reset in the middle of a fetch sets r_owe. The owed beat is accepted and dropped when it arrives, and the channel takes no new kick until then. Reset also drains the channel's FIFO, so descriptors prefetched for an aborted chain are never delivered.

Injection. An injected descriptor is a pulse with no backpressure. It waits in its channel's one-entry slot and is written into the FIFO, with no fetch, once the channel's fetch side is idle and the FIFO has room. An injection therefore never takes an entry an R beat needs. A descriptor that arrives while the slot is full, while the channel is held in reset, or that is EXTENDED is dropped and sets the channel's sticky inj_dropped bit. A drop is not a descriptor error.

15.2.5 MonBus

Completion and error packets go through a MON_FIFO_DEPTH-entry FIFO. Two behaviours differ from the per-channel engine:

- A refused kick address reports 0 in the response field. The per-channel engine reported its last latched read response, which was stale.
- An issue-side refusal and an R-side event can occur in the same cycle. The issue-side packet waits one cycle. If a second collision arrives while it waits, the waiting packet is dropped. Tests that count descriptor-engine packets must allow for this.

15.3 ctrlrd_engine_shared and ctrlwr_engine_shared

The two control engines share one structure: an unmodified `ctrlrd_engine` or `ctrlwr_engine` with three things around it. A control engine runs one request at a time, so nothing inside it has to change.

Table 2.10.4: Shared Control Engine Wrapper

Element	Behaviour
Round-robin pick	Offered to the engine only while it is idle, its last MonBus packet has drained, and no error remains from the previous owner. A stale error is cleared with a one-cycle engine reset before the next channel is taken.
Owner register	Set when the engine accepts a request. ARID or AWID carries the owner, and every R or B response belongs to the engine.
Per-channel view	The owner sees the engine's `*_idle` and `*_error` as a dedicated engine would show them. Every other channel sees idle 1 and error 0.
Channel reset	The owner's `cfg_channel_reset` resets the engine only while the owner's operation is in flight, so a channel held in reset cannot hold the engine for the others.
MonBus	Packets are re-tagged with the owner's channel and agent ID: 0x20 + channel for control reads, 0x28 + channel for control writes.

15.3.1 Parameters

Table 2.10.5: Shared Control Engine Parameters

Parameter	Default	Description
`NUM_CHANNELS`	8	Channels served.
`CHAN_WIDTH`	`$clog2(NUM_CHANNELS)` (1 minimum)	Channel ID width.

Parameter	Default	Description
ADDR_WIDTH	64	Address width.
AXI_ID_WIDTH	8	ARID or AWID width.
MON_AGENT_BASE	0x20 (read), 0x28 (write)	MonBus agent is MON_AGENT_BASE + channel.
MON_UNIT_ID	0x01	MonBus unit ID.

15.3.2 Port List

Table 2.10.6: Shared Control Engine Ports

Group	ctrlrd_engine_shared	ctrlwr_engine_shared
Requests (per channel)	ctrlrd_valid, ctrlrd_ready, ctrlrd_addr, ctrlrd_data, ctrlrd_mask	ctrlwr_valid, ctrlwr_ready, ctrlwr_addr, ctrlwr_data
Status (per channel)	ctrlrd_error, ctrlrd_idle	ctrlwr_error, ctrlwr_idle
Configuration	cfg_ctrlrd_max_try, cfg_channel_reset, tick_1us	cfg_channel_reset
AXI4 master	ar_*, r_* (32-bit data)	aw_*, w_* (32-bit data, 4-bit strobe), b_*
MonBus	i_mon_time, mon_valid, mon_ready, mon_packet, mon_timestamp	the same

15.3.3 Trade-off

A CTRL_READ that keeps polling holds the control-read engine, so other channels' control reads wait behind it, up to cfg_ctrlrd_max_try retries. Data transfers are not affected. The control-write engine keeps one write in flight, which the array-level serializer of the per-channel build also enforced.

15.4 Related Chapters

- Scheduler Group Array: instantiation, MonBus sources, and the `SHARED_ENGINES = 0` AXI arbiters.
- Descriptor Engine, Control-Read Engine, Control-Write Engine: the per-channel engines whose behaviour the shared engines keep.
- Tile Packet Header: the descriptor injection path.

16 Scheduler Group

Module: `scheduler_group.sv` **Location:** `projects/components/dma-ip/rapids/rtl/macro/` **Status:** Implemented **Last Updated:** 2026-09-30

16.1 Overview

The scheduler group is used only when `SHARED_ENGINES = 0`. The default build instantiates the schedulers directly and uses the Shared Engines instead. The scheduler group wraps one channel: a descriptor engine, the scheduler, the two control engines (control read and control write) and a MonBus arbiter that merges their monitor packets. It is the unit the scheduler group array instantiates once per channel.

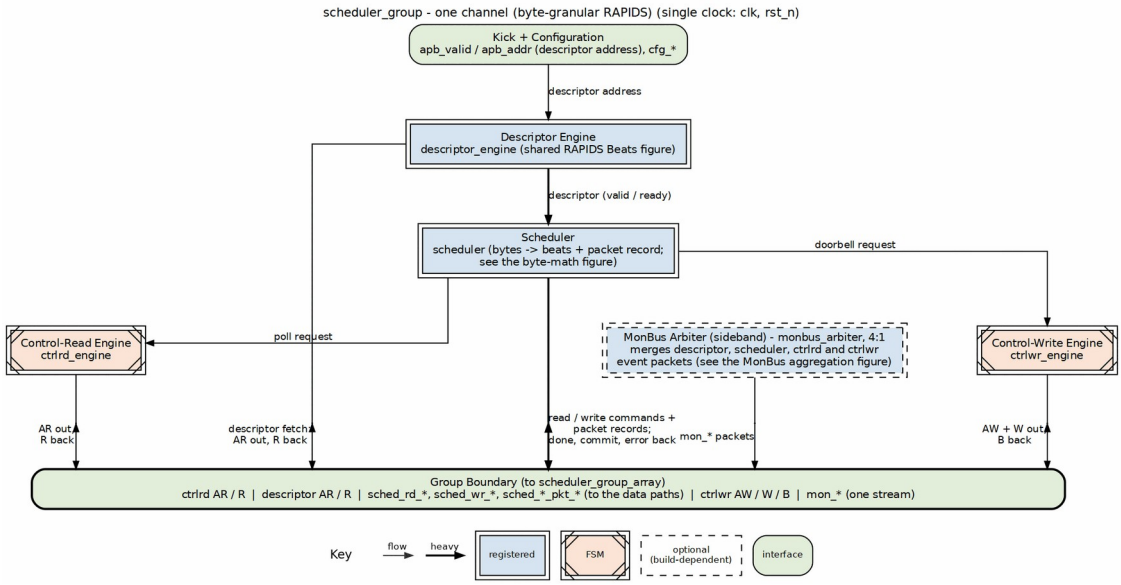
The structure is identical to the RAPIDS Beats group. The one change is the packet-record port set: the scheduler now publishes a per-direction record of the packet's byte length and start offset, and the group passes it through unchanged.

16.1.1 Key Features

- **One channel, four engines:** descriptor engine, scheduler, control read engine, control write engine.
- **Packet-record pass-through:** `sched_rd_pkt_*` and `sched_wr_pkt_*` connect the scheduler straight to the group boundary.
- **MonBus aggregation:** a four-client arbiter merges the four engines' packets.
- **Completion gating:** `cfg_sched_compl_enable` drops CORE Completion packets from the descriptor engine and the scheduler at the arbiter input and acknowledges the emitters, so nothing stalls (rapids ISSUE-005). Errors are never gated.
- **Status aggregation:** idle, FSM state and sticky error are exported.

16.1.2 Block Diagram

16.1.3 Figure 3.1.1: Scheduler Group Block Diagram



Scheduler group block diagram

Source: 07_scheduler_group.dot

16.2 Parameters

Table 3.1.1: Scheduler Group Parameters

Parameter	Default	Description
CHANNEL_ID	0	Channel identifier.
NUM_CHANNELS	8	Total channels.
CHAN_WIDTH	$clog2(NUM_CHANNELS)	Channel id width.
ADDR_WIDTH	64	Address width.
DATA_WIDTH	512	Data width in bits. Sets the beat size seen by the scheduler.
AXI_ID_WIDTH	8	AXI ID width for the descriptor and control masters.
DESC_MON_AGENT_ID	16	MonBus agent id of the descriptor engine.
SCHED_MON_AGENT_ID	48	MonBus agent id of the scheduler.
CTRLRD_MON_AGENT_ID	32	MonBus agent id of the control read engine.
CTRLWR_MON_AGENT_ID	33	MonBus agent id of the control write engine.
MON_UNIT_ID	1	MonBus unit id.
MON_CHANNEL_ID	0	Base MonBus channel id.
EN_READ	1	Enable the read direction of the scheduler.
EN_WRITE	1	Enable the write direction of the scheduler.
USE_ROW_COL_MAJOR_ADDRESSING	1	Descriptor engine addressing mode.
GEN_MON	1	Enable monitor packet generation.
OFF_W	$clog2(DATA_WIDTH/8)	Width of the packet-record byte offset. New in RAPIDS.

16.3 Port List

16.3.1 Clock, Reset and Kick-Off

Table 3.1.2: Clock, Reset and Kick-Off

Signal	Direction	Width	Description
clk	input	1	System clock.
rst_n	input	1	Active-low reset.
apb_valid	input	1	Channel kick-off.
apb_ready	output	1	Descriptor engine ready for kick-off.
apb_addr	input	AW	First descriptor address.

16.3.2 Configuration Interface

Table 3.1.3: Configuration Interface

Signal	Direction	Width	Description
cfg_channel_enable	input	1	Enable this channel.
cfg_channel_reset	input	1	Per-channel soft reset.
cfg_sched_timeout_cycles	input	32	Write-progress timeout window in cycles.
cfg_sched_timeout_limit	input	8	Consecutive timeout windows before a fatal error. 0 means never.
cfg_sched_timeout_enable	input	1	Enable timeout detection.
cfg_sched_err_enable	input	1	Enable error reporting.
cfg_sched_compl_enable	input	1	1 passes CORE Completion packets to the monitor bus; 0 drops them at the arbiter

Signal	Direction	Width	Description
			inputs.
cfg_sched_perf_enable	input	1	Enable performance monitoring.
cfg_ctrlrd_max_try	input	9	Control-read poll retry budget, 0 to 511.
tick_1us	input	1	1 us tick for control-read retry spacing.
cfg_desceng_prefetch	input	1	Enable descriptor prefetch.
cfg_desceng_fifo_thresh	input	4	Prefetch threshold.
cfg_desceng_addr0_base	input	AW	Valid descriptor address range 0 base.
cfg_desceng_addr0_limit	input	AW	Valid descriptor address range 0 limit.
cfg_desceng_addr1_base	input	AW	Valid descriptor address range 1 base.
cfg_desceng_addr1_limit	input	AW	Valid descriptor address range 1 limit.

16.3.3 Status

Table 3.1.4: Status Interface

Signal	Direction	Width	Description
descriptor_engine_idle	output	1	Descriptor engine idle.
scheduler_idle	output	1	Scheduler idle.
scheduler_state	output	7	Scheduler FSM state, one-hot.
sched_error	output	1	Scheduler error, sticky.

16.3.4 Descriptor AXI Master Interface

Table 3.1.5: Descriptor AXI Master Interface

Signal	Direction	Width	Description
`desc_ar_valid` / `desc_ar_ready`	output / input	1	AR handshake.
`desc_ar_addr`	output	AW	Descriptor fetch address.
`desc_ar_len`	output	8	Burst length minus one.
`desc_ar_size`	output	3	Burst size.
`desc_ar_burst`	output	2	Burst type.
`desc_ar_id`	output	IW	Transaction ID.
`desc_ar_lock`, `_cache`, `_prot`, `_qos`, `_region`	output	1, 4, 3, 4, 4	Attribute fields.
`desc_r_valid` / `desc_r_ready`	input / output	1	R handshake.
`desc_r_data`	input	256	Descriptor payload, fixed 256 bits.
`desc_r_resp`	input	2	Read response.
`desc_r_last`	input	1	Last beat.
`desc_r_id`	input	IW	Response ID.

16.3.5 Control Read and Control Write AXI Interfaces

The control read engine fetches a 32-bit semaphore word: `ctrlrd_ar_*` carries the AR channel (address, length, size, burst, id and the same attribute fields as above) and `ctrlrd_r_valid`, `ctrlrd_r_ready`, `ctrlrd_r_data[31:0]`, `ctrlrd_r_id`, `ctrlrd_r_resp` and `ctrlrd_r_last` the R channel. The control write engine writes a 32-bit word: `ctrlwr_aw_*` carries the AW channel, `ctrlwr_w_valid`, `ctrlwr_w_ready`, `ctrlwr_w_data[31:0]`, `ctrlwr_w_strb[3:0]` and `ctrlwr_w_last` the W channel, and `ctrlwr_b_valid`, `ctrlwr_b_ready`, `ctrlwr_b_id` and `ctrlwr_b_resp` the B channel. These are unchanged from RAPIDS Beats. See Control Read Engine and Control Write Engine.

16.3.6 Scheduler Data Interface

Table 3.1.6: Scheduler Data Interface

Signal	Direction	Width	Description
sched_rd_valid	output	1	Read request.
sched_rd_addr	output	AW	Working source byte address.
sched_rd_beats	output	32	Beats still to read.
sched_wr_valid	output	1	Write request.
sched_wr_ready	input	1	Write engine reports the last burst of the descriptor committed.
sched_wr_addr	output	AW	Working destination byte address.
sched_wr_beats	output	32	Beats still to write.
sched_rd_done_strobe	input	1	Read burst issued.
sched_rd_beats_done	input	32	Beats in that burst.
sched_wr_done_strobe	input	1	Write AW issued.
sched_wr_beats_done	input	32	Beats in that AW.
sched_wr_commit_strobe	input	1	Write B response received.
sched_wr_commit_beats	input	32	Beats committed by that B response.
sched_rd_error	input	1	Sticky read error from the read engine.
sched_wr_error	input	1	Sticky write error, including a packet-length error from the sink path.

16.3.7 Packet Record Interface

Table 3.1.7: Packet Record Interface

Signal	Direction	Width	Description
`sched_rd_pkt_valid`	output	1	One-cycle pulse: a read-side packet record is available.
`sched_rd_pkt_ready`	input	1	Consumer has room in its record queue.
`sched_rd_pkt_bytes`	output	32	Packet length in bytes.
`sched_rd_pkt_offset`	output	OFF_W	Byte offset of the packet start within its first memory beat.
`sched_wr_pkt_valid`	output	1	One-cycle pulse: a write-side packet record is available.
`sched_wr_pkt_ready`	input	1	Consumer has room in its record queue.
`sched_wr_pkt_bytes`	output	32	Packet length in bytes.
`sched_wr_pkt_offset`	output	OFF_W	Byte offset of the packet start within its first memory beat.

The group wires these signals directly to the scheduler. The read-side record goes to the source data path and the write-side record to the sink data path. The scheduler holds a descriptor in its fetch state while a required record queue is full, using the ready inputs. See Packet Record Interface.

16.3.8 MonBus Interface

Table 3.1.8: MonBus Interface

Signal	Direction	Width	Description
`i_mon_time`	input	`monbus_timestamp_t`	Shared monitor timebase.
`mon_valid`	output	1	Monitor packet valid.
`mon_ready`	input	1	Consumer ready.

Signal	Direction	Width	Description
mon_packet	output	monitor_packet_t	Monitor packet.
mon_timestamp	output	monbus_timestamp_t	Packet timestamp.

16.4 Internal Architecture

16.4.1 Descriptor Path

The descriptor engine fetches 256-bit descriptors from the address range the configuration allows and hands each one to the scheduler with its type, end-of-stream, end-of-list and error flags. The scheduler reports `scheduler_idle` back to the descriptor engine as `channel_idle`. No byte or beat arithmetic happens in the descriptor engine.

16.4.2 Control Engines

When a descriptor asks the scheduler to wait on or update a semaphore, the scheduler talks to the control engines through valid and ready handshakes with an address, data and mask. Both engines share the group's `AXI_ID_WIDTH` and each raises its own MonBus packets.

16.4.3 MonBus Aggregation

A four-client arbiter merges the descriptor engine, scheduler, control read and control write packets. `cfg_sched_compl_enable` gates only the CORE Completion packets of the descriptor engine and the scheduler: with it low, those packets are consumed at the arbiter input and never reach `mon_packet`. Error packets are never gated. `GEN_MON` masks every arbiter input when clear.

16.5 Differences from RAPIDS Beats

Table 3.1.9: Scheduler Group Delta

Item	RAPIDS Beats	RAPIDS
Parameter `OFF_W`	absent	present, `$clog2(DATA_WIDTH/8)`
`sched_rd_pkt_*`, `sched_wr_pkt_*` ports	absent	present, wired to the scheduler
Everything else		Unchanged

The beats chapter is Beats Scheduler Group. The scheduler is described in Scheduler.

Last Updated: 2026-09-30

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17 Scheduler Group Array

Module: `scheduler_group_array.sv` **Location:** `projects/components/dma-ip/rapids/rtl/macro/` **Status:** Implemented **Last Updated:** 2026-10-16

17.1 Overview

The scheduler group array runs one scheduler per channel and three engines that issue the array's AXI traffic: the descriptor fetch engine, the control-read engine and the control-write engine. `SHARED_ENGINES` (default 1) selects how the engines are built. With 1, the array has one of each engine for all of its channels, and each engine picks among the schedulers round-robin (see Shared Engines below and the Shared Engines chapter). With 0, every channel has its own three engines inside a scheduler group, and three AXI-side arbiters merge their masters (the original structure). It also owns the descriptor-AXI monitor and merges every channel's MonBus output with the monitor's into one stream.

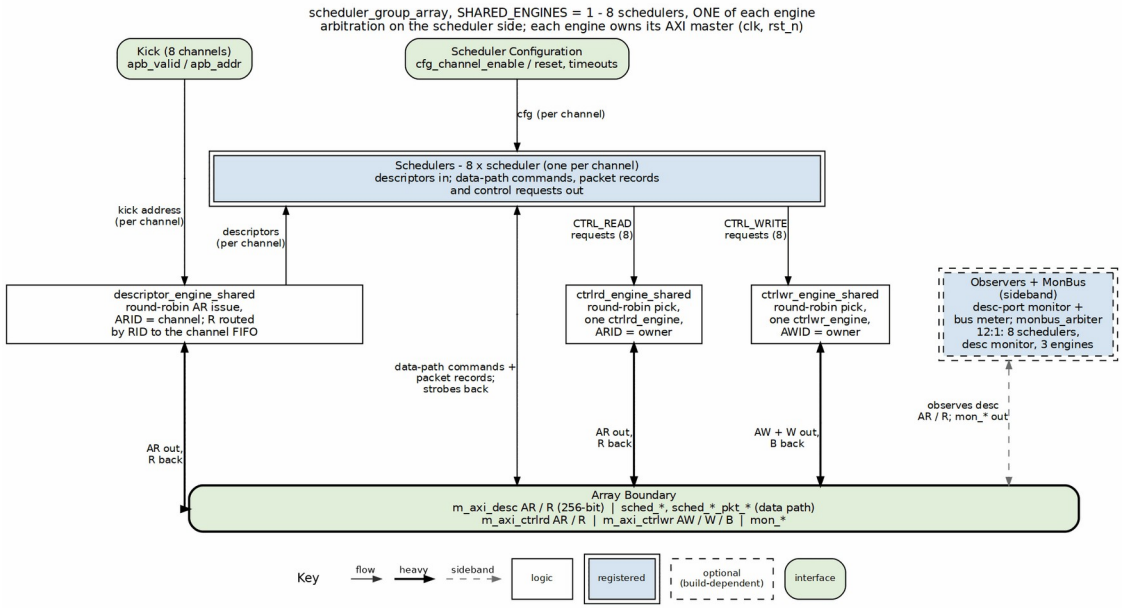
Compared with the RAPIDS Beats array, the sharing, arbitration and monitoring are unchanged. The array adds one parameter, `OFF_W`, and the per-channel packet-record ports, which it connects to each group by channel index.

17.1.1 Key Features

- **One group per channel:** `NUM_CHANNELS` scheduler groups.
- **Shared engines (`SHARED_ENGINES = 1`, default):** one `descriptor_engine_shared`, one `ctrlrd_engine_shared` and one `ctrlwr_engine_shared` per array, arbitrating on the scheduler side and owning their AXI masters.
- **Per-channel engines (`SHARED_ENGINES = 0`):** one scheduler group per channel, and three round-robin arbiters with grant acknowledgment on the AXI side (`u_desc_ar_arbiter`, `u_ctrlrd_ar_arbiter`, `u_ctrlwr_aw_arbiter`).
- **Descriptor-AXI monitor:** a lite AXI read monitor on the shared descriptor master, plus an always-on bus meter for the perf window.
- **Unified MonBus:** one round-robin arbiter over `NUM_CHANNELS + 4` sources (shared engines) or `NUM_CHANNELS + 1` (per-channel engines).
- **Direction gating:** `EN_READ` and `EN_WRITE` let the source array be read-only and the sink array write-only.
- **Per-channel packet records:** vector ports for the byte length and offset of each channel's packet.

17.1.2 Block Diagram

17.1.3 Figure 3.2.1: Scheduler Group Array Block Diagram



Scheduler group array block diagram

Source: 08_scheduler_group_array.dot

17.2 Parameters

Table 3.2.1: Scheduler Group Array Parameters

Parameter	Default	Description
NUM_CHANNELS	8	Number of channels.
CHAN_WIDTH	$clog2(NUM_CHANNELS)	Channel id width.
ADDR_WIDTH	64	Address width.
DATA_WIDTH	512	Data width in bits.
AXI_ID_WIDTH	8	AXI ID width.
DESC_MON_BASE_AGENT_ID	16	Base MonBus agent id of the descriptor engines.
SCHED_MON_BASE_AGENT_ID	48	Base MonBus agent id of the schedulers.
DESC_AXI_MON_AGENT_ID	8	MonBus agent id of the descriptor AXI monitor.
MON_UNIT_ID	1	MonBus unit id.
MON_MAX_TRANSACTIONS	16	Transaction table depth of the descriptor monitor.
EN_READ	1	Enable the read direction in every scheduler. Clear for a sink array.
EN_WRITE	1	Enable the write direction in every scheduler. Clear for a source array.
USE_AXI_MONITORS	1	0 omits the descriptor-AXI monitor hardware and ties its MonBus source to zero.
USE_ROW_COL_MAJOR_ADDRESSING	1	0 compiles out the extended addressing path.
GEN_MON	1	0 omits the per-channel completion and error emitters.
SHARED_ENGINES	1	1: one descriptor, control-read and control-write engine for

Parameter	Default	Description
		the array. 0: one of each per channel.
OFF_W	$clog2(DATA_WIDTH/8)	Width of the packet-record byte offset. New in RAPIDS.

17.3 Port List

Vector ports are NUM_CHANNELS wide (NC), indexed by channel.

17.3.1 Clock, Reset and Kick-Off

Table 3.2.2: Clock, Reset and Kick-Off

Signal	Direction	Width	Description
clk	input	1	System clock.
rst_n	input	1	Active-low reset.
apb_valid	input	NC	Per-channel kick-off.
apb_ready	output	NC	Per-channel ready.
apb_addr	input	NC x AW	First descriptor address per channel.
cfg_channel_enable	input	NC	Per-channel enable.
cfg_channel_reset	input	NC	Per-channel soft reset.

17.3.2 Global Scheduler and Descriptor Engine Configuration

These inputs are applied to every channel.

Table 3.2.3: Global Configuration

Signal	Direction	Width	Description
cfg_sched_enable	input	1	Master scheduler enable.
cfg_sched_timeout_cycles	input	32	Timeout window in cycles.
cfg_sched_timeout_limit	input	8	Consecutive timeouts before a fatal error. 0 means never.
cfg_sched_timeout_enable	input	1	Enable timeout detection.

Signal	Direction	Width	Description
cfg_sched_err_enable	input	1	Enable error reporting.
cfg_sched_compl_enable	input	1	Enable CORE Completion packets.
cfg_sched_perf_enable	input	1	Enable performance monitoring.
cfg_desceng_enable	input	1	Master descriptor engine enable.
cfg_desceng_prefetch	input	1	Enable descriptor prefetch.
cfg_desceng_fifo_thresh	input	4	Prefetch threshold.
cfg_desceng_addr0_base, _limit	input	AW	Valid descriptor address range 0.
cfg_desceng_addr1_base, _limit	input	AW	Valid descriptor address range 1.
cfg_ctrlrd_max_try	input	9	Control-read poll retry budget.
tick_1us	input	1	1 us tick for control-read retry spacing.

17.3.3 Descriptor Monitor Configuration and Status

Table 3.2.4: Descriptor Monitor Configuration and Status

Signal	Direction	Width	Description
cfg_desc_mon_enable	input	1	Enable the descriptor AXI monitor.
cfg_desc_mon_err_enable	input	1	Enable error packets.
cfg_desc_mon_perf_enable	input	1	Enable performance packets.
cfg_desc_mon_perf_run	input	1	Perf window control.
cfg_desc_mon_timeout_ena	input	1	Enable timeout packets.

Signal	Direction	Width	Description
ble			
cfg_desc_mon_timeout_cycles	input	32	Timeout threshold.
cfg_desc_mon_latency_thresh	input	32	Latency threshold.
cfg_desc_mon_pkt_mask	input	16	Packet type mask.
cfg_desc_mon_err_select	input	4	Error select.
cfg_desc_mon_err_mask, _timeout_mask, _compl_mask, _thresh_mask, _perf_mask, _addr_mask, _debug_mask	input	8 each	Per-class event masks.
cfg_sts_desc_mon_busy	output	1	Monitor busy.
cfg_sts_desc_mon_active_txns	output	8	Active transactions.
cfg_sts_desc_mon_error_count	output	16	Error count.
cfg_sts_desc_mon_txn_count	output	32	Transaction count.
cfg_sts_desc_mon_conflict_error	output	1	Conflict error.

17.3.4 Descriptor-AXI Perf Window

The perf counters come from an always-on bus meter on the descriptor R channel, not from the monitor. `cfg_desc_mon_perf_run` opens the window: counters clear on its rising edge, count while it is high and hold while it is low. `USE_AXI_MONITORS` does not gate the meter.

Table 3.2.5: Descriptor-AXI Perf Window

Signal	Direction	Width	Description
`sts_desc_mon_win_active`	output	1	Window open.
`sts_desc_mon_win_cycles`	output	32	Cycles in the window.
`sts_desc_mon_prod_cycles`	output	32	R valid and ready.
`sts_desc_mon_bp_cycles`	output	32	R valid, not ready.
`sts_desc_mon_starv_cycles`	output	32	R ready, not valid.
`sts_desc_mon_idle_cycles`	output	32	Neither valid nor ready.
`sts_desc_mon_beat_count`	output	32	R handshakes.
`sts_desc_mon_byte_count`	output	64	Bytes moved on R (32 per descriptor beat).
`sts_desc_mon_burst_count`	output	32	AR handshakes.

17.3.5 Status

Table 3.2.6: Status

Signal	Direction	Width	Description
`descriptor_engine_idle`	output	NC	Per-channel descriptor engine idle.
`scheduler_idle`	output	NC	Per-channel scheduler idle.
`scheduler_state`	output	NC x 7	Per-channel FSM state, one-hot.
`sched_error`	output	NC	Per-channel scheduler error, sticky.

17.3.6 Shared AXI Masters

The three shared masters carry the same signals as the per-group ports in Scheduler Group, prefixed `desc_axi_`, `ctrlrd_axi_` and `ctrlwr_axi_`. The descriptor master returns 256-bit data. The control read master returns 32-bit data. The control write master is 32 bits wide with a 4-bit

strobe. Each is a single interface: the arbiter selects one channel's request onto it and steers the response back by ID.

17.3.7 Per-Channel Scheduler Data Interface

Table 3.2.7: Per-Channel Scheduler Data Interface

Signal	Direction	Width	Description
`sched_rd_valid`	output	NC	Read request per channel.
`sched_rd_addr`	output	NC x AW	Working source byte address.
`sched_rd_beats`	output	NC x 32	Beats still to read.
`sched_wr_valid`	output	NC	Write request per channel.
`sched_wr_ready`	input	NC	Last burst of the descriptor committed.
`sched_wr_addr`	output	NC x AW	Working destination byte address.
`sched_wr_beats`	output	NC x 32	Beats still to write.
`sched_rd_done_strobe`	input	NC	Read burst issued.
`sched_rd_beats_done`	input	NC x 32	Beats in that burst.
`sched_wr_done_strobe`	input	NC	Write AW issued.
`sched_wr_beats_done`	input	NC x 32	Beats in that AW.
`sched_wr_commit_strobe`	input	NC	Write B response received.
`sched_wr_commit_beats`	input	NC x 32	Beats committed.
`sched_rd_error`	input	NC	Sticky read error.
`sched_wr_error`	input	NC	Sticky write error.

17.3.8 Per-Channel Packet Record Interface

Table 3.2.8: Per-Channel Packet Record Interface

Signal	Direction	Width	Description
`sched_rd_pkt_valid`	output	NC	Read-side record pulse per channel.
`sched_rd_pkt_ready`	input	NC	Read-side consumer has room.
`sched_rd_pkt_bytes`	output	NC x 32	Packet length in bytes.
`sched_rd_pkt_offset`	output	NC x OFF_W	Start offset in the first memory beat.
`sched_wr_pkt_valid`	output	NC	Write-side record pulse per channel.
`sched_wr_pkt_ready`	input	NC	Write-side consumer has room.
`sched_wr_pkt_bytes`	output	NC x 32	Packet length in bytes.
`sched_wr_pkt_offset`	output	NC x OFF_W	Start offset in the first memory beat.

Element `[ch]` of each vector connects to the matching port of `scheduler_group[ch]`. The array adds no logic between them. A source array uses only the read set and a sink array only the write set.

17.3.9 MonBus Interface

Table 3.2.9: MonBus Interface

Signal	Direction	Width	Description
`i_mon_time`	input	`monbus_timestamp_t`	Shared monitor timebase.
`mon_valid`	output	1	Packet valid.
`mon_ready`	input	1	Consumer ready.
`mon_packet`	output	`monitor_packet_t`	Packet.
`mon_timestamp`	output	`monbus_timestamp_t`	Timestamp.

17.4 Shared Engines

With `SHARED_ENGINES = 1` the per-channel descriptor and control engines are replaced by one of each for the whole array. The arbitration moves from the AXI side to the scheduler side: each engine picks the next channel to serve round-robin, and it owns its AXI master outright, with no arbiter behind it. ARID and AWID carry the channel. The schedulers, the array's ports and every interface above the array are unchanged.

Table 3.2.11: Shared Engines

Engine	What is shared	What stays per channel
`descriptor_engine_shared`	AR issuer, address-range comparators, the R path (each beat routed by RID straight into its channel's FIFO), MonBus reporter	State code, one pending fetch address, kick-in-progress flag, owed-beat flag, descriptor FIFO (and EXT chunk-1 FIFO)
`ctrlrd_engine_shared`	One `ctrlrd_engine`	Nothing: an owner register says which channel the engine serves
`ctrlwr_engine_shared`	One `ctrlwr_engine`	Nothing: an owner register says which channel the engine serves

Descriptor engine. Each channel keeps one fetch in flight and stays in order. Different channels' fetches overlap on the bus because each carries its own ARID. Each scheduler sees exactly the per-channel engine's behaviour: - the same kick policy, and refusal of a kicked address outside both ranges; - the extended-descriptor second fetch; - the prefetch throttle, up to the threshold plus one buffered; - autonomous chaining, the error pulse, and completion and error MonBus packets.

A fetch issues only when its channel's FIFO has a free entry, so a channel that stops consuming can never hold the shared R channel. A channel reset in the middle of a fetch records the owed beat. The beat is drained when it arrives, and the channel takes no new kick until then. Reset also drains the channel's FIFO, so descriptors prefetched for an aborted chain are not delivered afterwards. Once ARVALID is raised, the AR address and ID hold until accepted.

The shared descriptor engine's MonBus output differs from the per-channel engine's in two ways: - **Range refusal:** the error packet for a refused kicked address carries 0 in its response field. The per-channel engine carried whatever read response it had last latched, which was stale. - **Colliding events:** packets go through a small FIFO, and two events can occur in one cycle: a refusal on the issue side, and a completion or error on the R side. The issue-side packet waits one cycle. If another collision arrives while it waits, the waiting packet is dropped. The per-channel engine also dropped packets, whenever MonBus backpressured it. Tests that count descriptor-engine packets should allow for this.

Control engines. The engine runs one request at a time. - A request is offered only while the engine is idle, its last MonBus packet has drained, and no error remains from the previous owner.

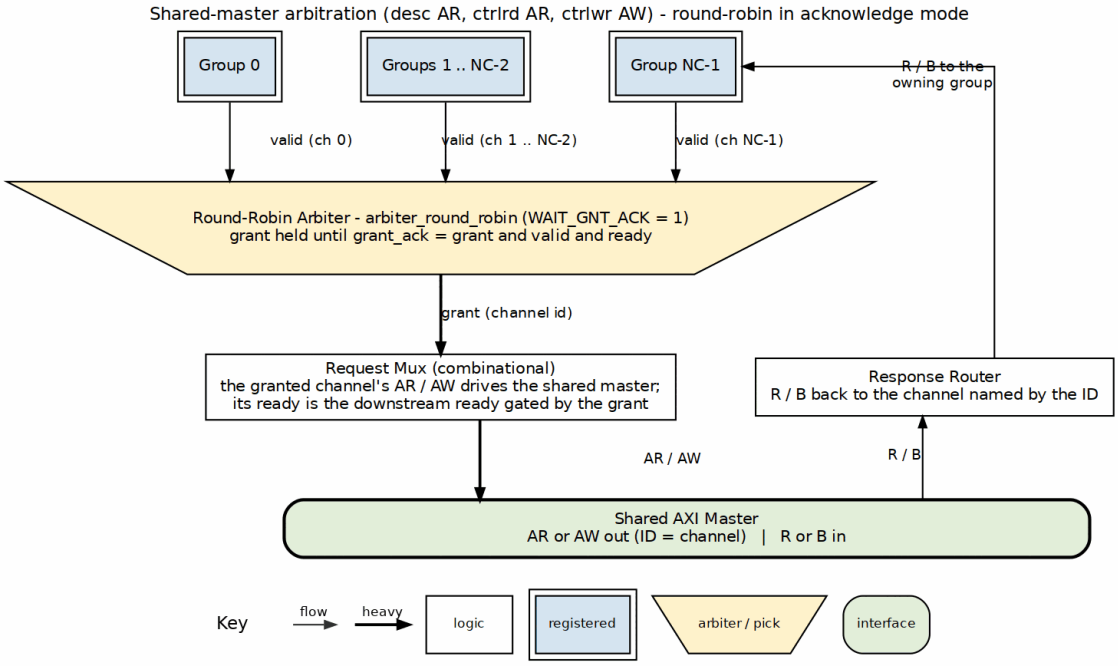
A stale error is cleared with a one-cycle engine reset before the next channel's request is taken. - The owner sees the engine's idle and error exactly as a dedicated engine would show them; every other channel sees idle 1 and error 0. - The owner's channel reset resets the engine only while the owner's operation is in flight, so a channel parked in reset cannot hold the engine for everyone else. - MonBus packets are re-tagged with the owner's channel and agent ID (0x20 + channel for control reads, 0x28 + channel for control writes).

Trade-off: a `CTRL_READ` that keeps polling holds the control-read engine, so other channels' control reads wait behind it. Data transfers are unaffected.

17.5 Shared Master Arbitration (SHARED_ENGINES = 0)

Each shared master has a round-robin arbiter over `NUM_CHANNELS` requesters in acknowledge mode. The grant is held until the granted channel's request is accepted (`grant_ack`), so a granted channel always sees its own valid and ready. The AR or AW ready seen by a channel is the downstream ready gated by that channel's grant.

17.5.1 Figure 3.2.2: Shared Master Arbitration



Shared master arbitration

Source: 11_shared_master_arbitration.dot

17.6 MonBus Aggregation

Table 3.2.10: MonBus Source Assignment

Source	Origin	Content
0 to NC-1	`scheduler_group[ch]` (per-channel engines) or `scheduler[ch]` (shared engines)	Per-channel engines: that channel's descriptor engine, scheduler, control read and control write packets. Shared engines: that channel's scheduler packets
NC	Descriptor AXI monitor	Descriptor fetch monitor packets
NC+1	Shared descriptor engine (shared engines only)	Every channel's descriptor completion and error packets
NC+2	Shared control-read engine (shared engines only)	Every channel's control-read packets
NC+3	Shared control-write engine (shared engines only)	Every channel's control-write packets

One round-robin arbiter with acknowledge mode merges the sources into `mon_*`: `NUM_CHANNELS + 4` with shared engines, `NUM_CHANNELS + 1` with per-channel engines. Each source carries its own timestamp alongside the packet.

17.7 Differences from RAPIDS Beats

Table 3.2.11: Array Delta

Item	RAPIDS Beats	RAPIDS
Parameter `OFF_W`	absent	present
`sched_rd_pkt_*`, `sched_wr_pkt_*` vectors	absent	present, wired by channel
Everything else		Unchanged

The beats chapter is Beats Scheduler Group Array.

Last Updated: 2026-09-30

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18 Sink Data Path

Module: snk_data_path.sv **Location:** projects/components/dma-ip/rapids/rtl/macro/
Status: Implemented **Last Updated:** 2026-09-30

18.1 Overview

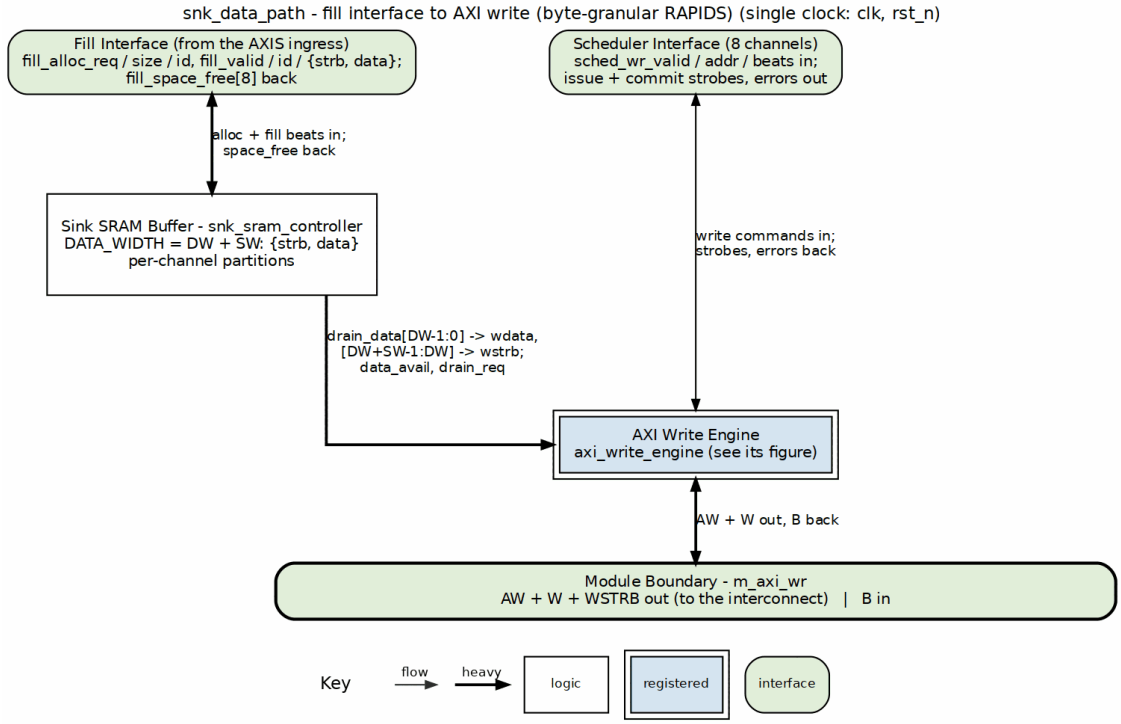
The sink data path joins the per-channel SRAM controller to the AXI write engine. Data enters through the fill interface, waits in the channel's SRAM partition and leaves as AXI4 write bursts.

In RAPIDS every SRAM word carries the byte enables of its data. The fill interface widens from `DW` to `DW + SW` bits, where `SW = DW/8`, and the write engine drives `m_axi_wstrb` from the stored enables. Everything else, including the allocation protocol, the scheduler interface and the burst rules, is as in RAPIDS Beats.

18.1.1 Key Features

- **Per-channel SRAM partitions:** flow control by allocation and space-free counts.
- **Byte enables stored with the data:** the SRAM is `DW + SW` bits wide.
- **AXI write engine:** 4 KB-safe, beat-aligned bursts. See AXI Write Engine.
- **Scheduler integration:** beat counts in, issue and commit strobes out.

18.1.2 Figure 3.3.1: Sink Data Path Block Diagram



Sink data path block diagram

Source: 09_snk_data_path.dot

18.2 Parameters

Table 3.3.1: Sink Data Path Parameters

Parameter	Default	Description
NUM_CHANNELS	8	Number of channels (alias NC).
ADDR_WIDTH	64	Address width (alias AW).
DATA_WIDTH	512	Data width in bits (alias DW). The board uses 256.
AXI_ID_WIDTH	8	AXI ID width (alias IW).
SRAM_DEPTH	512	Words per channel partition (alias SD).
SEG_COUNT_WIDTH	$clog2(SRAM_DEPTH) + 1	Space counter width (alias SCW).
PIPELINE	1	SRAM read pipeline setting.
AW_MAX_OUTSTANDING	8	AW commands the write engine may have in flight.
W_PHASE_FIFO_DEPTH	64	Depth of the write engine W-phase queue.
B_PHASE_FIFO_DEPTH	16	Depth of the write engine B-phase queue.
CIW	$clog2(NC)	Channel id width.
SW	DW / 8	Byte enables per beat. New in RAPIDS.

18.3 Interfaces

18.3.1 Fill Interface

Table 3.3.2: Fill Interface

Signal	Direction	Width	Description
`fill_alloc_req`	input	1	Space allocation request.
`fill_alloc_size`	input	8	Beats to allocate.
`fill_alloc_id`	input	CIW	Channel to allocate in.
`fill_space_free`	output	NC x SCW	Free space per channel.
`fill_valid`	input	1	Fill beat valid.
`fill_ready`	output	1	Ready for a fill beat.
`fill_id`	input	CIW	Channel of the fill beat.
`fill_data`	input	DW + SW	`{byte enables, data}`. Bits `[DW-1:0]` are data. Bits `[DW+SW-1:DW]` are enables.

18.3.2 Configuration

Table 3.3.3: Configuration

Signal	Direction	Width	Description
`cfg_axi_wr_xfer_beats`	input	8	Write burst size cap in beats, minus one. Renamed from `cfg_axi_wr_xfer`.
`cfg_channel_reset`	input	NC	Per-channel reset, level or pulse. Forwarded to the write engine and the SRAM controller wrapper.

18.3.3 Scheduler Interface

Table 3.3.4: Scheduler Interface

Signal	Direction	Width	Description
sched_wr_valid	input	NC	Channel requests writes.
sched_wr_ready	output	NC	Last burst of a descriptor committed.
sched_wr_addr	input	NC x AW	Working destination byte address.
sched_wr_beats	input	NC x 32	Beats still to write.
sched_wr_burst_len	input	NC x 8	Requested burst length. Not used for sizing.
sched_wr_done_strobe	output	NC	AW issued.
sched_wr_beats_done	output	NC x 32	Beats in that AW.
sched_wr_commit_strobe	output	NC	B response received.
sched_wr_commit_beats	output	NC x 32	Beats committed by that B.
sched_wr_error	output	NC	Sticky write error (bad B response).

18.3.4 AXI4 Write Master

The write master is `m_axi_aw*`, `m_axi_w*` and `m_axi_b*` with `IW`-bit IDs and `DW`-bit data. See AXI4 Interface Specification. `m_axi_wstrb` is `DW/8` bits and carries the stored byte enables.

18.4 Byte Enable Path

1. The upstream stage presents `{strb, data}` on `fill_data` with `fill_valid`.
2. The SRAM controller stores the whole `DW + SW` word. It is instantiated with `DATA_WIDTH = DW + SW`.
3. The drain side returns the same word as `drain_data`.
4. `drain_data[DW-1:0]` goes to `axi_wr_sram_data` and `drain_data[DW+SW-1:DW]` goes to `axi_wr_sram_strb` on the write engine.
5. The write engine drives `m_axi_wdata` and `m_axi_wstrb` from those two.

The SRAM depth counts words, so capacity in beats is unchanged. Only the width grows, by one bit per data byte.

18.4.1 Channel Reset

`cfg_channel_reset[ch]` reaches two blocks. The AXI Write Engine stops issuing for the channel, completes an open burst with null beats and clears its error flag. The SRAM controller wrapper (`snk_sram_controller`) holds one single-channel `sram_controller` per channel. Each instance's reset is a registered flop, `r_ch_rst_n[ch]`, which goes low one cycle after `cfg_channel_reset[ch]` rises, so the partition's pointers and counters return to empty. The shared STREAM `sram_controller` has no channel reset of its own and is not modified; giving each channel its own instance and reset is what makes the reset per channel. A reset of one channel leaves the instances of the other channels untouched.

18.5 Differences from RAPIDS Beats

Table 3.3.5: Sink Data Path Delta

Item	RAPIDS Beats	RAPIDS
`fill_data` width	`DW`	`DW + SW`
SRAM word width	`DW`	`DW + SW`
`m_axi_wstrb`	all ones	stored per-beat enables
Config port name	`cfg_axi_wr_xfer`	`cfg_axi_wr_xfer_beats`
Scheduler port names	`sched_wr`, `sched_wr_done`, `sched_wr_commit`	`sched_wr_beats`, `sched_wr_beats_done`, `sched_wr_commit_beats`
Fill allocation, drain, error export		Unchanged

The beats chapter is Sink Data Path. Its timing diagrams apply to the AW, W and B sequencing unchanged.

Last Updated: 2026-09-30

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19 Sink Data Path with AXIS (Ingress)

Module: `snk_data_path_axis.sv` **Location:** `projects/components/dma-ip/rapids/rtl/macro/` **Status:** Implemented **Last Updated:** 2026-09-30

19.1 Overview

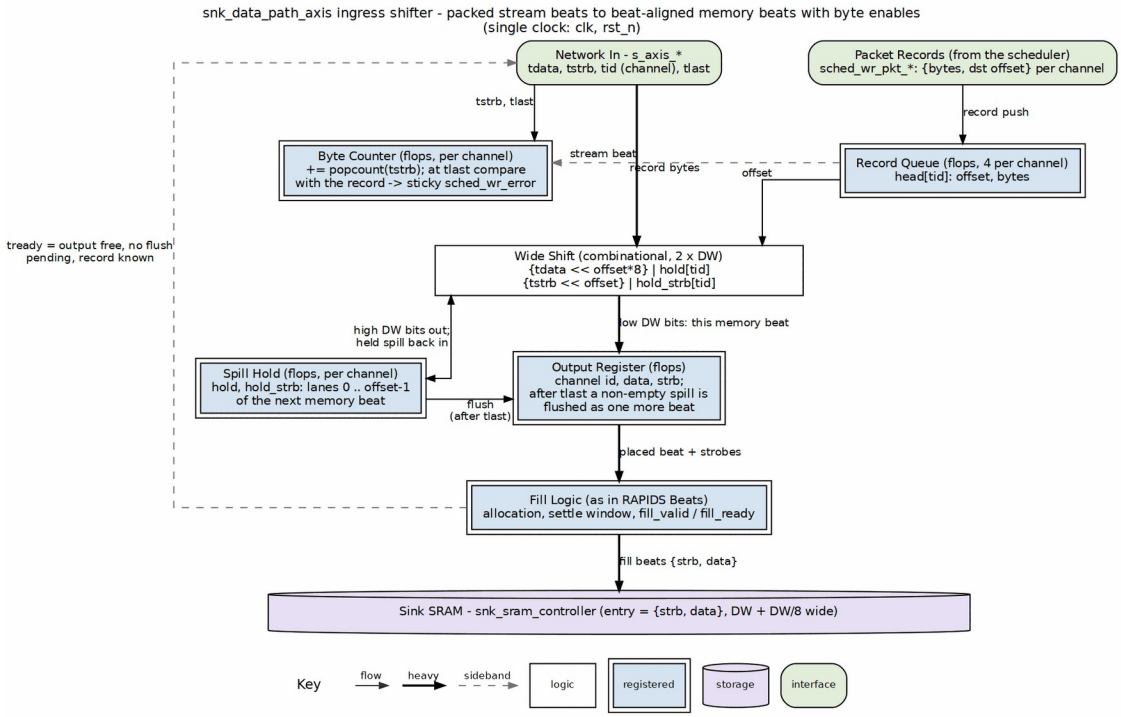
This module puts an AXI-Stream slave in front of the Sink Data Path and converts a packed byte stream into beat-aligned memory beats with byte enables. It is the block that makes the sink side byte-granular.

A packet arrives with its bytes packed from lane 0. Its destination in memory starts at a byte offset inside the first memory beat. The scheduler supplies that offset in a packet record. The module shifts each stream beat up by the offset, holds the bytes that spill past the beat boundary, and merges them into the next memory beat. After the last stream beat, a non-empty spill becomes one more, partial, memory beat.

19.1.1 Key Features

- **Barrel shift by the destination offset:** stream lane `l` lands in memory lane `offset + l`.
- **Byte enables follow the data:** the shifted strobe is stored beside the data and becomes WSTRB.
- **Per-channel state:** spill hold, received-byte count and length-error flag per channel, so packets of different channels interleave by beat.
- **Packet record queue:** four records per channel, fed by the scheduler.
- **Descriptor gating:** the stream is accepted only when the channel has a packet record.
- **Length check:** a packet whose byte count differs from the descriptor sets a sticky channel error.
- **Allocation logic unchanged from RAPIDS Beats**, including the settle wait.

19.1.2 Figure 3.4.1: Ingress Shifter



Ingress shifter

Source: 01_ingress_shifter.dot

19.2 Parameters

Parameters of Sink Data Path apply unchanged. The additions are:

Table 3.4.1: Additional Parameters

Parameter	Default	Description
`AXIS_ID_WIDTH`	8	`tid` width. The low `CIW` bits select the channel.
`AXIS_DEST_WIDTH`	4	`tdest` width. Not used to select a channel.
`AXIS_USER_WIDTH`	1	`tuser` width. Not used.
`SW`	`DW / 8`	Byte lanes per beat.
`OFF_W`	`$clog2(SW)`	Width of the packet-record offset.

19.3 Interfaces

19.3.1 AXI-Stream Slave

Table 3.4.2: AXI-Stream Slave

Signal	Direction	Width	Description
`s_axis_tdata`	input	DW	Packed data.
`s_axis_tstrb`	input	SW	Byte strobes. Contiguous from lane 0. Partial only on the `tlast` beat.
`s_axis_tlast`	input	1	Last beat of the packet.
`s_axis_tid`	input	AXIS_ID_WIDTH	Channel select.
`s_axis_tdest`	input	AXIS_DEST_WIDTH	Unused.
`s_axis_tuser`	input	AXIS_USER_WIDTH	Unused.
`s_axis_tvalid`	input	1	Beat valid.
`s_axis_tready`	output	1	Beat accepted.

`s_axis_tready` is high when the output register is free, no flush beat is pending and the addressed channel has a packet record queued. A channel's stream therefore stalls until its scheduler has fetched the descriptor. `s_axis_tready` is a single signal qualified by the TID of the beat on the bus, so a beat for a channel that has no record blocks the stream, and any channel's beats behind it, until that record arrives. Channels on one stream are not isolated from each other; a system that needs isolation gives each channel its own stream or guarantees the descriptor is queued before the packet is sent. `s_axis_tready` is also low for a beat whose TID names a channel in reset, and high without a record for a channel that is discarding the tail of a cut packet (see Channel Reset).

19.3.2 Scheduler Interface

The write request and completion ports are those of Sink Data Path. The packet record ports are new:

Table 3.4.3: Packet Record Ports

Signal	Direction	Width	Description
`sched_wr_pkt`	input	NC	Record pulse from the

Signal	Direction	Width	Description
_valid			scheduler.
sched_wr_pkt_ready	output	NC	Record queue not full.
sched_wr_pkt_bytes	input	NC x 32	Packet length in bytes.
sched_wr_pkt_offset	input	NC x OFF_W	Destination offset in the first memory beat.
cfg_channel_reset	input	NC	Per-channel reset, level or pulse. Clears the channel's queue, hold, byte count and length error, and discards the tail of a cut packet.

`sched_wr_error` is the OR of the write engine's B-response error and the packet length error of the channel. Both clear on `cfg_channel_reset` (see Channel Reset).

19.3.3 Other Ports

The AXI4 write master is that of Sink Data Path. `cfg_alloc_size` (8 bits) is the allocation segment per request, with 0 treated as 1. The debug outputs are `dbg_sram_bridge_pending`, `dbg_sram_bridge_out_valid`, `dbg_axis_beats_received` and `dbg_axis_packets_received`. `o_active_channel_id` and `o_active_channel_valid` carry the write engine's active channel to the per-channel bus meter.

19.4 Ingress Operation

19.4.1 Packet Records

Each channel has a four-entry queue of `{offset, bytes}`. A record is pushed on `sched_wr_pkt_valid` when the queue is not full and popped when the packet completes. The head record supplies the offset for every beat of the packet and the expected length at `tlast`.

19.4.2 Placement

For an accepted stream beat, with `off` the head record's offset:

- The wide word is `(tdata_m << off*8) | hold_data`, of width `2*DW`, where `tdata_m` is `tdata` with every lane whose `tstrb` bit is clear forced to zero.
- The wide strobe is `(tstrb << off) | hold_strb`, of width `2*SW`.
- The low half goes to the output register as the memory beat and its enables.
- The high half becomes the new hold for the channel.

The strobe mask matters on a partial last beat: the non-strobed lanes carry whatever the sender left there, and unmasked they would shift into the spill hold. A packet that ends without a spill never flushes the hold, so that junk would be ORed into the first memory beat of the channel's next packet. Masking by `tstrb` keeps the hold to real payload bytes.

The first memory beat has its low `off` strobe bits clear, so the bytes below the start address are not written.

19.4.3 Flush and Completion

On `tlast` the received-byte total is compared with the head record. If the wide strobe's high half is non-zero, a flush is scheduled and `s_axis_tready` drops until the hold has been emitted as the last memory beat. The packet then completes and its record pops. If the high half is empty the packet completes on the `tlast` beat itself.

19.4.4 Length Check

`r_pkt_rx_bytes` counts the popcount of `tstrb` over the packet. At `tlast`, a total that differs from the record's byte count sets the channel's sticky error. Nothing is padded or dropped. The error is visible on `sched_wr_error` and the scheduler treats it as fatal. It clears on `cfg_channel_reset`.

19.4.5 Channel Reset

`cfg_channel_reset[ch]` is a register level in the system (`SNK.CHANNEL_RESET.CH_RST` ORed with `GLOBAL_CTRL.GLOBAL_RST`), so a pulse and a held level must both work. The ingress registers it once into `r_rst_d1` and uses `w_rst[ch] = cfg_channel_reset[ch] | r_rst_d1[ch]`, which stretches a one-cycle pulse to two cycles and lets every clear see the reset on a cycle where it is asserted. The stretch is what settles an output-register beat and an allocation

that were launched on the cycle of the reset. The reset is written for one channel; every term is indexed by channel, so the other channels are not affected.

Table 3.4.4: Channel Reset Actions

State	Action on `w_rst[ch]`
Packet record queue	Write and read pointers cleared. A record arriving on the reset cycle is not pushed.
Spill hold, `r_pkt_rx_bytes`	Cleared.
`r_pkt_len_error`	Cleared. The channel's `sched_wr_error` contribution drops.
`r_pending_alloc`, `r_alloc_settle`	Cleared, so no allocation request of the old packet is issued.
Output register (`r_out_valid`), flush beat (`r_flush_valid`)	Dropped when the beat belongs to the channel, so no beat of the old packet enters the SRAM after the reset.
`s_axis_tready`	Low for a beat whose TID names a channel in reset.
`r_discard[ch]`	Set when a packet was part-way through the stream (received bytes non-zero and no flush of its own pending).

With `r_discard[ch]` set the ingress accepts the channel's beats without a record and drops them: `s_axis_tready` does not wait for a packet record, and no state is updated. `r_discard[ch]` clears on the dropped beat that carries `tlast`. The next beat is then an ordinary first beat of a new packet. The sender is not asked to do anything special: it finishes the packet it was sending and the tail is discarded. A packet whose beats had all been accepted (the flush beat aside) is not discarded.

Head-of-line blocking is unchanged. A beat for a channel with no record, and not discarding, still holds `s_axis_tready` low for every channel. After a reset the next packet for the channel waits for its descriptor as before.

The channel's write engine and SRAM partition are cleared by the blocks that own them; see AXI Write Engine and Sink Data Path. All of them settle within a few cycles of the reset, and the scheduler's own reset returns it to CH_IDLE, so a channel reset recovers the channel without `aresetn`.

19.4.6 Allocation

The output register holds one memory beat for the fill interface. Allocation follows the beats design:

- A channel requests `cfg_alloc_size` beats, or the remainder when less is free.
- The next allocation of a channel waits three cycles after its previous one so `fill_space_free` is current.
- A beat leaves the output register only when the fill interface accepts it.

19.5 Differences from RAPIDS Beats

Table 3.4.5: Ingress Delta

Item	RAPIDS Beats	RAPIDS
Stream beat to memory beat	one to one, full lanes	shifted by the offset, spill held
`s_axis_tstrb`	full only	partial on the `tlast` beat
Byte enables into SRAM	none	`{strb, data}` word
Packet record queue	absent	four per channel
Stream gating	buffers before the descriptor	waits for the packet record
Length mismatch	not checked	sticky channel error, cleared by channel reset
Channel reset	not applicable	per-channel clear, cut packet tail discarded

The beats chapter is Sink Data Path with AXIS.

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20 Sink SRAM Controller Specification

Module: `snk_sram_controller_beats.sv` **Location:** `projects/components/dma-ip/rapids/rtl/macro_beats/` **Status:** Implemented **Last Updated:** 2025-01-10

20.1 Overview

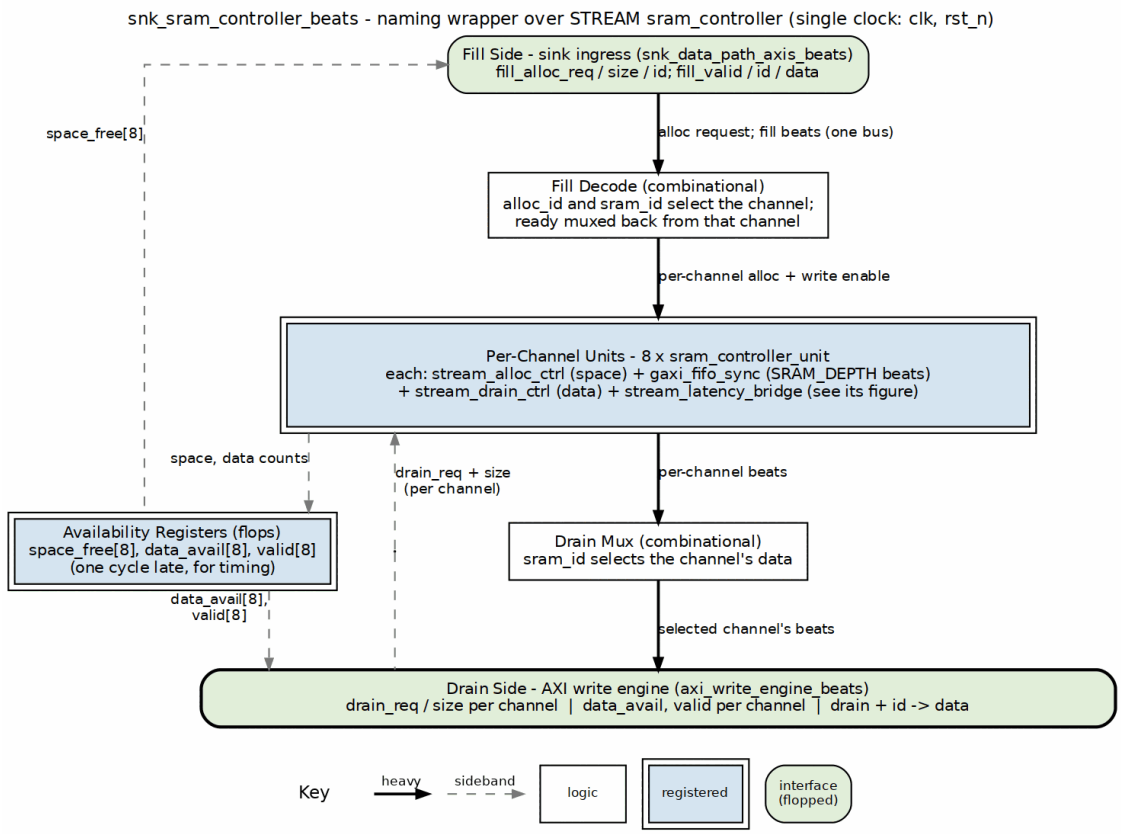
The Sink SRAM Controller is a naming wrapper, with no logic of its own, around STREAM's `sram_controller` – the one per-channel SRAM implementation both DMAs share since `bdf4e0dff`. It maps the RAPIDS `fill_*`/`drain_*` names onto STREAM's ports; the eight per-channel units (allocation counter, FIFO, drain counter, latency bridge) live inside STREAM's block. It manages 8 channel units, providing per-channel buffering with channel arbitration for the shared AXI write engine.

20.1.1 Key Features

- **One shared implementation:** instantiates STREAM's `sram_controller` once; it holds 8 `sram_controller_units`, each a `gaxi_fifo_sync` FIFO with `stream_alloc_ctrl`, `stream_drain_ctrl` and `stream_latency_bridge`
- **Channel Arbitration:** none inside the controller – the AXI write engine picks a channel and presents it on `drain_id`
- **Flow Control Integration:** per-channel `stream_alloc_ctrl` / `stream_drain_ctrl` virtual FIFOs; the drain counter's virtual depth is 2 x SRAM_DEPTH (stream BUG-011)
- **Space Tracking:** Reports available space per channel

20.1.2 Block Diagram

20.1.3 Figure 3.5.1: Sink SRAM Controller Block Diagram



Sink SRAM controller block diagram

Source: 15_snk_sram_controller.dot

20.2 Parameters

```
parameter int NUM_CHANNELS = 8;
parameter int DATA_WIDTH = 512;
parameter int SRAM_DEPTH = 512;                    // Per-channel depth
parameter int ADDR_WIDTH = $clog2(SRAM_DEPTH);
```

: Table 3.5.1: Sink SRAM Controller Parameters

20.3 Port List

20.3.1 Clock and Reset

Table 3.5.2: Clock and Reset

Signal	Direction	Width	Description
`clk`	input	1	System clock
`rst_n`	input	1	Active-low reset

20.3.2 Fill Interface (ID-Selected)

Table 3.5.3: Fill Interface (ID-Selected)

Signal	Direction	Width	Description
`fill_alloc_req`	input	1	Allocation request (single, ID-selected)
`fill_alloc_size`	input	8	Beats to allocate
`fill_alloc_id`	input	CIW	Transaction ID selects the channel
`fill_space_free`	output	NC x SCW	Available space per channel
`fill_valid`	input	1	Fill data valid
`fill_ready`	output	1	Ready for fill data
`fill_id`	input	CIW	Transaction ID selects the channel
`fill_data`	input	DW	Fill data

20.3.3 Arbitrated Drain Interface (to AXI Write Engine)

Table 3.5.4: Arbitrated Drain Interface

Signal	Direction	Width	Description
`drain_data_avail`	output	NC x SCW	Data available per channel
`drain_req`	input	NC	Drain reservation request per channel
`drain_size`	input	NC x 8	Beats to reserve

Signal	Direction	Width	Description
drain_valid	output	NC	Drain valid, registered (for arbitration)
drain_valid_comb	output	NC	Drain valid, combinational (gates m_axi_wvalid)
drain_read	input	1	Consumer read strobe
drain_id	input	CIW	Channel ID select for drain
drain_data	output	DW	Drain data (muxed from selected channel)

20.3.4 Debug Interface

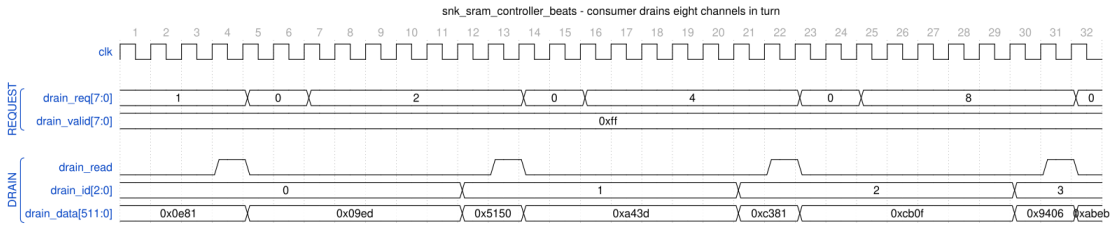
Table 3.5.5: Debug Interface

Signal	Direction	Width	Description
dbg_bridge_pending	output	NC	Bridge has a pending beat per channel
dbg_bridge_out_valid	output	NC	Bridge output valid per channel

20.4 Drain Selection

There is no arbiter inside the controller. Every channel unit raises its own `drain_valid[ch]` / `drain_size[ch]` when it has data; the AXI write engine picks a channel, drives it on `drain_id`, and the controller steers that channel's beats to the single `drain_*` port. Switching channels costs nothing beyond the consumer's own decision, and a channel keeps presenting `drain_valid` until it is empty.

20.4.1 Figure 3.5.2: Drain Selection by the Consumer



snk_sram_controller_beats - consumer drains eight channels in turn

Source: snk_sram_controller_drain_select.json, captured from `dv/tests/macro_beats/test_snk_sram_controller_beats.py` (`multi_channel`, 8 channels, 512-bit data, 512-entry SRAM, `REG_LEVEL=GATE`) with `WAVES=1`.

Reading it: all eight channels have data (`drain_valid` = 0xff) and the consumer walks them in turn. It raises one bit of `drain_req` at a time; the controller answers with `drain_id` naming that channel and `drain_data` presenting its head word on the same cycle, and the consumer's `drain_read` pulse pops it. Each channel's turn lasts nine cycles here because the test releases the request and re-arms the next one; the controller itself needs no gap between channels.

20.5 Integration with AXI Write Engine

```
snk_sram_controller_beats #(
    .NUM_CHANNELS(8),
    .DATA_WIDTH(512),
    .SRAM_DEPTH(512)
) u_snk_sram_ctrl (
    .clk                    (clk),
    .rst_n                  (rst_n),

    // Per-channel fill interface
    .fill_alloc_req         (fill_alloc_req),
    .fill_alloc_size        (fill_alloc_size),
    .fill_space_free        (fill_space_free),
    .fill_valid             (fill_valid),
    .fill_ready             (fill_ready),
    .fill_data              (fill_data),

    // Arbitrated drain interface
    .drain_req              (sram_drain_req),
    .drain_id               (sram_drain_id),
    .drain_size             (sram_drain_beats),

    // Status
    .drain_data_avail       (channel_data_avail),
    .dbg_bridge_pending     (channel_empty),
    .dbg_bridge_out_valid   (channel_full)
);
```

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21 Source Data Path Specification

Module: `src_data_path_beats.sv` **Location:** `projects/components/dma-ip/rapids/rtl/macro_beats/` **Status:** Implemented **Last Updated:** 2025-01-10

21.1 Overview

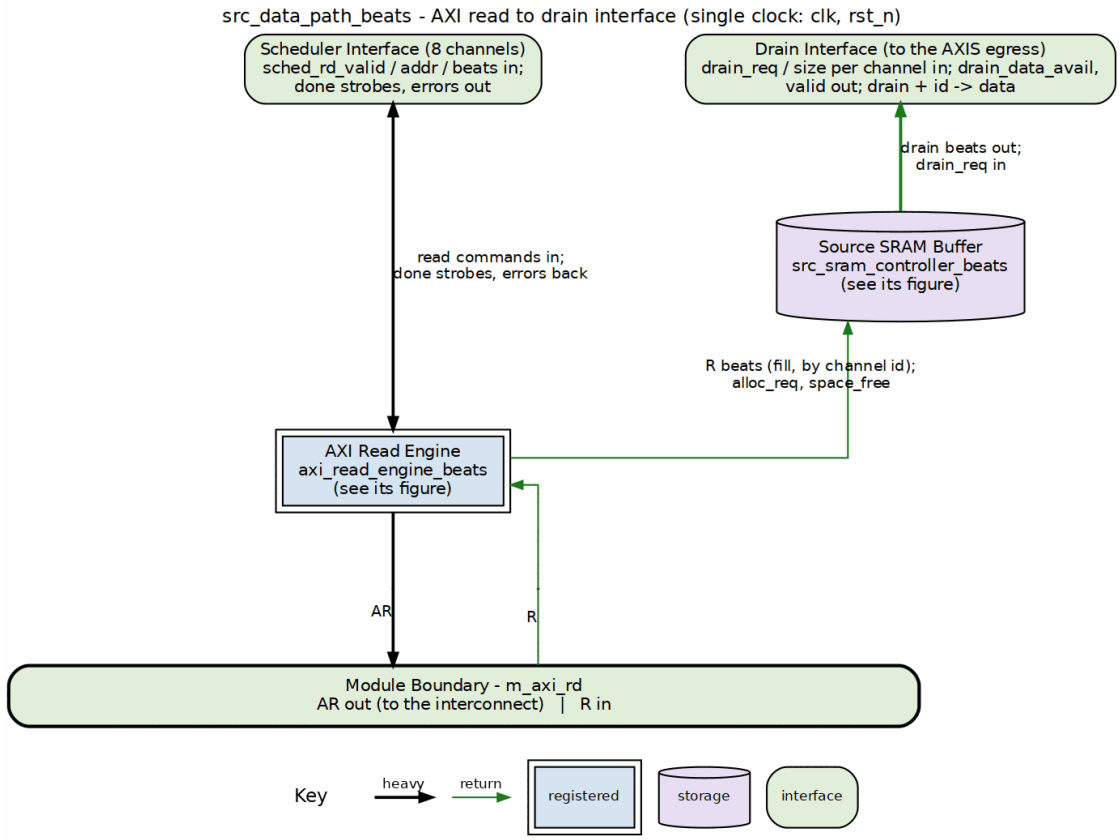
The Source Data Path integrates the AXI read engine and SRAM controller for memory-to-network transfers. Data is read from system memory via AXI4, buffered in SRAM, and delivered through the drain interface to the network.

21.1.1 Key Features

- **AXI Read Engine:** Streaming burst reads from system memory
- **8-Channel SRAM Controller:** Per-channel buffering with flow control
- **Beat-Level Tracking:** Precise flow control at beat granularity
- **Scheduler Integration:** Receives transfer requests from scheduler

21.1.2 Block Diagram

21.1.3 Figure 3.6.1: Source Data Path Block Diagram



Source data path block diagram

Source: 16_source_data_path.dot

21.2 Parameters

```
parameter int NUM_CHANNELS = 8;
parameter int ADDR_WIDTH = 64;
parameter int DATA_WIDTH = 512;
parameter int AXI_ID_WIDTH = 8;
parameter int SRAM_DEPTH = 512;
parameter int AR_MAX_OUTSTANDING = 8;
parameter int R_PHASE_FIFO_DEPTH = 64;
```

: Table 3.6.1: Source Data Path Parameters

21.3 Data Flow

21.3.1 Figure 3.6.2: Source Path Data Flow Sequence

```
1. Scheduler Read Request
   - sched_rd_valid = 1
   - sched_rd_addr = source address
   - sched_rd_beats = N
   - sched_rd_id = channel
        |
        v
2. Space Check (stream_alloc_ctrl inside STREAM's sram_controller)
   - Check src_drain_space_free[channel] >= N
   - Allocate N beats of space
        |
        v
3. AXI Read Transaction
   - axi_read_engine issues AR
   - m_axi_araddr = sched_rd_addr
   - m_axi_arlen = calculated from beats
        |
        v
4. R Channel Data
   - Data arrives on m_axi_rdata
   - Written to channel's SRAM partition
        |
        v
5. SRAM Write Complete
   - stream_drain_ctrl notified via the unit's latency bridge
   - Data available for drain
        |
        v
6. Drain Interface
   - src_drain_valid = 1
   - Data delivered to network master
        |
        v
7. Completion
   - sched_rd_done_strobe = 1
   - Scheduler notified of completion
```

21.4 Port List

21.4.1 Clock and Reset

Table 3.6.2: Clock and Reset

Signal	Direction	Width	Description
`clk`	input	1	System clock
`rst_n`	input	1	Active-low reset

21.4.2 Configuration

Table 3.6.3: Configuration

Signal	Direction	Width	Description
`cfg_axi_rd_xfer_beats`	input	8	AXI read transfer size in beats (all channels)

21.4.3 Scheduler Interface

Table 3.6.4: Scheduler Interface

Signal	Direction	Width	Description
`sched_rd_valid`	input	NC	Channel requests read
`sched_rd_addr`	input	NC x AW	Source addresses
`sched_rd_beats`	input	NC x 32	Beats remaining to read
`sched_rd_done_strobe`	output	NC	Burst completed (pulsed 1 cycle)
`sched_rd_beats_done`	output	NC x 32	Beats completed in burst
`sched_rd_error`	output	NC	Sticky error flag per channel

21.4.4 AXI Read Master Interface

Table 3.6.5: AXI Read Master Interface

Signal	Direction	Width	Description
`m_axi_arvalid`	output	1	AR channel valid

Signal	Direction	Width	Description
m_axi_arready	input	1	AR channel ready
m_axi_araddr	output	AW	Read address
m_axi_arlen	output	8	Burst length
m_axi_arsize	output	3	Burst size
m_axi_arburst	output	2	Burst type
m_axi_arid	output	ID_W	Transaction ID
m_axi_rvalid	input	1	R channel valid
m_axi_rready	output	1	R channel ready
m_axi_rdata	input	DW	Read data
m_axi_rresp	input	2	Read response
m_axi_rid	input	ID_W	Response ID
m_axi_rlast	input	1	Last beat

21.4.5 Drain Interface (Output to Network)

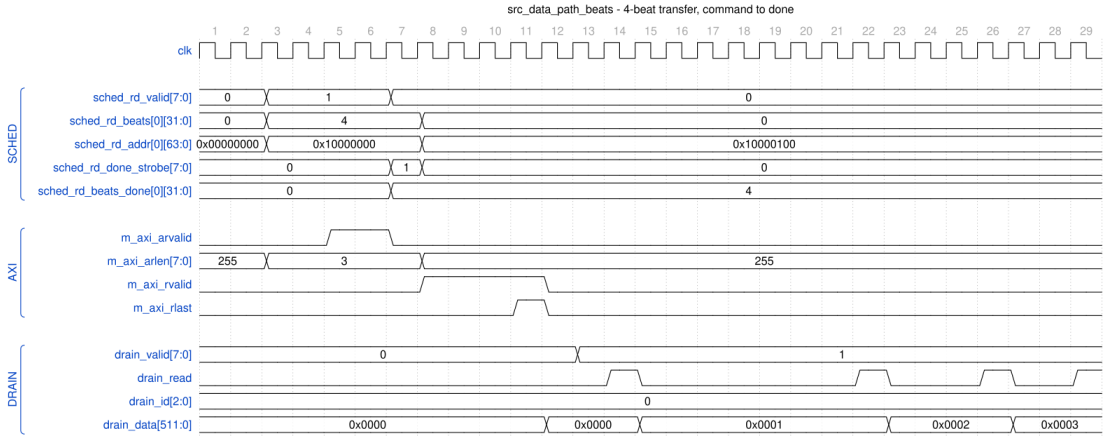
Table 3.6.6: Drain Interface

Signal	Direction	Width	Description
drain_data_avail	output	NC x SCW	Data available per channel
drain_req	input	NC	Drain reservation request per channel
drain_size	input	NC x 8	Beats to reserve
drain_valid	output	NC	Drain data valid per channel
drain_valid_comb	output	NC	Combinational per-channel valid (beat gate)
drain_read	input	1	Consumer read strobe
drain_id	input	CIW	Channel ID select for drain
drain_data	output	DW	Drain data (muxed from

Signal	Direction	Width	Description
			selected channel)

21.5 Timing Diagram

21.5.1 Figure 3.6.3: Source Path Transfer Timing



src_data_path_beats - 4-beat transfer, command to done

Source: src_data_path_transfer.json, captured inside `rapids_core_beats.u_src.u_source_data_path`, captured from `dv/tests/top_beats/test_rapids_core_beats.py` (source path, channel 0, 4 beats, 512-bit data, `REG_LEVEL=GATE`) with `WAVES=1`.

Reading it: the scheduler holds `sched_rd_valid[0]` with 4 beats at 0x10000000. The read engine issues the AR (`arlen = 3`) two cycles later; the four R beats land from cycle 8, `rlast` on the fourth, and the cycle after it the engine pulses `sched_rd_done_strobe[0]` with `beats_done = 4` – the scheduler advances its address to 0x10000100 on that strobe. The data reaches the source SRAM two cycles after the last R beat (`drain_valid[0]` rises) and the AXIS side drains it one word per `drain_read` pulse at the pace the egress consumer allows.

21.6 Integration Example

```
src_data_path_beats #(
    .NUM_CHANNELS(8),
    .ADDR_WIDTH(64),
    .DATA_WIDTH(512),
    .SRAM_DEPTH(512)
) u_source_path (
    .clk                    (clk),
    .rst_n                  (rst_n),

    // Scheduler interface
    .sched_rd_valid         (sched_rd_valid),
    .sched_rd_addr          (sched_rd_addr),
    .sched_rd_beats         (sched_rd_beats),
    .sched_rd_done_strobe   (sched_rd_done_strobe),
    .sched_rd_beats_done    (sched_rd_beats_done),
    .sched_rd_error         (sched_rd_error),

    // AXI read master
    .m_axi_arvalid          (src_axi_arvalid),
    .m_axi_arready          (src_axi_arready),
    .m_axi_araddr           (src_axi_araddr),
    .m_axi_arlen            (src_axi_arlen),
    .m_axi_rvalid           (src_axi_rvalid),
    .m_axi_rready           (src_axi_rready),
    .m_axi_rdata            (src_axi_rdata),
    .m_axi_rlast            (src_axi_rlast),

    // Drain interface
    .drain_valid            (src_drain_valid),
    .drain_data             (src_drain_data),
    .drain_id               (src_drain_id),
    .drain_data_avail       (src_data_avail)
);
```

Last Updated: 2025-01-10

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22 Source Data Path with AXIS (Egress)

Module: `src_data_path_axis.sv` **Location:** `projects/components/dma-ip/rapids/rtl/macro/` **Status:** Implemented **Last Updated:** 2026-09-30

22.1 Overview

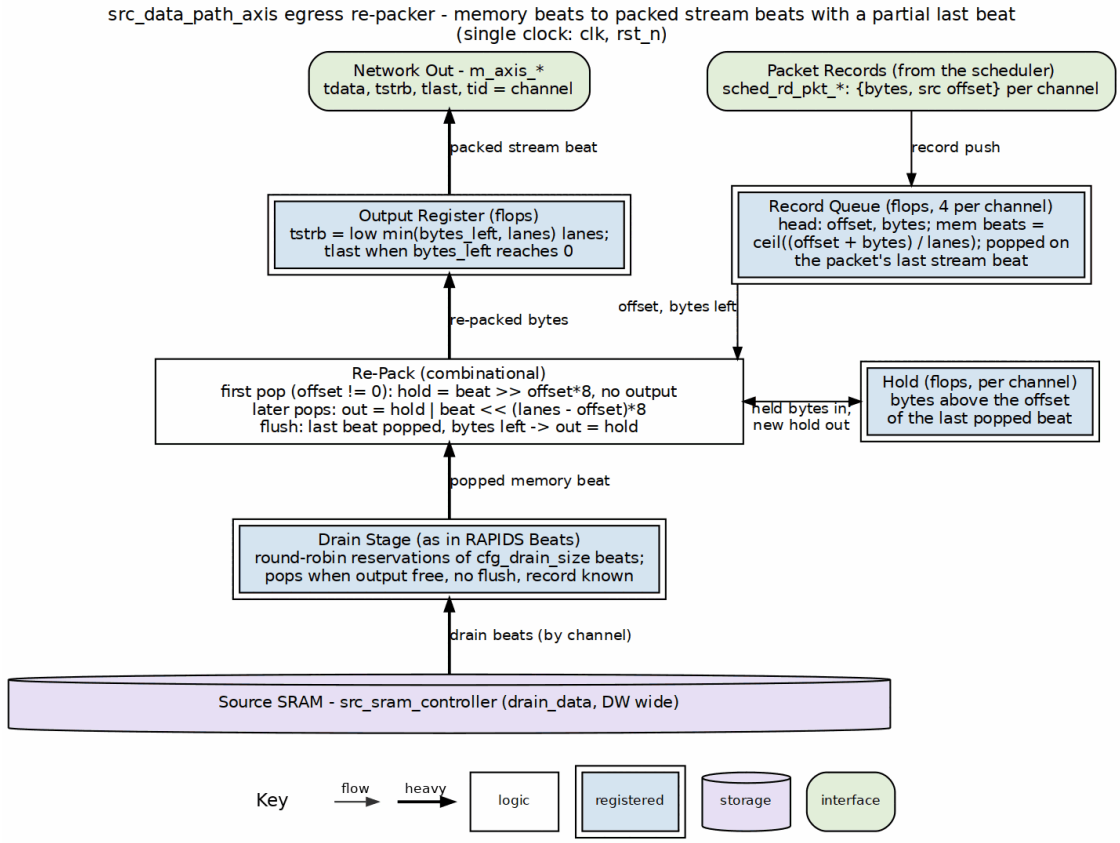
This module puts an AXI-Stream master behind the source data path. It pops beat-aligned memory beats from the SRAM and re-packs them into stream beats that start at lane 0, so a packet leaves as exactly the bytes its descriptor named.

A descriptor's source address may start in the middle of a beat. The scheduler supplies that offset and the byte length in a packet record. The module shifts each memory beat down by the offset, holds the bytes that belong to the next stream beat, and marks the last stream beat with a contiguous strobe and `tlast`. One descriptor produces one AXI-Stream packet.

22.1.1 Key Features

- **Barrel shift by the source offset:** memory lane `offset + l` goes out on stream lane `l`.
- **Partial last beat:** `tstrb` is the low `min(bytes_left, SW)` lanes and `tlast` marks the final beat.
- **Packets follow descriptors,** not drain reservations.
- **Two-stage reservation and drain,** unchanged from RAPIDS Beats. The drain stage advances on memory-beat pops, not on stream handshakes.
- **Per-channel state:** hold, bytes-left and beats-left per channel, so reservations of different channels interleave.
- **Packet record queue:** four records per channel, fed by the scheduler.

22.1.2 Figure 3.7.1: Egress Shifter



Egress shifter

Source: 02_egress_shifter.dot

22.2 Parameters

Table 3.7.1: Parameters

Parameter	Default	Description
NUM_CHANNELS	8	Channels (alias NC).
ADDR_WIDTH	64	Address width (alias AW).
DATA_WIDTH	512	Data width in bits (alias DW). The board uses 256.
AXI_ID_WIDTH	8	AXI ID width (alias IW).
SRAM_DEPTH	512	Words per channel partition.
SEG_COUNT_WIDTH	$clog2(SRAM_DEPTH) + 1	Space counter width.
PIPELINE	1	SRAM read pipeline setting.
AR_MAX_OUTSTANDING	8	ARs the read engine may have in flight.
AXIS_ID_WIDTH	8	tid width.
AXIS_DEST_WIDTH	4	tdest width.
AXIS_USER_WIDTH	1	tuser width.
SW	DW / 8	Byte lanes per beat.
OFF_W	$clog2(SW)	Width of the packet-record offset.

22.3 Interfaces

22.3.1 Configuration

Table 3.7.2: Configuration

Signal	Direction	Width	Description
`cfg_axi_rd_xfer_beats`	input	8	Read burst size cap in beats, minus one.
`cfg_drain_size`	input	8	Beats per drain reservation. 0 is treated as 1.
`cfg_channel_reset`	input	NC	Per-channel reset, level or pulse. Clears the channel's queue, hold, egress counters, drain reservations and error flag.

22.3.2 Scheduler Interface

Table 3.7.3: Scheduler Interface

Signal	Direction	Width	Description
`sched_rd_valid`	input	NC	Channel requests reads.
`sched_rd_addr`	input	NC x AW	Working source byte address.
`sched_rd_beats`	input	NC x 32	Beats still to read.
`sched_rd_pkt_valid`	input	NC	Record pulse from the scheduler.
`sched_rd_pkt_ready`	output	NC	Record queue not full.
`sched_rd_pkt_bytes`	input	NC x 32	Packet length in bytes.
`sched_rd_pkt_offset`	input	NC x OFF_W	Source offset in the first memory beat.
`sched_rd_done_strobe`	output	NC	AR issued.
`sched_rd_bea`	output	NC x 32	Beats in that AR.

Signal	Direction	Width	Description
ts_done			
sched_rd_error	output	NC	Sticky read error.

22.3.3 AXI-Stream Master

Table 3.7.4: AXI-Stream Master

Signal	Direction	Width	Description
m_axis_tdata	output	DW	Packed data.
m_axis_tstrb	output	SW	Contiguous from lane 0. Partial only on the tlast beat.
m_axis_tlast	output	1	Last beat of the packet.
m_axis_tid	output	AXIS_ID_WIDTH	Channel.
m_axis_tdest	output	AXIS_DEST_WIDTH	Channel.
m_axis_tuser	output	AXIS_USER_WIDTH	Zero.
m_axis_tvalid	output	1	Beat valid.
m_axis_tready	input	1	Consumer ready.

22.3.4 Other Ports

The AXI4 read master is that of the AXI Read Engine, with IW-bit IDs and DW-bit data. The debug outputs are dbg_rd_all_complete, dbg_r_beats_rcvd, dbg_sram_writes, dbg_arb_request, dbg_sram_bridge_pending, dbg_sram_bridge_out_valid, dbg_axis_beats_sent and dbg_axis_packets_sent.

22.4 Egress Operation

22.4.1 Reservation and Drain

Unchanged from RAPIDS Beats. A round-robin decision per cycle selects a channel whose available data, less reservations still in flight, reaches `cfg_drain_size` beats. It may also select one whose fill is finished and has any data left. Each decision is a one-cycle `drain_req` of up to `cfg_drain_size` beats. Up to four reservations queue ahead of the drain stage, which loads the next entry on the last beat of the current one.

22.4.2 Pop

A memory beat is popped when all of these hold:

- The drain stage is active on channel `c` and has a beat.
- The output register is free or being taken.
- No flush is pending.
- The channel's packet record is known.

Each pop counts against the reservation block, so the drain stage advances on pops.

22.4.3 Shift and Hold

With `off` the head record's offset:

Table 3.7.5: Egress Shift Rules

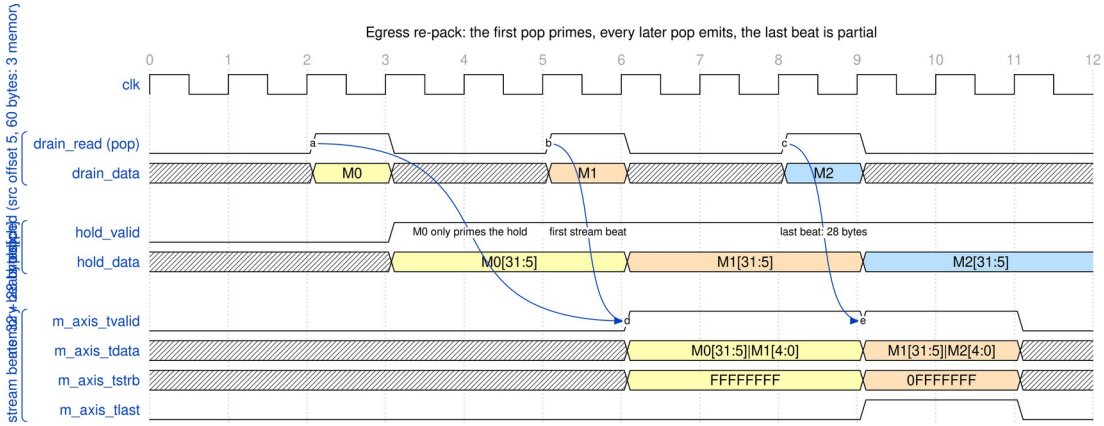
Situation	Action
`off = 0`	Every pop emits a stream beat.
`off > 0`, first pop	The pop only primes the hold with `beat >> off*8`. It emits only when the packet fits in that single memory beat.
`off > 0`, later pops	Emit `hold | (beat << (SW - off)*8)`. The new hold is `beat >> off*8`.
Last memory beat popped, bytes left	The flush emits the hold alone.

The memory beat count of a packet is `(off + bytes + SW - 1) >> OFF_W`, computed in 33 bits.

22.4.4 Strobe and Last

Each emitted beat carries `n = min(bytes_left, SW)` bytes. `tstrb` is the low `n` lanes. `tlast` is set on the beat where `bytes_left` equals `n`. That beat also retires the packet record and clears the channel's hold, started flag and counters.

22.4.5 Waveform 3.7: Egress Re-pack



Egress re-pack

Source: 01_egress_repack.json

Offset 5, 60 bytes and a 256-bit beat give three memory beats. The first pop, M0, only primes the hold. The second pop emits the first stream beat with 32 bytes. The third pop emits the last beat with 28 bytes, `tstrb` `0FFFFFFF` and `tlast`.

22.5 Channel Reset

`cfg_channel_reset[ch]` is a register level, so a pulse and a held level must both work. As in the sink ingress the data path forms `w_rst[ch] = cfg_channel_reset[ch] | r_rst_d1[ch]`, a two-cycle stretch of a pulse. The read engine clears its error flag and settles the reads in flight (see AXI Read Engine); the data path clears everything that would otherwise let the old descriptor keep streaming:

Table 3.7.6: Channel Reset Actions

State	Action on `w_rst[ch]`
Packet record queue	Pointers cleared. A record arriving on the reset cycle is not pushed.
Egress state	`r_eg_started`, `r_hold_valid`, `r_hold_data`, `r_eg_bytes_left` and `r_eg_mem_left` cleared.
Drain reservation	Grants for the channel are masked. `r_drain_tminus1[ch]` is cleared.
Reservation queue	Entries that belong to the channel are zeroed. An entry of size 0 loads the drain stage idle, so the drain stage skips it. A reservation being drained for the channel is dropped.
Output register	A beat already in `r_out` completes, so `m_axis` stays stable.

The channel's SRAM partition is cleared by `src_sram_controller`, which holds one single-channel `sram_controller` per channel with a registered per-channel reset `r_ch_rst_n[ch]`. The shared STREAM controller has no channel reset of its own and is not modified. Reads of the old descriptor that are already on the bus are discarded by the read engine, so no stale data reaches the cleared partition.

A packet cut by the reset is left unterminated. If the reset lands after the first beat of a packet has gone out and before its `tlast` beat, the data path does not send a `tlast` and does not send a null terminator. The downstream receiver sees the packet stop. A packet that had not started, or whose `tlast` beat was already in `r_out` when the reset hit, is unaffected. A receiver that needs to resynchronize after a channel reset treats the reset as the end of the packet.

Other channels are unaffected: every term above is indexed by channel, the drain and reservation stages skip only the channel in reset, and their beats keep flowing through the shared `m_axis`.

22.6 Differences from RAPIDS Beats

Table 3.7.7: Egress Delta

Item	RAPIDS Beats	RAPIDS
Stream beat from memory beat	one to one, full lanes	shifted by the offset, hold merged
`m_axis_tstrb`	all ones	contiguous, partial on the last beat
`m_axis_tlast`	on the drain block boundary	on the last byte of the descriptor
Drain advance	stream handshakes	memory-beat pops
Packet framing	one per drain block	one per descriptor
Packet record queue	absent	four per channel
Channel reset	not applicable	per-channel clear of every stage, cut packet left unterminated
Reservation stage, allocation of drain		Unchanged

The beats chapter is Source Data Path with AXIS.

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23 Source SRAM Controller Specification

Module: `src_sram_controller_beats.sv` **Location:** `projects/components/dma-ip/rapids/rtl/macro_beats/` **Status:** Implemented **Last Updated:** 2025-01-10

23.1 Overview

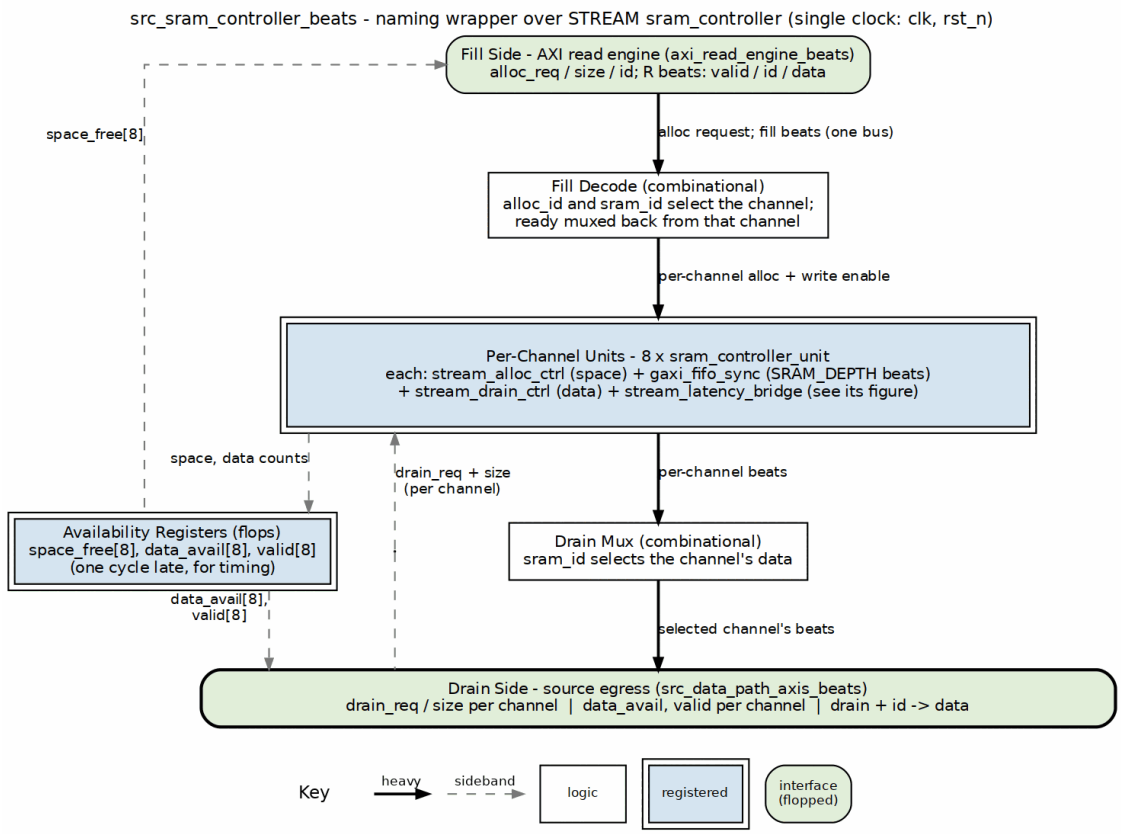
The Source SRAM Controller is a naming wrapper, with no logic of its own, around STREAM's `sram_controller` – the one per-channel SRAM implementation both DMAs share since `bdf4e0dff`. It maps the RAPIDS `fill_*`/`drain_*` names onto STREAM's ports; the eight per-channel units (allocation counter, FIFO, drain counter, latency bridge) live inside STREAM's block. It manages 8 channel units for the source data path, providing per-channel buffering with channel arbitration for data delivery to the network.

23.1.1 Key Features

- **One shared implementation:** instantiates STREAM's `sram_controller` once; it holds 8 `sram_controller_units`, each a `gaxi_fifo_sync` FIFO with `stream_alloc_ctrl`, `stream_drain_ctrl` and `stream_latency_bridge`
- **Channel Arbitration:** none inside the controller – the AXIS egress arbiter picks a channel and presents it on `drain_id`
- **Flow Control Integration:** per-channel `stream_alloc_ctrl` / `stream_drain_ctrl` virtual FIFOs; the drain counter's virtual depth is 2 x SRAM_DEPTH (stream BUG-011)
- **Data Availability Tracking:** Reports available data per channel

23.1.2 Block Diagram

23.1.3 Figure 3.8.1: Source SRAM Controller Block Diagram



Source SRAM controller block diagram

Source: 18_src_sram_controller.dot

23.2 Parameters

```
parameter int NUM_CHANNELS = 8;
parameter int DATA_WIDTH = 512;
parameter int SRAM_DEPTH = 512;                    // Per-channel depth
parameter int ADDR_WIDTH = $clog2(SRAM_DEPTH);
```

: Table 3.8.1: Source SRAM Controller Parameters

23.3 Port List

23.3.1 Clock and Reset

Table 3.8.2: Clock and Reset

Signal	Direction	Width	Description
`clk`	input	1	System clock
`rst_n`	input	1	Active-low reset

23.3.2 Fill Interface (ID-Selected, from AXI Read Engine)

Table 3.8.3: Fill Interface (ID-Selected)

Signal	Direction	Width	Description
`fill_alloc_req`	input	1	Allocation request (single, ID-selected)
`fill_alloc_size`	input	8	Beats to allocate
`fill_alloc_id`	input	CIW	Transaction ID selects the channel
`fill_space_free`	output	NC x SCW	Available space per channel
`fill_valid`	input	1	Fill data valid
`fill_ready`	output	1	Ready for fill data
`fill_id`	input	CIW	Transaction ID selects the channel
`fill_data`	input	DW	Fill data

23.3.3 Arbitrated Drain Interface (to Network)

Table 3.8.4: Arbitrated Drain Interface

Signal	Direction	Width	Description
`drain_data_avail`	output	NC x SCW	Data available per channel
`drain_req`	input	NC	Drain reservation request per channel
`drain_size`	input	NC x 8	Beats to reserve

Signal	Direction	Width	Description
drain_valid	output	NC	Drain data valid per channel
drain_valid_comb	output	NC	Combinational per-channel valid (beat gate)
drain_read	input	1	Consumer read strobe
drain_id	input	CIW	Channel ID select for drain
drain_data	output	DW	Drain data (muxed from selected channel)

23.3.4 Debug Interface

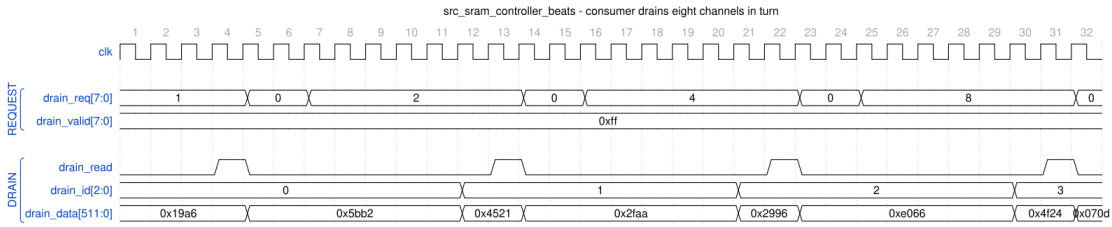
Table 3.8.5: Debug Interface

Signal	Direction	Width	Description
dbg_bridge_pending	output	NC	Bridge has a pending beat per channel
dbg_bridge_out_valid	output	NC	Bridge output valid per channel

23.4 Drain Selection

There is no arbiter inside the controller. Every channel unit raises its own `drain_valid[ch]` / `drain_size[ch]` when it has data; the AXIS egress arbiter picks a channel, drives it on `drain_id`, and the controller steers that channel's beats to the single `drain_*` port. Switching channels costs nothing beyond the consumer's own decision, and a channel keeps presenting `drain_valid` until it is empty.

23.4.1 Figure 3.8.2: Drain Selection by the Consumer



src_sram_controller_beats - consumer drains eight channels in turn

Source: src_sram_controller_drain_select.json, captured from `dv/tests/macro_beats/test_src_sram_controller_beats.py` (`multi_channel`, 8 channels, 512-bit data, 512-entry SRAM, `REG_LEVEL=GATE`) with `WAVES=1`.

Reading it: the same selection contract as the sink controller (Figure 3.5.2). With `drain_valid` = 0xff the consumer asks for one channel at a time through `drain_req`; `drain_id` and `drain_data` follow the request on the same cycle and `drain_read` pops the word. The nine-cycle turn per channel is the test's pacing, not the controller's.

23.5 Comparison: Sink vs Source SRAM Controller

Table 3.8.6: Sink vs Source Controller Comparison

Aspect	Sink Controller	Source Controller
Fill Source	Network ingress	AXI read data
Drain Destination	AXI write engine	Network egress
Fill Interface	Per-channel from network	Per-channel from AXI R
Drain Interface	Arbitrated SRAM read	Arbitrated drain to network
Primary Use	Network -> Memory	Memory -> Network

23.6 Integration Example

```
src_sram_controller_beats #(
    .NUM_CHANNELS(8),
    .DATA_WIDTH(512),
    .SRAM_DEPTH(512)
) u_src_sram_ctrl (
    .clk                    (clk),
    .rst_n                  (rst_n),

    // Per-channel fill interface (from AXI read engine)
    .fill_alloc_req         (fill_alloc_req),
    .fill_alloc_size        (fill_alloc_size),
    .fill_space_free        (fill_space_free),
    .fill_valid             (fill_valid),
    .fill_ready             (fill_ready),
    .fill_data              (fill_data),

    // Arbitrated drain interface (to network)
    .drain_valid            (src_drain_valid),
    .drain_data             (src_drain_data),
    .drain_id               (src_drain_id),

    // Status
    .drain_data_avail       (channel_data_avail),
    .dbg_bridge_pending     (channel_empty),
    .dbg_bridge_out_valid   (channel_full)
);
```

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24 RAPIDS Core, Sink Half and Source Half

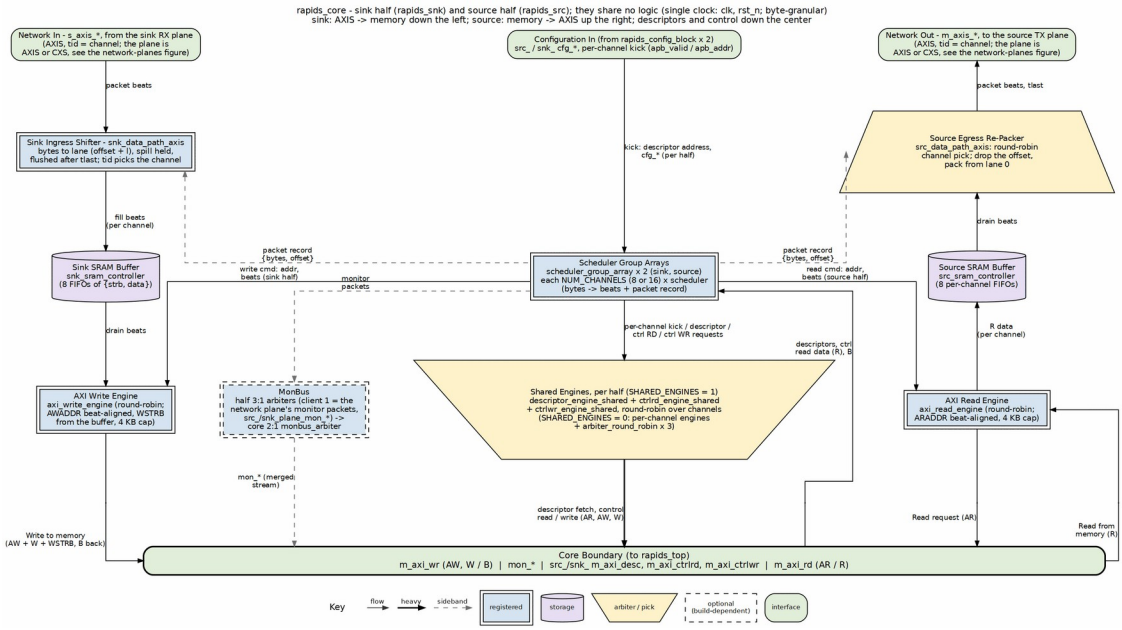
Module: `rapids_core.sv`, `rapids_snk.sv`, `rapids_src.sv` **Location:** `projects/components/dma-ip/rapids/rtl/macro/` **Status:** Implemented **Last Updated:** 2026-09-30

24.1 Overview

`rapids_core` is a thin structural wrapper over two independent halves. `rapids_src` moves memory to the network. `rapids_snk` moves the network to memory. The halves share no logic and no signals. The core only presents one boundary and merges the two MonBus streams.

The structure, the port naming convention and the register and monitor plumbing are those of RAPIDS Beats. This chapter records what the byte-granular RTL contains and how it differs. The full port tables are in RAPIDS Core Beats, with the port renames below applied.

24.1.1 Figure 3.9.1: Core Block Diagram



Core block diagram

Source: 04_rapids_core.dot

24.2 Byte-Granular Content

24.2.1 Packet Record Nets

Each half declares the packet record nets of its direction and connects them between its scheduler group array and its AXIS data path:

Table 3.9.1: Packet Record Nets per Half

Half	Nets	Producer	Consumer
rapids_src	sched_rd_pkt_valid, _ready, _bytes, _offset	array read set	src_data_path_axis
rapids_snk	sched_wr_pkt_valid, _ready, _bytes, _offset	array write set	snk_data_path_axis

The offset nets are $clog2(DW/8) bits wide. The unused direction of each array is tied off: valid, bytes and offset are left open and ready is held at all ones so nothing stalls. The record nets are internal, so the core port list has no packet record ports.

24.2.2 Port Renames

The core ports that carry the beat-count unit are renamed. Everything else keeps its Beats name.

Table 3.9.2: Core Port Renames

RAPIDS Beats	RAPIDS
cfg_axi_rd_xfer	cfg_axi_rd_xfer_beats
cfg_axi_wr_xfer	cfg_axi_wr_xfer_beats
src_dbg_r_rcvd	src_dbg_r_beats_rcvd
src_dbg_axis_sent	src_dbg_axis_beats_sent
snk_dbg_axis_received	snk_dbg_axis_beats_received

24.2.3 Data Ports

Table 3.9.3: Data Ports

Interface	Signals	Notes
Source AXIS	m_axis_*	tstrb is contiguous, partial

Interface	Signals	Notes
		on the `tlast` beat.
Sink AXIS	`s_axis_*`	`tid` selects the channel. Partial `tstrb` only on `tlast`.
Source AXI read	`m_axi_rd_*`	ARADDR is beat-aligned, bursts stop at 4 KB.
Sink AXI write	`m_axi_wr_*`	`wstrb` carries the stored byte enables.

24.3 Parameters

Table 3.9.4: Core Parameters

Parameter	Default	Description
NUM_CHANNELS	8	Channels.
ADDR_WIDTH	64	Address width.
DATA_WIDTH	512	Data width in bits. The board uses 256.
AXI_ID_WIDTH	8	AXI ID width.
SRAM_DEPTH	512	Words per channel.
PIPELINE	1	Write engine pipelining.
AR_MAX_OUTSTANDING, AW_MAX_OUTSTANDING	8	Outstanding bursts.
W_PHASE_FIFO_DEPTH, B_PHASE_FIFO_DEPTH	64, 16	Write engine queues.
AXIS_ID_WIDTH, AXIS_DEST_WIDTH, AXIS_USER_WIDTH	8, 4, 1	AXIS side widths.
DESC_MON_BASE_AGENT_ID	16	Descriptor engine agents.
SCHED_MON_BASE_AGENT_ID	48	Scheduler agents.
DESC_AXI_MON_AGENT_ID	8	Descriptor AXI monitor.
MON_UNIT_ID, MON_MAX_TRANSACTIONS	1, 16	Monitor unit id and table depth.
USE_AXI_MONITORS	1	0 omits the monitor hardware.
USE_ROW_COL_MAJOR_ADDRESSING	1	0 compiles out extended addressing.
GEN_MON	1	0 omits the completion and error emitters.
SHARED_ENGINES	1	1: one descriptor, control-read and control-write engine per half (Shared Engines). 0: one of each per channel.

Parameter	Default	Description
TILE_HDR, TILE_SRC_COORD	0, 14'h3000	Tile packet header on both network ports (Tile Packet Header). Needs SHARED_ENGINES = 1.
TILE_CREDITS	0	End-to-end tile credits: 64 destination counters (each node's internal and external ID) in the source half, the advertisement window and the legal-destination mask (Tile Packet Header). Needs TILE_HDR = 1.

The network endpoint and its monitor-lite moved to the top's network planes (rapids TASK-024), so SNK_AXIS_MON_AGENT_ID, SRC_AXIS_MON_AGENT_ID and ACLK_MHZ are top parameters now. The core takes each plane's monbus packets on src_plane_mon_* / snk_plane_mon_*. See docs/rapids_planes.md.

24.3.1 Channel Reset Routing

Each half has its own configuration block, and that block drives the half's per-channel reset from its CHANNEL_RESET.CH_RST field ORed with GLOBAL_CTRL.GLOBAL_RST. The core exposes it as src_cfg_channel_reset and snk_cfg_channel_reset, and each half forwards it twice: to its scheduler group array (scheduler, descriptor engine, control engines) and to its byte data path (src_data_path_axis or snk_data_path_axis, through which it reaches the AXI engine and the SRAM wrapper). A reset therefore reaches every block that holds a sticky flag or an in-flight beat of the channel, and a channel reset recovers every error class without aresetn. Table 3.9.5 lists the blocks reached.

Table 3.9.5: Channel Reset Routing

Half	Blocks reached	Chapter
rapids_snk	scheduler, axi_write_engine, snk_sram_controller, snk_data_path_axis	Sink Data Path AXIS
rapids_src	scheduler, axi_read_engine, src_sram_controller, src_data_path_axis	Source Data Path AXIS

The two halves are independent: a reset of a sink channel does not touch the source channel with the same index, so software resets both directions of a channel by writing both halves' reset fields. A reset of one channel never disturbs another channel of the same half.

24.4 Differences from RAPIDS Beats

Table 3.9.6: Core Delta

Item	RAPIDS Beats	RAPIDS
Half structure, monitor merge		Unchanged
Packet record nets inside each half	absent	present
Data path instances	`*_axis_beats`	byte-granular `*_axis`
Beat-unit port names	no suffix	`_beats` suffix, see Table 3.9.2

The Beats chapters for the halves are folded into the core chapter there.

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25 RAPIDS Registers Specification

Module: `rapids_regs.sv` (generated) **Source:** `projects/components/dma-ip/rapids/rtl/macro_beats/rapids_regs.rdl`, `rapids_mon_regs.rdl` **Generated:** `projects/components/dma-ip/rapids/regs/generated/rtl/rapids_regs.sv` **Status:** Implemented (PeakRDL-generated)

25.1 Overview

`rapids_regs` is the PeakRDL-generated register block for the RAPIDS Beats top-level. It exposes configuration and status through a single command/response CPU interface and drives the hardware via a `hwif_out` structure that `rapids_config_block` maps onto the core/monitor `cfg_*` signals.

A single addrmap (one APB slave) contains two regfiles:

- **Base config/status** registers at `0x100`-`0x3FF`.
- A nested **`rapids_mon_regs`** monitor regfile at base `0x1000`, decoded under `hwif_out.MON.*` (all DAXMON / RDMON / WRMON / MON_FIFO and per-monitor performance registers).

Because the monitor regfile lives at `0x1000`, the register-block address decode requires at least **13 bits** (`cpuif_addr` is a 13-bit compare in the generated RTL). The APB address bus feeding the block (`APB_ADDR_WIDTH`) must therefore be >= 13 to reach the monitor regfile.

25.2 Base Register Map (0x100-0x3FF)

Table 3.10.1: Base Register Map

Offset	Name	Access	Description
0x100	GLOBAL_CTRL	RW	GLOBAL_EN[0], GLOBAL_RST[1] (level; write 0 to release)
0x104	GLOBAL_STATUS	RO	SYSTEM_IDLE[0] (all channels idle)
0x108	VERSION	RO	MINOR[7:0]=0x5A, MAJOR[15:8], NUM_CHANNELS[23:16]=0x08
0x120	CHANNEL_ENABLE	RW	CH_EN[7:0] per-channel enable
0x124	CHANNEL_RESET	RW	CH_RST[7:0] per-channel reset (level; write 0 to release)
0x140	CHANNEL_IDLE	RO	CH_IDLE[7:0] – not driven at the top level, reads 0
0x144	DESC_ENGINE_IDLE	RO	Per-channel descriptor-engine idle
0x148	SCHEDULER_IDLE	RO	Per-channel scheduler idle (excludes CH_ERROR)
0x150-0x16C	CH_STATE[0..7].STATE	RO	Per-channel FSM state (stride 0x4)
0x170	SCHED_ERROR	RO	Per-channel sticky scheduler error
0x174	AXI_RD_COMPLETE	RO	Per-channel read-complete flags (SRC half only; reads 0 in SNK)
0x178	AXI_WR_COMPLETE	RO	Per-channel write-complete flags – not driven, reads 0
0x200	SCHED_TIMEOUT_CYCLES	RW	TIMEOUT_CYCLES[31:0] write-progress window (reset 1000)

Offset	Name	Access	Description
0x204	SCHED_CONFIG	RW	SCHED_EN, TIMEOUT_EN, ERR_EN, COMPL_EN, PERF_EN
0x208	SCHED_TIMEOUT_LIMIT	RW	LIMIT[7:0] consecutive-timeout escalation limit (reset 4, 0 = never)
0x220	DESCENG_CONFIG	RW	DESCENG_EN[0], PREFETCH_EN[1], FIFO_THRESH[5:2] (reset 8)
0x224	DESCENG_ADDR0_BASE	RW	Descriptor address range 0 base [31:0]
0x228	DESCENG_ADDR0_LIMIT	RW	Descriptor address range 0 limit [31:0]
0x22C	DESCENG_ADDR1_BASE	RW	Descriptor address range 1 base [31:0]
0x230	DESCENG_ADDR1_LIMIT	RW	Descriptor address range 1 limit [31:0]
0x240	CTRL_CONFIG	RW	CTRLRD_MAX_TRY[8:0] control-read poll retry budget (0-511, reset 16)
0x2A0	AXI_XFER_CONFIG	RW	AXI transfer sizing configuration
0x2B0	PERF_CONFIG	RW	Performance profiler configuration
0x2C0	OBS_CTRL	RW	Observation mux control (channel/category select)
0x2C4	OBS_FLAGS	RO	Observation flags
0x2C8	OBS_DATA0	RO	Observation data word 0
0x2CC	OBS_DATA1	RO	Observation data word 1
0x300-0x31C	CH_SCRATCH[n].SCRATCH	RW	Eight software bits per channel, SCRATCH[7:0] (reset 0); storage only, for flags or semaphores between the host and, in rapids-g, the half's gatling.

Offset	Name	Access	Description
			Not atomic across masters: give each bit one writer
0x35C	PERF_CH_SEL	RW	Performance channel select
0x378	HIST_SEL	RW	Histogram bucket select
0x37C	HIST_DATA	RO	Histogram bucket data
0x380	HIST_TOTAL	RO	Histogram total count

25.2.1 Key Register Fields

SCHED_TIMEOUT_LIMIT (0x208) – `LIMIT[7:0]` sets the number of consecutive write-progress timeout windows a channel tolerates before the recoverable timeout escalates to a fatal, sticky `CH_ERROR`. `0` = never escalate (pure soft timeout). Total time to escalate is approximately `LIMIT x TIMEOUT_CYCLES`. Reset value is 4.

DESCENG_CONFIG (0x220) – `PREFETCH_EN[1]` and `FIFO_THRESH[5:2]` control the now-functional descriptor prefetch: prefetch off = on-demand chaining (~1 ahead), prefetch on = buffer up to `FIFO_THRESH` descriptors ahead.

CTRL_CONFIG (0x240) – `CTRLRD_MAX_TRY[8:0]` bounds the control-read poll retry budget (0-511, reset 16) fed to every channel's `ctrlrd_engine`. A `CTRL_READ` descriptor polls its gate address once per retry; if the masked value never matches within `CTRLRD_MAX_TRY` attempts the engine raises `ctrlrd_error` so a never-satisfied gate cannot hang the channel. See the Control-Read / Control-Write Engine specs (Sections 2.8 / 2.9) and the Scheduler control interface (Section 2.1).

25.3 Monitor Register Map (rapids_mon_regs @ 0x1000)

The monitor regfile is instantiated at base `0x1000` under `hwif_out.MON.*`. It holds the MonBus FIFO status, the three AXI-monitor configuration groups (DAXMON = descriptor monitor, RDMON = read monitor, WRMON = write monitor), and per-monitor performance counters.

Table 3.10.2: Monitor Register Map (base 0x1000)

Offset	Name	Access	Description
0x1000	`MON.MON_FIFO_STATUS`	RO	MonBus capture/error FIFO status
0x1004	`MON.MON_FIFO_COUNT`	RO	MonBus FIFO occupancy
0x10C0	`MON.DAXMON_ENABLE`	RW	Descriptor-monitor enables (incl. `COMPRESS_EN`)
0x10C4	`MON.DAXMON_TIMEOUT`	RW	Descriptor-monitor timeout cycles
0x10C8	`MON.DAXMON_LATENCY_THRESH`	RW	Descriptor-monitor latency threshold
0x10CC	`MON.DAXMON_PKT_MASK`	RW	Descriptor-monitor packet mask
0x10D0	`MON.DAXMON_ERR_CFG`	RW	Descriptor-monitor error select/mask
0x10D4-0x10DC	`MON.DAXMON_MASK1/2/3`	RW	Descriptor-monitor category masks
0x10E0-0x10FC	`MON.RDMON_*`	RW	Source-egress AXIS monitor-lite config (same layout as DAXMON; SRC instance, rapids TASK-015)
0x1100-0x111C	`MON.WRMON_*`	RW	Sink-ingress AXIS monitor-lite config (same layout as DAXMON; SNK instance, rapids TASK-015)
0x1150-0x1178	`MON.DAXMON_PERF_*`	RO/RW	Descriptor-monitor performance counters
0x1180-0x11A8	`MON.RDMON_PERF_*`	RO/RW	Read-monitor performance counters

Offset	Name	Access	Description
0x11B0-0x11D8	MON.WRMON_PERF_*	RO/RW	Write-monitor performance counters
0x11E0-0x11EC	MON.{RD,WR}MON_PERF_CH_*	RO	Per-channel producer/backpressure/starve/idle
0x11F0	MON.RDMON_PERF_CH_OVERFLOW	RO	Read-monitor per-channel overflow
0x11F4	MON.WRMON_PERF_CH_OVERFLOW	RO	Write-monitor per-channel overflow

Each *_ENABLE register carries MON_EN, ERR_EN, COMPL_EN, TIMEOUT_EN (and COMPRESS_EN on the write monitor); each *_PERF_* group provides window, producer, backpressure, starve, idle, beat, byte (lo/hi), and burst counters.

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26 RAPIDS Config Block Specification

Module: `rapids_config_block.sv` **Location:** `projects/components/dma-ip/rapids/rtl/macro_beats/` **Status:** Implemented

26.1 Overview

`rapids_config_block` is a combinational adapter that maps the register block's `hwif_out` structure onto the flat `cfg_*` signals consumed by `rapids_core_beats` and the AXI monitors. It contains no state; every output is a direct assignment (with a small number of width expansions and global-enable gates).

The monitor registers arrive under `hwif_out.MON.*`; base configuration and status arrive at the top level of `hwif_out`.

26.2 Mapping Groups

26.2.1 Base Configuration

Routes scheduler, channel, and descriptor-engine configuration to the core:

- `cfg_channel_enable = CHANNEL_ENABLE.CH_EN & {8{GLOBAL_CTRL.GLOBAL_EN}}` (per-channel enable gated by the global master enable).
- `cfg_sched_timeout_cycles = SCHED_TIMEOUT_CYCLES.TIMEOUT_CYCLES` (32-bit).
- `cfg_sched_timeout_limit  = SCHED_TIMEOUT_LIMIT.LIMIT` (8-bit, direct).
- `cfg_sched_{timeout,err,compl,perf}_enable = SCHED_CONFIG.{TIMEOUT,ERR,COMPL,PERF}_EN`. `COMPL_EN` is the one the scheduler group acts on for its monitor packets: 0 drops the scheduler's and descriptor engine's CORE Completion packets at the group (emitters still acknowledged); `ERR_EN` and `PERF_EN` reach the group but gate nothing today (errors always reported, perf not implemented).
- `cfg_desceng_*` from `DESCENG_CONFIG` (enable / prefetch / FIFO threshold).
- Descriptor address ranges (`cfg_desceng_addr{0,1}_{base,limit}`) are zero-extended from the 32-bit register fields to the 64-bit `ADDR_WIDTH`.

26.2.2 Monitor Configuration (hwif_out.MON.*)

Three parallel groups map to the descriptor-read monitor and, since rapids TASK-015, the two AXIS data-path monitor-lites (one per half):

Table 3.11.1: Monitor Register-to-Config Mapping

Register group	cfg_* target
`MON.DAXMON_*`	`cfg_desc_mon_*` (descriptor monitor)
`MON.RDMON_*`	`cfg_rdeng_mon_*` (source-egress AXIS monitor-lite; wired in the SRC half only)
`MON.WRMON_*`	`cfg_wreng_mon_*` (sink-ingress AXIS monitor-lite; wired in the SNK half only)

The `*_ENABLE` bits map onto the AXIS monitor-lite's class enables: `MON_EN` (gated by `GLOBAL_EN`), `ERR_EN` (error, channel-change and TSTRB checks), `COMPL_EN` (one Completion/STREAM_END per

packet), PERF_EN (stream start/pause/resume and Credit/BACKPRESSURE stall reports), TIMEOUT_EN. *_TIMEOUT is MICROSECONDS in its low 16 bits (0 = never), *_LATENCY_THRESH the stall CYCLES that raise a backpressure report, and *_PKT_MASK is a MASK: bit[type] = 1 masks (drops) that packet type at the monitor-lite and again at the monbus group. It resets to all-masked (0xFFFF); the host clears the bits of the classes it wants to see (rapids BUG-008 corrected the description, which used to read “1 = enable”; the polarity and the reset are as intended).

Within each group the enable/timeout/latency/mask fields map 1:1, e.g. `cfg_desc_mon_enable = MON.DAXMON_ENABLE.MON_EN & GLOBAL_CTRL.GLOBAL_EN`, `cfg_desc_mon_timeout_cycles = MON.DAXMON_TIMEOUT.TIMEOUT_CYCLES`, and the category masks (`timeout`/`compl`/`thresh`/`perf`/`addr`/`debug`) pass through directly. Key enables are gated by `GLOBAL_CTRL.GLOBAL_EN`; masks pass through ungated.

26.2.3 Performance and Observation

PERF_CONFIG, PERF_CH_SEL, and OBS_CTRL map to the profiler and channel-observation-mux `cfg_*` signals.

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27 RAPIDS Top

Module: `rapids_top.sv`, `rapids_cxs_top.sv`, `rapids_8ch_top.sv`, `rapids_8ch_cxs_top.sv`, `rapids_16ch_top.sv`, `rapids_16ch_cxs_top.sv` (all generated), `rapids_top_base.sv` **Location:** `projects/components/dma-ip/rapids/rtl/top/` **Status:** Implemented **Last Updated:** 2026-10-17

27.1 Overview

`rapids_top` is the synthesizable top level. It wraps RAPIDS Core with an APB4 register interface, two config blocks, the register file, per-channel bus meters and the MonBus AXI-Lite group. It has the same hierarchy as RAPIDS Beats. Only the beat-count unit names differ.

The byte-granular behavior is entirely below the top: the packet record nets are internal to the core halves, so the top port list has no new signals. The full port list is in Top-Level Port List and the top integration description is in RAPIDS Beats Top.

Both tops are generated by `bin/gen_plane_tops.py` from `rapids_top_base`: `rapids_top` adds the AXIS network planes and `rapids_cxs_top` the CXS planes, one CXS link per direction with `CXSSRCID` carrying the channel. The CXS planes carry `tstrb` in `CXSUSER`, so the byte-granular partial last beats and unaligned starts cross the link intact; that caps `DATA_WIDTH` at 1024. The planes are shared with RAPIDS Beats: see Network Planes and the CXS Interface Specification.

27.1.1 Channel-Count Variants

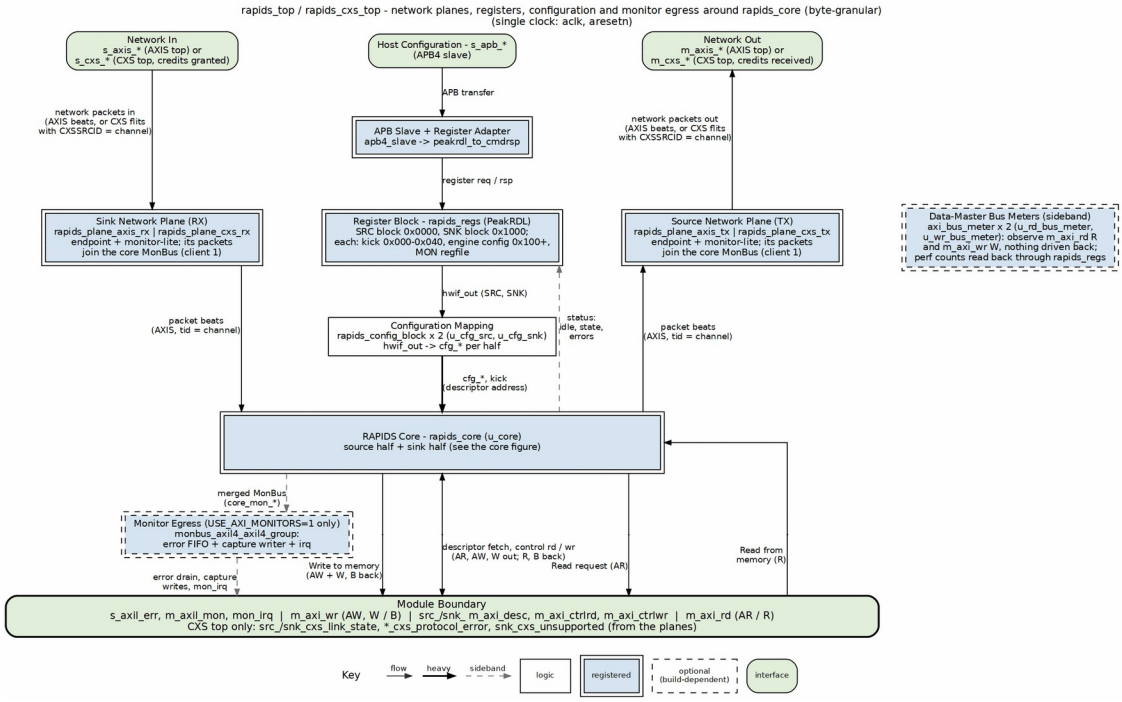
The generator also emits fixed-width variants that pair `NUM_CHANNELS` with the register map built for it:

Table 3.12.1: Top Variants

Top	NUM_CHANNELS	Register map	Filelist
`rapids_top`, `rapids_cxs_top`	8 (parameter)	8-channel	`rtl/filelists/top/rapids_top.f`, `rapids_cxs_top.f`
`rapids_8ch_top`, `rapids_8ch_cxs_top`	8	8-channel	`rapids_8ch_top.f`, `rapids_8ch_cxs_top.f`
`rapids_16ch_top`, `rapids_16ch_cxs_top`	16	16-channel (`regs/generated_16ch/`)	`rapids_16ch_top.f`, `rapids_16ch_cxs_top.f`

`rapids_top_base` stops elaboration if `NUM_CHANNELS` is larger than the register map compiled with it. The 16-channel map is described in the HAS 16-Channel Register Map.

27.1.2 Figure 3.12.1: Top Hierarchy



RAPIDS top integration

Source: 10_rapids_top.dot

27.2 APB Address Map

The APB address is 13 bits. Bit 12 selects the half.

Table 3.12.2: APB Address Map

Range	Content
`0x0000` to `0x003F`	Source staged descriptor addresses.
`0x0040`	Source kick-off, one single-pulse bit per channel.
`0x0040` to `0x0FFF`	Source configuration registers.
`0x1000` to `0x103F`	Sink staged descriptor addresses.
`0x1040`	Sink kick-off, one single-pulse bit per channel.
`0x1040` to `0x1FFF`	Sink configuration registers.

The monitor register windows sit at `+0x800` of each half's 4 KB space. With `USE_MON_REGS` clear the windows answer with an APB error, so a host can tell a build without monitors from a zero setting.

27.3 Parameters

Table 3.12.3: Top Parameters

Parameter	Default	Description
NUM_CHANNELS	8	Channels.
DATA_WIDTH	512	Data width in bits. The board uses 256.
ADDR_WIDTH	64	Address width.
AXI_ID_WIDTH	8	AXI ID width.
SRAM_DEPTH	4096	Words per channel at the top.
APB_ADDR_WIDTH, APB_DATA_WIDTH	13, 32	APB widths.
MON_MAX_TRANSACTIONS	16	Monitor table depth.
SNK_AXIS_MON_AGENT_ID, SRC_AXIS_MON_AGENT_ID	9, 10	Network-plane monitor agent ids (AXIS or CXS plane).
ACLK_MHZ	100	Clock in MHz for the network-plane monitor tick.
USE_AXI_MONITORS	1	0 omits the monitors and the MonBus AXI-Lite group.
USE_ROW_COL_MAJOR_ADDRESSING	1	0 compiles out extended addressing.
GEN_MON	1	0 omits the completion and error emitters.
SHARED_ENGINES	1	1: one descriptor, control-read and control-write engine per half. 0: one of each per channel. See Shared Engines.
TILE_HDR, TILE_SRC_COORD	0, 14'h3000	Tile packet header; needs SHARED_ENGINES = 1 and DATA_WIDTH of at least 256. See Tile Packet Header.
TILE_CREDITS	0	End-to-end tile credits and the legal-destination mask (SRC.TILE_ADV_WINDOW, SRC.TILE_LEGAL_DEST_LO /

Parameter	Default	Description
		_HI, SRC.TILE_CREDIT_STATUS); needs TILE_HDR = 1. See Tile Packet Header.
USE_MON_REGS	USE_AXI_MONITORS != 0	Wire the monitor register windows.
AR_MAX_OUTSTANDING, AW_MAX_OUTSTANDING	8	Outstanding bursts.
PIPELINE	1	Write engine pipelining.
AXIS_ID_WIDTH, AXIS_DEST_WIDTH, AXIS_USER_WIDTH	8, 4, 1	AXIS side widths.

Clock and reset are aclk and aresetn. cam_clear synchronously clears the MonBus group transaction tables.

Each half's configuration block drives its per-channel reset vector (src_cfg_channel_reset, snk_cfg_channel_reset) from SRC/SNK.CHANNEL_RESET.CH_RST and GLOBAL_CTRL.GLOBAL_RST. The top wires each vector to its half of the core (see RAPIDS Core), and it reaches every block that can hold a channel error: a channel reset recovers every error class, including the AXI response and sink packet length errors, and aresetn is needed only at power-up. The top has no port change.

27.4 Differences from RAPIDS Beats

Table 3.12.4: Top Delta

Item	RAPIDS Beats	RAPIDS
Module	`rapids_beats_top`	`rapids_top`
Core instance	`rapids_core_beats`	`rapids_core`
Register file, APB map, config blocks		Unchanged
Bus meter beat counters	`r_rd`, `r_wr`	`r_rd_beats`, `r_wr_beats`
Config nets	`*_xfer`	`*_xfer_beats`
Port list		Unchanged

The register-file field names `RD_XFER_BEATS` and `WR_XFER_BEATS` are the same in both designs. Their meaning is beats in both, since bursts are beat-based.

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28 Network Planes Specification

Modules: `rapids_plane_axis_rx.sv`, `rapids_plane_axis_tx.sv`, `rapids_plane_cxs_rx.sv`, `rapids_plane_cxs_tx.sv` **Location:** `projects/components/dma-ip/rapids/rtl/plane/` **Generator:** `projects/components/dma-ip/rapids/bin/gen_plane_tops.py` **Status:** Implemented (rapids TASK-024, TASK-025)

28.1 Overview

A network plane is the endpoint between the core's network-side AXIS stream and the pins: the RX plane feeds the sink half's ingress, the TX plane takes the source half's egress. Each plane carries the monitor-lite for its port. Both RAPIDS trees (byte-granular `rtl/top`, beat-granular `rtl/top_beats`) share the same four planes.

Every plane has the same parameters and the same inner interface, so a top selects AXIS or CXS by module name alone:

- `fub_axis_*`: the core-facing AXIS stream (`tdata`, `tstrb`, `tlast`, `tid` = channel, `tdest`, `tuser`)
- `cfg_mon_*` in, `sts_mon_*` out: the `SNK.WRMON_*` / `SRC.RDMON_*` monitor CSRs, from `rapids_config_block`
- `monbus_*` out: client 1 of the half's 3:1 `monbus_arbiter`, entering the core on `snk_plane_mon_*` / `src_plane_mon_*`

Table 3.13.1: RAPIDS Network Planes

Plane	Contents	Outer pins
`rapids_plane_axis_rx`	`axis4_slave_monlite` (skid stage, `SKID_DEPTH` = 4, + `axis_monitor_lite` tap)	`s_axis_*`
`rapids_plane_axis_tx`	`axis4_master_monlite` (skid stage + tap)	`m_axis_*`
`rapids_plane_cxs_rx`	`cxs_slave_monlite` (`u_endpoint`) + `cxs_to_axis` (`u_cxs_to_axis`)	`s_cxs_*`, `snk_cxs_*` status
`rapids_plane_cxs_tx`	`axis_to_cxs` (`u_axis_to_cxs`) + `cxs_master_monlite` (`u_endpoint`)	`m_cxs_*`, `src_cxs_*` status

The monitor counters of every plane are at `u_snk_plane.u_endpoint` and `u_src_plane.u_endpoint` in the top.

28.2 Generated Tops

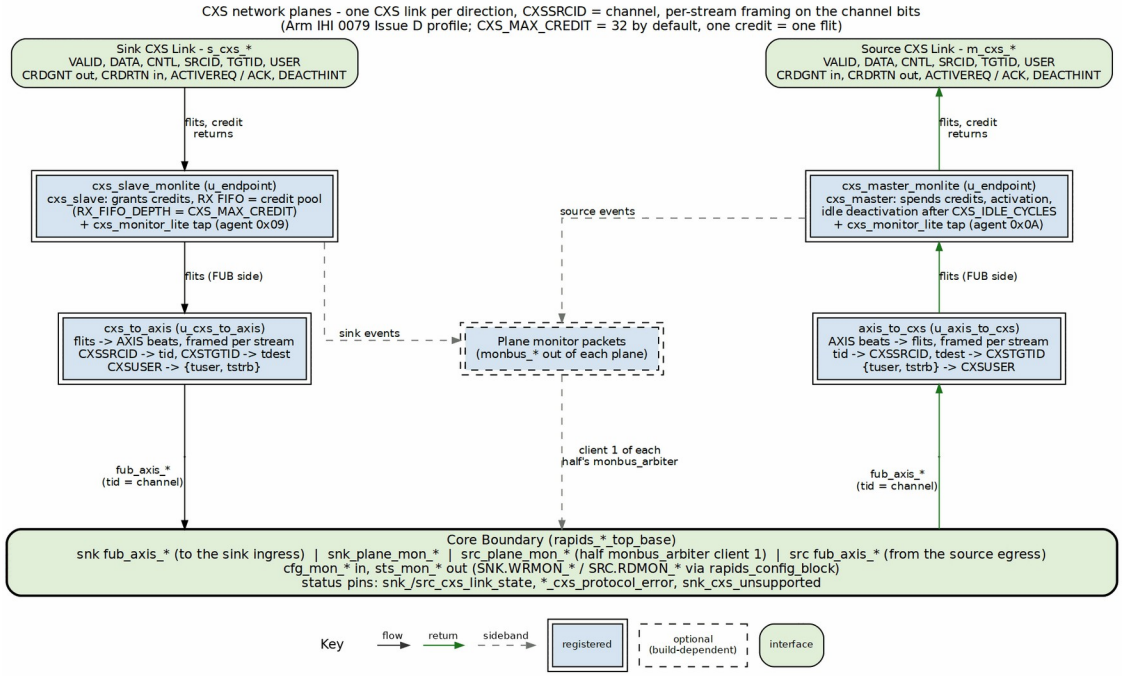
Table 3.13.2: Generated Tops

Top	Tree	Base	Planes
rapids_top	byte	rapids_top_base	AXIS
rapids_cxs_top	byte	rapids_top_base	CXS
rapids_beats_top	beats	rapids_beats_top_base	AXIS
rapids_beats_cxs_top	beats	rapids_beats_top_base	CXS

gen_plane_tops.py copies the base's parameter and port text verbatim, then instantiates the base (u_base) and the two planes (u_snk_plane, u_src_plane). The AXIS tops therefore keep exactly the interface they had before the split, and existing instantiations are unchanged. Run the generator with --check to fail on a stale top.

28.3 CXS Planes

28.3.1 Figure 3.13.1: CXS Network Planes



CXS network planes

Source: 23_cxs_planes.dot

28.3.2 Receive (sink) plane

`cxs_slave_monlite` is `cxs_slave` with a `cxs_monitor_lite` tap. `cxs_slave` grants credits up to `CXS_MAX_CREDIT`, runs the activation handshake, and holds received flits in a receive FIFO of `RX_FIFO_DEPTH` (default `CXS_MAX_CREDIT`) entries. The FIFO is the credit pool: a credit is only granted for a free entry, so a flit always has space. `cxs_to_axis` turns flits back into AXIS beats: `CXSSRCID` becomes `tid`, `CXSTGTID` becomes `tdest`, and `CXSUSER` is split into `{tuser, tstrb}`. When the core stops taking beats the FIFO fills, and the slave stops granting credits; that is the CXS form of the sink's `tready` backpressure.

`cxs_to_axis` is combinational and stateless, so it can only represent the flits `axis_to_cxs` makes: at most one START and one END per flit, with a packet starting at byte 0. Any other flit (a transmitter that packs two packets into one flit) is still passed on, with `tlast` from END[0], and sets the sticky `snk_cxs_unsupported`.

28.3.3 Transmit (source) plane

axis_to_cxs packs AXIS beats into flits (tid -> CXSSRCID, tdest -> CXSTGTID, {tuser, tstrb} -> CXSUSER) and cxs_master_monlite sends them, one credit per flit. The master requests activation when it has a flit to send and deactivates the link after CXS_IDLE_CYCLES idle cycles.

28.3.4 Per-stream framing

axis_to_cxs and both endpoints' control checkers track an open packet per stream, keyed on the low CXS_STREAM_SRCID_BITS = CIW bits of CXSSRCID (the channel); cxs_to_axis needs no state because each beat carries its own tid. The byte-granular source arbitrates per drain block while a packet spans the whole descriptor, so with two or more channels active the egress switches channel in the middle of a packet. Per-link framing (one open packet per link) rejects that traffic; per-stream framing accepts it, as IHI 0079D 2.5.2 allows.

28.3.5 Parameters

Table 3.13.3: CXS Plane Parameters

Parameter	Default	Description
CXS_MAX_CREDIT	32	Credits granted by the receiver. One credit is one flit; the grant -> flit -> free round trip is about 3 cycles
CXS_RX_FIFO_DEPTH	CXS_MAX_CREDIT	Receive FIFO depth (the credit pool); need not be a power of two
CXS_CREDIT_BEYOND_SPEC	CXS_MAX_CREDIT > 63	Permits more credits than IHI 0079D allows; the far end must match
CXS_IDLE_CYCLES	32	Idle cycles before the transmitter deactivates the link
CXS_IW, CXS_DESTW, CXS_UW, CXS_CNTL_PW	derived	Port widths; do not override

DATA_WIDTH must be at most 1024: CXSUSER is at most 128 bits and carries tstrb (DATA_WIDTH/8 bits). tuser is dropped from CXSUSER when DATA_WIDTH/8 + AXIS_USER_WIDTH > 128.

28.3.6 Monitoring

The CXS planes drive cxs_monitor_lite from the same cfg_mon_* CSRs as the AXIS planes and use the same agent IDs (sink 0x09, source 0x0A). Packets are tagged PROTOCOL_CXS; the top's

monitor group filters them through its AXIS mask slot. A PACKET_END completion carries the channel in `event_data[55:48]`; `channel_id` holds `{PRCLTYPE[2:0], TGTID[5:0]}`. The lite's output queue is 4 entries deep; when events outrun it the lite drops and counts them (an Error / EVENT_DROPPED packet), so a check on completions counts delivered plus dropped.

28.4 Proposed Changes

A proposal (2026-10-16, not implemented) puts a tile packet header at the front of every packet: destination and source coordinates, payload type (`cfg` / `prog` / `mem`), length and target address. It is described once in the compute-engine tile's `TILE_SPEC.md` Section 11; CXS carries it as the first flit. The link signals, credits and per-stream framing on this page do not change, because `CXSSRCID` keeps the channel.

28.5 Verification

Table 3.13.4: CXS Top Test Cells

Suite	Cells
`dv/tests/top/test_rapids_cxs_top.py`	source, sink, source_bytes, sink_bytes, source_concurrent, sink_concurrent, cxs_mon, ext_addressing
`dv/tests/top_beats/test_rapids_beats_cxs_top.py`	source, sink, source_concurrent, sink_concurrent, cxs_mon, ext_addressing

The concurrent cells start four channels with one `KICK_ENABLE` write, so channels interleave on the link. Every cell also requires a clean link: no BFM or wire-monitor violation and the DUT's protocol-error flags low. The core testbenches drive packets on `plane_mon_*` and require them on the MonBus output.

Last Updated: 2026-10-16

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29 Tile Packet Header

Modules: `rapids_tile_hdr_rx.sv` and `rapids_tile_credits.sv` (new), with header and credit logic in `scheduler.sv`, `src_data_path_axis.sv` and `descriptor_engine_shared.sv` **Location:** `projects/components/dma-ip/rapids/rtl/fub/`, `rtl/macro/` **Status:** Header behind `TILE_HDR` = 1, end-to-end credits behind `TILE_CREDITS` = 1 (both default 0) **Last Updated:** 2026-10-18

29.1 Overview

With TILE_HDR = 1, RAPIDS is the I/O of a compute-engine tile. Every packet on the source port and the sink port starts with one header beat. The header and its packing are defined in the tile's TILE_SPEC Section 11.5; this section describes the RAPIDS hardware that builds and reads it. The network planes do not change: on a CXS link the header is the data of the first flit of the packet, which is what TILE_SPEC 11.6 asks of CXS, and an AXIS plane carries it as the first beat in the same way.

TILE_HDR needs SHARED_ENGINES = 1, because cell-issued descriptors are written into the shared descriptor engine's queues, and DATA_WIDTH >= 256, so that a descriptor arrives in one beat. Both are elaboration-time checks.

29.2 Header Beat

The header occupies bits [127:0] of its beat (`rapids_pkg::tile_hdr_pack`) and its strobe is the low 16 bytes. Bits [127:119] are zero. A coordinate is 15 bits, `{ext, level, X, Y, Z}`: `ext` picks the node's internal ID (0, the node itself) or its external ID (1, what is attached to it, such as a cell).

Table 3.14.1: Header Fields and Their Sources

Bits	Field	Source half fills it from
[14:0]	Destination coordinate `{ext, level, X, Y, Z}`	descriptor `dst_addr[63:49]`
[29:15]	Source coordinate	`TILE_SRC_COORD` (level 3, the tile I/O)
[32:30]	`ptype`	descriptor [215:213]
[33]	`op` (0 write, 1 read)	descriptor [216]
[65:34]	Length in bytes	descriptor `length`
[114:66]	Target address	descriptor `dst_addr[48:0]`
[118:115]	Tag	the issuing channel

29.2.1 Header Beat Chaining on Narrow Fabric Links (2026-10-19)

RAPIDS ports are 256 bits or wider, so the 128-bit tile header is one payload beat and the existing `DATA_WIDTH` >= 256 elaboration check stays in force. On narrower fabric links the header occupies the first `ceil(128 / DATA_WIDTH)` payload beats, LSB-first, per TILE_SPEC §11.6. That chaining rule is the contract toward narrow delta links (delta `DATA_WIDTH` = 64 today → 2 beats); consumers on those links parse the header per TILE_SPEC §11.6, and RAPIDS does not insert a width converter for the header.

29.3 Source Half

Scheduler. It latches `ptype` and `op` with the descriptor and passes `{op, ptype, dst_addr}` (`sched_rd_pkt_tile`) with the packet record it already sends the source data path. A read request (`op` = 1) loads zero read beats, so no memory is read, and completes like any data descriptor once its record is queued. A source descriptor whose `ptype` is not `cfg`, `prog` or `mem` is refused as an invalid descriptor (CH_ERROR): `credit` and `desc` packets are built by the tile, never by a descriptor.

Source data path. The packet-record queue stores the tile fields beside the byte length and offset. Before a packet's first data beat, the egress emits its header from the head record:

- The output register takes, in priority order, a flush beat, a header beat, then a data pop.
- A data packet's header goes only when its first memory beat is at the drain stage, so header and payload leave back to back. Only the channel the drain stage is parked on can need a header, so it is served first.
- A header-only packet (read request) can go at any time. When it competes with a data pop, the two alternate, and header-only channels are scanned round-robin, so neither a run of read requests nor a long data packet starves the other.
- The header-only record is popped when its header leaves.

29.4 Sink Half: rapids_tile_hdr_rx

The module sits between the sink plane and snk_data_path_axis. Packets are framed per stream, and the stream is the TID (CXSSRCID, the channel), so the “next beat is a header” state is kept per stream.

Table 3.14.2: Sink-Port Packet Handling

Header	Action
cfg / prog / mem, op = 0	Header dropped; payload beats to the sink data path unchanged, delivered by stream ID
desc, op = 0	First payload beat [255:0] goes out on desc_inj_* with the tag as the source channel; never stored in the sink
credit	Consumed and counted (TILE_COUNTS.CREDIT_COUNT); {source coordinate, op, count, class} goes to the source half’s counters on credit_* (used when TILE_CREDITS = 1)
Anything else	Consumed; sets UNSUPPORTED: a read request to RAPIDS, a reserved ptype, a desc with no payload or a tag past NUM_CHANNELS, extra desc payload, or DATA_WIDTH < 256

Only data payload waits on the sink data path’s tready. Header, desc, credit and dropped beats are taken at once, so a full sink channel never holds a descriptor or a credit behind it. A data packet whose tag differs from its stream ID is still delivered by stream ID and sets TAG_MISMATCH.

29.5 Cell-Issued Descriptors: `descriptor_engine_shared` (`INJ_EN`)

`desc_inj_*` is a single-cycle pulse with no backpressure. Each channel has a one-descriptor slot:

- **Capture.** A descriptor for a channel whose slot is empty is captured. The chain decision (non-zero next pointer, not last, valid, in range) is taken once at capture, so one set of comparators serves every channel.
- **Write.** The slot is written into the channel's descriptor FIFO when the channel's fetch side is idle (`S_IDLE`), no R beat is owed from a fetch a channel reset aborted, and the FIFO has room. Idle means no fetch is in flight, so the R path never writes that FIFO in the same cycle and the injection never takes the entry an outstanding fetch needs.
- **Start and chain.** An idle channel starts on the descriptor as if kicked; a busy one runs it after what is already queued. A chaining descriptor moves the channel to `S_PEND` with its next pointer, and the chain continues from memory.
- **Kicks** wait while the slot is full.
- **Drops.** A descriptor that arrives while the slot is full, while the channel is held in reset, or of type EXT (its second chunk has no way in) is dropped and sets the channel's sticky drop bit. A drop is not a descriptor error: the channel and the work it holds carry on.
- **Hold (`INJ_HOLD`, `TILE_CREDITS` = 1).** `inj_*` becomes a valid/ready handshake from the descriptor intake queue (below). A descriptor for a channel whose slot is full waits (`inj_ready` low) instead of being dropped; one for a channel in reset, or an EXT one, is still taken and dropped.

29.6 End-to-End Credits: rapids_tile_credits (TILE_CREDITS)

With TILE_CREDITS = 1 (it needs TILE_HDR = 1), a packet leaves the source port only when its destination has room for it (TILE_SPEC 11.9, 11.10). The credits are end to end, between RAPIDS and the cells and control cores. A delta node only transports: it never originates a packet, so it never advertises or returns credits, and it consumes what is addressed to it on arrival.

Counters. The source half keeps 64 counters in flits, one per destination index {ext, n} (rapids_pkg::tile_dest_index): the internal and the external ID of every node. Only the external IDs carry credits: a node's internal ID is legal by topology, and a packet to it is never held for credits or charged. An external ID with nothing behind it never advertises and so is never legal.

Table 3.14.3: Credit Counter Index

Index	Destination
0-15	Cell-level node (internal), {Y[1:0], X[1:0]}
16-19	Tetrad node (internal), {Y[0], X[0]} of its group
20-21	Root node 0, 1 (internal)
32-47	Cell: the cell-level node's external ID
48-51	Tetrad control core: the tetrad node's external ID
others	Spare: never advertised, always illegal

A coordinate outside this map (the tile I/O itself, a cell past 4x4, a tetrad past 2x2, a root past 1) has no index. Every external counter starts at 0. A class-0 credit packet from an external ID fills it: op = 1 (set) replaces the count and op = 0 (add) adds to it, saturating. Class-1 credits (descriptor entries), credits carrying a node's internal ID and credits from a coordinate with no index are ignored.

Advertisement window. rapids_tile_credits counts cycles from reset up to SRC.TILE_ADV_WINDOW (default 4096). When the window closes, every node index is legal and every external counter still at 0 marks an illegal destination, and the 64-bit result is saved in SRC.TILE_LEGAL_DEST_LO (internal IDs) and _HI (external IDs) until the next reset. SRC.TILE_CREDIT_STATUS.ADV_DONE reports the close. A later advertisement updates the count but never makes a destination legal.

Scheduler. A tile data descriptor with a non-zero length waits in CH_FETCH_DESC until the window closes. Its destination must then be legal (an index whose bit is set in the mask), or the

descriptor is refused as invalid (CH_ERROR) and nothing is sent. A read request to a node's internal ID is refused the same way, with or without `TILE_CREDITS`: a node never replies.

Source data path. Each channel asks for its head packet's flits, the header and `ceil(bytes / beat)` data beats or the header alone for a read request, against its destination (`tile_acq_*`). `rapids_tile_credits` grants one request per cycle, round-robin, and only when the counter covers the whole packet. It charges the counter in the same cycle and pulses `tile_acq_gnt` the next. A request to a node is granted without a count check and charges nothing. Until the grant, neither the packet's header nor any further SRAM reservation of the channel goes ahead, so a channel waiting for credits does not hold the shared drain stage. A reservation made under the previous packet can still run into this one's first beat, and the drain stage then waits there for the grant. Counters are per destination, so concurrent channels to one cell share its buffer.

Limits. A channel reset that discards a packet whose credits are already charged does not refund them. With `TILE_CREDITS` a channel reset acts only while the tile is quiescent (below), when no packet is in flight against a charge, so the case does not arise under the quiescent rule.

29.7 Straps Toward RAPIDS (TILE_CREDITS)

What a cell or control core sends into the sink port is charged against a strap RAPIDS reserves, and RAPIDS returns the credit as it consumes (TILE_SPEC 11.9, 11.11). The straps are fixed by construction, not granted: a sender starts from its strap after reset. They are Hive-level parameters that the cells and cores must be built with too.

Table 3.14.4: Straps toward RAPIDS

Sender (external slot = index - 32)	Strap	Reserved in RAPIDS	Returned when
Cell, slots 0-15	TILE_LAND_FLITS flits (default 16), header included	A landing space of TILE_LAND_FLITS beats	Each beat leaves the landing space; the header when it is decoded
Control core, slots 16-19	TILE_DESC_PER_CORE entries (default 2), credit class 1	A share of the intake queue (4 x TILE_DESC_PER_CORE deep)	The descriptor moves into the descriptor queue

Header decode (`rapids_tile_hdr_rx`, STRAPS). The sender is the header's source coordinate as an external slot (`rapids_pkg::tile_ext_slot`). Data payload goes on with the sender's cell slot to the landing space; the header flit returns at once. A packet the decode drops (a read request into RAPIDS, a reserved `ptype`) returns every beat. A `desc` packet refused at its header returns its entry. A data packet from a sender with no landing space (not a cell) is dropped and flagged `UNSUPPORTED`, and returns nothing: it was charged to no strap. Credit packets are never charged.

Landing space (`rapids_tile_landing`, in `rapids_snk`). One FIFO of `TILE_LAND_FLITS` beats per cell, holding payload by sender. A cell under its strap can never fill its FIFO, so the sink port never waits on a landing space. The drain moves one whole packet at a time into `snk_data_path_axis`, round-robin over the cells. A packet is a candidate once its `tlast` beat has landed, so the drain never waits on the network, and only when its channel can take all of it now: `snk_data_path_axis` `chan_can_take` says the channel is out of reset, has the packet's record (its descriptor is running), no spill is waiting, and `TILE_LAND_FLITS + 4` beats are free (the packet, a spill and what the previous packet may still hold in the shifter). A packet for a disabled channel is drained and thrown away. A cell whose channel has no descriptor yet holds only its own landing space; every other cell's data keeps flowing. Each beat that leaves, delivered or thrown away, returns one flit. A beat arriving for a full landing space (a cell past its strap) waits and sets `TILE_CREDIT_STATUS.LAND_OVERRUN`. The per-channel sink SRAM must be at least `TILE_LAND_FLITS + 4` beats (an elaboration check).

Descriptor intake (`rapids_tile_desc_intake`, in `rapids_core`). `desc` packets go into a FIFO `4 x TILE_DESC_PER_CORE` deep with their channel and sender, and leave in order through the `INJ_HOLD` handshake into `descriptor_engine_shared`. A descriptor for a busy channel at the head waits there; the schedulers' two-deep descriptor skid keeps that short. When an entry leaves,

its class-1 credit returns. A `desc` arriving at a full queue (a core past its strap) is dropped, sets `TILE_CREDIT_STATUS.INTAKE_OVERRUN`, and still returns its entry: no path consumes a descriptor silently.

Credit returns (`rapids_tile_credit_tx`, in `rapids_core`). Pending returns are kept per external slot and class and taken round-robin into one output register as soon as it is free; whatever arrives while a return waits is coalesced into that slot's next one, and nothing is held back to build a batch. A return is a header-only credit packet (`ptype` = `credit`, `op` = `add`, count in the length, class in target bits [1:0]) addressed to the sender's source coordinate as last seen in a header, on its own interface, `m_tile_credit_*` (valid/ready, the 128-bit header): a return never waits behind data. `TILE_CREDIT_STATUS.RET_COUNT` counts them.

Resync (TILE_SPEC 11.10). A write of `SRC.TILE_RESYNC.GO` requests one; it is taken only while the top-level `tile_quiescent` input is high and waits for it otherwise (`TILE_CREDIT_STATUS.RESYNC_BUSY`). It then sends one `set` per legal sender in the saved mask: `TILE_LAND_FLITS` (class 0) to each cell and `TILE_DESC_PER_CORE` (class 1) to each control core, replacing that slot's pending `add`. Sets go before adds. RAPIDS sends one `set` per sender rather than the multicast TILE_SPEC 11.10 describes: it has no multicast header to build. The answering advertisements refill RAPIDS's own counters as at boot.

Quiescent gating. With `TILE_CREDITS`, a channel reset of either half (`CHANNEL_RESET`) acts only while `tile_quiescent` is high; a reset bit written while it is low has no effect until it rises.

Stalled-data checks (`rapids_tile_stall_mon`). Data that stops moving is a monitor error (TILE_SPEC 11.9, hazard rules). Each half carries one check, on the spare client 2 of its MonBus arbiter:

Table 3.14.5: Stalled-data checks

Half	Pending	Moves on	Agent
Sink	The channel's sink SRAM holds data not yet written out (an allocated, unfilled segment does not count)	A W beat of the channel	`0x0B`
Source	The channel has read bursts outstanding (AR accepted, RLAST not seen; ARID low bits = channel)	An R beat for the channel	`0x0C`

A channel pending for `SRC.TILE_STALL_CYCLES` cycles (default 1000, 10 us at 100 MHz; 0 turns both off) with no movement raises one error packet (`PktTypeError`, `PROTOCOL_CORE`, `RAPIDS_EVENT_STALL`; event data bit 32 = 0 sink / 1 source, bits [31:0] the threshold) and sets its bit in that half's `TILE_STALLED`, and raises nothing more until the data moves or drains.

29.8 Ordering Across the Fabric (open: delta DELTA-013)

Delta is moving to multi-path route tables, with failover as a table write (delta HAS ch07, Resilient Routing: Open Hazards). A request and its reply can then take different paths with different latencies, so delta guarantees the order of flits within one packet and nothing between packets. RAPIDS must not rely on fabric order. Under correct configuration, four places would; O4 has an interim rule (2026-10-19), the rest are open:

Table 3.14.6: RAPIDS ordering hazards under multi-path routing

#	Where	Mechanism	Candidate direction
O1	Read replies	The tag is the sink channel only. Two reads outstanding on one sink channel, to different cells or to one cell over two paths, can reply in either order, and the sink lands the bytes in arrival order	A transaction ID per read, echoed in the reply, and the sink places by ID (AXI-ID style), or one read outstanding per sink channel
O2	Credit `set` vs `add`	A resync `set` (TILE_SPEC 11.10) can overtake, or be overtaken by, an `add` sent before it, and the counter ends wrong	A sequence or epoch on credit packets; an `add` older than the last `set` is discarded
O3	Write, then a dependent packet	A cell's data write and a later packet that depends on it (a `desc` whose source is that data, a flag) can arrive in either order	Completion is signalled (the credit return says the data was consumed), never implied by send order

#	Where	Mechanism	Candidate direction
O4	Writes from one cell to one sink channel	The sink lands a channel's data in arrival order, so a cell's later write that overtakes an earlier one lands in the wrong place	Interim rule: delta may interleave and reorder packets (delta HAS, Interface Assumptions), so the cell sends its next packet only once RAPIDS has returned every flit it has sent (its strap is whole). RAPIDS never returns a flit it has not received, so a whole strap proves every earlier packet landed and drained. No RAPIDS change, at the cost of one round trip per packet. Pencilled in (rapids TASK-033): tagged, indexed intake, below

The flits of one packet stay in order. The header-then-payload framing of the source and sink halves also needs each CXS plane port to deliver a packet's flits together; whether delta keeps that at its plane ports once it interleaves is open.

29.9 Interleaved and Out-of-Order Intake (pencilled in, rapids TASK-033)

Not built. Validated in a cycle model only (`dv/models/tile_intake_model.py`, `dv/tests/models/test_tile_intake_model.py`). It lets delta interleave and reorder packets into the sink port with no send rule beyond the straps.

The tag. Every flit, data flits included, carries a sideband tag beside its data: `{sender, pos, head}`. `sender` is the source's 5-bit external slot (`rapids_pkg::tile_ext_slot`: cells 0-15, control cores 16-19). `head` marks a header flit. `pos` is the flit's place in the space RAPIDS holds for that sender and class. About 10 bits; delta carries it on AXIS `tuser` and CXS `user` (delta HAS, Interface Assumptions).

Table 3.14.7: Tagged intake by class (pencilled in)

Class	Sender	Packet	pos	RAPIDS storage	Leaves in
0, data	Cells	Header + up to `TILE_LAND_FLITS` - 1 beats	Flits sent, header included, mod `TILE_LAND_FLITS`	The landing space as a ring, written at `pos`, a valid bit per slot	Ring order: the packet at the read pointer, once its header and every flit to its last are present
1, desc	Control cores	Header + one descriptor beat	Packets sent mod `TILE_DESC_PER_CORE`	One intake entry per `pos`	Entry order

Debug never rides the fabric. Each cell emits one MonBus on AXI4-Lite, each tetrad merges its cells and its node into one AXI4-Lite MonBus, and the tile merges the tetrads and the root nodes into one MonBus (owner 2026-10-19, TILE_SPEC 5). So nodes never send into RAPIDS, and a MonBus class on the fabric (pencilled in earlier the same day) is retired. A header's class-0 flit returns when its packet drains, not on arrival, because its slot holds the packet's channel and length. The strap keeps a sender's flits in flight below the space, so a `pos` is never reused while its slot is full. Resync clears the rings and entries and both sides' `pos`.

Model results (one flit per cycle into RAPIDS, random packet sizes, fabric latency 2-120 cycles): both classes delivered exactly once and in send order at `TILE_LAND_FLITS` 16 and 8, every strap whole at the end. Controls: today's arrival-order landing corrupts as soon as a cell has two packets in flight (strap-only and the retired one-flit rule both fail); the wait-for- every-credit rule is correct

on it. Mutations caught: draining any complete packet instead of the one at the read pointer; `pos` counting payload only. Payload throughput, tagged intake vs wait-for-every-credit:

Table 3.14.8: Model payload flits per cycle (link peak 1; headers take the rest)

Cells	Fabric latency 2-10	Fabric latency 8-40
1	0.36 vs 0.28	0.21 vs 0.18
2	0.65 vs 0.48	0.42 vs 0.32
4	0.85 vs 0.77	0.70 vs 0.55
16	0.89 vs 0.89	0.88 vs 0.88

A cell’s own ceiling is its strap over the round trip, so the gain is largest with few active cells; with many, the link saturates either way.

29.10 Status Registers

SNK.TILE_STATUS (UNSUPPORTED, TAG_MISMATCH), SRC.TILE_DESC_DROPPED (DROPn) and SNK.TILE_COUNTS, plus, for TILE_CREDITS, SRC.TILE_ADV_WINDOW (RW), SRC.TILE_LEGAL_DEST_LO / _HI, SRC.TILE_CREDIT_STATUS (ADV_DONE, RESYNC_BUSY, LAND_OVERRUN, INTAKE_OVERRUN, RET_COUNT), SRC.TILE_RESYNC (GO, singlepulse), SRC.TILE_STALL_CYCLES (RW), SRC / SNK.TILE_STALLED and TILE_STRAPS (see the Register Map). TILE_STRAPS is the last config word of each half, 0x7FC below the monitor block (SRC 0x07FC, SNK 0x17FC, on the 8- and 16-channel maps alike), so the Hive finds it without the rest of the map: LAND_FLITS [15:0] = TILE_LAND_FLITS and DESC_PER_CORE [31:16] = TILE_DESC_PER_CORE, both 0 when TILE_CREDITS = 0. The W1C fields are live mirrors of stickies owned by the RTL. rapids_top_base decodes a write to either register from the register block's own request and sends a one-cycle clear (wr_data & wr_biten) to the owner, where a same-cycle event wins. The same values are on the top-level sts_tile_* ports.

29.11 Verification

`test_rapids_cxs_top.py` builds `rapids_cxs_top` (and `rapids_16ch_cxs_top`) with `TILE_HDR` = 1:

Table 3.14.7: Tile Header Test Cells

Cell	Proves
`tile_source_write`	Header fields from the descriptor, ahead of the payload
`tile_source_read`	`op` = 1 is a header-only packet with no memory read
`tile_sink_write`	Header stripped, payload to memory
`tile_cell_descriptors`	Idle, busy and chaining injections on four channels (and channels 12-15 at 16)
`tile_desc_overflow`	A slot overflow drops, flags `DROPn` (and clears it by W1C); the running transfer and queued work finish
`tile_source_bad_ptype`	A source descriptor with `ptype` `credit` is refused
`tile_unsupported`	`credit` counted, a read request to RAPIDS flagged and cleared by W1C, the stream still carries data
`tile_credits[window]`	`TILE_CREDITS`: every node legal unadvertised, node credits ignored; the cells and core that advertise are legal; a zero and a late advertisement stay illegal; a descriptor kicked during the window waits for it
`tile_credits[no_oversend]`	A destination takes only what it advertised and returned; one flit short holds the packet; class-1 and unindexed credits change nothing
`tile_credits[shared_counter]`	Two channels to one cell share its counter (also on `rapids_16ch_cxs_top`)
`tile_credits[node]`	`cfg` writes to an unadvertised tetrad node go out uncharged; its control core stays illegal; a read request to the node is refused

Cell	Proves
`tile_credits[illegal]`	Descriptors to an unadvertised cell or core, a zero-advertised cell, an out-of-map or the tile-I/O destination are refused; one to a legal cell and one to that cell's node still go
`tile_straps[land]`	A cell streams past its strap into one channel: every packet lands, the cell keeps going only on the returns, every flit (header included) comes back, the first promptly
`tile_straps[isolation]`	A cell whose channel has no descriptor holds only its own landing space: another cell's packets land meanwhile; the waiting data drains once the descriptors arrive (also on `rapids_16ch_cxs_top`)
`tile_straps[discard]`	A disabled channel's data, a read request into RAPIDS and a reserved `ptype` are consumed and return every flit; a core's data packet returns none; the channel works afterwards
`tile_straps[coalesce]`	Returns held by `m_tile_credit_ready` leave as the held one plus one coalesced `add`
`tile_straps[intake]`	Core descriptors under the strap all run, in order per channel, none dropped, one class-1 entry back each
`tile_straps[intake_overrun]`	A flood past the intake queue: `INTAKE_OVERRUN`, every entry returned, the descriptor queue drops none, the survivors run in send order
`tile_straps[resync]`	A resync waits for `tile_quiescent`, then sets exactly the legal cells (`TILE_LAND_FLITS`) and core (`TILE_DESC_PER_CORE`)
`tile_straps[quiescent_reset]`	A channel reset written while not quiescent does nothing; written while quiescent it clears the channel's error
`tile_strap_regs[off]` / `[default]` / `[alt]`	`TILE_STRAPS` sits at 0x7FC in both halves and reads 0 / 16, 2 / 8, 1 (also on

Cell	Proves
	rapids_16ch_cxs_top)
tile_straps_alt[land] / [isolation] / [intake]	The straps hold in a TILE_LAND_FLITS = 8, TILE_DESC_PER_CORE = 1 build, with the senders modelling the same straps
tile_straps[stall_sink] / [stall_source]	Held WREADY / late R data raise one MonBus stall record (agent, channel, kind, threshold) and the TILE_STALLED bit, which clears when the data moves; the data then lands / goes out correctly

test_descriptor_engine_shared.py inject_owed_beat covers the owed-beat guard. Every cell was mutation-checked.

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30 AXI4 Interface Specification

Module: `axi_read_engine.sv` (source AXI read master), `axi_write_engine.sv` (sink AXI write master) **Location:** `projects/components/dma-ip/rapids/rtl/fub/` **Status:** Implemented
Last Updated: 2026-09-30

30.1 Overview

RAPIDS drives two AXI4 masters: a read master on the source half and a write master on the sink half. The channel signals, the outstanding-transaction limits and the response handling are those of RAPIDS Beats and are specified in AXI4 Interface Specification (shared with RAPIDS Beats). This chapter records only what the byte-granular design adds.

The AXI side stays beat-oriented. Every burst starts at a beat-aligned address and moves whole beats. A descriptor that starts or ends inside a beat is handled by the byte lanes of the data path, not by narrow AXI transfers. Both engines are specified in AXI Read Engine and AXI Write Engine.

30.2 Byte Address to Beat Mapping

The scheduler splits a byte address into a beat address and a byte offset. `OFF_W` is `$clog2(DATA_WIDTH/8)`, which is 5 for the 256-bit board build.

Table 4.1.1: Address Split

Quantity	Value
Offset	`addr[OFF_W-1:0]`
First beat address	`addr` with the low `OFF_W` bits cleared
Beats for a descriptor	`(offset + length + lanes - 1) >> OFF_W`, zero for zero length

The engines receive the working byte address and the remaining beat count. `ARADDR` and `AWADDR` are driven with the low `OFF_W` bits at zero, so every burst is beat-aligned.

30.3 Burst Rules

Table 4.1.2: Burst Rules

Rule	Read master	Write master
Burst start	beat-aligned	beat-aligned
Burst size	full bus width	full bus width
4 KB boundary	no burst crosses one	no burst crosses one
Burst cap	smaller of the configured burst and the beats left to the 4 KB boundary	same

The number of beats left to the boundary is computed from the beat-aligned address, so a byte offset never lets a burst run past it.

30.4 Write Strobes

The write master carries a byte enable with every beat. The data path stores the strobe next to the data in the sink SRAM, and the write engine drives it onto `WSTRB`. Interior beats of a packet are fully strobed. The first beat and the last beat of a descriptor that start or end inside a beat carry a partial strobe, so only the bytes of the descriptor are written and neighbouring bytes in memory are untouched.

The read master has no byte enables. It always reads whole beats and the source data path discards the bytes outside the descriptor.

Table 4.1.3: Byte-Granular Write Signal

Signal	Width	Meaning
`m_axi_wstrb`	`DATA_WIDTH/8`	Byte lanes written by the beat, from the stored strobe.

30.5 Limitations

- TYPE=EXT descriptors remain beat-aligned in address and length. This is a permanent limitation.
- A descriptor never issues a narrow AXI transfer. Sub-beat behavior is entirely in the strobes.

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31 AXI-Stream Interface Specification

Module: `src_data_path_axis.sv` (source master), `snk_data_path_axis.sv` (sink slave)
Location: `projects/components/dma-ip/rapids/rtl/macro/` **Status:** Implemented **Last Updated:** 2026-09-30

31.1 Overview

RAPIDS has one AXI-Stream master on the source half and one AXI-Stream slave on the sink half. Both carry packed bytes. Unlike RAPIDS Beats, the stream is no longer a train of full beats: each descriptor is one packet, the last beat of a packet may be partial, and TSTRB says which byte lanes are valid. The handshake, the width parameters and the monitor tap are those of RAPIDS Beats, see AXI-Stream Interface Specification (shared with RAPIDS Beats). The two data paths are specified in Source Data Path AXIS and Sink Data Path AXIS.

31.2 Packet Contract

Table 4.2.1: Packet Contract

Property	Rule
Packet	One descriptor is one packet, ending with TLAST.
Packing	Bytes are packed from lane 0 upward. The descriptor start offset in memory does not appear on the stream.
TSTRB	Contiguous from lane 0. All ones on every beat but the last.
Partial beat	Only the TLAST beat may be partial, carrying the remaining bytes.
Packet length	Equals the descriptor length in bytes.
Zero length	A zero-length descriptor produces no packet.

31.3 Source Master Signals

Table 4.2.2: Source Master Signals

Signal	Value
`m_axis_tdata`	Re-packed bytes from the drain shifter.
`m_axis_tstrb`	Low `min(bytes_left, DATA_WIDTH/8)` lanes set.
`m_axis_tlast`	Set when the remaining byte count equals the bytes in this beat.
`m_axis_tid`	Channel number.
`m_axis_tdest`	Channel number.
`m_axis_tuser`	Zero.

The source path pops memory beats and re-packs them, so a stream beat is not tied one to one to a memory beat. The first pop of an offset descriptor only primes the hold register and produces no output unless the packet fits in that single beat. When the last memory beat leaves bytes in the hold register, one extra flush beat carries them.

31.4 Sink Slave Signals

Table 4.2.3: Sink Slave Signals

Signal	Use
`s_axis_tdata`	Payload, shifted into the per-channel hold by the descriptor offset.
`s_axis_tstrb`	Valid byte lanes. Only the `TLAST` beat may be partial.
`s_axis_tid`	Channel select, low `CIW` bits.
`s_axis_tdest`, `s_axis_tuser`	Unused.
`s_axis_tlast`	Ends the packet.
`s_axis_tready`	Deasserted while the output register is busy, a flush beat is pending, or the selected channel has no packet record.

The sink stalls a packet until the scheduler has issued the record of its descriptor. A packet for a channel whose record queue is empty is not accepted, so no byte enters the SRAM before the destination offset is known. The queue holds four records per channel.

The spill of a shifted beat is held per channel. After `TLAST`, if the hold register still holds bytes, `TREADY` drops for one flush beat that writes them to the SRAM.

31.5 Length Check

At TLAST the sink compares the bytes received with the length in the head record of the channel. A mismatch sets the sticky per-channel `r_pkt_len_error` bit, which is ORed with the write engine error into `sched_wr_error`. Nothing is padded or dropped silently. The bytes received are counted from the lanes set in TSTRB on each beat.

Last Updated: 2026-09-30

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32 MonBus Interface Specification

Last Updated: 2025-01-10

32.1 Overview

The MonBus (Monitor Bus) is a 64-bit packet-based interface for reporting internal events, status, and performance data. All RAPIDS subsystems generate MonBus packets that are aggregated and output through a unified interface.

32.2 MonBus Signal Interface

32.2.1 Signal Table

Table 4.3.1: MonBus Interface Signals

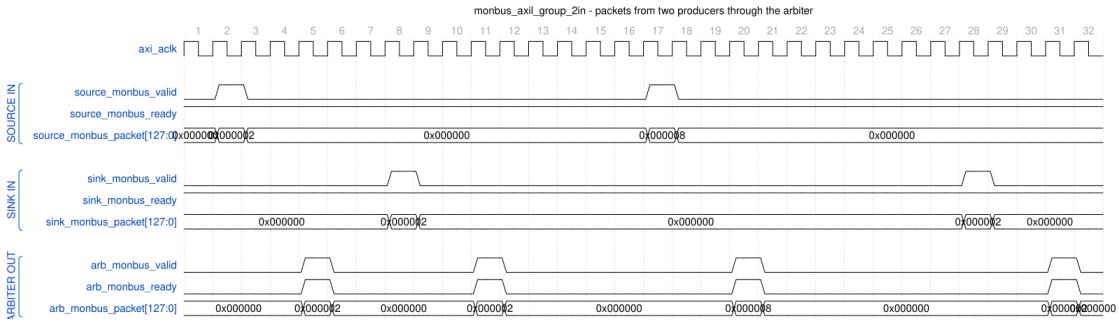
Signal	Direction	Width	Description
`monbus_pkt_valid`	output	1	Packet valid
`monbus_pkt_ready`	input	1	Consumer ready
`monbus_pkt_data`	output	64	Packet data

32.2.2 Handshaking Protocol

Standard valid/ready handshaking: - Data transfers when `valid & ready` both high - Producer holds `valid` until `ready` seen - Consumer asserts `ready` when able to accept

32.2.3 Timing Diagram

32.2.4 Figure 4.3.1: MonBus Packet Transfer Timing



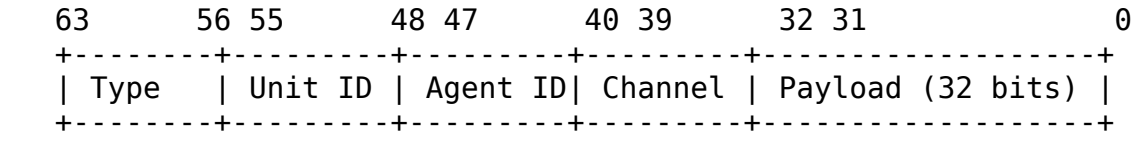
monbus_axil_group_2in - packets from two producers through the arbiter

Source: monbus_group_packet_transfer.json, captured from `dv/tests/macro/test_monbus_axil_group.py` (`basic_flow`, error FIFO 64 x 32, `TEST_LEVEL=full`) with `WAVES=1`. The 128-bit packets are shown as their top 24 bits (`[127:104]`: packet type nibble, then the protocol field at `[108:105]`, so `0x000002` is AXIS and `0x000008` is CORE).

Reading it: a MonBus link is a single-beat valid/ready handshake with the packet presented alongside `valid`. The group holds `ready` high on both inputs, so the source and sink packets are accepted the cycle they are offered, and each one reappears on the arbiter output three cycles later as another one-cycle `valid`/`ready` pair. Between packets the links idle low; nothing is held across transfers.

32.3 MonBus Packet Format

32.3.1 64-bit Packet Structure



32.3.2 Figure 4.3.2: MonBus Packet Format

Table 4.3.2: MonBus Packet Fields

Field	Bits	Description
Type	[63:56]	Packet type code
Unit ID	[55:48]	Reporting unit identifier
Agent ID	[47:40]	Source agent within unit
Channel	[39:32]	Channel number (0-7)
Payload	[31:0]	Type-specific payload

32.4 Packet Types

32.4.1 Type Code Definitions

Table 4.3.3: MonBus Packet Types

Type Code	Name	Description
0x01	DESC_FETCH	Descriptor fetch started
0x02	DESC_COMPLETE	Descriptor fetch completed
0x03	SCHED_START	Scheduler started transfer
0x04	SCHED_DONE	Scheduler transfer complete
0x10	AXI_RD_START	AXI read burst started
0x11	AXI_RD_DONE	AXI read burst completed
0x12	AXI_WR_START	AXI write burst started
0x13	AXI_WR_DONE	AXI write burst completed
0x20	SRAM_FILL	SRAM fill event
0x21	SRAM_DRAIN	SRAM drain event
0x30	ERROR	Error event
0x31	TIMEOUT	Timeout event
0xFF	HEARTBEAT	Periodic heartbeat

32.4.2 Payload Formats by Type

32.4.2.1 DESC_FETCH / DESC_COMPLETE (0x01, 0x02)

Table 4.3.4: Descriptor Packet Payload

Bits	Description
[31:16]	Descriptor address [47:32]
[15:0]	Descriptor address [31:16]

32.4.2.2 SCHED_START / SCHED_DONE (0x03, 0x04)

Table 4.3.5: Scheduler Packet Payload

Bits	Description
[31:16]	Transfer ID
[15:0]	Beat count

32.4.2.3 AXI_RD/WR_START/DONE (0x10-0x13)

Table 4.3.6: AXI Event Packet Payload

Bits	Description
[31:24]	AXI ID
[23:16]	Burst length
[15:0]	Reserved

32.4.2.4 ERROR (0x30)

Table 4.3.7: Error Packet Payload

Bits	Description
[31:24]	Error code
[23:16]	Error source
[15:0]	Error details

32.5 MonBus Source Aggregation

32.5.1 RAPIDS Core MonBus Sources

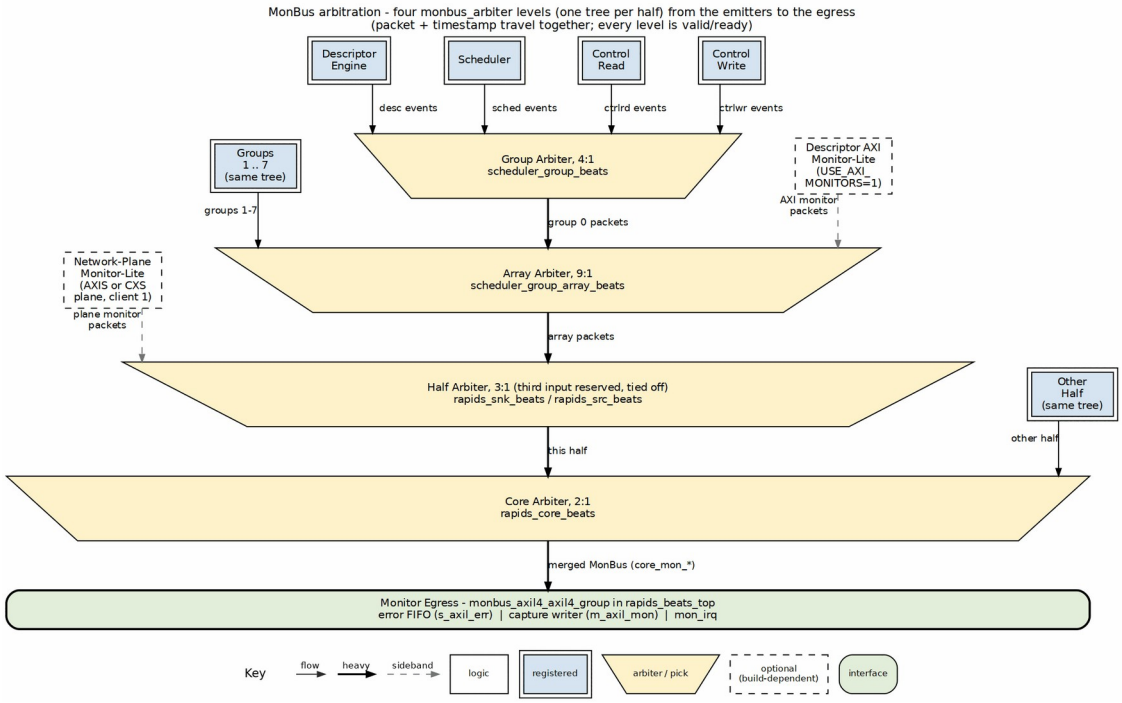
Table 4.3.8: MonBus Source Assignment

Source ID	Agent	Description
0-7	Scheduler[0:7]	Per-channel scheduler events
8-15	DescEngine[0:7]	Per-channel descriptor events
16-23	SnkSRAM[0:7]	Sink SRAM controller events
24-31	SrcSRAM[0:7]	Source SRAM controller events
32	AXI Write Engine	Write transaction events
33	AXI Read Engine	Read transaction events
34	Error Handler	Error events

Scheduler and DescEngine COMPLETION packets are subject to the half's `SCHED_CONFIG.COMPL_EN` (reset 1): the scheduler group drops them ahead of its arbiter when the bit is clear and acknowledges the emitter, so the data path is unaffected. Their ERROR packets are always forwarded (rapids ISSUE-005).

32.5.2 Aggregation Hierarchy

32.5.3 Figure 4.3.3: MonBus Aggregation Tree



MonBus aggregation tree

Source: 21_monbus_aggregation.dot

32.6 Integration Guidelines

32.6.1 Consumer Requirements

MonBus consumers must: 1. Provide sufficient ready cycles to prevent packet loss 2. Handle bursty traffic (multiple sources may have packets simultaneously) 3. Process or buffer packets at wire rate

32.6.2 Recommended Consumer Patterns

```
// Pattern 1: Direct FIFO buffering
gaxi_fifo_sync #(
    .DATA_WIDTH(64),
    .DEPTH(256)
) u_gaxi_fifo_sync (
    .axi_aclk               (axi_aclk),
    .axi_aresetn            (axi_aresetn),
    .wr_valid               (wr_valid),
    .wr_ready               (wr_ready),
    .wr_data                (wr_data),
    .rd_ready               (rd_ready),
    .count                  (count),
    .rd_valid               (rd_valid),
    .rd_data                (rd_data)
);
```

32.7 Timing Requirements

Table 4.3.9: MonBus Timing Parameters

Parameter	Value	Description
Min Ready	1 cycle	Must assert ready within N cycles
Max Latency	Configurable	Before packet dropped
Packet Rate	Variable	Depends on activity

Last Updated: 2025-01-10

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33 Packet Record Interface Specification

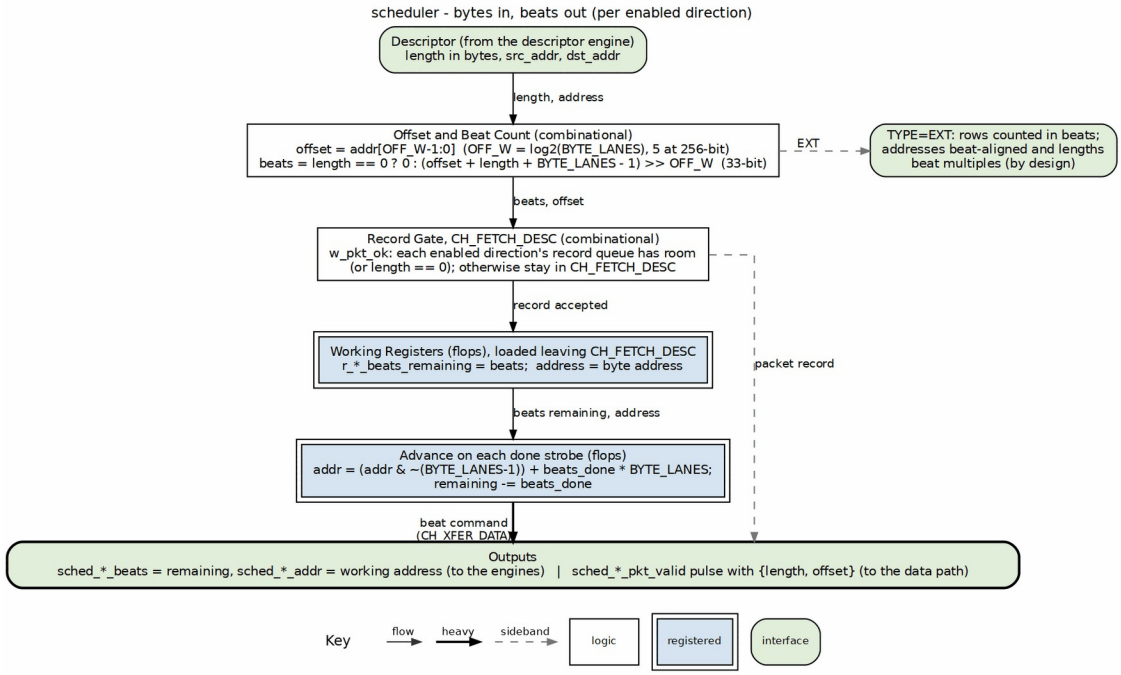
Module: `scheduler.sv` (producer), `src_data_path_axis.sv` and `snk_data_path_axis.sv` (consumers) **Location:** `projects/components/dma-ip/rapids/rtl/fub/` and `rtl/macro/`
Status: Implemented **Last Updated:** 2026-09-30

33.1 Overview

The packet record interface is new in the byte-granular design. It tells a data path how many bytes a descriptor covers and where in its first beat the data starts. RAPIDS Beats has no equivalent, because every transfer there is a whole number of aligned beats.

The scheduler is the producer. It emits one record per data descriptor. The source or sink data path is the consumer and holds the records in a small queue per channel. The record interface is internal to each half: it does not appear on the `rapids_core` or `rapids_top` ports.

33.1.1 Figure 4.4.1: Scheduler Byte Math



Scheduler byte math

Source: assets/graphviz/03_scheduler_byte_math.dot

33.2 Signals

There is one set per direction. The source half uses `sched_rd_pkt_*` and the sink half uses `sched_wr_pkt_*`. Each signal is a vector with one entry per channel.

Table 4.4.1: Packet Record Signals

Signal	Direction (scheduler view)	Width per channel	Description
`sched_*_pkt_valid`	output	1	One-cycle pulse: a record is presented.
`sched_*_pkt_ready`	input	1	The consumer queue has room.
`sched_*_pkt_bytes`	output	32	Descriptor length in bytes.
`sched_*_pkt_offset`	output	`OFF_W`	Address offset inside the first beat.

`OFF_W` is `$clog2(DATA_WIDTH/8)`, 5 for a 256-bit build.

33.3 Protocol

The scheduler raises `valid` for one cycle, in the cycle it leaves the descriptor fetch state. It does not leave that state until every enabled direction can take the record, so `ready` is a pre-condition of the pulse, not a wait state after it. Both directions of a channel see exactly one pulse per data descriptor.

A consumer accepts a record when `valid` and its queue is not full. It drives `ready` low when the queue is full, which is what holds the scheduler in its fetch state.

Table 4.4.2: Record Rules

Rule	Detail
Records per descriptor	One, for data descriptors of nonzero length.
Zero-length descriptors	No record.
Non-data descriptors	No record.
Order	Records are consumed in the order pushed, per channel.
Queue depth	Four records per channel in each data path.
Pop	The record pops on the last beat of its packet.

The offset is taken from the source address in the read direction and from the destination address in the write direction. The length is the same for both. When both directions are enabled, each data path receives its own offset.

33.4 Unused Direction

A scheduler group array instantiated for only one direction ties the other direction off inside the half. The unused `valid`, `bytes` and `offset` outputs are left open and the unused `ready` inputs are held at all ones, so the scheduler never waits on a queue that does not exist.

33.5 Consumer Behavior

- The source data path uses the record to compute the number of memory beats, to decide when a pop emits a stream beat, to set `TSTRB` and `TLAST`, and to retire the record.
- The sink data path uses the record to decide how far to shift each incoming beat, to check the received length at `TLAST`, and it holds off `TREADY` for a channel until the record exists.

Last Updated: 2026-09-30

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34 CXS Interface Specification

Applies to: `rapids_beats_cxs_top`, `rapids_cxs_top` **Protocol:** Arm IHI 0079 (CXS) Issue D, one fixed profile **Last Updated:** 2026-10-16

34.1 Overview

The CXS tops replace the two AXIS network ports with one CXS link per direction: the **sink port** `s_cxs_*` (a CXS receiver) into the sink half and the **source port** `m_cxs_*` (a CXS transmitter) out of the source half. One link carries every channel; the channel rides in `CXSSRCID`. The planes that implement the links are described in the Network Planes chapter (Section 3.13).

34.2 Link Profile

Table 4.4.1: CXS Link Profile

Property	Value
Issue	D (the issue with SRCID, TGTID and USER)
`CXSMAXPKTPERFLIT`	2
Packet start alignment	16 bytes; RAPIDS always starts a packet at byte 0
Packet mapping	One AXIS packet (`tlast`-delimited) is one CXS packet
`CXSSRCID`	The channel (AXIS `tid`)
`CXSTGTID`	AXIS `tdest`
`CXSUSER`	`{tuser, tstrb}`; `tuser` dropped when `DATA_WIDTH/8 + AXIS_USER_WIDTH > 128`
Stream framing	Per stream on the channel bits of `CXSSRCID`
Link control	Explicit credit return, activation, deactivation
Parity	Off

34.3 Signal Table

Table 4.4.2: CXS Interface Signals

Signal	Sink port	Source port	Width	Description
{s,m}_cxs_valid	in	out	1	Flit valid
{s,m}_cxs_data	in	out	DATA_WIDTH	Flit data
{s,m}_cxs_cntl	in	out	CXS_CNTL_PW	START / END / pointers
{s,m}_cxs_srcid	in	out	AXIS_ID_WIDTH	Channel
{s,m}_cxs_tgtid	in	out	AXIS_DEST_WIDTH	tdest
{s,m}_cxs_user	in	out	CXS_UW	{tuser, tstrb}
{s,m}_cxs_crdgnt	out	in	1	Credit grant (receiver to transmitter)
{s,m}_cxs_crdrtn	in	out	1	Credit return (transmitter to receiver)
{s,m}_cxs_activereq	in	out	1	Transmitter requests the link
{s,m}_cxs_activeack	out	in	1	Receiver acknowledges
{s,m}_cxs_deacthint	out	in	1	Receiver asks the transmitter to deactivate
snk_cxs_link_state, src_cxs_link_state	out	out	2	{ACTIVEREQ, ACTIVEACK}
snk_cxs_protocol_e	out	out	1	Sticky protocol violation

Signal	Sink port	Source port	Width	Description
rror, src_cxs_protocol_error				
snk_cxs_unsupported	out	-	1	Sticky: a flit AXIS framing cannot carry

34.4 Credit Flow

One credit permits one flit (IHI 0079D 2.1). The receiver grants up to `CXS_MAX_CREDIT` (default 32) credits, one per free entry in its receive FIFO, so every flit sent against a credit has space. The transmitter spends one credit per flit and returns the credits it still holds on deactivation. Credits are a single pool for the link; a grant carries no ID and cannot be reserved for a channel.

The grant -> flit -> free round trip between the RAPIDS endpoints is about 3 cycles, so 32 credits sustains one flit per cycle with margin for pipeline stages or a clock crossing on the link. More than 63 credits is beyond the spec and requires `CXS_CREDIT_BEYOND_SPEC` on both ends.

34.5 Link Activation

Table 4.4.3: Link Activation Sequence

Step	Transmitter (`m_cxs_*`)	Receiver (`s_cxs_*`)
1	Raises `ACTIVEREQ` when it has a flit to send (and only once the previous activation's `ACTIVEACK` has fallen: 4-phase)	-
2	-	Raises `ACTIVEACK`; starts granting credits
3	Sends flits, one credit each	Frees an entry and regrants as the core drains
4	After `CXS_IDLE_CYCLES` idle cycles in RUN (or on `DEACTHINT`) drops `ACTIVEREQ`, then returns every held credit on `CRDRTN`, one per cycle	-
5	-	Drops `ACTIVEACK` once its credits are back

34.6 Backpressure

On the sink port, a full receive FIFO means no further grants: the transmitter stalls for lack of credit. This is the CXS form of the AXIS sink's `s_axis_tready = 0`, including the byte-granular sink's hold until a channel's descriptor has been fetched. On the source port, the plane stalls the core's egress (`fub_axis_tready = 0`) while it has no credit.

34.7 Proposed Changes

A proposal (2026-10-16, not implemented) puts a tile packet header at the front of every packet: destination and source coordinates, payload type (`cfg` / `prog` / `mem`), length and target address. It is described once in the compute-engine tile's `TILE_SPEC.md` Section 11; CXS carries it as the first flit. The link signals, credits and per-stream framing on this page do not change, because `CXSSRCID` keeps the channel.

35 rapids-g Overview

Status: Implemented (rapids TASK-034) **Last Updated:** 2026-10-22

35.1 What rapids-g Is

rapids-g is RAPIDS with its scheduler state machines replaced by firmware. Each half of the core (source and sink) has one gatling, the four-context barrel RV32I core (`projects/components/riscv-ip/gatling-rv32i`), running one firmware image for all of that half's 4, 8 or 16 channels. The firmware makes every decision RAPIDS's state machine makes and does the EXT address arithmetic RAPIDS's `scheduler` does in hardware:

- taking a descriptor, validating it, starting it, finishing it, and following a chain;
- the EXT run bases, `base + ((i0 * stride_0) & wrap_0) + ((i1 * stride_1) & wrap_1)`, computed by repeated addition (gatling has no multiplier) and pushed ahead into a small queue per direction.

Everything that happens at the engines' rate stays hardware: the counters the done strobes decrement, the address advance within a run, the timeouts, the sticky errors, the packet records and the MonBus packets. That hardware is `scheduler_core`, the same module RAPIDS's `scheduler` uses (Section 2.1), so the two designs share one datapath.

rapids-g is register-compatible with RAPIDS: the same register map, the same descriptor format, the same host software, the same network planes.

Two capabilities RAPIDS does not have: the firmware pre-validates the next descriptor while the current one runs, so the hardware starts it the moment the current one completes (Section 5.2); and each half's gatling can read and write its own half's registers – including the eight per-channel `CH_SCRATCH` bits it shares with the host – through the host's register command queue (Section 5.3).

35.2 Structure

Table 5.1.1: rapids and rapids-g, level by level

Level	rapids	rapids-g	How the -g module is made
fub	`scheduler` = state machine + ceilings + 2 run-base generators + `scheduler_core`	`scheduler_g` = command port + ceilings + base FIFOs + `scheduler_core`	hand-written (`rtl/fub_g/`)
macro	`scheduler_group_array`	`scheduler_group_array_g`: the same, with `scheduler_g` and one `rapids_seq_g`	generated
macro	`rapids_src`, `rapids_snk`, `rapids_core`	`rapids_src_g`, `rapids_snk_g`, `rapids_core_g`	generated
top	`rapids_top_base`, `rapids_top`, `rapids_cxs_top`	`rapids_g_top_base`, `rapids_g_top`, `rapids_g_cxs_top`	generated
sequencer	–	`rapids_seq_g`: gatling + instruction ROM + scratch + window decoder	hand-written (`rtl/macro_g/`)
firmware	–	`rtl/fw_g/sched_g.S`	assembled into `rtl/macro_g/generated/fw/sched_g_fw.svh`

`bin/gen_rapids_g.py` derives every generated file from the RAPIDS file of the same name (Section 5.4); a change to RAPIDS reaches rapids-g by rerunning it.

35.3 Channels per gatling

By default each hart serves one channel at 4 and 8 channels (`NUM_HARTS` = `NUM_CHANNELS`): 4 or 8 harts, one slot each, every hart issuing one instruction every `NUM_HARTS` clocks. At 16 channels the default is 8 harts with two slots each (channels h and h + 8), measured best overall (Section 5.5; rapids TASK-036); 16 harts x 1 and 4 harts x 4 are also built and tested. `NUM_HARTS` = 4 at 8 channels gives each hart two slots, channels h and h + 4. The generated array refuses any other channel count, and `SHARED_ENGINES` = 0, at elaboration. An 8-channel core therefore has two gatlings, one per half, each sequencing eight schedulers; a 16-channel core has two, each sequencing sixteen.

35.4 What stays hardware, and why

Two transitions are interlocks, not policy, and stay hardware so they take effect in the cycle RAPIDS's `scheduler` would take them:

- a channel reset forces CH_IDLE;
- a fatal fault – a descriptor, engine or control error, an escalated timeout, or the serving hart halting – forces CH_ERROR, which drops the engine valids at once.

Every other transition is a firmware command, checked against an edge table (Section 5.2).

36 scheduler_g

Module: `scheduler_g.sv`, `scheduler_g_base_fifo.sv` **Location:** `projects/components/dma-ip/rapids/rtl/fub_g/` **Status:** Implemented (rapids TASK-034) **Last Updated:** 2026-10-22

36.1 Overview

`scheduler_g` has the whole port list of RAPIDS's `scheduler` plus a sequencer window. Inside it is `scheduler_core` (Section 2.1), the same two byte-to-beat ceilings as `scheduler`, and small pieces in place of its state machine and run-base generators:

- a state register written by command, with an edge table and the two interlocks;
- pre-validation: the firmware judges the next descriptor while this one runs, and the hardware takes the approved edges itself;
- the beat totals, computed in hardware with `scheduler`'s own ceiling and loaded from the skid head on the cycle it is taken, and the bases-done flags, written by the firmware;
- two four-entry run-base FIFOs (`scheduler_g_base_fifo`), one per direction.

36.2 Sequencer Port

Table 5.2.1: Sequencer port

Signal	Dir	Description
`seq_wr_en`, `seq_wr_addr[4:0]`, `seq_wr_data[31:0]`	in	a word write into the window (registered in `rapids_seq_g`)
`seq_rd_addr[4:0]`, `seq_rd_data[31:0]`	in / out	a word read; combinational here, registered in `rapids_seq_g`
`seq_attn`	out	the firmware has something to do on this channel
`seq_fault`	in	the hart serving this channel has halted: a fatal fault
`seq_alg_max`	in	the firmware image's highest next-address selector (`FW_ALG_MAX`, rapids TASK-035), a constant from `rapids_seq_g`

36.3 Register Window

Table 5.2.2: The window, by word

Word	Read	Write
0	STATUS	CMD: `[6:0]` next state (0 = none), `[8]` read bases done, `[9]` write bases done, `[10]` STAGE, `[11]` ARM, `[12]` ARM to NEXT_DESC, `[13]` GEN_OK, `[14]` DECLINE, `[15]` STAGE_ARM, `[16]` STAGE_ARM to NEXT_DESC
1-2	SRC_LO, SRC_HI	1: FW_ERROR (sends an error packet, data `{word, PKT_LO}`); 2: reserved
3-4	DST_LO, DST_HI	3: RD_BASE_LO, 4: RD_BASE_HI (the HI write pushes `{HI, LO}`)
5	LENGTH (bytes)	WR_BASE_LO
6	NEXT_PTR	WR_BASE_HI (pushes)
7-12	chunk 1 words 0-5 (`descriptor_ext_t`)	7: PKT_LO (the next packet's low data word); 8: FW_DEBUG (sends a debug packet, data `{word, PKT_LO}`); 9-12: reserved
13	FIFO: `[2:0]` read bases queued, `[6:4]` write bases queued, `[15:8]` CMD_ERR, `[16]` firmware packet pending	–
14	CFG: `[15:0]` BYTE_LANES, `[19:16]`	–

Word	Read	Write
	OFF_W, [20] EN_READ, [21] EN_WRITE, [22] EXT built, [23] TILE_HDR built	
15-18	the skid head, before it is taken: src [31:0], dst [31:0], length, descriptor [223:192] (valid, opcode, type, tile op)	–
19-20	the beat totals in use (hardware)	–
21	HEAD_SUM: the skid head's [0] valid (and its next-address selector one the image runs), [1] last, [2] chain (next_ptr != 0), [3] open (not yet staged or declined, in a state where STAGE and DECLINE are legal)	–
22	ALG: the latched descriptor's next-address selector, chunk 0 [224:217]	–
23	ALG_MAX: seq_alg_max	–

Table 5.2.3: STATUS

STATUS bits	Field
[6:0]	state (channel_state_t)
[7]	seq_attn
[8], [9], [10]	descriptor at the skid head, channel enable, channel reset active
[11], [13:12]	descriptor valid, opcode

STATUS bits	Field
[14], [15], [16]	last, gen_irq, chain (`next_ptr != 0`)
[17], [18], [19]	tile_bad, pkt_ok, exec_complete
[20], [21]	start pending, cmd_err (sticky)
[22], [23], [24], [25]	EXT, header only, read per-beat, write per-beat
[26], [27]	read bases done, write bases done
[28], [29], [30], [31]	armed, head staged, EXT set-up needed, head declined

36.4 Commands and the Edge Table

A CMD write names the next state. The table below is the only way state changes outside the interlocks. A command it does not list is ignored, sets the sticky `cmd_err` and counts in CMD_ERR (saturating at 255; a channel reset clears both).

Table 5.2.4: Legal edges

From	Command	Condition	Effect
IDLE	FETCH_DESC	a descriptor at the head, channel enabled	the head is taken on the same edge
NEXT_DESC	FETCH_DESC	a descriptor at the head	taken
FETCH_DESC	XFER_DATA	descriptor valid, not tile_bad	start pending; taken the cycle `pkt_ok` holds, which is the cycle the packet records fire
XFER_DATA	COMPLETE	exec_complete	
COMPLETE	IDLE or NEXT_DESC	–	
FETCH_DESC, XFER_DATA, COMPLETE, NEXT_DESC	ERROR	–	

CH_ERROR is left only by a channel reset, as in RAPIDS's `scheduler`. The pending start is what keeps the packet records a single handshake: `scheduler_core` raises them only with `start_req`, so a FETCH_DESC held by the firmware never offers a record early.

A command that arrives after a channel reset it was not written for – the firmware was part-way through a handler when the reset came – is ignored by the same table and counted. The equivalence tests expect CMD_ERR to stay 0 on every channel that is never reset.

36.5 Pre-validation and the Automatic Edges

Owner direction, 2026-10-21: “Pre validation is a must.” While descriptor N transfers, the firmware:

- ARMs N: complete it on `exec_complete`, then go to NEXT_DESC (`[12]` = 1) or IDLE. Legal once per descriptor, in XFER_DATA.
- judges the descriptor at the skid head: valid -> STAGE; not valid -> DECLINE. Legal while a descriptor is at the head and not yet judged, from XFER_DATA through NEXT_DESC.
- with STAGE, may set STAGE_ARM (`[15]`) and its NEXT_DESC choice (`[16]`): the head’s own chain outcome, decided from HEAD_SUM. The hardware then takes that head already armed. STAGE_ARM without STAGE is illegal and counted.

STAGE_ARM is how the firmware keeps up with short descriptors. Without it, every descriptor needs an ARM written inside its own XFER_DATA, and the firmware’s path to that write is about 25 instructions – 200 clocks at 8 harts, where a 64-beat descriptor transfers in under 90. Each such descriptor finished before it was armed and went the hand path: 672 clocks per descriptor on a 64-beat sink chain (Section 5.5). With STAGE_ARM, a descriptor needs one firmware write, made any time before its predecessor completes.

With N armed, the hardware then takes, in the cycles RAPIDS’s `scheduler` would: XFER_DATA -> COMPLETE (on `exec_complete`) -> IDLE or NEXT_DESC -> FETCH_DESC (the staged head, its totals, the start already requested) -> XFER_DATA (when `pkt_ok`). The latched descriptor is still checked (valid, tile_bad) as on the hand path. Anything not approved stops the sequence and raises attention: an unarmed descriptor, an unstaged or declined head, a disabled channel.

Three ordering rules make it safe, each the fix of a race the equivalence bench found:

Rule	Why
An EXT descriptor completes automatically only once both directions’ bases-done bits are set	the firmware writes them after its last push, so no push or done bit of this descriptor can land after the next is taken
GEN_OK (set-up done) is written before the set-up	a descriptor cannot complete before its done bits, which the set-up writes after GEN_OK, so a GEN_OK can never land on the next descriptor
The firmware judges both directions’ refills on one STATUS snapshot	a fresh STATUS read after the first direction’s last done bit may already describe the next descriptor

A head the hardware takes is flagged (`[30]`) when it is EXT, and the firmware sets up its run bases in XFER_DATA; run 0 uses the descriptor's own address, so only short first runs wait for them.

36.6 Firmware Packets

The firmware can put its own packets on the channel's MonBus port, from the channel's agent:

Table 5.2.5: Firmware packets

Write	Packet type	Event code	Event data
FW_DEBUG (word 8)	`PktTypeDebug`	`RAPIDS_EVENT_FW_DEBUG` (0x09)	`{written word, PKT_LO}`
FW_ERROR (word 1)	`PktTypeError`	`RAPIDS_EVENT_FW_ERROR` (0x0A)	`{written word, PKT_LO}`

`scheduler_core`'s own packets are one-cycle pulses and always win the cycle; a firmware packet leaves in the first cycle with none and `mon_ready` high. One packet waits at a time (FIFO `[16]`); a send while one waits is dropped and counted in CMD_ERR, like a push to a full base FIFO. A firmware error packet only reports: it does not move the channel to CH_ERROR, which stays a CMD or an interlock. Neither packet type is a completion, so `cfg_sched_compl_enable` does not filter them.

36.7 Attention

`seq_attn` is high when the firmware has work on the channel, and is what the sequencer's SUMMARY register reports:

- IDLE with the channel enabled and an unstaged descriptor at the head; NEXT_DESC with an unstaged descriptor at the head;
- FETCH_DESC with no start pending, or turned tile_bad;
- XFER_DATA not yet armed, EXT set-up needed, an EXT base FIFO with room and bases still to come, or an unjudged head;
- COMPLETE not armed (the hand path).

Nothing raises attention in CH_ERROR, during a channel reset, or on a fault.

36.8 Run-Base FIFOs

`scheduler_g_base_fifo` holds four 64-bit bases (the depth of `stream_run_addr_gen`'s own queue). A LO write is held; the HI write pushes `{HI, LO}`. The FIFO is flushed whenever the channel is outside FETCH_DESC and XFER_DATA, and on a channel reset, so a base can never outlive its descriptor. A push to a full FIFO is dropped and counted in CMD_ERR. `scheduler_core` takes a base only between runs, exactly as from the generator.

---

37 rapids_seq_g and the Scheduler Firmware

Module: `rapids_seq_g.sv`; firmware `rtl/fw_g/sched_g.S` **Location:** `projects/components/dma-ip/rapids/rtl/macro_g/` **Status:** Implemented (rapids TASK-034, TASK-037, TASK-035, TASK-036, TASK-038) **Last Updated:** 2026-10-23

37.1 rapids_seq_g

One `gatling_core`, used unchanged through its own filelist, with the three memories it needs, each answering in one stage as gatling requires:

Table 5.3.1: rapids_seq_g

Block	What	Size
Instruction memory	the firmware image, from `sched_g_fw.svh`; one read port serves every hart (the barrel fetches for one context per cycle). A ROM, or with `FW_WRITABLE` = 1 the same block RAM with a second port the host loads through (below)	`IMEM_WORDS` = 1024
Scratch	per-hart memory, indexed `{context, word}`; the run-base generator state of each slot	`SCRATCH_WORDS` = 128 per hart (256 at four slots)
Window decoder	routes a data access to the slot's `scheduler_g` by context and slot	–

Parameters: `NUM_CHANNELS` (4, 8 or 16), `NUM_HARTS` (4, 8 or 16, dividing `NUM_CHANNELS` with at most four slots per hart; default `NUM_CHANNELS`, one hart and one slot per channel; hart h serves channels h + `NUM_HARTS` x s), `IMEM_WORDS`, `SCRATCH_WORDS` (a power of 2, at least 64 per slot; a smaller scratch would alias slot 1 onto slot 0, and elaboration refuses it), `FW_WRITABLE` (0 = ROM, 1 = loadable; Loading Firmware Live).

37.1.1 Data Memory Map

Table 5.3.2: Data memory map

Address	Region
0x0000_0000	instruction memory (fetch only; a data read returns 0)
0x1000_0000	slot 0 window (Section 5.2),

Address	Region
	word-indexed
0x1000_0100 + 0x100 s	slot s window (s < SLOTS)
0x1000_1000	SUMMARY: `[SLOTS-1:0]` `seq_attn` of this hart's slots
0x1000_1004	SLOTS: slots per hart (1, 2 or 4)
0x1000_3000	register mailbox (below), aliased by context
0x2000_0000	scratch, aliased by context

Contexts stay implicit (gatling PRD 7): every hart runs the same image, and "slot 0" is a different channel for each, chosen by the context the access comes from. Reads are registered (address in X, data in W); window writes are registered into `scheduler_g` one cycle after X, so a hart's next instruction, `NUM_HARTS` (at least four) cycles later, always sees the effect of its own write. Each hart issues one instruction every `NUM_HARTS` clocks: at the default 8 harts an instruction costs a hart 8 clocks, which is the unit of every firmware cost in Section 5.5.

37.1.2 Register Mailbox

Each hart can read and write its own half's PeakRDL registers – the same `rapids_regs` the host programs, config and status alike – through a mailbox in its data memory map at 0x1000_3000, aliased by context:

Table 5.3.3: Register mailbox, per hart

Offset	Word	What
+0x00	REG_ADDR	byte offset within the half, `[11:0]`
+0x04	REG_WDATA	write data
+0x08	REG_CTRL	`[0]` go, `[1]` write; a store with go set posts the access
+0x0C	REG_STATUS	`[0]` busy, `[1]` done, `[2]` error (the response's PSLVERR)
+0x10	REG_RDATA	read data, valid once done

The firmware posts an access and polls REG_STATUS: a single-stage barrel cannot stall on a bus. A store to REG_* while the hart's own access is busy is ignored, so a hart polls for done before it

posts the next one. The scheduler firmware does not use the mailbox today; the path is there for firmware that reads or adjusts its half's configuration, and `test_rapids_g_regpath` proves it (below).

37.1.2.1 A Serialized Path

The register path is serialized, end to end. The register block has one command/response path (`peakrdl_to_cmdrsp` in front of `rapids_regs`), and exactly one access is on it at any time. One agent – the host's APB, or one hart of one gatling – is granted the path, sends its one command, and **owns the path until its response is complete**: the grant is released only by the response handshake. No other agent's command enters the path in between, there is no pipelining and no second outstanding access, and a response is never returned to anyone but the agent that owns the path. This is the same one-outstanding discipline `apb4_slave` keeps on the host path alone, and the MON-window guard behind the arbiter depends on it: it registers its select when a command is accepted and steers the response on that registered select.

Two levels of round-robin arbitration feed that one path, and both hold their grant for the whole access:

Level 1, the harts of one gatling (`rapids_seq_g`). Each hart has its own mailbox state, indexed by context: `r_mb_addr[h][11:0]`, `r_mb_wdata[h]`, `r_mb_wr[h]`, and `r_mb_busy[h]` – set by the REG_CTRL store, and the hart's request – with `r_mb_done[h]`, `r_mb_err[h]` and `r_mb_rdata[h]` for the completion. An `arbiter_round_robin_simple_ack` takes `request = r_mb_busy` and grants one hart (`w_mb_gnt_valid`, `w_mb_gnt_id`). That hart's access goes out once on the gatling's one master port – `reg_cmd_valid = w_mb_gnt_valid && !r_mb_sent && r_mb_busy[gnt]`, with `reg_cmd_pwrite`, `reg_cmd_paddr[11:0]` and `reg_cmd_pwdata` from the granted hart – and `r_mb_sent` records that it has gone. `reg_rsp_ready = w_mb_gnt_valid && r_mb_sent`. The response handshake clears the hart's busy, latches done, `err <- reg_rsp_pslverr` and `rdata <- reg_rsp_prdata`, and is the arbiter's ack: only then does the grant move to the next requesting hart.

Up the tree. The port is `seq_reg_*` on `scheduler_group_array_g`, `rapids_src_g` / `rapids_snk_g` and `rapids_core_g` (`src_seq_reg_*`, `snk_seq_reg_*`), carried up unchanged by `gen_rapids_g.py`, and arrives in the top base as `w_src_seq_reg_*` and `w_snk_seq_reg_*`. Each carries `cmd_valid`, `cmd_ready`, `cmd_pwrite`, `cmd_paddr[11:0]`, `cmd_pwdata`, `rsp_valid`, `rsp_ready`, `rsp_prdata`, `rsp_pslverr`.

Level 2, the host against the two gatlings (`rapids_g_reg_arb`, included by `rapids_g_top_reg_arb.svh`). The generator renames `apb4_slave`'s command/response outputs to `host_cmd_*` / `host_rsp_*`, and three masters meet:

Table 5.3.4: Register-path masters

Master	Index	Address into the 13-bit map
host APB, `host_cmd_*` / `host_rsp_*`	0	`host_cmd_paddr`, the whole map
SRC gatling,	1	`{1'b0,`

Master	Index	Address into the 13-bit map
w_src_seq_reg_*		w_src_seq_reg_cmd_paddr}: pinned to SRC, 0x0000-0x0FFF
SNK gatling, w_snk_seq_reg_*	2	{1'b1, w_snk_seq_reg_cmd_paddr}: pinned to SNK, 0x1000-0x1FFF

A second arbiter_round_robin_simple_ack takes request = m_cmd_valid[2:0] and grants one master (w_gnt_valid, w_gnt, w_gnt_id). The granted master's command passes through once – s_cmd_valid = w_gnt_valid && !r_cmd_sent && m_cmd_valid[gnt], m_cmd_ready[gnt] = s_cmd_ready && !r_cmd_sent – and r_cmd_sent holds the grant open for the response. The response goes back only to the owner: m_rsp_valid[gnt] = s_rsp_valid && r_cmd_sent and s_rsp_ready = r_cmd_sent && m_rsp_ready[gnt] (m_rsp_prdata and m_rsp_pslverr are shared wires, meaningful only with that master's valid). w_ack = (s_rsp_valid && s_rsp_ready) ? w_gnt : 0 releases the grant and clears r_cmd_sent. The arbiter's slave side is apb_cmd_* / apb_rsp_*, which runs on through the MON-window guard (an error response for paddr[11] when USE_MON_REGS = 0) to peakrdl_to_cmdrsp and rapids_regs.

Table 5.3.5: One gatling access, end to end

Step	Owner of the path	Signals
a hart posts	–	REG_CTRL store sets r_mb_busy[h]
level 1 grants that hart	the hart, within its gatling	w_mb_gnt_id = h; reg_cmd_valid
level 2 grants that gatling	the hart, end to end	w_gnt_id; s_cmd_valid -> apb_cmd_valid; r_cmd_sent, r_mb_sent set
the register block answers	still the hart	apb_rsp_valid -> m_rsp_valid[gnt] -> reg_rsp_valid
the response is accepted	released at both levels	w_ack (level 2), reg_rsp_valid && reg_rsp_ready (level 1); r_mb_done[h]

What the serialization costs and guarantees:

- **Ordering.** Accesses from all agents are totally ordered by grant; an access completes before the next one starts, so a read after a write – from any agent – sees the write.
- **Fairness.** Under full contention the host gets at least every third access and each gatling the same; inside a gatling each of its harts gets at least one of every `NUM_HARTS` of that gatling's turns (1 in 24 of the path at 8 harts).
- **Latency.** An access holds the path from command to response, including `peakrdl_to_cmdrsp`'s two-cycle request, so a busy gatling delays a host access by an access time, and a host access delays a waiting hart the same way. That is invisible at UART speeds and small at full APB rate; the mailbox is not a per-descriptor path.
- **No atomicity across accesses.** Each access is atomic; a read-modify-write is two accesses another agent can come between. The per-channel `CH_SCRATCH[n].SCRATCH` bits (0x300 in each half) are for the host and the gatling to share flags or semaphores: give each bit one writer.

`test_rapids_g_regpath` (dv/tests/top_g) replaces the scheduler image with a test program in which every hart reads SCHED_TIMEOUT_CYCLES of its own half through the mailbox and writes SCHED_TIMEOUT_LIMIT, while the host programs both halves and reads them back over APB through the same path. It checks that each gatling sees its own half's value, that its writes land in its own half, that no mailbox access reported an error, and that every host read during the contention returned the value the host expects.

37.1.3 Faults

`seq_fault[c]` is the halted bit of the hart serving channel c, so a firmware trap is a fatal fault for that hart's channels: they enter CH_ERROR and send the scheduler's error packet. A halted hart restarts from address 0 once every channel it serves is in channel reset.

37.2 Firmware

`sched_g.S` is RV32I with no stack and no data section; state that outlives one step lives in the slot's scratch. It is an **executive** over a **policy table** (rapids TASK-037). The executive polls and dispatches and is the same for every policy; the table says what each (state, event) pair does. Both firmware defects found on 2026-10-22 came from mixing the two (Section 5.5): the poll and the handlers disagreed about who owned an open head, and a handler that re-dispatched itself kept a hart for several descriptors. Neither can recur in this shape.

37.2.1 The Executive

Run to completion, no preemption: the barrel's harts are the threads. Each pass of the poll gives a slot at most one handler, and the handler returns to the poll. The image reads SLOTS at start-up and takes one of two polls.

One slot per hart (the default, `NUM_HARTS` = `NUM_CHANNELS`). The poll reads HEAD_SUM first. HEAD_SUM = 13 (open, valid, `next_ptr` != 0, not last) is a chained head: the poll writes STAGE | STAGE_ARM | STAGE_ARM_NEXT, held in a register, so a running chain costs four instructions from the read to the CMD write. This chain hook is the one policy entry that lives in the poll itself. Any other open head is judged inline (`judge_head`: stage it armed with its own chain outcome, or decline it). With no open head the poll reads SUMMARY; the idle pass is seven instructions. When SUMMARY shows attention, the poll reads HEAD_SUM again, so a head that arrived between the two reads is judged inline too, then calls `step`.

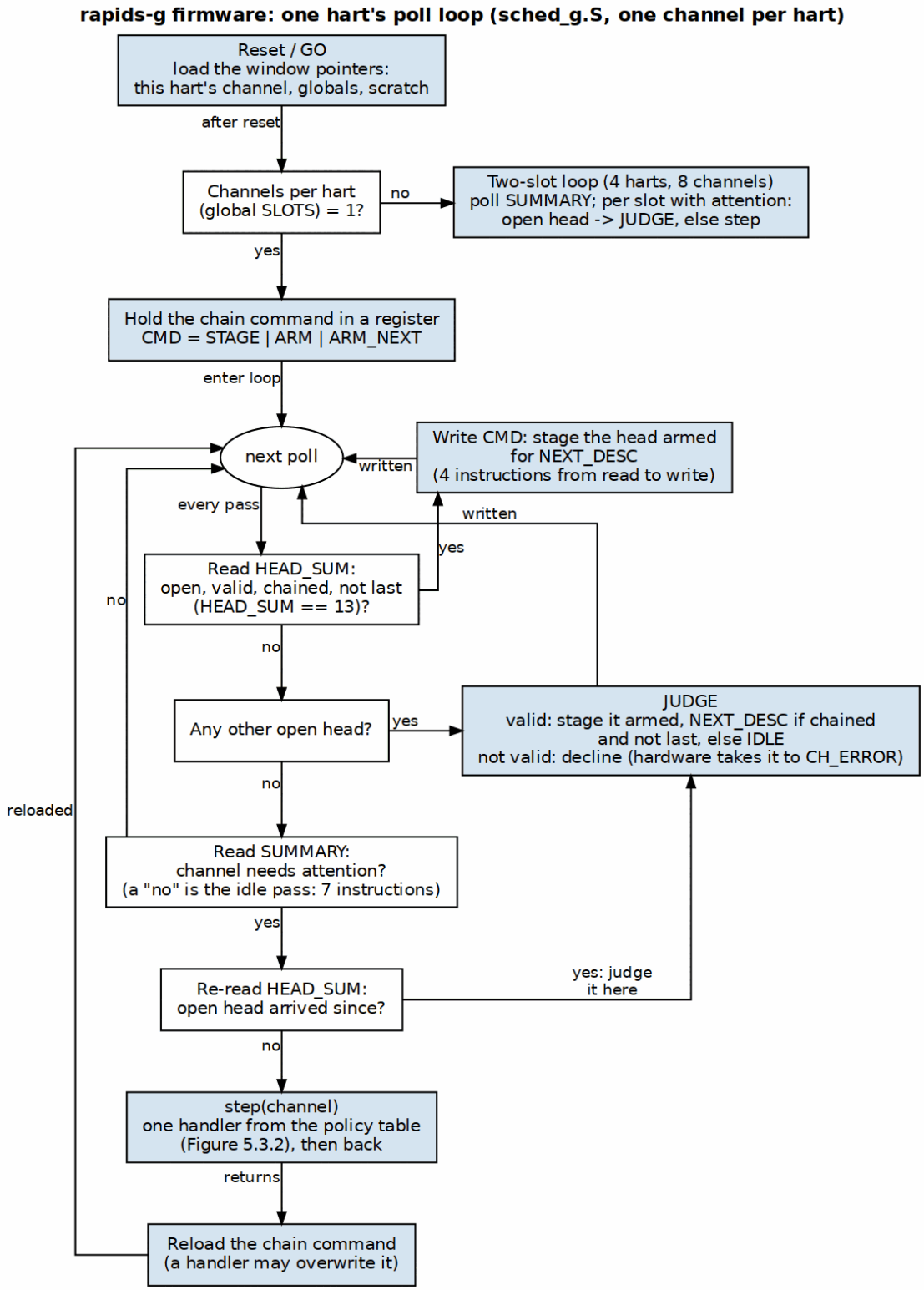
Two slots per hart (`NUM_HARTS` = 4 at 8 channels, or 8 at 16, the 16-channel default). The poll reads SUMMARY first, so the idle pass is two instructions. For each slot with attention it reads HEAD_SUM and, before anything else, takes the same chain hook: HEAD_SUM = 13 is staged armed for NEXT_DESC straight from the read, with the stage word held in `tp` (reloaded after every `step`, which may clobber it). Any other open head is judged inline; otherwise the poll calls `step`. Unrolled for the two slots.

A chain is serial: the hardware takes about 20 clocks per one-beat descriptor (FETCH_DESC, XFER_DATA, COMPLETE, NEXT_DESC), then waits in NEXT_DESC with the next head open until the firmware's stage command lands, so every instruction between the head opening and the CMD write is paid per descriptor. A cycle trace of the earlier two-slot polls (Section 5.5) showed it: the image before 2026-10-23 judged every head through SUMMARY and JUDGE, about 22 instructions (about 180 clocks a descriptor); a first fix that put the two HEAD_SUM checks ahead of SUMMARY alternated between a 36-clock wait (the check caught the head) and a 156-clock one (the head opened just after its check and the pass fell to JUDGE), about 120 on average. Checking SUMMARY first and the chain hook inside each slot's branch never misses: about 99 clocks a descriptor on the board, against about 44 with one slot per hart, which tests HEAD_SUM in a four-instruction loop.

Four slots per hart (`NUM_HARTS` = 4 at 16 channels, rapids TASK-036). The same poll as a loop over SUMMARY's bits, the slot's offset in `sp` (window and scratch share the 0x100 slot stride).

`step` reads STATUS and jumps through `ptab`, indexed by the 6-bit one-hot state: four instructions from the STATUS read to the state's row. `ptab` has 64 entries, so no value of the field can index past it; the five states with work have a jump to their row, and every other entry (CH_ERROR, which only a channel reset leaves, and the values that are not states) returns. The image is linked at 0 and `ptab` sits below 2 KB, so the jump is `jalr` off the index register alone and needs no base register (`gen_rapids_g_fw.py` refuses an image where it does not fit).

37.2.2 Figure 5.3.1: One Hart's Poll



rapids-g firmware poll loop

Source: `assets/graphviz/12_rapids_g_fw_poll_loop.dot`

37.2.3 The Policy Table

A state's row picks one handler by event, highest priority first. A handler does one step – one or two CMD writes, or one run-base set-up or top-up – and returns. No handler loops, re-reads STATUS to carry on, or calls `step`: the slot's next step is the next pass of the poll.

Table 5.3.6: The policy table

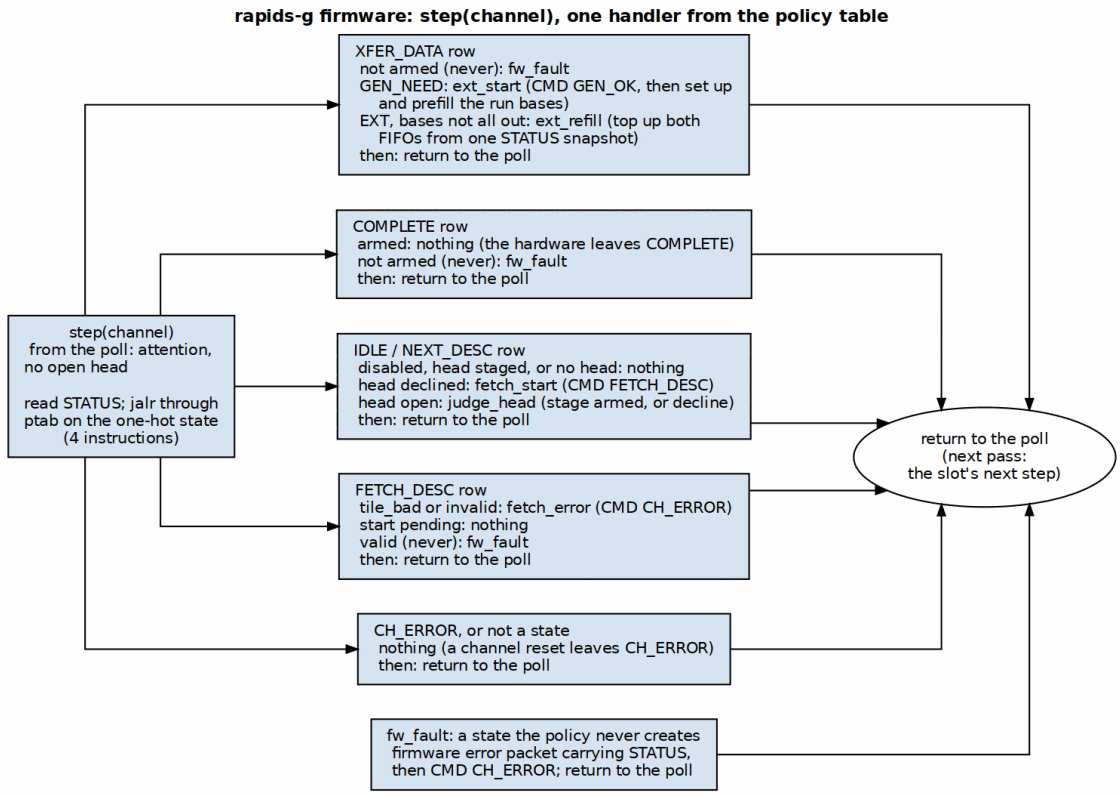
State	Event, in priority order	Handler	What it does
any	an open head (in the poll)	`judge_head`	valid: STAGE armed with its chain outcome; not valid: DECLINE
IDLE (enabled), NEXT_DESC	head staged, or no head	–	nothing: the hardware takes a staged head
	head declined	`fetch_start`	CMD FETCH_DESC; FETCH_DESC then takes it to CH_ERROR
	head open	`judge_head`	as above
FETCH_DESC	tile_bad, or descriptor invalid	`fetch_error`	CMD CH_ERROR
	start already pending	–	nothing
	valid (never: see below)	`fw_fault`	firmware error packet carrying STATUS; CMD CH_ERROR
XFER_DATA	not armed (never)	`fw_fault`	as above
	GEN_NEED (an EXT descriptor the hardware took)	`ext_start`	CMD GEN_OK, then set up and prefill the run bases (`ext_init`)
	EXT, bases not all out	`ext_refill`	top up both directions' base FIFOs from one

State	Event, in priority order	Handler	What it does
			STATUS snapshot
COMPLETE	armed	–	nothing: the hardware leaves COMPLETE next cycle
	not armed (never)	`fw_fault`	as above

The rows marked “never”. Every valid head is staged armed, so the hardware takes it, completes it and leaves COMPLETE on its own; the firmware takes a head itself (FETCH) only when it declined it, and a declined head is invalid by the same test the latch applies. A descriptor can therefore never be in XFER_DATA or COMPLETE unarmed, nor valid in FETCH_DESC by the firmware’s hand. TASK-034’s firmware still carried a hand path for those states (arm in XFER_DATA, route out of COMPLETE, start from FETCH_DESC); the mutation set showed no test could reach it (MAS 5.5), because nothing can. It is replaced by `fw_fault`: a state the policy never creates stops the channel visibly instead of being guessed at, and a channel reset recovers it.

The run-base routines (`ext_init`, `refill`) are a layer below the table: they are where the next-address algorithm lives, and the descriptor’s selector picks one (rapids TASK-035, below).

37.2.4 Figure 5.3.2: step(), the Policy Table



rapids-g firmware step and policy table

Source: `assets/graphviz/13_rapids_g_fw_service.dot`

37.2.5 EXT Run Bases

The positions, the step and the 16-bit index wrap are exactly `stream_run_addr_gen`'s: run-contiguous when `stride_0` equals the beat size (one base per run of `inner_count` beats, rows 1, 2, ...), otherwise per beat (every beat after the first, inner index fastest). A base is `base + ((i0 * s0) & m0) + ((i1 * s1) & m1)` in 64 bits. Without wrap masks (the common case) the firmware keeps the address itself and a base is one 64-bit add (`cur += s0` along a row, `row += s1` at a row's end); with masks it keeps the two products as running sums and masks them for each base. Whenever an EXT descriptor's FIFO has room, attention brings the firmware back to top it up; the last base sets the direction's bases-done bit.

37.2.6 Next-Address Selector

The descriptor's selector (chunk 0 `[224:217]`, HAS Section 5.1) chooses the algorithm. The image declares the highest one it implements with `.equ ALG_MAX`; `gen_rapids_g_fw.py` writes it into `sched_g_fw.svh` as `FW_ALG_MAX`, and `rapids_seq_g` drives it to every `scheduler_g` as `seq_alg_max`. The check is hardware, the same in both places it is made: `scheduler_core` latches a descriptor whose selector is above it (or non-zero without TYPE=EXT) as invalid, and

scheduler_g folds the same test into HEAD_SUM's valid bit, so the firmware declines such a head and the chain hook (HEAD_SUM = 13) is untouched. A selector is refused from reset, before the firmware runs an instruction.

Table 5.3.7: Next-address algorithms in the image

Selector	Set-up	Top-up
0	gen_init: positions, step and masks as stream_run_addr_gen	refill (masked) or refill_fast (no wrap)
1, Z-order	gen_init_z: unit = a beat or a run of inner_count beats; S0 = stride_0 or inner_count << OFF_W; position k = 1	refill_z

refill_z keeps x x S0 and y x stride_1 as running sums, like the masked path, and steps the position in Morton order without a multiplier: k -> k + 1 clears k's trailing ones and sets the bit above them, and bit 2i is x's bit i, bit 2i + 1 is y's, so each cleared bit subtracts the stride shifted by i and the set bit adds it – about two bits a step on average. The generator block's mode word tells refill which path to take (1 fast, 0 masked, -1 Z-order), so selector 0 costs one instruction more per masked top-up and nothing on the fast path.

37.2.7 Building the Image

bin/gen_rapids_g_fw.py runs riscv-none-elf-as / ld (text at 0) / objcopy -O binary and writes FW_WORDS and FW_IMAGE[] into rtl/macro_g/generated/fw/sched_g_fw.svh, which rapids_seq_g includes, so neither simulation nor synthesis needs a file path. --check fails when the include is stale. Never edit the include.

37.2.8 Loading Firmware Live

With FW_WRITABLE = 1 (rapids TASK-038) the instruction memory is the same 1024 x 32 block RAM with its second port used for a load: write and read back. It powers up holding the generated image, so a writable build that is never loaded runs exactly as the ROM build. The host reaches it through four registers in each half, at the same offsets in the 8- and 16-channel maps:

Table 5.3.8: Firmware load registers, per half

Offset	Register	Fields
0x3A0	FW_CTRL	HOLD[0] RW: hold every hart of the half in reset, and allow FW_DATA writes; WRITABLE[1] RO:

Offset	Register	Fields
		1 on a FW_WRITABLE build; ALG_MAX[15:8] RW: the loaded image's highest next-address selector
0x3A4	FW_ADDR	ADDR[15:0] RW: the word index; advances by one after each FW_DATA write and each FW_RDATA read
0x3A8	FW_DATA	DATA[31:0]: a write stores the word at FW_ADDR, only while HOLD = 1
0x3AC	FW_RDATA	DATA[31:0] RO: the word at FW_ADDR (0 on a ROM build)

A load is: FW_CTRL HOLD = 1 with the image's ALG_MAX; FW_ADDR = 0; one FW_DATA write per word; FW_ADDR = 0 and one FW_RDATA read per word to compare; FW_CTRL HOLD = 0. Releasing the hold restarts the harts from address 0 on the new image, like `seq_restart`, so hold only while every channel of the half is idle. While HOLD = 1 the next-address cap every `scheduler_g` applies (`seq_alg_max`) follows FW_CTRL.ALG_MAX and keeps that value after the release, so a loaded image carries its own selector set. On a ROM build and on RAPIDS the registers exist, WRITABLE reads 0 and the rest has no effect.

The top base's glue (`rapids_g_top_fw.svh`) turns each register access into one store or one advance. Behind `peakrdl_to_cmdrsp`, `swmod` and `swacc` are a two-cycle level, not a pulse (the handbook's generated-RTL note); the glue takes each access on its rising edge, one flop late so the field storage has the written value. The first version counted every cycle: it stored each word twice and advanced FW_ADDR by two, which a readback walking the same counter could not see below word 512 – the self-test's direct address probe now catches it.

Host side (`flows-rapids/host/run_characterization.py`): `--fw <sched.S>` assembles the source with the same generator that writes the ROM include, loads both halves, reads every word back and releases, and records the image's SHA-256 in the results (on a writable build the bitstream no longer names the firmware). `--fw-selftest` first proves the path: a FW_DATA write without HOLD is refused; a one-word image of 0x00000000, an illegal instruction, halts every hart, so every channel goes to CH_ERROR (the harts run what was loaded); direct probes across the memory read the right word and advance by one; loading the real image and a channel reset bring every channel back to CH_IDLE. On a ROM build it checks the opposite, that the same

writes change nothing. A full load is about 2,600 UART operations, too many for a 100 ms simulation beside a measurement, so a simulation can instead build the candidate in (`TEST_FW_IMAGE`, rendered by `gen_rapids_g_fw.py --src/--out`) and check it with `--fw-expect`, which reads back the first and last eight words; the memory holds the same image at GO either way.

The flow is: iterate a candidate live on a writable bitstream, check it in simulation built in, compare simulation and board, and once it is final commit it as `sched_g.S`, so the ROM build carries it. The two-slot poll above was developed this way, two candidates loaded on one bitstream; the committed image is byte-identical (SHA-256 `00b7003b...`) to the one measured on the board.

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38 The Generated rapids-g Tree

Generator: `bin/gen_rapids_g.py` **Status:** Implemented (rapids TASK-034) **Last Updated:** 2026-10-21

Above `scheduler_g` the rapids-g RTL is RAPIDS's, so it is derived rather than copied:

Table 5.4.1: Generated files

Generated	From	Edits
`rtl/macro_g/generated/scheduler_group_array_g.sv`	`rtl/macro/scheduler_group_array.sv`	rename; `scheduler` becomes `scheduler_g`; the `seq_*` ports wired; `scheduler_group_array_g_seq.svh` (one `rapids_seq_g`) included in front of the scheduler loop; `$fatal` unless `SHARED_ENGINES` = 1 and `NUM_CHANNELS` is 4, 8 or 16
`rtl/macro_g/generated/rapids_src_g.sv`, `rapids_snk_g.sv`	`rtl/macro/rapids_src.sv`, `rapids_snk.sv`	rename; the array becomes `scheduler_group_array_g`
`rtl/macro_g/generated/rapids_core_g.sv`	`rtl/macro/rapids_core.sv`	rename; the halves become `_g`
`rtl/top_g/generated/rapids_g_top_base.sv`	`rtl/top/rapids_top_base.sv`	rename; the core becomes `rapids_core_g`; `apb4_slave`'s cmd/rsp outputs renamed `host_*`; `rapids_g_top_reg_arb.svh` (host + both gatlings into the register command queue) included after it
`rtl/top_g/generated/rapids_g_top.sv`,	the -g base	the thin-top derivation of `bin/gen_plane_tops.py`

Generated	From	Edits
`rapids_g_cxs_top.sv`		
`rtl/top_g/generated/rapids_g_16ch_top.sv`, `rapids_g_16ch_cxs_top.sv`	the -g base	the same, `NUM_CHANNELS` = 16; compiled with `rtl/filelists/top_g/rapids_g_16ch{,_cxs}_top.f`, which select the 16-channel register map (rapids TASK-036)

The register-mailbox master (`seq_reg_*`) is added to every level's port list after `rst_n` and connected at every instance (array -> half -> core as `src_` / `snk_` -> top base), so the RAPIDS files need no knowledge of it.

Every edit matches its anchor exactly once or the generator fails, so a RAPIDS change that moves an anchor is a failed run, not a wrong file. `python3 bin/gen_rapids_g.py --check` exits 1 when any generated file is stale. The filelists are `rtl/filelists/fub_g/`, `macro_g/` and `top_g/`; they pull gatling through `$GATLING_ROOT/rtl/filelists/top/gatling_core.f`, and the 8-channel register map as `top/rapids_top_base.f` does.

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39 rapids-g Verification and Measurements

Status: Implemented (rapids TASK-034) **Last Updated:** 2026-10-22

39.1 Verification

Table 5.5.1: rapids-g suites

Suite	What it shows
`dv/tests/fub_g/test_rapids_g_equiv.py`	RAPIDS's `scheduler` and `scheduler_g` + firmware side by side (`rapids_g_equiv_tb_top`), 4, 8 and 16 channels, 256 and 512 bits, one, two and four channels per hart (`NUM_HARTS`: 8 on 4, 16 on 16, 8 and 4), same descriptors, engine models with different latencies per side. Every engine command, packet record, MonBus packet and state transition must match in order. The address sequences are also checked against an independent golden model (`RapidsGEquivTB.golden_cmds`: the byte-to-beat ceiling and `dma_address_gen`'s formula). Cells: linear, ext, reset_ext, backpressure, error_reset, reset_any (a channel reset in every state), hart_fault (every hart halted by an illegal word patched over its SUMMARY poll, found by encoding, faults its channels and restarts), alg (next-address selectors: unrunnable ones invalid on both sides, selector 1 on gatling against the golden model's Z-order and refused by RAPIDS), perf.
`dv/tests/fub_g/test_scheduler_g_edges.py`	`scheduler_g` alone with the testbench as its firmware: every command and flag in every state, 57 checks; legal edges taken, the rest ignored and counted, CH_ERROR left only by a reset; the automatic COMPLETE -> IDLE -> FETCH -> XFER of an armed

Suite	What it shows
	descriptor with a staged head, and the stop at a declined head.
`dv/tests/top_g/ test_rapids_g_regpath.py`	Both gatlings run a test program that reads and writes their own half's registers (`SCHED_TIMEOUT_CYCLES`, `SCHED_TIMEOUT_LIMIT`, `CH_SCRATCH`) through the mailbox while the host hammers the same command queue: every hart reads its own half's value, each gatling's writes land in its own half only, no host read is wrong.
`dv/tests/macro_g/`	Every RAPIDS `scheduler_group_array` test on `scheduler_group_array_g`, at 4, 8 and 16 channels.
`dv/tests/top_g/`	Every RAPIDS `rapids_top` and `rapids_cxs_top` test, unchanged, on `rapids_g_top` and `rapids_g_cxs_top`; the 16-channel cells on `rapids_g_16ch_top` and `rapids_g_16ch_cxs_top`.

The test files are RAPIDS's own: the -g runners load the RAPIDS cocotb tests by module name (`macro_g`) or re-host the RAPIDS wrappers (`top_g/rapids_g_rehost.py`), so rapids-g is held to exactly RAPIDS's checks. They caught a firmware race on the way: a head that arrived between the tight loop's HEAD_SUM and SUMMARY reads went down `service`'s slower path, and `tile_source_bad_ptype` saw its channel error after the test's wait on three of eight seed bases; the loop now re-reads HEAD_SUM before `service`.

Mutation (re-run on the executive image, rapids TASK-037, and the selector, TASK-035; each from clean against the equivalence bench, or the edge test where it owns the logic):

Table 5.5.2: Mutation, executive image

Mutant	Where	Result
JUDGE never chains	firmware	caught at two slots per hart (8 on 4, 16 on 8); with one slot a chained head always takes the chain hook, so JUDGE

Mutant	Where	Result
		never sees one
route out of an unarmed COMPLETE	firmware	survived: unreachable (below); the code is now `fw_fault`
stride sign extension	firmware	caught
wrap mask dropped	firmware	caught
XFER_DATA's `ptab` entry returns	firmware	caught
GEN_OK not written	firmware	caught
Z-order: y subtracts	firmware	caught (`alg`)
selector ignored (always selector 0)	firmware	caught (`alg`)
edge-table hole: COMPLETE always legal	`scheduler_g`	caught by the edge test; the bench cannot reach it, because the firmware never commands COMPLETE
fault interlock removed	`scheduler_g`	caught
HEAD_SUM ignores the selector	`scheduler_g`	caught (`alg`)
the latch ignores the selector	`scheduler_core`	caught (`alg`)

The first run left three survivors, and each said something different. JUDGE's chain decision is live only when a hart serves two slots, a configuration the bench did not build until it gained its `NUM_HARTS` axis. The edge-table hole belongs to the edge test. The unarmed-COMPLETE route, with the rest of the old hand path, cannot run at all: every valid head is staged armed, and the firmware takes a head itself only when it declined it, which the latch also calls invalid (Section 5.3). That code became the `fw_fault` row rather than a handler no test can reach.

Two RAPIDS bugs were found on the way and fixed in RAPIDS: rapids BUG-016 (a held CH_FETCH_DESC sent DESC_START every cycle) and rapids BUG-017 (stale EXT run bases after a channel reset; also in RAPIDS Beats and STREAM).

39.2 Performance on the Board

rapids-8ch and rapids-g-8ch ran the same 45-point `gcompare` profile on the Genesys 2 harness (`projects/fpga-systems/Genesys2/dma-ip/rapids/flows-rapids`): 8 channels, DATA_WIDTH 256, 100 MHz, peak **3,200 MB/s per direction**. A figure is the bytes moved over the hardware timer's cycles from the first request to the last write, descriptor fetch and launch included. Every point is golden-checked and passes on both designs. The point names read `<group>_ch<N>_b<B>[_d<D>][_il]`: `beat` is one descriptor per channel and `chain` a linked chain, `ch<N>` the active channels, `b<B>` beats (32 bytes) per descriptor, `d<D>` descriptors per channel (absent: one), and `il` an interleaved sink. Without `il` the sink stream feeds the channels one after another, so only one channel holds data at a time; the source reads memory for every channel at once in every point.

Records: `reports/board/flow_results/byte_perf_gcompare_fsm_il_rc.json` (rapids-8ch) and `byte_perf_gcompare_gfin_rc.json` (rapids-g-8ch).

The simulation of the board build (`make sim-gcompare`, run in seven chunks) matches the board cycle for cycle: 85 of 85 windows on each design.

The executive image with the next-address selector (rapids TASK-037 and TASK-035; `byte_perf_gcompare_gexe_rc.json`, 2026-10-23) passes all 45 points on the board, matches its simulation on 85 of 85 windows, and is identical to the 2026-10-22 image (gfin) on all 85: every gcompare point is a linear descriptor whose head the poll stages itself, the path the rewrite did not lengthen, so the board numbers in Table 5.5.3 stand unchanged. That build takes 111,738 LUTs, 65,246 registers and 58 block RAM tiles and closes 100 MHz at +0.061 ns. rapids-8ch's simulation, on the shared datapath with the selector check, still matches its own board run on 85 of 85. Matching needed one change to the host program. The shared descriptor engine's round-robin pointer (`descriptor_engine_shared`, `r_rr_ptr`) is reset only by `aresetn`, so it carries from one point to the next. A chunk that starts cold in simulation served the channels in a different order from the board, where the previous point had run, and took 16 clocks longer (8 on rapids-8ch). Each chunk after the first now runs the previous chunk's last point first, unrecorded (`byte_perf.chunk_warmup`). The RTL is unchanged.

Table 5.5.3: Genesys 2 gcompare, MB/s (peak 3,200; – = not measured: the source side has no interleaved variant)

Point	rapids-8ch sink	rapids-g-8ch sink	rapids-8ch source	rapids-g-8ch source
beat_ch1_b1	54	14	50	14
beat_ch1_b4	197	56	191	56
beat_ch1_b16	624	210	648	212
beat_ch1_b64	1,575	701	1,613	709
beat_ch1_b256	2,544	1,693	2,568	1,703
beat_ch1_b1024	3,006	2,617	3,015	2,624

Point	rapids-8ch sink	rapids-g-8ch sink	rapids-8ch source	rapids-g-8ch source
beat_ch1_b4096	3,149	3,031	3,152	3,033
beat_ch1_b32768	3,194	3,178	3,194	3,178
beat_ch2_b1	77	29	74	28
beat_ch2_b4	281	111	281	110
beat_ch2_b16	853	394	890	398
beat_ch2_b64	1,896	1,151	1,941	1,160
beat_ch2_b256	2,731	2,214	2,754	2,223
beat_ch2_b1024	3,068	2,879	3,075	2,883
beat_ch2_b4096	3,166	3,113	3,168	3,114
beat_ch2_b32768	3,196	3,189	3,196	3,189
beat_ch4_b1	99	45	100	45
beat_ch4_b4	363	174	368	180
beat_ch4_b16	1,056	589	1,095	621
beat_ch4_b64	2,122	1,517	2,161	1,569
beat_ch4_b256	2,840	2,505	2,857	2,540
beat_ch4_b1024	3,102	2,993	3,107	3,005
beat_ch4_b4096	3,175	3,145	3,176	3,149
beat_ch4_b32768	3,197	3,193	3,197	3,194
beat_ch8_b1	116	66	121	66
beat_ch8_b4	425	252	436	250
beat_ch8_b16	1,198	806	1,237	811
beat_ch8_b64	2,257	1,837	2,291	1,843
beat_ch8_b256	2,897	2,699	2,911	2,703
beat_ch8_b1024	3,119	3,058	3,123	3,059
beat_ch8_b4096	3,179	3,163	3,180	3,164
beat_ch8_b32768	3,197	3,195	3,198	3,195
chain_ch1_b1_d16	149	73	278	73
chain_ch1_b64_d16	2,214	1,875	3,015	1,788
chain_ch1_b1024_d16	3,113	3,103	3,188	3,177

Point	rapids-8ch sink	rapids-g-8ch sink	rapids-8ch source	rapids-g-8ch source
chain_ch1_b8192_d4	3,186	3,180	3,194	3,188
chain_ch8_b1_d16	164	74	1,177	419
chain_ch8_b64_d16	2,276	1,975	3,123	3,059
chain_ch8_b1024_d16	3,121	3,117	3,195	3,191
chain_ch8_b8192_d4	3,190	3,188	3,198	3,195
beat_ch8_b16_il	1,048	726	–	–
beat_ch8_b64_il	2,098	1,717	–	–
chain_ch8_b1_d16_il	822	418	–	–
chain_ch8_b64_d16_il	2,709	2,662	–	–
chain_ch8_b1024_d16_il	3,166	3,162	–	–

The firmware's cost per descriptor sets the gap. Its history on the hardest point, a one-channel chain of 64-beat sink descriptors:

Table 5.5.4: Firmware cost per chained descriptor (the 16-channel rows' clocks are the point's measured cycles / 16)

Firmware	chain_ch1_b64_d16 sink	Clocks per descriptor
first 8-hart image (one hart per channel)	1,118 MB/s	about 288
one-slot tight chain loop, HEAD_SUM re-read before `service`	1,833 MB/s (board)	about 90
seven-instruction idle pass (this image)	1,875 MB/s	about 90
rapids-8ch, for reference	2,214 MB/s	–
two slots per hart (16 channels), SUMMARY then JUDGE	1,122 MB/s (board)	184
two slots, HEAD_SUM checks ahead of SUMMARY	1,546 MB/s (board)	134
two slots, SUMMARY then the chain hook per slot (this image)	2,008 MB/s (board)	103
rapids-16ch, for reference	2,232 MB/s	93

Before STAGE_ARM (Section 5.2) the same point cost 672 clocks a descriptor. The tight loop (Section 5.3) follows a chained head in four instructions; the shorter idle pass recovered a 1-2% dip on the eight-channel single-descriptor sink points (`beat_ch8_b4` to `b256`).

39.3 Scheduler Limit

Measured with `cocotb_test_perf`: engines that take every burst in one cycle and never stall, so the scheduler is the only limit; 8 channels and 8 harts, DATA_WIDTH 256, cycles at 100 MHz. The bench runs both directions on every channel, twice the load of one RAPIDS half (each half moves one direction).

Table 5.5.5: Cycles to finish, schedulers alone

Scenario	rapids-8ch	rapids-g-8ch, 10-22	rapids-g-8ch, executive	executive / rapids-8ch
16 one-beat descriptors, one channel	131	2,371	2,371	18x
16 one-beat descriptors on each channel	131	2,316	2,316	18x
16 64-beat descriptors on each channel	1,255	2,389	2,389	1.9x
4 64 KB descriptors on each channel	9,248	9,529	9,412	1.02x
2 1 MB descriptors on each channel	84,277	84,420	84,443 (*)	1.002x
EXT, runs of 4, 64 beats, 4 per channel	407	40,433	39,201	96x
EXT, per beat, 64 beats, 4 per channel	1,043	135,848	130,086	125x
EXT, runs of 256, 1 MB, 1 per channel	29,574	72,329	70,641	2.4x

The executive image (rapids TASK-037) is 2-4% faster on every EXT row and unchanged on the linear ones: an XFER_DATA step costs four instructions of dispatch and returns, where `service` re-read STATUS and re-judged the head before returning. (*) The scenarios run in sequence in one simulation, so when the 64 KB row finishes sooner the 1 MB row starts at another phase of the barrel; run alone from the same cycle, both images take 84,420 (and 9,454 for 64 KB).

With one hart per channel the channels no longer share a hart, so one busy channel and eight busy channels cost the same (2,371 and 2,316). On 4 harts the same rows were 7,215 and 14,139. Pre-validation hides the firmware behind the descriptor in flight whenever the transfer outlasts it: two 1 MB descriptors per channel finish 0.17% behind rapids-8ch, and what remains is each channel's first descriptor. Descriptors shorter than the firmware's work still pay it in full. EXT descriptors are limited by base generation: a base costs the firmware about 15-35 instructions,

each NUM_HARTS clocks, against one cycle in `stream_run_addr_gen`, so with both directions on every channel even 256-beat runs cannot be fed at the engines' rate. Custom gatling instructions for DMA (`gatling_dma`) are the lever for that case.

39.3.1 Sixteen Channels

rapids-g-16ch (rapids TASK-036): the same bench at 16 channels, DATA_WIDTH 256, with each way of spreading the 16 channels over one gatling's harts, measured on the committed image (the two-slot poll with the chain hook, Section 5.3).

Table 5.5.6: Cycles to finish at 16 channels

Scenario	rapids-16ch	16 harts x 1	8 harts x 2 (default)	4 harts x 4
16 one-beat descriptors, one channel	131	4,731	3,131	1,619
16 one-beat descriptors on each channel	131	4,620	5,383	5,627
16 64-beat descriptors on each channel	1,255	4,700	5,469	5,704
4 64 KB descriptors on each channel	9,248	9,609	9,585	9,579
2 1 MB descriptors on each channel	84,277	84,610	84,612	84,612
EXT, runs of 4, 64 beats, 4 per channel	407	78,362	77,998	77,714
EXT, per beat, 64 beats, 4 per channel	1,043	260,192	257,831	256,406
EXT, runs of 256, 1 MB, 1 per channel	29,574	140,577	139,688	139,184

Sixteen harts make every instruction cost a hart 16 clocks, which a lone busy channel pays in full (4,731); four harts are best for one channel and on EXT, but lose 22% to sixteen when all sixteen channels run one- or 64-beat descriptors, each hart walking four slots. Eight harts with two slots each are 16% behind sixteen there, within 0.6% of the best on EXT, 1.5x faster than sixteen harts for a single channel, and keep half the hart state; `rapids_seq_g` defaults to them at 16 channels. Against the image this table first carried (the two-slot poll before the chain hook), the eight-hart column is about 10% slower on the one- and 64-beat rows and 1.5% on EXT: this bench's descriptors are independent (`next_ptr` = 0), so none takes the hook, and the SUMMARY-first pass costs a few instructions per head. What the hook buys is on chains, which this bench does not run:

Tables 5.5.9 and 5.5.10. The 1 MB rows are within 0.4% of rapids-16ch, the 64 KB rows within 3.6%.

39.4 Sixteen Channels on the Genesys 2

rapids-g-16ch and rapids-16ch in the full Genesys 2 harness (rapids TASK-036), `make bitstream-perf CHANNELS=16 ROW_COL=1`, 2026-10-23. The first rapids-g-16ch build placed at 198,555 LUTs (97.4%, every slice) and did not route: the harness's two 2048-entry descriptor RAMs had fallen to distributed RAM (25.6k LUTs each), because `sdpram_core`'s byte-enabled write port will not map to block RAM. The harness writes whole descriptors, so it now builds them with `USE_WSTRB` = 0, as the STREAM harness does; the 8-channel bitstreams take the same saving, not yet re-measured.

Table 5.5.8: 16-channel bitstreams

Bitstream	LUTs	WNS at 100 MHz	Board == simulation
rapids-16ch	135,389 (66.4%)	+0.192 ns	109/109 windows identical
rapids-g-16ch, ROM	151,472 (74.3%)	+0.148 ns	109/109 windows identical
rapids-g-16ch, `FW_WRITABLE`	152,813 (75.0%)	+0.034 ns	109/109 identical, the committed image loaded live

The 16-channel gcompare profile is 57 points: the 8-channel profile's single descriptors and chains, plus 16-channel single descriptors, chains at 1, 8 and 16 channels in size-major order, and the interleaved sink at 16 (`byte_perf.build_points`); in simulation it runs as 18 chunks to keep each under the 100 ms cap. "ROM" is the image before the two-slot chain hook; "v2" is the committed image (Section 5.3), loaded live on the writable bitstream with `--fw`.

Table 5.5.9: Genesys 2 gcompare at 16 channels, MB/s (peak 3,200; – = no source variant)

Point	rapids-16ch sink	rapids-g ROM sink	rapids-g v2 sink	rapids-16ch source	rapids-g ROM source	rapids-g v2 source
beat_ch1_b1	1,600	1,600	1,600	356	356	356
beat_ch1_b256	2,643	1,736	1,625	3,103	3,103	3,103

Point	rapids-16ch sink	rapids-g ROM sink	rapids-g v2 sink	rapids-16ch source	rapids-g ROM source	rapids-g v2 source
beat_ch8_b16	1,241	813	788	3,012	3,012	3,012
beat_ch8_b256	2,913	2,704	2,686	3,188	3,188	3,188
beat_ch16_b1	133	70	67	2,048	247	230
beat_ch16_b16	1,309	974	938	3,103	2,626	2,498
beat_ch16_b256	2,935	2,820	2,810	3,194	3,194	3,194
beat_ch16_b32768	3,198	3,197	3,197	3,200	3,200	3,200
chain_ch1_b1_d16	178	19	36	397	19	36
chain_ch1_b64_d16	2,232	1,122	2,008	3,175	1,208	2,190
chain_ch8_b1_d16	169	21	38	2,677	154	284
chain_ch8_b64_d16	2,278	1,236	2,152	3,197	3,197	3,197
chain_ch16_b1_d16	168	20	36	3,091	172	346
chain_ch16_b	2,280	1,232	2,084	3,198	3,198	3,198

Point	rapids-16ch sink	rapids-g ROM sink	rapids-g v2 sink	rapids-16ch source	rapids-g ROM source	rapids-g v2 source
64_d16						
chain_ch16_b1024_d16	3,121	3,119	3,120	3,200	3,200	3,200
chain_ch16_b8192_d4	3,190	3,189	3,190	3,200	3,200	3,200
beat_ch16_b16_il	1,309	839	813	–	–	–
beat_ch16_b64_il	2,351	1,879	1,845	–	–	–
chain_ch16_b1_d16_il	948	155	292	–	–	–
chain_ch16_b64_d16_il	2,772	2,702	2,738	–	–	–
chain_ch16_b1024_d16_il	3,172	3,166	3,169	–	–	–

The two-slot chain hook roughly doubles the short chains (19 to 36 MB/s on one-beat descriptors, 1,122 to 2,008 on 64-beat ones) and costs single descriptors one to three instructions: 1-7% on the points where the firmware is the limit, nothing from 256 beats per channel up.

39.4.1 Where rapids-g Catches Up

The sweep that sizes the gap, in simulation (`make sim-gscale CHANNELS=8|16`; the same simulation matches the board on all 109 gcompare windows): single descriptors and 16-descriptor interleaved-sink chains on every channel of the build, in 2x size steps, rows in beats per descriptor (32 bytes each). Each cell is rapids cycles / rapids-g cycles (1.00 = parity). Columns: `chain` is the 16-descriptor interleaved-sink chain (8 channels run one slot per hart; at 16, ROM is the image before the chain hook and v2 the committed one); `sink`, `source` and `il sink` are a single descriptor per channel, plain or with the sink streams interleaved.

Table 5.5.10: rapids / rapids-g cycles, every channel busy

Beats	chain 8ch	chain 16ch ROM	chain 16ch v2	sink 8ch	sink 16ch v2	source 8ch	source 16ch v2	il sink 8ch	il sink 16ch v2
1	0.52	0.17	0.32	0.57	0.53	0.55	0.49	0.57	0.53
2	0.49	0.22	0.41	0.57	0.53	0.56	0.51	0.58	0.53
4	0.60	0.41	0.68	0.59	0.56	0.57	0.54	0.61	0.56
8	0.90	0.76	0.93	0.63	0.62	0.61	0.60	0.66	0.62
16	0.95	0.93	0.97	0.67	0.72	0.66	0.70	0.71	0.67
32	0.97	0.96	0.98	0.74	0.82	0.73	0.82	0.76	0.73
64	0.98	0.97	0.99	0.81	0.88	0.80	0.88	0.83	0.80
128	0.99	0.99	0.99	0.88	0.92	0.88	0.92	0.89	0.87
256	0.99	0.99	1.00	0.93	0.96	0.93	0.96	0.93	0.92
512	1.00	1.00	1.00	0.96	0.98	0.96	0.98	0.96	0.96
1024	1.00	1.00	1.00	0.98	0.99	0.98	0.99	0.98	0.98
2048	–	–	–	0.99	0.99	0.99	0.99	0.99	0.99
4096	–	–	–	0.99	1.00	0.99	1.00	1.00	0.99

How much a channel must move per descriptor before rapids-g is within 10%, 5% and 1% of rapids, from the sweep above (DATA_WIDTH 256, so 32 bytes a beat; the smallest size from which every larger size in the sweep stays inside the bound). A chain row is bytes per descriptor in a 16-descriptor chain, so the channel moves sixteen times that in the run:

Table 5.5.11: Bytes per descriptor for rapids-g to be within 10%, 5% and 1% of rapids

Shape	Channels	Within 10%	Within 5%	Within 1%
single descriptor, sink	8	8 KB	16 KB	64 KB
single descriptor, sink	16	4 KB	8 KB	64 KB

Shape	Channels	Within 10%	Within 5%	Within 1%
single descriptor, source	8	8 KB	16 KB	128 KB
single descriptor, source	16	4 KB	8 KB	64 KB
single descriptor, interleaved sink	8	8 KB	16 KB	64 KB
single descriptor, interleaved sink	16	8 KB	16 KB	128 KB
16-descriptor chain, interleaved sink	8	512 B	512 B	4 KB
16-descriptor chain, interleaved sink	16	256 B	512 B	4 KB

Chains close the gap at 512 bytes (the 8-channel chain reads 0.90 at 256 B in Table 5.5.10 but is 0.899) and are within 1% at 4 KB on both builds; a single descriptor per channel needs 8-16 KB to be within 5% and 64-128 KB to be within 1%, because its whole firmware cost (fetch, judge, stage and complete, with no next head to overlap) is paid once per descriptor against a data phase that only grows with the size. With one gatling per half serving 16 channels the expectation was that chains need twice the size to catch the FSM, since each hart serves two channels: on the ROM image they did (0.95 at 16 beats on 8 channels, at 32 on 16). With the chain hook in the two-slot poll they catch up at 16 on both, and are ahead of the 8-channel build from 4 beats; one- and two-beat chains remain behind it (0.32-0.41 against 0.49-0.52), the cost of walking two slots per pass. Single descriptors are at most 0.04 behind the 8-channel build at any size (and ahead of it on the plain sink from 16 beats, where rapids-16ch slows too): the firmware's work is per descriptor, so more channels do not move it.

39.5 Area and Timing

`rapids_core` and `rapids_core_g`, out of context on the Genesys 2 part (xc7k325t-2), DATA_WIDTH 256, `SHARED_ENGINES` 1, `USE_AXI_MONITORS` 0, `USE_ROW_COL_MAJOR_ADDRESSING` 1, default harts (8 at 8 channels, 8 at 16), placed and routed at 100 MHz, measured 2026-10-23 on the executive image with the next-address selector:

Table 5.5.7: Out-of-context results

	rapids-8ch	rapids-g-8ch	rapids-16ch	rapids-g-16ch
LUTs	69,184	77,975 (+12.7%)	126,820	141,067 (+11.2%)
Registers	56,105	58,837	101,808	104,738
Block RAM tiles	64	70	128	134
DSPs	64	0	128	0
WNS at 100 MHz	+0.684 ns	+0.533 ns	+0.079 ns	+0.050 ns

The selector check costs RAPIDS 411 LUTs at 8 channels (68,773 on 2026-10-22). At 16 channels each `scheduler_g` is about 1,630 LUTs against 1,430 for `scheduler`: its window read mux and base FIFOs in fabric outweigh the run-base generators, whose multipliers are DSPs on the RAPIDS side; over 32 schedulers that is most of the difference. Both 16-channel builds close 100 MHz, with the thinner margin (+0.050 ns) the cost of the larger part fill. The register arbiter (`rapids_g_reg_arb`) is in the top, outside `rapids_core_g`, and not in these numbers. The 2026-10-22 8-channel figures below are the first image's. Per half, `rapids_seq_g` is about 1.95k LUTs (gatling about 1.39k of it); the sink half is 38,693 LUTs against 34,943 for `rapids_snk`, and the source half 37,180 against 32,746 for `rapids_src`. The 4-hart build measured 76,350 LUTs, 56,645 registers, 69 block RAM tiles and +0.249 ns: eight harts cost about 600 LUTs and 2k registers, and gatling's own path is no longer the worst. The worst path of `rapids_core_g` is now in the source half, from a scheduler's source address (`gen_schedulers[3].u_scheduler/u_core/r_src_addr_reg`) to the AXI read engine's outstanding counter enable, 20 levels. The 64 DSPs of `rapids_core` are the run-base generators' index-by-stride multipliers, which the firmware replaces with additions.

In the full Genesys 2 harness of Table 5.5.3, rapids-g-8ch takes 111,738 LUTs (54.8%), 65,246 registers, 58 block RAM tiles and no DSPs and closes 100 MHz at +0.061 ns on the executive image (110,662 LUTs at +0.072 ns on 2026-10-22); rapids-8ch takes 100,007 LUTs at +0.357 ns (2026-10-22 build).

40 RAPIDS Beats Architecture MAS

Version: 0.14 **Date:** 2026-10-20 **Purpose:** Module Architecture Specification for RAPIDS Phase 1 “Beats” Architecture

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40.1 Document Organization

Note: All chapters linked below for automated document generation.

40.1.1 Front Matter

- Document Information

40.1.2 Chapter 1: Overview

- Architecture Overview
- Top-Level Port List
- Clocks and Reset

40.1.3 Chapter 2: FUB (Functional Unit Blocks)

Control Path: - Scheduler - Descriptor Engine

AXI Engines: - AXI Read Engine - AXI Write Engine

Flow Control: - Beats Alloc Control - Beats Drain Control - Beats Latency Bridge

Control Engines: - Control-Read Engine - Control-Write Engine

40.1.4 Chapter 3: Macro (Integration Blocks)

Scheduler Integration: - Beats Scheduler Group - Beats Scheduler Group Array

Sink Data Path (Network to Memory): - Sink Data Path - Sink Data Path AXIS - Sink SRAM Controller

Source Data Path (Memory to Network): - Source Data Path - Source Data Path AXIS - Source SRAM Controller

Top-Level Integration: - RAPIDS Core Beats - RAPIDS Registers - RAPIDS Config Block - RAPIDS Beats Top - Network Planes

40.1.5 Chapter 4: Interfaces

- Interface Overview
- AXI4 Interface Specification
- AXIS Interface Specification
- MonBus Interface Specification
- CXS Interface Specification

40.2 Quick Reference

40.2.1 FUB Modules (fub_beats/)

FUB Module Summary

Module	File	Purpose	Status
scheduler_beats	`scheduler_beats.sv`	Transfer coordinator, descriptor processing	Implemented
descriptor_engine_beats	`descriptor_engine_beats.sv`	Descriptor fetch/parse (256-bit)	Implemented
axi_read_engine_beats	`axi_read_engine_beats.sv`	AXI read master, streaming pipeline	Implemented
axi_write_engine_beats	`axi_write_engine_beats.sv`	AXI write master, streaming pipeline	Implemented
alloc_ctrl_beats	`alloc_ctrl_beats.sv`	Space allocation tracking (virtual FIFO); kept and tested, no longer in the SRAM path since `bdf4e0dff`	Implemented
drain_ctrl_beats	`drain_ctrl_beats.sv`	Data availability tracking (virtual FIFO); kept and tested, no longer in the SRAM path	Implemented
latency_bridge_beats	`latency_bridge_beats.sv`	Registered-FIFO to valid/ready skid bridge; kept and tested, no longer in the SRAM path	Implemented
ctrlrd_engine	`ctrlrd_engine.sv`	Control-read consumer gate (poll-until-match, retry budget)	Implemented
ctrlwr_engine	`ctrlwr_engine.sv`	Control-write producer doorbell	Implemented

Module	File	Purpose	Status
		(single-beat write)	

40.2.2 Macro Modules (macro_beats/)

Macro Module Summary

Module	File	Purpose	Status
scheduler_group_beats	`scheduler_group_beats.sv`	Scheduler + Descriptor Engine wrapper	Implemented
scheduler_group_array_beats	`scheduler_group_array_beats.sv`	8-channel scheduler array with arbitration	Implemented
snk_data_path_beats	`snk_data_path_beats.sv`	Network-to-memory path integration	Implemented
snk_data_path_axis_beats	`snk_data_path_axis_beats.sv`	AXIS variant of sink path	Implemented
src_data_path_beats	`src_data_path_beats.sv`	Memory-to-network path integration	Implemented
src_data_path_axis_beats	`src_data_path_axis_beats.sv`	AXIS variant of source path	Implemented
snk_sram_controller_beats	`snk_sram_controller_beats.sv`	Sink SRAM: naming wrapper around STREAM's `sram_controller` (one instance, per-channel FIFOs inside it)	Implemented
src_sram_controller_beats	`src_sram_controller_beats.sv`	Source SRAM: naming wrapper around STREAM's `sram_controller`	Implemented
rapids_src_beats / rapids_snk_beats	`rapids_src_beats.sv`, `rapids_snk_beats.sv`	One half each: scheduler array + data path + monbus merge	Implemented

Module	File	Purpose	Status
rapids_core_beats	`rapids_core_beats.sv`	Both halves + the core monbus arbiter	Implemented
rapids_regs	`regs/generated/rtl/rapids_regs.sv` (generated from `rtl/macro_beats/rapids_regs.rdl`)	PeakRDL register block (base + MON regfile @ 0x1000)	Implemented
rapids_config_block	`rapids_config_block.sv`	Maps register `hwif_out` to core/monitor `cfg_*`	Implemented
rapids_beats_top	`rapids_beats_top.sv`	Top-level: APB slave, kickoff, monitors, MonBus AXI-Lite group	Implemented

40.3 Architecture Comparison: RAPIDS vs STREAM

The “beats” architecture is a Phase 1 implementation of RAPIDS that shares concepts with STREAM but targets network-to-memory (and vice versa) transfers rather than memory-to-memory.

RAPIDS Beats vs STREAM Comparison

Feature	STREAM	RAPIDS Beats
Primary Use	Memory-to-memory DMA	Network-to-memory accelerator
Data Paths	Single bidirectional	Separate sink/source paths
Network Interface	None	AXIS master/slave
SRAM Buffering	Shared buffer	Separate sink/source buffers
Descriptor Format	256-bit	256-bit (compatible)
Channel Count	8	8
Flow Control	alloc_ctrl/ drain_ctrl	STREAM `stream_alloc_ctrl`/`stream_drain_ctrl` inside the shared `sram_controller` (`alloc_ctrl_beats`/`drain_ctrl_beats` kept, out of the path)

40.4 Clock and Reset Summary

40.4.1 Clock Domains

Clock Domains

Clock	Frequency	Usage
`aclk`	100-500 MHz	Primary - all RAPIDS logic, AXI/AXIS interfaces

40.4.2 Reset Signals

Reset Signals

Reset	Polarity	Type	Usage
`aresetn`	Active-low	Async assert, sync deassert	Primary - all RAPIDS logic

See: Clocks and Reset for complete timing specifications

40.5 Interface Summary

40.5.1 External Interfaces

External Interfaces

Interface	Type	Width	Purpose
AXI4 (Descriptor)	Master	256-bit	Descriptor fetch
AXI4 (Sink Write)	Master	512-bit (param)	Sink data write to memory
AXI4 (Source Read)	Master	512-bit (param)	Source data read from memory
AXIS (Sink)	Slave	512-bit (param)	Network data ingress
AXIS (Source)	Master	512-bit (param)	Network data egress
MonBus	Master	64-bit	Monitor packet output

40.5.2 Internal Buses

Internal Buses

Interface	Width	Purpose
MonBus	64-bit	Internal monitoring bus
Descriptor Bus	256-bit	Descriptor distribution
SRAM Data Bus	512-bit	SRAM read/write data

40.6 Related Documentation

- **PRD.md** - Product requirements and overview
- **CLAUDE.md** - AI development guide
- **RAPIDS Beats HAS** - Architecture specification (the original pre-beats rapids_spec/ tree was retired)

40.7 Specification Conventions

40.7.1 Signal Naming

- **Clock:** `aclk`
- **Reset:** `aresetn` (active-low)
- **Valid/Ready:** Standard AXI/AXIS handshake
- **Registers:** `r_` prefix (e.g., `r_state`, `r_counter`)
- **Wires:** `w_` prefix (e.g., `w_next_state`, `w_grant`)

40.7.2 Parameter Naming

- **Uppercase:** `NUM_CHANNELS`, `DATA_WIDTH`, `ADDR_WIDTH`
- **Derived:** `CHAN_WIDTH = $clog2(NUM_CHANNELS)`

40.7.3 Figure and Table Numbering

- **Figures:** `Figure X.Y.Z: Title` where X=chapter, Y=section, Z=figure number
- **Tables:** Caption line `: Table Title` after table markdown

Last Updated: 2026-07-13 **Maintained By:** RAPIDS Architecture Team